

Introduction to Econometrics

Course notes | MGT581 | EPFL

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Preface

You are looking at the course notes for *Introduction to Econometrics* (MGT581) at EPFL. Please note that there might be typos. In doubt, please reach out to me or the teaching assistants.

These notes are here to help; they do not systematically replace the coverage of content by the textbooks recommended in the syllabus.

Time stamp

This version: September 29, 2026. (Compare this date with the latest version available on moodle.)

Additional readings

Here's a short list of references if you want to deepen your knowledge in topics covered in this course.

Probability theory. A good overview is Casella and Berger (2021). Another one is DeGroot and Schervish (2002) but is less advanced. There exist many shorter introductions that cover the background needed for this course (e.g., Rozanov 1969). Stachurski (2016) has good chapters on the topic. For measure theory, see Billingsley (1995) or Tao (2011) (this will be overkill for the content of this course, except if you plan to go further into research). On asymptotic theory, see Van der Vaart (2000) or DasGupta (2008).

Econometrics. Recent textbooks include Verbeek (2018), Stock and Watson (2011), and Stachurski (2016). Hayashi (2000) and Davidson and MacKinnon (2004) are a bit older but good. Wooldridge (2002) covers panel data. Peter M. Aronow and Miller (2019) is a good modern treatment. B. Hansen (2022a) is extensive and pedagogically friendlier than Greene (2008), which is itself a good reference on the mechanics of econometrics. Goldberger (1991) is now quite dated, but it does a good job building up intuition, as does Kennedy (2003).

Causal inference. S. L. Morgan and Winship (2014) covers causal inference and quasi-experimental methods. A more technical approach to causal inference is Imbens and Rubin (2015). Gailmard (2014) contains both probability theory material as well as inference. Pearl (2009) introduces (directed acyclic) graphs, going in quite some details. J. Angrist and Pischke (2008) builds bridges between econometric theory and causal inference. M. Huber (2023) connects causal inference to machine learning.

Next steps. For other topics in econometrics and applied statistics, see Theodore W. Anderson (2003) (multivariate analysis), Hamilton (2020) (time series), Gelman and Hill (2007), Jackman (2009), and McElreath (2018) (Bayesian modeling), Hastie et al. (2009), Murphy (2022), Chernozhukov et al. (2024) (machine learning), Grimmer, Roberts, and Stewart (2022) (text-as-data).

Thanks

Thanks to Jeanne Fernandez, Federico Lanfranchi, Christopher Soriano, and Aidas Venckunas for catching several typos. All remaining errors are mine.

1 Foundations

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Overview

We review the key concepts from probability and measure theory that serve as the basis for this course. Some of these concepts will be used repeatedly. Others belong to the “nice to know” category.

Readings

Textbooks that cover foundational material in mathematical statistics include Casella and Berger (2021), DeGroot and Schervish (2002), or Durrett (2019). B. Hansen (2022b) covers probability theory for econometrics. For measure theory, a reference is Billingsley (1995). For asymptotic theory, see Van der Vaart (2000). In French, LeGall’s lecture notes *Intégration, Probabilités et Processus Aléatoire* (available online) are good, but generally beyond what we will cover. Amir Dembo’s course notes from his PhD-level probability course are also excellent ([link](#)).

1.1 Probability

1.1.1 The building blocks of probability

Probabilities are a core concept in econometrics. They will pop up repeatedly. For instance:

- *Sampling*. Eg, the probability of selecting a representative sample from a population.
- *Expected values*. Eg, the expected profit from a new product.
- *(Conditional) probabilities*. Eg, the probability of obtaining a contract for your business given your compliance (or not) with human rights rules.

- *Inference.* Eg, the probability of obtaining a given sample of data.

Ultimately, what we want to figure out is something like: *what is the probability that a given event about occurs?* But this leaves a lot of questions open: what's a probability? What's an event? To answer these questions rigorously, we need to tool up a little bit. Implicitly or explicitly, our analysis rests on a **probability space** $(\Omega, \mathcal{F}, \mathbb{P})$. Let's unpack it.

The first term (Ω) is a *sample space*, which refers to the set of *outcomes* that we are interested in. For instance, an outcome could be the annual sales of a firm (a non-negative number), the share of votes a politician receives in an election (a number between 0 and 100), or whether a patent is granted (granted or not granted).

Definition 1.1 (Sample space). Ω (upper-case omega) is the *sample space*, that is, the set of potential outcomes. Let $\omega \in \Omega$ (lower-case omega) be an arbitrary *outcome* from the sample space.

Next, we convert *outcomes* into *events*. An event is true/false statement about the realization of outcomes. For instance: did firm i have sales between CHF1m and CHF2m? If yes, then the event occurred ("true"). An **event space** is the set of events that we want to study.

Definition 1.2 (Event space). $\mathcal{F} \subseteq 2^\Omega$ is the *event space*, that is, the set of events. This set of event is a collection of subsets of Ω . Let $A \in \mathcal{F}$ be an arbitrary *event* from the event space.

The event space organizes the set of events that are admissible and to which probabilities will be assigned. As we will see shortly, we will typically want to impose restrictions on the event space, so that it doesn't create problematic situation. One very common set of restrictions are so-called σ -algebras ("sigma-algebra").

Definition 1.3 (σ -algebra). We say that the set $\mathcal{F}$ of subsets of Ω is a σ -algebra if:

1. $A \in \mathcal{F} \rightarrow A^c \in \mathcal{F}$. If you can assign a probability to A , you can also assign it to non- A ("closed under the taking of complements").
2. $A_1, A_2, \dots \in \mathcal{F} \rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$. You can assign probabili-

Events are also distinct from outcomes in the following way: an event might occur after one of several outcomes are realized. For instance, an event could be: a right-wing politician wins an election with two candidates; the outcomes that yield this even include all the potential election results between 50% and 100% of the vote share. That's why the event space is defined as a set of subsets of outcomes 2^Ω .

ties to any combination of events (“closed under the taking of unions”).

3. $\Omega \in \mathcal{F}$

Note. (i) Conditions (1) and (3) imply that $\emptyset \in \mathcal{F}$. (ii) Applying **De Morgan’s law** to (2), we can show that a σ -algebra is closed under countable *intersections* in addition to unions. (iii) A σ -algebra doesn’t itself have to be countable.

⚠ Warning

From now on, the event space $\mathcal{F}$ will always be assumed to be a σ -algebra.

I’ll explain why we use σ -algebras shortly. Before this, let’s define **probability measures**, which is the quantity we are most interested in.

Definition 1.4 (Measurable space and probability measure). A pair $(\Omega, \mathcal{F})$ of sample space and σ -algebra is a *measurable space*. A *probability measure* $\mathbb{P}$ is a set function $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$, such that:

1. $\mathbb{P}[\emptyset] = 0$
2. $\mathbb{P}[A] \geq 0$ for all A .
3. $\mathbb{P}[\cup_i A_i] = \sum_i \mathbb{P}[A_i]$ for any countable collection of disjoint sets of events $A_i \in \mathcal{F}$.
4. $\mathbb{P}[\Omega] = 1$.

Informally, a measure is a way to assign a size to a set. The size of “nothing” (empty set) is set to zero. The size of something has to be non-negative. And when you add disjointed sets, the size should be the sum of the size of individual set. With *probability* measures, what we add is that the measure of the entire set of all events is 1.

⚠ Warning

If you’ve taken measure theory courses, you will notice that a *probability measure* is simply a *measure* (i.e., a set function meeting conditions (1) to (3)), to which we add the requirement that it is normalized so that $\mathbb{P}[\Omega] = 1$.

Informally, a probability measure is a (set) function such that the probability of nothing happening is 0, the probability of something happening is 1, the probability of any given event is located between 0 and 1, and the probability of the union of events equals the sum of their individual probabilities. This definition is also referred to as the **Axioms of Probability** or **Kolmogorov axioms** (Kolmogorov 1956).

See Example 1.1 for examples of sample spaces, outcomes, and events.

Example 1.1 (Sample spaces, outcomes, events). Here are examples of a sample space, with corresponding outcomes and events.

Throwing a dice one time (countable sample space):

- **Sample space** $\Omega := \{1, 2, 3, 4, 5, 6\}$
- **Arbitrary outcome** $\omega_4 = 4$
- **Event** A : even outcomes from the dice, i.e. $A = \{2, 4, 6\}$.
Did the dice yield an even number? Yes/no?

Annual sales (in USD) of Nvidia (uncountable sample space):

- $\Omega := \mathbb{R}_+$.
- $\omega_i = 26.974$ b (USD).
- A : annual sales in billions, e.g., $[26 - 27)$. *Did Nvidia make sales in the $[26b - 27b)$ interval? Yes/no?*

Or written slightly differently:

$$\Omega := \{\omega | \omega \in \mathbb{R}_+\}.$$

Throwing a dart (uncountable sample space):

- $\Omega :=$ surface of a dart board.
- $\omega_{i,j} =$ hitting point $\{i, j\}$.
- A : hitting a specific *area* (e.g., the bull's eye). *Did the dart hit the bull's eye? Yes/no?*

Definition 1.5 (Probability space). A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$, that is, a sample space, a σ -algebra event space, and a probability measure.

⚠ Warning

- $(\Omega, \mathcal{F})$ is a *measurable space*.
- $\mathbb{P}$ is a probability measure.
- $(\Omega, \mathcal{F}, \mathbb{P})$ is a *probability space*.

Figure 1.1 illustrates (informally) the relation between Ω , $\mathcal{F}$, and $\mathbb{P}$.

1.1.2 Building a σ -algebra

You might be puzzled by imposing that $\mathcal{F}$ must be a σ -algebra: why do we need this and what is it in the first place? And if our goal is to compute probabilities, why not just attach them to ω ?

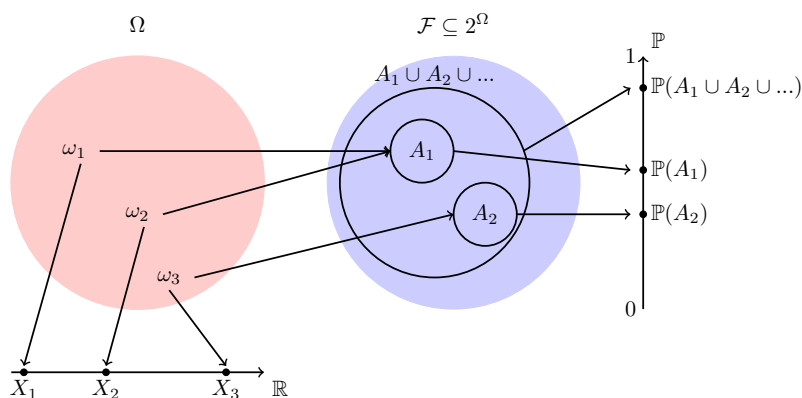


Figure 1.1: Probability space. Ω contains three outcomes that map onto two events. Each set of events maps onto a probability, i.e. a real number on the $[0,1]$ interval. At the bottom, outcomes map onto a variable X defined on a segment of the real line.

There are two connected reasons for this. First, we are interested in *events* A , not *outcomes* ω . So we must build the set of events over which probabilities must be defined. This is $\mathcal{F}$, namely a set of subsets of Ω (which is why I wrote $\mathcal{F} \subseteq 2^\Omega$) that form the list (set) of events we are interested in.

This explains why we want a set of events. But why specifically a σ -algebra? Well, we could accidentally construct a pathological $\mathcal{F}$. For instance, we could define the set of events of two throws of a coin as: $\mathcal{F} := \{HH, HT, TH\}$ and then apply a probability measure to it. But note that TT is missing!

Thus, we will need an extra step to ensure that the set of events $\mathcal{F}$ is well-behaved and that we can assign probabilities to each of the event it contains. Imposing a σ -algebra ensures that we don't accidentally create a non-sensical set of events.

How to build a σ -algebra depends on the nature of the outcomes Ω we are interested in. To keep things simple, we can broadly distinguish between two types of Ω . The easy case is when Ω is finite and countable. In that case, we could build a σ -algebra from the power set of Ω (sometimes also written $\mathcal{P}(S)$).

For instance, let's suppose we are tossing two coins. For now, let's say that we don't yet know what kind of events we are interested in. Then, the power set of Ω allows us to take into account all possible

A set is countable if its elements can be placed in a one-to-one correspondence with natural numbers $\mathbb{N}$. Remember: countable is not the same as finite. A set can be countably infinite (e.g., $\mathbb{N}$ or $\mathbb{Q}$). Uncountable sets (like $\mathbb{R}$ or $[-1, 1]$) cannot be put into one-to-one correspondence with $\mathbb{N}$.

events:

$$\begin{aligned}\mathcal{F} = 2^\Omega = & \{\emptyset, \{(H, H)\}, \{(H, T)\}, \{(T, H)\}, \{(T, T)\}, \\ & \{(H, H), (H, T)\}, \\ & \{(H, H), (T, H)\}, \\ & \{(H, H), (T, T)\}, \\ & \{(H, T), (T, H)\}, \\ & \{(H, T), (T, T)\}, \\ & \{(T, H), (T, T)\}, \\ & \{(H, H), (H, T), (T, H)\}, \\ & \{(H, H), (H, T), (T, T)\}, \\ & \{(H, H), (T, H), (T, T)\}, \\ & \{(H, T), (T, H), (T, T)\}, \\ & \underbrace{\{(H, H), (H, T), (T, H), (T, T)\}}_{\Omega}\end{aligned}$$

But $\mathcal{F}$ could be smaller. For instance, we could decide that we are interested in the event of getting two identical tosses (A_1) versus not (A_2). We would define:

$$\begin{aligned}A_1 &:= \{(H, H), (T, T)\} \\ A_2 &:= \{(H, T), (T, H)\}\end{aligned}$$

and the corresponding σ -algebra:

$$\begin{aligned}\mathcal{F} := & \{\emptyset, \\ & \{(H, H), (T, T)\}, \\ & \{(H, T), (T, H)\}, \\ & \underbrace{\{(H, H), (T, T), (H, T), (T, H)\}}_{\Omega}\end{aligned}$$

As you can see, each event A_1 and A_2 and their complement are in $\mathcal{F}$ and so is their union (which coincides with Ω), as is the empty set. It is therefore a σ -algebra.

Example 1.2 (Illustration of a σ -algebra). Consider a dice thrown a single time. A σ -algebra set could be as follows. Suppose the event we are interested in is whether a throw is even ($A = \{2, 4, 6\}$). Then:

$$\mathcal{F} = \{\underbrace{\{1, 2, 3, 4, 5, 6\}}_{\Omega}, \underbrace{\{2, 4, 6\}}_A, \underbrace{\{1, 3, 5\}}_B\}$$

This qualifies as a σ -algebra:

1. A and A^c are included in $\mathcal{F}$. A is defined as $\{2, 4, 6\}$ and A^c is $\{1, 3, 5\}$ (i.e. B).
2. The union of all sets is included. Here, the union of $\{2, 4, 6\}$ and $\{1, 3, 5\}$ is there (trivially, since it's Ω).
3. $\Omega \in \mathcal{F}$ (that's the set $\{1, 2, 3, 4, 5, 6\}$).

Obviously, a more comprehensive σ -algebra than this one would be 2^Ω , which contains $2^6 = 64$ elements (including the empty set). But we might have reasons to restrict ourselves (e.g., if we don't see the dice outcome ourselves and someone only tells us if it's even or odd).

The next step is to assign probabilities to each sets.

Things get a bit more complicated with continuous Ω . For instance, it could correspond to people's height. We cannot use the same strategy (the sum of the probability of all events would go to infinity instead of 1). Instead, we will use intervals on Ω . We restrict $\mathcal{F}$ (the set of admissible events) to only include subsets that have a determinable length. That's one of the benefits of restricting ourselves to σ -algebra! Limiting ourselves to a σ -algebra guarantees that we will be able to assign well-defined probabilities to each event in $\mathcal{F}$.

One particularly useful class of σ -algebras whenever we deal with $\mathbb{R}^N$ (or subsets thereof) is the **Borel σ -algebra** (or **Borel field**) of $\mathcal{F}$ (typically written $\mathcal{B}(\Omega)$). $\mathcal{B}(\mathbb{R})$ is the σ -algebra *generated by* all open intervals (a, b) with $a < b$ on $\mathbb{R}$ (i.e., the smallest σ -algebra containing all such intervals). Since it includes all unions, intersections, and complements (per definition of a σ -algebra), it means that it includes every closed, open, and semi open sets on the real line. That is, it contains $[a, b]$, $(a, b]$, $[a, b)$, and (a, b) for all a, b . (Adapt accordingly if we go in higher dimensions $\mathbb{R}^N$.)

With Borel fields, we can define probabilities even for uncountable sample spaces Ω , including $\mathbb{R}$.

Here it gets a bit complicated, but the short answer is that not all subsets on $\mathbb{R}$ (which is what we are using in this example) have a measurable *length* (here: the 'distance' between a and b). As a result, probabilities cannot be defined for these. Those interested in this should read more about measurement theory (Tao 2011) and problems such as the Banach-Tarski paradox.

1.1.3 Building a probability measure

We now have our set of events, $\mathcal{F}$. We can then assign probabilities to each element of $\mathcal{F}$ using a probability measure, as defined above. For instance, we could write: $\mathbb{P}[\text{event} = A]$ as the probability that the event $A \in \mathcal{F}$ occurs. But what's a probability? From Kolmogorov's axioms, we know that a probability function must meet.

That doesn't tell us how to conceptualize and quantify the probability of an event.

The most intuitive way to think about a probability is to define it as a ratio of relative frequencies (Rozanov 1969):

$$\mathbb{P}(A) = \frac{N(A)}{N},$$

where N is the number of outcomes and $N(A)$ is the number of outcomes leading to the event A . We can also think of this with an empiricist's view as a limit: the relative frequency of A as the number of trials goes toward infinity:¹

$$\mathbb{P}(A) = \lim_{n \rightarrow \infty} \frac{n(A)}{n},$$

where lower case n denotes the number of trials.

Once we think of probabilities as ratios, it becomes natural to think about counting the numerator and denominator. That's where textbooks introduce combinatorial analysis. See for instance Rozanov (1969, 1.2).

1.1.4 Implications and additional concepts

We can generate several implications and additional concepts from these building blocks.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Assume $A, B \in \mathcal{F}$ and $\mathbb{P}(\cdot) \in (0, 1)$ (probability of events are neither 0 nor 1). If $A \subset B$, then:

1. $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$
2. $\mathbb{P}(A) \leq \mathbb{P}(B)$

Note also:

3. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$
4. $\mathbb{P}(\emptyset) = 0$

¹There are other views about what probabilities are. Some prominent mathematicians and statisticians, such as Frank Ramsey (1926), studied probabilities as a *subjective* belief rather than some kind of (asymptotic) property of the world. See also Savage (1951), and De Finetti (1975).

Next, we can define the concept of **conditional probability**. For this, we write $A \cap B$ as the *intersection* of A and B , meaning that both events must occur for this statement to be true. We $A \cup B$ to denote the *union* of A and B , meaning that *either* event has to occur for this statement to be true. By definition we can write:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B|A) = \mathbb{P}(B) \cdot \mathbb{P}(A|B).$$

Then, for two events A , B , the conditional probability of A given B is:

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

Recall that we assume non-zero probabilities here.

This is often a source of confusion. $\mathbb{P}(A|B)$ is different from $\mathbb{P}(B|A)$. For instance, consider the probability of speeding, given that you are a man, versus the probability of being a man, given that you are speeding. See the notion of the **Prosecutor's Fallacy**. From there, we can then recover **Bayes' law**:

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)}$$

One type of conditional probability that we will be interested in is: *given* the data we obtained (our sample), what is the probability of a parameter of interest being equal to a certain value (or, more generally, range of values). For instance, denote $\mathbf{X}$ a set of data and τ an unknown parameter (or vector of unknown parameters) that says something about $\mathbf{X}$. We may wonder what the latter's conditional probability distribution is:

$$\mathbb{P}(\tau|\mathbf{X}).$$

And sometimes we may do the opposite: given a dataset, what is the probability of a given parameter.

$$\mathbb{P}(\mathbf{X}|\tau).$$

This logic underpins null hypothesis testing.

Next, we say that two events A , B are **independent** if and only if:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B).$$

Note that if A and B are independent, then $\mathbb{P}(A|B) = \mathbb{P}(A)$. And more generally, for independent events $A_1, A_2, \dots, A_N \in \mathcal{F}$:

$$\mathbb{P}(\cap_{n=1}^N A_n) = \prod_{n=1}^N \mathbb{P}(A_n).$$

Intuitively, the notion of independence means that knowing whether B happened or not doesn't move the needle regarding our belief that A happened. For instance, knowing whether it rained in New York doesn't give me information regarding whether I decided to bike in Lausanne. However, knowing whether it rained in Geneva does change the probability that I decided to bike to EPFL.

1.1.5 Probability distributions

Probabilities are fairly straightforward when events correspond to discrete outcomes (e.g., did a coin yield tail?). Things get more complicated with events built from continuous outcomes (e.g., what is the probability that a firm earns CHF1.3437b?).

When the sample space is (a subset of) the real line $\mathbb{R}$, we saw that it was convenient to restrict the σ -algebra $\mathcal{F}$ to a Borel field $\mathcal{B}(\mathbb{R})$. Still, this doesn't tell us what the probability of any subset of the real line is. Now: a probability measure defined over $\mathcal{B}(\mathbb{R})$ is called a **distribution** (or sometimes a **law**). For a variable X built from a continuous outcome, a distribution tells us *what the probability that the event $X = x$ occurs is*.

Some distributions are more common than others. For instance, the normal (Gaussian) probability density function (PDF) is:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

We generally write more compactly $X \sim \mathcal{N}(\mu, \sigma^2)$. We shall shortly talk more about this, but note that this distribution function is entirely parametrized by two variables: μ and σ^2 , which correspond to the mean (or first moment) and the variance (or second central moment) of the distribution.

It will sometimes be useful to work with the **cumulative distribution function** (CDF), which integrates the PDF:

$$\Phi(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^x \exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right) dt$$

Here, we integrate from minus infinity upward to some value x_i . This tells us how much mass under the curve is below x_i . Of course, if we integrate from minus to plus infinity, we must obtain 1 (since $\mathbb{P}(\Omega) = 1$). The term in front of the integral is precisely ensuring that the integral sums up to one.

For a historical account of the normal distribution, tracing back to Abraham de Moivre, see Hald (2008).

Notice that both expressions become considerably easier if we center X (so that $\mu = 0$) and if we standardize it (so that $\sigma^2 = 1$). In that case, we talk of the **standard normal distribution**, i.e. $X \sim N(0, 1)$.

There are many other distributions that we will run into (F distribution, χ^2 distribution, etc.). At this stage, no need to dig deeper into them, since doing so is not particularly enlightening.

1.2 Random variables

A random variable is a measurable function X that maps from the outcome space Ω to a measurable space $\mathcal{M}$, a set equipped with its own σ -algebra (for $\mathbb{R}$, this is the Borel σ -algebra $\mathcal{B}$):

$$X : \Omega \rightarrow \mathcal{M}$$

Oftentimes, $\mathcal{M}$ will be the real line $\mathbb{R}$.

Outcomes versus random variables

A random variable X is conceptually distinct from an outcome Ω , though sometimes the two hard to distinguish. An outcome is a realization of something we are interested in. The random variable attached a real value to it.

Sometimes, the two look the same. If the outcome is the height of individuals, then an arbitrary outcome might $\omega = 1.75$ and the corresponding random variable will be $X = 1.75$. In such a case, $X(\Omega) = \Omega$.

But that doesn't have to be the case. For instance, Ω could be the throw of two dices, and X could be their sum. In that case, $\Omega := \{(i, j) : i, j \in 1, \dots, 6\}$, and $X \in \{2, 3, \dots, 12\}$.

Other (more common) example. If you are doing a consumer survey and ask them how much they like your brand, we could have: $\Omega := \{\text{Strongly dislike, Dislike, Neutral, Like, Strongly like}\}$. You could in principle then attach real values to each, such that $X \in \{-2, -1, 0, 1, 2\}$. Whether these numbers mean something is a different matter.

Note. Whether X is **continuous** or **discrete** is not decided by $\mathcal{M}$. Saying that we use $\mathbb{R}$ doesn't affect the nature of X in that way. Whether it's continuous or discrete is a property of the *distribution* (law) of X . The variable is **discrete** if its distribution sits on a countable set (it

has a probability mass function), and **continuous** if its distribution has a density with respect to the Lebesgue measure. A discrete variable can still map into $\mathbb{R}$ (e.g. a coin flip taking values $\{0, 1\} \subset \mathbb{R}$). Conversely, a variable mapping into $\mathbb{N}$ is necessarily discrete.

We can link random variables back to σ -algebras.

⚠ Valid random variables

For a random variable X to be valid, it must be **measurable** with respect to $(\Omega, \mathcal{F})$. What this means: for any (measurable) set of values B in $\mathcal{M}$, the **event** that X takes a value in B must belong to $\mathcal{F}$. This is precisely what lets us assign it a probability: $\mathbb{P}[X \in B]$ is only defined when that event lives in $\mathcal{F}$.

For instance, suppose $\Omega = \mathbb{R}$ with $X(\omega) = \omega$, and consider two events: $X < 0$ (call it A_1) and $X \geq 0$ (A_2). For X to be measurable, the events that pin down whether $X < 0$ or $X \geq 0$ must be in $\mathcal{F}$. Here, it means that the sets $(-\infty, 0)$ and $[0, \infty)$ must belong to $\mathcal{F}$. This is indeed the case when $\mathcal{F}$ is the Borel σ -algebra.

Notice that x doesn't have to yield an actually *variable* variable. For instance, x could generate a constant c . More generally, this function x can take all kinds of shape. For instance, consider a famous case (in some circles at least), namely Bernoulli's St. Petersburg paradox. Someone tosses a coin until a head is obtained, at which stage the game ends. Suppose that we want to know the number of tails until head. Suppose that tails equal zero and heads 1. The random variable is (Stachurski 2016, 90):

$$x(\omega) = x(\text{toss \#1, toss \#2, ...}) = \min\{n \in \mathbb{N} : \text{toss}_n = 1\}$$

1.2.1 Describing random variables

Random variables are drawn from an underlying **probability density function** (PDFs). As we saw, some are well known: the normal distribution, the Student's t distribution, etc. Other variables could follow arbitrary PDFs.

Be this as it may, we often find it useful to describe random variables based on their **moments**. These are parameters of the underlying PDF that help us describe them.

The **first moment** is **expectation** of a variable (aka μ). Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, the first moment is defined as:

$$\begin{aligned}\mathbb{E}[X] &:= \mu_x = \sum_{i=1}^J x_i \cdot \mathbb{P}[X = x_i] && \text{discrete RV} \\ \mathbb{E}[X] &:= \mu_x = \int_{-\infty}^{\infty} x \cdot f_X(x) dx && \text{continuous RV}\end{aligned}$$

where $\mathbb{P}[X = x_i]$ is the probability of event $X = x_i$ occurring and $f_X(x)$ is the probability density function of X .

Note here that this is why we care a lot about well-defined probabilities (and thus why we talked about σ -algebra): without well-defined probabilities, we cannot compute any moment of X . And as we will see in a second, we care *a lot* about moments in econometrics (and any discipline that tries to make causal claims from data). See box.

We can extend this definition to expectations of functions (and similarly for discrete functions):

$$\mathbb{E}[g(X)] = \int g(x) \cdot f_X(x) dx$$

We could have written: $\mathbb{P}(A)$, where $A := \{\omega \in \Omega : X(\omega) = x_i\}$ to highlight that it's the probability of an event, not of x itself.

Probabilities and expectations can be connected in another way, namely **Markov's inequality**. For a non-negative random variable X and any $\delta > 0$:

$$\mathbb{P}(X \geq \delta) \leq \frac{\mathbb{E}[X]}{\delta}$$

⚠ Properties of expectations

Expectations have several properties that derive from their definition.

- For a constant a :

$$\mathbb{E}[a] = a$$

- For any random variables, $X, Y, \dots, Z$:

$$\mathbb{E}[X + Y + \dots + Z] = \mathbb{E}[X] + \mathbb{E}[Y] + \dots + \mathbb{E}[Z]$$

- For a, b constant:

$$\mathbb{E}[a \cdot f(X) + b \cdot g(X)] = a\mathbb{E}[f(X)] + b\mathbb{E}[g(X)]$$

As a result:

$$\mathbb{E}[aX + b] = \mathbb{E}[aX] + \mathbb{E}[b] = a\mathbb{E}[X] + b$$

- For two *independent* random variables X, Y :

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

- For an event A (where $A \in \{0, 1\}$), note that:

$$\begin{aligned}
\mathbb{E}[A] &= \sum_{i \in \{0,1\}} i \mathbb{P}[A = i] && \text{by def} \\
&= 0 \cdot \mathbb{P}[A = 0] + 1 \cdot \mathbb{P}[A = 1] && \text{expanding} \\
&= \mathbb{P}[A = 1] \\
&= \mathbb{P}[A]
\end{aligned}$$

Thus, the expected value of a dichotomous event is equal to its probability.

Note. We talk of **centered moment** if we compute the moment of a centered variable, i.e. $X - \mathbb{E}[X]$. A centered first moment equals 0 by construction.

Note. Extending the definition of expected value $\mathbb{E}[A]$ to *conditional* expected value $\mathbb{E}[A|B]$ is reasonably straightforward. For instance:

$$\mathbb{E}[X|Y = y] = \sum_{j=1}^J x_j \cdot \mathbb{P}(X = x_j | Y = y),$$

where the last term is a conditional probability.

Definition 1.6 (Law of Iterated Expectations). Let X and Y be random variables on a common probability space. Then:

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | Y]].$$

More generally, for a third random variable Z :

$$\mathbb{E}[X | Z] = \mathbb{E}[\mathbb{E}[X | Y, Z] | Z].$$

The intuition behind the LIE is that to compute the unconditional mean of X , we can first compute the conditional mean for each value of Y , then average those conditional means over the distribution of Y . The inner expectation is taken over X given Y ; the outer expectation is taken over Y . LIE will appear repeatedly when we decompose treatment effects, derive identification conditions, and work with regression.

Theorem 1.1 (Law of Total Variance). *Let X and Y be random*

variables on a common probability space with $\mathbb{V}[Y] < \infty$. Then:

$$\mathbb{V}[Y] = \mathbb{E}[\mathbb{V}[Y | X]] + \mathbb{V}[\mathbb{E}[Y | X]].$$

The total variance of Y decomposes into two parts: (1) the average variance of Y *within* each value of X (unexplained variation), and (2) the variance of the conditional mean of Y *across* values of X (explained variation). The second term is zero if knowing X gives no information about $\mathbb{E}[Y]$. The LTV is a bit like the Law of Iterated Expectations, but for variance. Later, it will be helpful to understand why controlling for covariates (“control variables”) reduces standard errors from estimates obtained with ordinary least squares (the second term is attenuated by conditioning).

The first moment tells us “where” the random variable X is “anchored” on the real line. This will be a quantity that we will be extremely interested in. For instance, we could be interested in the expected salary of individuals *given* that they obtained a college degree:

$$\mathbb{E}[\text{Salary} | \text{College degree}].$$

We might then be interested in figuring out what the gains of college are:

$$\delta = \mathbb{E}[\text{Salary} | \text{College degree}] - \mathbb{E}[\text{Salary} | \text{No college degree}].$$

Or we might be interested in the expected effect of a new product on a firm’s profit:

$$\delta = \mathbb{E}[\text{Profit} | \text{Innovation}] - \mathbb{E}[\text{Profit} | \text{No innovation}].$$

As we will see, we can interpret (under certain conditions) δ as an **average treatment effect** (here: of college, respectively of an innovation). We will spend a lot of the semester understanding how to **estimate** quantities like these in a credible manner. Because here’s the problem: *these moments are unobserved **population** parameters*. We never observe the entire universe of observations. Thus, we will often work with **samples** from which we will estimate these parameters. Even $\mathbb{E}[X]$ must be estimated! This will create all kinds of complicated problems.

Distributions have often more than one moment. There are two standard types:

- The k th **raw moment** of X is $\mathbb{E}[X^k]$.
- The k th **central moment** of X is $\mathbb{E}[(X - \mathbb{E}[X])^k]$.

The two coincide when X is already centered (i.e., $\mathbb{E}[X] = 0$), but differ in general. Thus, the **second central moment** (aka the **variance**, $\mathbb{V}$, or σ^2) is:

$$\mathbb{V}[X] := \sigma_X^2 := \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2,$$

which directly gives us the standard deviation:

$$\text{SD}(X) := \sigma_X := \sqrt{\mathbb{E}[(X - \mathbb{E}[X])^2]}.$$

Both quantities are non-negative by construction: since $(X - \mathbb{E}[X])^2 \geq 0$ everywhere, its expectation satisfies $\mathbb{V}[X] \geq 0$, with equality only for a degenerate (constant) random variable. The variance tells us how spread out the distribution of X is. This will be very helpful when we think about the uncertainty surrounding our estimates of treatment effects. Thanks to this, we will be able to engage in **hypothesis testing** and build **confidence intervals**.

Notice that if the random variable is centered (i.e., transformed such that $\mathbb{E}[X] = 0$), then this simplifies to:

$$\mathbb{V}[X] = \mathbb{E}[X^2].$$

One result that is can be helpful to make sense of this is **Jensen's Inequality**.

Theorem 1.2 (Jensen's Inequality). *Let X be a random variable and $g : \mathbb{R} \rightarrow \mathbb{R}$ a function.*

- If g is **convex** (e.g., $g(x) = x^2$, $g(x) = e^x$), then:

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)].$$

- If g is **concave** (e.g., $g(x) = \log x$, $g(x) = \sqrt{x}$), then:

$$g(\mathbb{E}[X]) \geq \mathbb{E}[g(X)].$$

Equality holds in both cases if and only if X is degenerate (constant almost surely) or g is linear on the support of X .

The intuition behind Jensen's Inequality is pretty straightforward: for a convex function, the function of the mean lies below the mean of the function, because convexity means the function curves upward. That's why a variance will always be strictly positive.

We can continue like this. The **skewness** of a distribution (3rd moment) tells us how asymmetric the distribution of X is. If skewness

is zero, then the distribution is symmetrical. If skewness is positive, then the distribution has a long right tail.

The **kurtosis** (4th moment) tells us something about the mass of the tails of the distribution. If it's large, then we have heavy tails ("leptokurtotic" distribution).

Skewness and kurtosis often generate less interest. But some audiences will value them. For instance, the consequences of fat tails can be dramatic in stock markets.

Common distributions (laws) generally have a finite number of moments. The normal distribution, for instance, is entirely characterized by its first and second moments — all higher moments exist and are finite, but they are fully determined by μ and σ^2 (e.g., the 4th moment of $N(0, 1)$ equals 3). Some distributions, like Cauchy, don't have any finite moments at all. Here, note the difference between **populations** and **samples**: we can always compute sample analogues to 1st, 2nd, etc. moments. For instance, you can always compute the mean of a variable in your sample (e.g., the height of students enrolled in my class). However, this *sample* quantity might not correspond to any *population* quantity if the population quantity is undefined. For instance, if height was distributed Cauchy in the universe of all possible individuals, then its first moment would not be defined (because Cauchy variables don't have finite means). We could use our calculators to compute the mean of our sample, but it would not correspond to any meaningful population value.

1.3 Association between random variables

We are often interested in the relation between two (or more) variables. A starting point is the **covariance**, which is simply a co-first moment:

$$\text{Cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

The problem with covariance is that its scale is not intuitive. We could end up with an estimate $\widehat{\text{Cov}}(X, Y) = 8$; is this large? Small?

Thus, we often scale the covariance by the product of the standard deviations of X and Y to generate the **correlation coefficient** (aka ρ) between X and Y :

$$\text{Cor}(X, Y) := \rho =: \frac{\mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]}{\sigma_x \sigma_y} \in [-1, 1].$$

Note that if X and Y are centered, this boils down to:

$$\begin{aligned}\text{Cov}(X, Y) &= \mathbb{E}[XY] \\ \text{Cor}(X, Y) &= \frac{\mathbb{E}[XY]}{\sqrt{\mathbb{E}[X^2]\mathbb{E}[Y^2]}}\end{aligned}$$

And note that if X and Y are independent, then the covariance (and thus the correlation) is zero. Independence implies $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$, so:

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[X]\mathbb{E}[Y] = 0.$$

Note that we don't observe the covariance or the correlation. These are population parameters (they are unobserved quantities). We will need to estimate them from a sample of observations, where population expectations are replaced with sample means. We can do this as follows (for covariance; it's similar for correlations).

Theorem 1.3. *Suppose a sample of iid observations of Y and D . Then, the sample covariance estimator:*

$$\widehat{\text{Cov}}(Y, D) = \frac{\sum_i (y_i - \bar{y})(d_i - \bar{d})}{n - 1}$$

is an unbiased estimator of $\text{Cov}(Y, D)$.

where $\hat{\cdot}$ will refer either to the estimate of a population parameter or to predicted values (the context will make it clear). Why is this a good estimator? It's because it's **unbiased**. We will see this later. For now, just know that an *unbiased* estimator $\hat{\gamma}$ is one that satisfies $\mathbb{E}[\hat{\gamma}] = \gamma$: the expected value of $\hat{\gamma}$ (which is often a random variable) is γ .

1.4 Asymptotics

We are often interested in understanding how certain quantities of interest behave as the number of data points we have increases toward infinity. That's what we mean by **asymptotics**.

Here, a few key theorems and laws will be very helpful. But first, let's define the idea of **convergence**.

1.4.1 Convergences

The concept of **convergence** will often come up, especially when we will cover the problem of asymptotics..

There are different types of convergence.

Definition 1.7 (Convergence in probability). A sequence of random variables $\{X_n\}$ converges in probability to a random variable X if:

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| \geq \varepsilon) = 0 \quad \forall \varepsilon > 0.$$

In other words, the probability of getting a value X_n further than ε from X goes toward zero as n goes to infinity. The *weak law of large numbers* is a converges in probability result.

This is also written as:

$$X_n \xrightarrow{p} X.$$

Definition 1.8 (Convergence in distribution (weak convergence)). A sequence of random variables $\{X_n\}$ converges in distribution to a random variable X if the CDF of X_n , denoted $F_n(x)$, converges to the CDF of X , denoted $F^*(x)$:

$$\lim_{n \rightarrow \infty} F_n(x) = F^*(x) \quad \forall x \text{ where } F^*(x) \text{ is continuous}$$

This is also reported as:

$$X_n \xrightarrow{d} X.$$

As we will see shortly, the *central limit theorem* is a convergence in distribution result.

Definition 1.9 (Almost sure convergence (strong convergence)). A sequence of random variables $\{X_n\}$ converges almost surely to a random variable X if:

$$\mathbb{P} \left(\lim_{n \rightarrow \infty} X_n = X \right) = 1.$$

This is denoted as:

$$X_n \xrightarrow{a.s.} X.$$

The *strong law of large numbers* is an almost sure convergence result.

Note:

1. **Almost sure convergence** implies **convergence in probability**.
2. **Convergence in probability** implies **convergence in distribution**.
3. **Convergence in mean square** implies **convergence in probability**

Note: the reverse implications do not necessarily hold.

1.4.2 (Weak) Law of large numbers

The **weak law of large numbers** (WLLN) is going to be a useful result. WLLN is a result of the principle of convergence in probability. To obtain it, a useful result is **Chebyshev's Inequality**.

Theorem 1.4 (Chebyshev's Inequality). *Let X be a random variable with mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \mathbb{V}[X] < \infty$. Then for any $k > 0$:*

$$\mathbb{P}(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

The probability that X deviates from its mean by more than k is bounded above by σ^2/k^2 . Wider distributions (larger σ^2) allow more probability mass far from the mean; larger thresholds k tighten the bound at rate $1/k^2$. Why this matters is that Chebyshev's inequality provides a bound using only the mean and variance. It underpins the proof of the WLLN below: applying it to $\bar{X}_n$ (which has mean μ and variance σ^2/n) gives $\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2) \rightarrow 0$.

Theorem 1.5 (Weak law of large numbers). *Assume we have a sequence of observations from a random variable that are independent and identically distributed (iid) $X_1, X_2, \dots$ with first moment $\mathbb{E}[X] = \mu$ and $\mathbb{V}[X] = \sigma^2 < \infty$. Define the mean of X up to n as $\bar{X}_n = 1/n \sum_{i=1}^n X_i$. Then:*

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) = 0, \quad \forall \varepsilon > 0$$

Note: In other words, the probability that the sample average is further from μ (true mean) than ε goes to zero.

Or, put another way, the absolute value of the error (the gap between $\bar{X}_n$ and μ) is increasingly less likely to be greater than ε . Or put yet another way: the probability of getting an inaccurate average value from our sample is less and less likely as n increases. This result will be useful when we discuss the topic of **consistency**. Informally, we will be interested in figuring out whether a quantity of interest (such as an estimated treatment effect) is increasingly likely to be close to the true value of that quantity.

Note: Finite variance is a sufficient condition here (this version follows from Chebyshev's inequality). A weaker form (Khinchine's WLLN) requires only that $\mathbb{E}[X] = \mu < \infty$.

Theorem 1.6 (Strong law of large numbers (Kolmogorov)). *Under the same iid condition as the WLLN, with $\mathbb{E}[|X|] < \infty$, the convergence holds almost surely:*

$$\bar{X}_n \xrightarrow{a.s.} \mu.$$

Almost sure convergence implies convergence in probability but the converse isn't true. So the SLLN is strictly stronger than the WLLN. The WLLN shows the probability of a large deviation goes away as n increases. The SLLN says the realized sequence of sample averages itself converges, with probability one. For most econometric applications we are interested in the WLLN will be enough.

1.4.3 Central limit theorem

The **central limit theorem** (CLT) offers a result that neighbors the WLLN. The idea here is that we could, hopefully, characterize the *distribution* of the difference between $\bar{X}_n$ and μ . That's what the CLT does. There exists several variants of the CLT. We will primarily use the **Lindeberg–Lévy CLT**.

Theorem 1.7 (Lindeberg–Lévy CLT). *Suppose we have a sequence of iid random variables $X_1, X_2, \dots$ with first moment $\mathbb{E}[X] = \mu$ and second central moment $\mathbb{V}[X] = \sigma^2 < \infty$. Then:*

$$\sqrt{n} \frac{(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$$

where $\xrightarrow{d}$ denotes convergence in distribution.

Or, equivalently:

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2).$$

Right now, it might not seem clear why we would care about this result (aside from its aesthetic elegance). What we will find out later is that it will help us tremendously with hypothesis testing.

Theorem 1.8 (Multivariate CLT). *Let $\mathbf{X}_1, \mathbf{X}_2, \dots$ be iid k -dimensional random vectors with mean vector $\mu = \mathbb{E}[\mathbf{X}]$ and finite covariance matrix $\Sigma = \mathbb{V}[\mathbf{X}]$. Then:*

$$\sqrt{n}(\bar{\mathbf{X}}_n - \mu) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \Sigma).$$

Note: The univariate CLT explains scalar sample averages; the multivariate CLT explains vector averages. With ordinary least squares (OLS), which we will see later, the key quantity $\frac{1}{\sqrt{n}} \sum_i \mathbf{x}_i \varepsilon_i$ is a k -vector, and its asymptotic normality is a consequence of the multivariate CLT. The asymptotic distribution of $\hat{\beta}$ derived in Ch. 4 thus relies on this theorem.

Now, here's another result that will come in handy: **Slutsky's Theorem**.

Theorem 1.9 (Slutsky's Theorem). *Consider two random variables X and Y . Let $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$, where c is a constant. Then:*

$$X_n + Y_n \xrightarrow{d} X + c, \quad X_n \cdot Y_n \xrightarrow{d} c \cdot X.$$

Why is this useful? Well, recall that the WLLN converges in probability and the CLT converges in distribution. Slutsky's Theorem allows us to combine the two together.

1.4.4 Moment-generating function ■

The CLT may appear magical, so it's helpful to understand how we can prove it. For this, we will need a particular function: the **moment-generating function**.

Definition 1.10 (Moment-generating function). For a random variable X , the moment generating function (MGF) is

$$M_X(t) = E[e^{tX}]$$

defined for all $t > 0$ in some open interval around 0 (from $-t$ to t) where the expectation is finite.

Note M_X is a *function* of t , not a scalar. Now, MGFs are useful for several reasons, including:

1. *MGF generates moments.* Using a Taylor expansion of e^{tX} gives $M_X(t) = \sum_n \frac{t^n}{n!} E[X^n]$, so $M_X^{(n)}(0) = E[X^n]$.
2. *MGF turns sums into products.* If X, Y are independent, $M_{X+Y}(t) = M_X(t) \cdot M_Y(t)$.
3. *Uniqueness.* If two distributions have MGFs that agree on an open interval around 0, they are the *same* distribution.

i Proving the CLT with moment-generating functions

Here's how the MGF can be used to prove the CLT.

Let $X_1, \dots, X_n$ be iid with first moment (mean) μ , second moment (variance) σ^2 , and MGF existing near 0. We don't make any assumptions about the distribution of X aside from assuming that both the first and second moments are finite. (Which distributions does this rule out?)

1. **Standardize.** Let $Y_i = \frac{X_i - \mu}{\sigma}$, so $E[Y_i] = 0$, $E[Y_i^2] = 1$.

Define:

$$Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i$$

where $1/\sqrt{n}$ is used to rescale the sum (almost like an average).

2. **Transform the sum into a product.** By independence and the iid assumption, the sum becomes a product when we use MGF instead of the original variable:

$$M_{Z_n}(t) = \left[M_Y\left(\frac{t}{\sqrt{n}}\right) \right]^n$$

3. **Taylor expand $M_Y(s)$, which is the only unknown quantity.** We don't know $M_Y(s)$ (where s is used to distinguish it from t which we already used for M_{Z_n}). We can make use of $E[Y_i] = 0$, $E[Y_i^2] = 1$. Near $s = 0$:

$$M_Y(s) = 1 + \frac{s^2}{2} + c_3 s^3 + c_4 s^4 + \dots$$

where $c_k = E[Y^k]/k!$ (as we will see in one second, we don't need higher moments).

4. Substitute $s = t/\sqrt{n}$:

$$M_Y\left(\frac{t}{\sqrt{n}}\right) = 1 + \frac{t^2}{2n} + \frac{c_3 t^3}{n^{3/2}} + \frac{c_4 t^4}{n^2} + \dots$$

5. Take logs, multiply by n , let $n \rightarrow \infty$. Using the approximation that $\log(1+x) \approx x$ for small x :

$$n \log M_Y\left(\frac{t}{\sqrt{n}}\right) = \frac{t^2}{2} + \underbrace{\frac{c_3 t^3}{\sqrt{n}}}_{\rightarrow 0} + \underbrace{\frac{c_4 t^4}{n}}_{\rightarrow 0} + \dots \rightarrow \frac{t^2}{2}$$

Every term after the quadratic goes to zero regardless of the actual values of $c_3, c_4, \dots$. Note that this holds regardless of the original distribution. The only assumptions we made was that μ and σ^2 are finite.

6. Math $M_{Z_n}(t)$ to the standard normal. So $M_{Z_n}(t) \rightarrow e^{t^2/2}$, which is exactly the MGF of a standard normal (which you might not have known, but you can verify it by yourself). By the continuity theorem (MGF convergence $\Rightarrow$ distribution convergence), $Z_n \xrightarrow{d} \mathcal{N}(0, 1)$.

Things to note: only the mean and variance of the original distribution survive into the limit. Everything else (skewness, kurtosis, and other higher moments) disappear, which is caused by the $1/\sqrt{n}$ scaling we did in step 1. That's why the sum of iid random variables ends up as a standard normal.

Theorem 1.10 (Continuous Mapping Theorem). *Let $g : \mathbb{R}^k \rightarrow \mathbb{R}^m$ be continuous at every point in the support of X . Then for any sequence $\{X_n\}$:*

- $X_n \xrightarrow{p} X \implies g(X_n) \xrightarrow{p} g(X)$.
- $X_n \xrightarrow{d} X \implies g(X_n) \xrightarrow{d} g(X)$.
- $X_n \xrightarrow{a.s.} X \implies g(X_n) \xrightarrow{a.s.} g(X)$.

The intuition is that when we apply a continuous transformation, we preserve the type of convergence. We will use this result later when the CMT is used alongside Slutsky's theorem and the WLLN.

2 Causal inference and the linear model

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Overview

We explain the fundamental problem of causal inference and why econometrics has become focused on the identification of causal effects.

Readings

Imbens and Rubin (2015) provides an in-depth introduction to causal effects from an econometric standpoint. S. L. Morgan and Winship (2014) does the same in a slightly less technical manner. Paul, Hall, and Collins (2004) offer philosophical insights into causality. J. Angrist and Pischke (2008) explains how identification issues arise and how they are (or were) solved in economics.

2.1 Treatment effects

A large part of econometrics consists in identifying causal effects. You can think of two ways to do so.

1. **Test hypotheses:** we use a discrete rule to assess whether some treatment has a causal effect (is there an effect: yes/no).
2. **Quantify:** we want to measure the strength of the causal effect of a cause on an effect (how strong/weak is the effect).

The notion of **cause** itself is tricky. Here, I follow Rubin (1974) and think of causal effects as **potential outcomes**.¹ The idea is that the causal effect of a treatment is a contrast between an outcome under one state of the treatment and an outcome under a different state of the treatment.

Note: people also talk about (cause): independent variable, treatment, intervention, X or D ; and (effect) outcome, dependent variable, Y . Terminology will vary with time, norms, and disciplines, but they ultimately all capture the idea that there are some things that, when exogenously modified, change other things.

For an individual unit i , we will be interested in a relation between the outcome Y under treatment D versus control ($D \in \{0, 1\}$), holding fixed the distribution of treatments across all other units ($\mathbf{d}_{-i}$):

$$Y(D_i = 1, \mathbf{d}_{-i}) \text{ versus } Y(D_i = 0, \mathbf{d}_{-i})$$

In general, we will be interested in the difference between the two.

Definition 2.1 (Unit treatment effect). We define the **unit treatment effect** (UTE) of D as:

$$\text{UTE} = Y(D_i = 1, \mathbf{d}_{-i}) - Y(D_i = 0, \mathbf{d}_{-i}) \quad (2.1)$$

Note that the treatment effect here may differ for each individual unit $i \in \{1, 2, \dots, N\}$.

Note: the UTE, like other quantities defined later, is an **estimand** — the population quantity we want to learn about. The function of the data that produces an estimate of the estimand is the **estimator** (Lundberg, Johnson, and Stewart 2021). In this course, we follow an *estimand-first* approach: we first precisely define what we want to estimate, and only then ask how to estimate it. This prevents data availability or computational convenience from quietly driving the research question.

Note also that it is here a function of $\mathbf{d}_{-i}$, which is the vector of treatment for all units aside from i . Consider an experimental setting

See Lewis (1974) for a philosophical take on causation and counterfactuals. The idea is that there are “possible worlds” and causation is the ‘what-if’ relation from a cause to an effect that creates that counterfactual world. See also Paul, Hall, and Collins (2004).

The “versus” comes from Rubin. Also, $\mathbf{d}_{-i}$ is a vector of length $N - 1$.

¹The potential outcome framework is also referred to as the Neyman-Rubin Causal model (Peter M. Aronow and Miller 2019, 237) in reference to a paper published in 1923 by Jerzy Neyman and translated in Splawa-Neyman (1990), which uses the idea of *potential* outcome (in his article: potential agricultural yield) and in reference to the work by Donald Rubin (Rubin 1974, 1986).

with five units. As the experimenter, you end up with 5 joint decisions to treat or not each unit. This could look like:

$$\mathbf{d} = [0, 0, 0, 1, 1]$$

In other words, the UTE for unit 1 is here a function of how all units 1 to N are assigned to treatment or control. At this stage, I cannot rule out that the my treatment effect could look different if, for instance, the researcher had randomly allocated treatments like:

$$\mathbf{d} = [0, 1, 0, 0, 1]$$

We will want to simplify this from here on. To do so, we will make our first two assumptions (Imbens and Rubin 2015, 10). The first one is that units do not interfere with each other (whether B is treated doesn't affect the potential outcome of A), which is referred to as the *no interference and no spillover* assumption (Imbens and Rubin 2015, 10) (Assumption 1).

2.2 Assumption: No interference, no spillover

Assumption 1 (No interference, no spillover). *No interference and no spillovers. The potential outcome of unit i does not vary with the treatment assignment to other units $-i$.*

The second assumption is the *no hidden variation* assumption (Assumption 2). Here, the idea is that a treatment value is uniquely defined. Let's take an example: if I study the effect of giving households access to solar electricity on their wellbeing, the treatment $D = 1$ is the same for everyone. We don't have different dosages or different types of technology lumped together in $D = 1$.

2.3 Assumption: No hidden variation

Assumption 2 (No hidden variation). *No hidden variation. Potential outcomes are uniquely defined by treatment values, such that there are unique values attached to $Y(0)$, $Y(1)$.*

Together, the *no interference/spillover* combined with the *no hidden variation* assumptions are referred to as the **Stable Unit Treatment Value** assumption, or **SUTVA**.

2.4 Assumption: SUTVA

Assumption 3 (SUTVA). *Stable unit treatment value assumption (SUTVA): no interference/spillover and no hidden variations of treatment are jointly met.*

Note: Consider a study on income discrimination by race, with the “treatment” being black and the control group being white. For simplicity, let’s assume that there are only black and white people in the population we are interested in. Try to think about what would happen to potential outcomes if (a) everybody was, say, black vs. (b) everybody was white vs. (c) some distribution in-between. It’s not clear that the potential outcomes would remain the same in all cases. It may well be that potential outcomes depend on how many people get the ‘treatment.’ Thus, SUTVA might not be met here.

If we assume SUTVA, we can write:

$$\begin{aligned} Y_i^1 - Y_i^0 &= \delta_i \\ &\equiv \text{UTE} \qquad \text{with SUTVA} \end{aligned}$$

Y_i^1 could also have been written $Y(D_i = 1)$ or $Y_i(1)$. It’s the vector of outcomes when all units are treated. Same for Y^0 .

The next table gives us a sense of what δ_i could look like for an imaginary sample of 5 units.

i	Y^0	Y^1	δ_i	D
1	4	8	4	0
2	2	8	6	0
3	5	6	1	0
4	5	7	2	1
5	4	9	5	1

Table 2.1: Hypothetical outcomes and treatment effects for 5 units.

A special case – though one that’s often implicitly (or unknowingly) assumed – is one in which δ_i is *homogenous* (or invariant or constant) across units. This contrasts with *heterogeneous* treatment effects that vary with i . Assuming homogeneity is a strong assumption, since it means that the treatment has the same effect for all units in the population. Yet relaxing it — allowing for **heterogeneous** treatment effects — creates complications when estimating various quantities of interest.

Note that there are two types of heterogeneity: *systematic* and *truly random*. Systematic heterogeneity is predictable: e.g., women may respond differently than men to a treatment, in which case we can model it by interacting gender with D (more on this later). Truly random heterogeneity is residual individual-level variation in δ_i that cannot be explained by observed covariates.

i	Y^0	Y^1	δ_i	D
1	4	8	4	0
2	4	8	4	0
3	5	9	4	0
4	5	9	4	1
5	4	8	4	1

Table 2.2: Hypothetical outcomes for 5 units with homogenous treatment effects.

This is not the only function we might be interested in. People interested in relative risks could study the ratio of outcomes or of probability of outcomes

$$\frac{Y_i^1}{Y_i^0}$$

We could also focus on transformations, such as the likelihood ratio of Y :

$$\frac{\mathbb{P}(Y_i^1 = 1)}{\mathbb{P}(Y_i^0 = 1)}$$

But the most common case is that we are interested in δ or a variant thereof.

2.5 The problem

The problem is that we never observe the UTE because we don't simultaneously observe the two outcomes jointly. The data we have looks like Table 2.3.

This is the **fundamental problem of causal inference** (Rubin 1974, 690; Holland 1986). Instead we see a partial story. As a result, we cannot recover the UTE without generating a counterfactual outcome.

i	Y^0	Y^1	Y^{observed}	δ_i
1	4		4	?
2	2		2	?
3	5		5	?
4		7	7	?
5		9	9	?

i	Y^0	Y^1	Y^{observed}	δ_i
---	-------	-------	-----------------------	------------

Table 2.3: Observed data if units 1 to 3 were untreated and 4-5 are treated.

2.6 Average treatment effects (ATE)

One thing that stands out from the previous table is: while we don't have the counterfactual for individual units, we do have data for both Y^0 and Y^1 . One could wonder: couldn't we get some kind of *average* treatment effect (or *expected* causal effect)?

Note: moving from *unit* to *average* causal effect feels like moving the goal posts. Yet average effects might be more interesting. UTE tell us what happened to a specific unit. But if we are interested in policy recommendations, we will be interested in saying what will happen *on average* if the treatment D is implemented.

Definition 2.2 (Average treatment effect). Define the **average treatment effect** (ATE) as

$$\begin{aligned}\mathbb{E}[\delta] &= \mathbb{E}[Y^1 - Y^0] \\ &= \mathbb{E}[Y^1] - \mathbb{E}[Y^0]\end{aligned}\tag{2.2}$$

Five things to note.

1. Virtually all the problems in causal inference come from finding a way to generate reliable predictions for the missing counterfactual observations (or at least their expected value). The mathematical expectation $\mathbb{E}[\cdot]$ is an (unobserved) population parameter. It will thus have to be **estimated**.
2. Estimands that are about moments of the population are only defined if the population has these moments. If Y is Cauchy-distributed, for instance, it doesn't make sense to talk about $\mathbb{E}[Y]$ since it's undefined.
3. There are many ways within and outside quantitative methods to generate counterfactuals. Some historians try to spell out counterfactuals ("what if the UK had lost the Battle of

Britain?”). Same with writers (Philip K. Dick’s *Man in the High Castle*). War games are attempts to generate counterfactuals in strategic settings. And nothing prevents people to do this with qualitative methods.

4. We want to capture the *average* treatment effect. This raises an ambiguity: is it a constant that’s identical across all units, or is it a variable? If it’s the latter, then we talk of **heterogeneous treatment effects**. Estimating how treatment effects vary across subgroups (CATEs as a function of covariates $\mathbf{X}$) is a major research area. Modern tools such as causal forests (Wager and Athey 2018) and double/debiased machine learning (Chernozhukov, Chetverikov, et al. 2018) provide ways to deal with this. We return to them later.
5. There are many things we could be interested in aside from $\mathbb{E}[\delta]$. We could be interested in the change in distribution, in the change of quintile, etc. For instance, if I look at the effect of a drug, I might be interested in finding out whether the spread of wellness outcome increases or not. In addition, we can also be interested in other transformations (Colnet et al. 2023).

$$\begin{aligned}
\text{Risk ratio} &:= \frac{\mathbb{E}[Y^1]}{\mathbb{E}[Y^0]} \\
&= \frac{\mathbb{P}[Y^1 = 1]}{\mathbb{P}[Y^0 = 1]} \\
\text{Excess Risk Ratio} &:= \frac{\mathbb{E}[Y^1] - \mathbb{E}[Y^0]}{\mathbb{E}[Y^0]} \\
\text{Survival ratio} &:= \frac{\mathbb{P}[Y^1 = 0]}{\mathbb{P}[Y^0 = 0]} \\
\text{Odds ratio} &:= \frac{\frac{\mathbb{P}[Y^1=1]}{\mathbb{P}[Y^1=0]}}{\frac{\mathbb{P}[Y^0=1]}{\mathbb{P}[Y^0=0]}} \\
&= \frac{\mathbb{P}[Y^1 = 1]}{\mathbb{P}[Y^1 = 0]} \frac{\mathbb{P}[Y^0 = 0]}{\mathbb{P}[Y^0 = 1]}
\end{aligned}$$

2.7 Conditional average treatment effects (CATE)

There are other estimand we might be interested in. In particular, we might be interested in the average treatment effect for a specific subgroup. This is a **conditional average treatment effect** (CATE).

Definition 2.3 (Conditional ATE). The **conditional average treatment effect** (CATE) is the ATE *within* an arbitrary subgroup $g \in \mathcal{G}$:

$$\text{CATE}(g) = \mathbb{E}[Y^1 - Y^0 \mid G = g]$$

Among an infinite set of potential CATEs, there are two that are especially of interest. We can define the **average treatment effect on the treated** (ATT) and **average treatment on the control** (ATC) as (Equation 2.3):

$$\begin{aligned} \mathbb{E}[Y^1 \mid D = 1] - \mathbb{E}[Y^0 \mid D = 1] &\equiv \text{ATT} \\ \mathbb{E}[Y^1 \mid D = 0] - \mathbb{E}[Y^0 \mid D = 0] &\equiv \text{ATC} \end{aligned} \quad (2.3)$$

The ATT is the effect of the treatment among those in the population who get the treatment (and likewise for the ATC).

Example. Suppose we are interested in student performance at school. Imagine that the treatment is having a laptop at school and the outcome is test scores. ATT would be the effect of the laptop among students are typically have a laptop. ATC is the effect of a laptop among those what typically don't have a laptop. The two might be different if, on average, δ differs between the two groups (i.e. $\mathbb{E}[\delta \mid D = 1] \neq \mathbb{E}[\delta \mid D = 0]$). For instance, students who buy laptops are also those who benefit more from having a computer and may have a larger average δ .

To see the difference between the three estimands (ATE, ATT, ATC), we can use our earlier sample (Table 2.4). There are differences, though they are quite small.

i	Y^0	Y^1	δ_i	D
1	4	8	4	0
2	2	8	6	0
3	5	6	1	0
4	5	7	2	1
5	4	9	5	1
ATE	= 3.6		(18/5)	
ATT	= 3.5		(7/2)	
ATC	= 3.7		(11/3)	

Table 2.4: (True) ATE, ATT, and ATC for this population.

At this stage, we haven't really any reason to say why the three estimands would differ or not. Suffice to say that there is not a unique and identical δ_i for all i . Instead, the expected value of δ might vary depending on the subset of the population we are looking at.

Note: A useful connection between the ATE and CATE follows directly from the **Law of Iterated Expectations** (Chapter 1). Averaging the CATE over the marginal distribution of covariates $\mathbf{X}$ recovers the ATE:

$$\text{ATE} = \mathbb{E}[\text{CATE}(\mathbf{X})] = \mathbb{E}[\mathbb{E}[Y^1 - Y^0 \mid \mathbf{X}]].$$

This means that if we can estimate the CATE for each covariate profile $\mathbf{X} = x$, we can recover the ATE by averaging. It also foreshadows a key strategy for identification: conditioning on observed covariates $\mathbf{X}$ can eliminate baseline bias, and then the LIE lets us aggregate back to the population-level ATE.

2.8 Estimation: bias and inconsistency

The problem to be solved is: how can we estimate population estimands (ATE, ATT, ATC) from a random sample? Given that we always miss half the data on the outcome, this will generally be a challenge in observational settings.

Note: we talk about *observational* data when the treatment assignment isn't controlled by the researcher. In contrast, *experimental* data is generated under a design controlled by the researcher. A *randomized* experiment refers to the type of design used by the researcher (here: random assignment of the treatment).

Intuitively, one way to estimate the ATE is to compute the estimate $\tilde{\delta}$

$$\tilde{\delta} = \frac{1}{N_1} \sum_{i: D_i=1} Y_i - \frac{1}{N_0} \sum_{i: D_i=0} Y_i, \quad N_1 = \sum_i D_i, \quad N_0 = N - N_1.$$

This is simply the difference between the mean of the treatment group and the mean of the control group.

The question is: will $\mathbb{E}[\tilde{\delta}] = \text{ATE}$? This will often not be the case in observational setting.

Realized Y	D = 0	D = 1	Overall
Y^0	a	b	$\mathbb{E}[Y^0] = a(1 - \pi) + \pi b$
Y^1	c	d	$\mathbb{E}[Y^1] = c(1 - \pi) + \pi d$

Table 2.5: Decomposition of the average treatment effect. c and b cannot be observed. Note for instance that d should be read as: $d = \mathbb{E}[Y^1|D = 1]$ (same for the other entries). π is the share of the population that receives the treatment.

The reason is that the naive estimate misses three key components, as shown in Table 2.5.

- a: untreated outcome for units that aren't typically treated (observable).
- d: treated outcome for units that are typically treated (observable).
- b and c: counterfactual cases (unobservable).
- π : population share of treated units.

We can visualize the problem in Figure Figure 2.1.

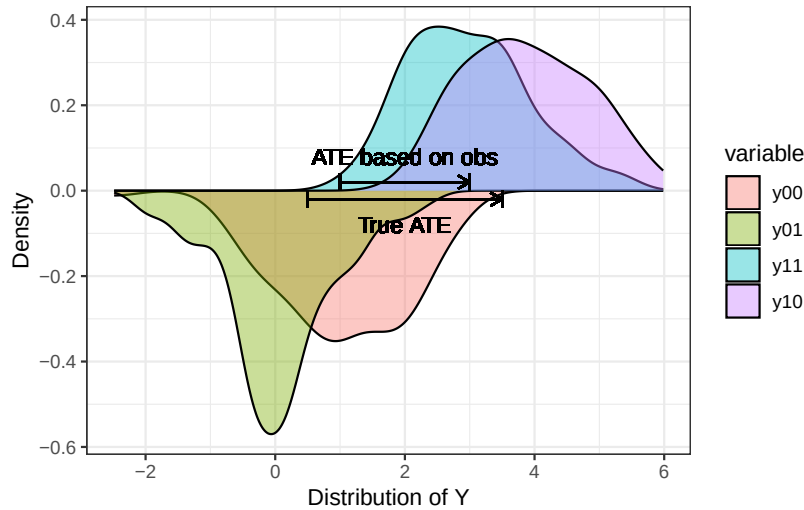


Figure 2.1: Example of bias generated by unobserved counterfactuals. $E[Y^0|D = 0] = 1, E[Y^0|D = 1] = 0, E[Y^1|D = 0] = 4, E[Y^1|D = 1] = 3$. Assume all group to be of equal size. Naive ATE: 2. True ATE: 3

We can then revisit our definition of ATE:

$$\begin{aligned}
\text{ATE} &\equiv \mathbb{E}[\delta] = \mathbb{E}[Y^1] - \mathbb{E}[Y^0] \\
&= \underbrace{\left[\pi \mathbb{E}[Y^1|D=1] + (1-\pi) \overbrace{\mathbb{E}[Y^1|D=0]}^{\text{unobservable}} \right]}_{\text{weighted average of } Y^1} \\
&\quad - \underbrace{\left[\pi \overbrace{\mathbb{E}[Y^0|D=1]}^{\text{unobservable}} + (1-\pi) \mathbb{E}[Y^0|D=0] \right]}_{\text{weighted average of } Y^0} \\
&= [\pi d + c(1-\pi)] - [a(1-\pi) + \pi b] \quad \text{above notation}
\end{aligned}$$

where π is the proportion of units that are typically treated.

What this decomposition tells us is that $\mathbb{E}[Y^1]$ isn't just the outcome of the treated units. It is the (weighted) mean of the (mean) outcomes of the treated units (which can be observed) and the (mean) outcomes of the control units *had they been treated* (which cannot be observed). The weight is π . And the same applies to $\mathbb{E}[Y^0]$.

“Typically” since we could have problems such as units accidentally not taking the treatment, non-compliers, etc.

Note the difference between the ATE and $\tilde{\delta}$: the equation of $\tilde{\delta}$ only includes 2 out of the 5 terms in the ATE. It is missing b , c , and π . Putting the two side by side makes this clear:

$$\begin{aligned}
\text{ATE} &= \mathbb{E}[Y^1] - \mathbb{E}[Y^0] \\
&= [\pi d + c(1-\pi)] - [a(1-\pi) + \pi b] \\
&\quad \text{versus} \\
\tilde{\delta} &= \mathbb{E}[Y^1|D=1] - \mathbb{E}[Y^0|D=0] \\
&= d - a
\end{aligned}$$

We can already intuitively guess that $\tilde{\delta} = \text{ATE}$ if $d = c$ and $a = b$.

But let's dig deeper. We can characterize the biases that ignoring these unobservable values creates. To do so, we can rearrange the earlier decomposition:

Later, we will see how panel data allows us to deal with systematic differences in baseline via the use of difference-in-difference models.

$$\begin{aligned}
\pi d + (1-\pi)c - \pi b - (1-\pi)a &= \mathbb{E}[\delta] && \text{by def} \\
\pi d + c - \pi c - \pi b - a + \pi a &= \mathbb{E}[\delta] && \text{distributing} \\
-a &= \mathbb{E}[\delta] - c - \pi d + \pi c + \pi b - \pi a && \text{moving around} \\
d - a &= \mathbb{E}[\delta] + d - c - \pi d + \pi c + \pi b - \pi a && \text{add } d \text{ both sides} \\
d - a &= \mathbb{E}[\delta] + (b - a) + [(1-\pi)((d - b) - (c - a))] && \text{magic}
\end{aligned}$$

Note that we can rewrite these last quantities as follows (see Table 5):

$$\begin{aligned}
d - a &= \tilde{\delta} \\
d - b &= \mathbb{E}[Y^1 - Y^0 | D = 1] \\
&= \mathbb{E}[\delta | D = 1] \\
c - a &= \mathbb{E}[Y^1 - Y^0 | D = 0] \\
&= \mathbb{E}[\delta | D = 0]
\end{aligned}$$

Thus we can now rewrite this more explicitly (Expression 2.4):

$$\begin{aligned}
\underbrace{\mathbb{E}[Y^1 | D = 1] - \mathbb{E}[Y^0 | D = 0]}_{\text{naive difference } \tilde{\delta}} &= \underbrace{\mathbb{E}[\delta]}_{\text{ATE}} \\
&+ \underbrace{[\mathbb{E}[Y^0 | D = 1] - \mathbb{E}[Y^0 | D = 0]]}_{\text{baseline bias}} \\
&+ (1 - \pi) \underbrace{[\mathbb{E}[\delta | D = 1] - \mathbb{E}[\delta | D = 0]]}_{\text{differential average treatment effect bias}}
\end{aligned} \tag{2.4}$$

There are thus two reasons why a difference between a control group and a treatment group could lead to an inaccurate estimate of ATE:

1. **Baseline bias.** The two groups are different at baseline. The potential Y^0 for those who are treated isn't the same as the (possibly observed) one for the control. This can be caused by confounding and leads to **endogeneity** and cause a **selection** effect (Heckman, Urzua, and Vytlačil 2006). To see this, define $Y_i^0 = \mathbb{E}[Y^0] + \varepsilon_i^0$. We sometimes talk about *selection* because units that have (say) a high ε_i could self-select into the treatment arm, thus creating correlation between ε and D . In regression analysis, this leads to biased estimates.
2. **Differential treatment effect bias.** The average treatment effect isn't the same for the treated and the control. On average, δ could be different for the two groups. This is a problem of **essential heterogeneity** (Heckman, Urzua, and Vytlačil 2006).

Note: when we talk about **identification**, we generally talk about a research design in which the biases (such as baseline and differential treatment effects, but also others connected to measurement error, etc.) are plausibly eliminated in order to recover the estimand of interest (Joshua D. Angrist and Krueger 1999).

This connects to the classical **Heckman selection problem** (Heckman and Vytlačil 2005), where treatment assignment is correlated with potential outcomes due to self-selection. The baseline bias term here is the selection bias in Heckman's framework; the π -weighted differential treatment effect bias has no direct analogue there but matters here when the treatment share is non-trivial.

Note: a fourth important estimand is the **Local Average Treatment Effect** (LATE) — the ATE for *compliers*, i.e., units who change their treatment status in response to an **instrumental variable**. No need to worry about this just yet.

💡 Does living near the sea increase life expectancy?

A recent study by Cao, Martins, and Wu (2025) tests the effect of living close to the ocean (and other water bodies) on life expectancy. The estimate a linear model at the individual level:

$$\text{Life expectancy}_i = \alpha + \beta \mathbb{I}(\text{Living within 50km of the ocean}) + \dots + \varepsilon_i,$$

where $\mathbb{I}$ is an indicator that equals 1 if an individual lives in a neighborhood that is within 50km of the ocean, and 0 otherwise. That is their treatment.

Their estimate of β is positive ($\hat{\beta} = 0.3$, 95% confidence interval: 0.26 to 0.39). They conclude that living near the ocean causes an increases life expectancy. But can you think of reasons why this may be misleading?

This is why we need to make two assumptions (and implement a design that makes them likely to hold) to recover the ATE (or only one to recover the ATT or ATC):

1. Same potential treated Y : $\mathbb{E}[Y^1|D = 1] = \mathbb{E}[Y^1|D = 0]$
2. Same potential control Y : $\mathbb{E}[Y^0|D = 1] = \mathbb{E}[Y^0|D = 0]$

We then have three cases.

1. If both assumptions hold: we can recover the ATE (which turns out to be equal to the ATT and ATC).
2. If only the first assumption holds, then we can recover:

$$\begin{aligned} \tilde{\delta} &= \mathbb{E}[Y^1|D = 1] - \mathbb{E}[Y^0|D = 0] \\ &= \mathbb{E}[Y^1|D = 0] - \mathbb{E}[Y^0|D = 0] \quad \text{by assumption} \\ &\equiv \text{ATC} \end{aligned}$$

3. If only the second assumption holds, then we can recover:

$$\begin{aligned} \tilde{\delta} &= \mathbb{E}[Y^1|D = 1] - \mathbb{E}[Y^0|D = 0] \\ &= \mathbb{E}[Y^1|D = 1] - \mathbb{E}[Y^0|D = 1] \quad \text{by assumption} \\ &\equiv \text{ATT} \end{aligned}$$

These conditions have formal names worth knowing.

Definition 2.4 (Unconfoundedness (ignorability)). The treatment D is **unconfounded** (or **ignorable**) conditional on covariates $\mathbf{X}$ if:

$$(Y^0, Y^1) \perp D \mid \mathbf{X}.$$

That is, conditional on $\mathbf{X}$, treatment assignment is independent of the potential outcomes.

 The various names of unconfoundedness

We often call Definition 2.4 the **ignorability assumption**.

When $\mathbf{X}$ is empty (i.e., no conditioning is needed), this becomes simple independence $(Y^0, Y^1) \perp D$. This holds by design in a randomized experiment (which we discuss later). In observational settings, where it's rare (though not impossible; we will see *quasi-experiments* later) to obtain unconditional independence, it requires that all confounders (i.e., variables that jointly affect treatment assignment and the outcome) are (a) observed and (b) included in $\mathbf{X}$.

Unconfoundedness is also sometimes called **selection on observables**, because it asserts that once observed covariates are controlled for, there is no remaining selection bias. It eliminates the baseline bias term $[\mathbb{E}[Y^0|D=1] - \mathbb{E}[Y^0|D=0]]$ and the differential treatment effect bias from the decomposition in Equation 2.4.

Definition 2.5 (Overlap). There is **overlap** (or **common support**) if every covariate profile has a strictly positive probability of receiving either treatment status:

$$0 < \mathbb{P}(D = 1 \mid \mathbf{X} = x) < 1 \quad \text{for all } x \text{ in the support of } \mathbf{X}.$$

 Assumption: overlap (or common support)

Intuition: To estimate the treatment effect for units with covariate profile x , we need observations from *both* treatment and control at x . If some subgroup is never treated ($\mathbb{P}(D = 1|\mathbf{X} = x) = 0$), we have no treated observations to compare against the control there. If some subgroup is always treated ($\mathbb{P}(D = 1|\mathbf{X} = x) = 1$), we have no untreated counterfactual. In both cases, the unobservable quantities $b = \mathbb{E}[Y^0|D = 1, \mathbf{X} = x]$ and $c = \mathbb{E}[Y^1|D = 0, \mathbf{X} = x]$ from our bias table cannot be

estimated, even in principle.

Going back to Figure 2.1: overlap ensures that for every covariate profile, we can find comparable units on both sides of the treatment, making the counterfactual densities estimable.

The quantity $p(x) := \mathbb{P}(D = 1 | \mathbf{X} = x)$ is called the **propensity score** (Rosenbaum and Rubin 1983). Checking that propensity scores are bounded away from 0 and 1 is a standard diagnostic in observational studies.

Unconfoundedness + overlap together are sometimes called the **strong ignorability** assumption (Rosenbaum and Rubin 1983), and together they are sufficient to non-parametrically identify the ATE.

Example: Consider the effect of getting a solar panel on household income. Can we assume that the expected income of those who got the panel would have been the same *had they not obtained a panel* than the expected income of those who haven't bought one? If so, the naive $\tilde{\delta}$ recovers the ATT.

2.9 Linear model

Let us set biases aside and see how the causal model discussed above can be connected to the workhorse of econometrics: the **linear model** (i.e., a model that's linear in parameters). Compactly, we write:

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon$$

This notation can be traced back (at least) to Aitken (1936).²

By “linear,” we mean linear *in parameters*. The linear model admits nonlinearity in *variables* (e.g., by adding a squared term of X , taking the log of X , or taking its square root).

The model links a set of variables (commonly referred to as *independent variables*, *treatments*, *interventions*, *explanatory variables*, *right-hand side variables*, *regressors*, etc.) to another variable (*dependent variable*, *outcome*, *left-hand side variable*, etc.). To account for stochasticity in any kind of **data-generating process**, we include an *error term* ε whose exact nature will be defined below.

By convention, the independent variable(s) are generally referred to as $\mathbf{X}$ and the dependent variable as $\mathbf{y}$. Sometimes, when we are

²Replace $\mathbf{X}$ with $\mathbf{P}$ and β with a (Aitken uses y for $\hat{y}$).

interested in a particular (generally dichotomous) treatment, we will use D .

Typically, we will either (a) be interested in measuring the **causal effect** of a subset of the variables in $\mathbf{X}$, or (b) be interested in **forecasting** y . In general, in this class, we will focus on (a).

2.10 Foundations of the linear model

What are the foundations of the linear model? Suppose you came up with a theory that states that a treatment D increases Y . In the counterfactual language, this means that Y would have had another value had D had itself another value. Let us first begin with the case in which $D \in \{0, 1\}$ is a dichotomous variable. Then:

1. Let's say we are interested in the expected value of Y , i.e. $\mathbb{E}[Y]$.
2. We are asking whether $\mathbb{E}[Y]$ is affected by D ; we are then therefore interested in the conditional expectation $\mathbb{E}[Y|D]$.
3. If $D = 0$, we write:

$$\mathbb{E}[Y|D] = \mathbb{E}[Y|D = 0]$$

Likewise, if $D = 1$:

$$\mathbb{E}[Y|D] = \mathbb{E}[Y|D = 1]$$

4. Thus:

$$\mathbb{E}[Y|D] = \mathbb{E}[Y|D = 0] + D(\mathbb{E}[Y|D = 1] - \mathbb{E}[Y|D = 0])$$

5. Let us define

$$\alpha := \mathbb{E}[Y|D = 0]$$

and

$$\beta := (\mathbb{E}[Y|D = 1] - \mathbb{E}[Y|D = 0])$$

We can rewrite:

$$\mathbb{E}[Y|D] = \alpha + \beta D$$

6. Lastly, let us add Y on both sides (because $\mathbb{E}[Y|D]$ is not directly observable, whereas Y is).

$$Y = \alpha + \beta D + (Y - \mathbb{E}[Y|D])$$

7. Let us define $\varepsilon := (Y - \mathbb{E}[Y|D])$ as the difference between a given value of Y (y_i) and the expected value of Y at D . By construction, this error satisfies the **zero conditional mean condition**:

$$\mathbb{E}[\varepsilon|D] = \mathbb{E}[Y - \mathbb{E}[Y|D] | D] = \mathbb{E}[Y|D] - \mathbb{E}[Y|D] = 0.$$

This property — $\mathbb{E}[\varepsilon|D] = 0$ — holds here by definition. In observational data, however, it will typically fail whenever D is correlated with unobserved determinants of Y , which is precisely the source of the baseline bias described earlier.

We find the **linear model** :

$$\begin{aligned} \mathbb{E}[Y|D] &= \alpha + \beta D && \text{from step 5} \\ Y &= \alpha + \beta D + \varepsilon && \text{from step 6} \\ Y &= \mathbb{E}[Y|D] + \varepsilon && \text{by plugging the 1st eq in the 2nd} \end{aligned}$$

The last equation corresponds to the **conditional expectation function** (CEF) decomposition theorem (see (J. Angrist and Pischke 2008), p.32).

Generalizing this to a matrix of variables $\mathbf{X}$ (which may include D), we obtain the expression we started with, namely:

$$\begin{aligned} \mathbb{E}[\mathbf{y}|\mathbf{X}] &= \mathbf{X}\beta \\ \mathbf{y} &= \mathbf{X}\beta + \varepsilon \\ \mathbf{y} &= \mathbb{E}[\mathbf{y}|\mathbf{X}] + \varepsilon \end{aligned}$$

Note that the first expression gives us a natural interpretation of β_k : the partial derivative of $\mathbb{E}[\mathbf{y}|\mathbf{X}]$ with respect to X_k is β_k , which is thus the marginal effect of X_k on the conditional mean of $\mathbf{y}$:

$$\frac{\partial \mathbb{E}[\mathbf{y}|\mathbf{X}]}{\partial X_k} = \beta_k.$$

The parameter β_k , then, measures the effect of a small change in X_k on the expected value of Y .

Here, we note two things:

1. The (population) CEF may or may not be linear. In what follows, we will generally assume it is. If it's not exactly linear, then $\mathbf{X}\beta$ will *approximate* it. How well it does so is case-specific.

2. A model that is **saturated** is linear by construction. The model is saturated if $\mathbf{X}$ is entirely composed of dichotomous variables and their interactions. In such a case, each conditional expectation $\mathbb{E}[Y|D = d, X = x, Z = z, \dots]$ has its own unique parameter or combination of parameters.

Here's an example of a saturated model. Suppose we have a model in which we want to test whether the average price of a cars sold is higher for consumers with more/less formal education education, where education is defined as a categorical variable (with 4 values):

Education_{*i*} ∈ {Less than High School, High School, College, PhD}.

We could then estimate:

$$\text{Price of car sold}_i = \alpha + \beta \text{High School}_i + \gamma \text{College}_i + \lambda \text{PhD}_i + \varepsilon_i,$$

where each variable is a dichotomous indicator. Then, each value of education has its own estimate:

- $\mathbb{E}[\text{Price}|\text{Less than HS}] = \alpha$
- $\mathbb{E}[\text{Price}|\text{High School}] = \alpha + \beta$
- $\mathbb{E}[\text{Price}|\text{College}] = \alpha + \gamma$
- $\mathbb{E}[\text{Price}|\text{PhD}] = \alpha + \lambda$

You can also construct saturated models with more than one variable. You would then need to include a number of interaction effects. For instance, if we included gender in addition to education, we would need to interact it with every dummy above. See more later on interaction effects.

When the treatment is **continuous**, we cannot discretize it in a way that guarantees linearity. We might then want to start with a flexible formulation such as:

$$\mathbf{y} = f(\mathbf{X}) + \varepsilon.$$

We would then impose restrictions on $f()$ ranging from a fully parametric approach (in which we replace $f()$ with any desired function) to semi- and non-parametric specifications. More on this later.

3 Ordinary least squares: point estimation

This version: September 29, 2026

Overview

We go through the mechanics of point estimation of an unknown parameter β using ordinary least squares (OLS).

Readings

People interested in least squares and point estimation may refer to: B. Hansen (2022a) (ch.4), Stachurski (2016) (ch.11-12), Verbeek (2018) (ch.2.1-2.3, 2.8-2.9, and 3), Hayashi (2000) (ch1.1-1.2) Stock and Watson (2011) (ch.4 and 6), Wooldridge (2009) (ch.2-3). See S. L. Morgan and Winship (2014) (ch.6) for the connection between least squares and causal inference.

3.1 Estimators

The vector β is a set of unknown and unobservable parameters. These parameters capture the relation between $\mathbf{X}$ and $\mathbf{y}$. At this stage, this relation could be correlational or it could be causal. We don't know just yet. There are two problems here:

1. Can we map from β to some causal estimand of interest?
2. Since we don't know β , we will have to estimate it from data, i.e., from a sample. How?

One way to think of what econometrics is to see this as a problem of drawing inferences from the data about these parameters, $\mathbb{P}(\beta|\mathbf{X})$. To make inferences, we need first an **estimator**, namely a rule that takes data as input and produces estimates as output. There exists several such methods, including various forms of **least squares** (ordinary

least squares, weighted least squares, etc.), **methods of moments**, **maximum likelihood**, **Bayesian estimators**, and so forth. We will see a few of them, starting with ordinary least squares.

3.2 Ordinary least squares

We start with the linear model (see box for notation):

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon$$

💡 Notation

I generally try to use the following notation. For a sample that contains n units (e.g., consumers, firms, countries, etc.) and k right-hand side parameters, we write:

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n \times 1} \quad \mathbf{X} = \begin{bmatrix} 1 & x_{1,2} & z_{1,3} & \cdots & w_{1,k} \\ 1 & x_{2,2} & z_{2,3} & \cdots & w_{2,k} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n,2} & z_{n,3} & \cdots & w_{n,k} \end{bmatrix}_{n \times k}$$

$$\beta = \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \vdots \\ \tau_k \end{bmatrix}_{k \times 1} \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}_{n \times 1}$$

Here, the letters x , z , etc. refer to random variables (I number them here to help count them, but I'll stop doing it from now on). For instance, in a dataset on firms, x could be the number of employees, z annual spending on R&D, etc. The column of 1's is included to serve as a constant.

Note that $\mathbf{X}$ is a stacked matrix of vectors $\mathbf{x}'_i$, where each vector represents the data for unit i . In that sense, $\mathbf{X}$ is like a spreadsheet with each unit being a separate row, and each variable being a column. We sometimes call it the **design matrix**.

We then write (dropping the number for the variables):

$$\begin{aligned} y_i &= \alpha + \beta x_i + \gamma z_i + \dots + \tau w_i + \varepsilon_i && \text{unit } i \\ y_i &= \mathbf{x}'_i \beta + \varepsilon_i && \text{matrix notation} \\ \mathbf{y} &= \mathbf{X}\beta + \varepsilon && \text{matrix notation} \end{aligned}$$

When we move to data and estimated quantities, we write:

$$\begin{aligned} y_i &= \hat{\alpha} + \hat{\beta}x_i + \hat{\gamma}z_i + \dots + \hat{\tau}w_i + u_i && \text{observed } y \\ \hat{y}_i &= \hat{\alpha} + \hat{\beta}x_i + \hat{\gamma}z_i + \dots + \hat{\tau}w_i && \text{predicted } y \\ \mathbf{y} &= \mathbf{X}\hat{\beta} + \mathbf{u} && \text{same in matrix} \\ \hat{\mathbf{y}} &= \mathbf{X}\hat{\beta} \end{aligned}$$

where u_i is called the **residual** and represents the difference between the observed value of y for unit i and its predicted value $\hat{y}_i$. Note that the residuals are not exactly the same as the error term. To see this (B. Hansen 2022a, 173):

$$u_i = y_i - \mathbf{x}'_i\hat{\beta} = (\mathbf{x}'_i\beta + \varepsilon_i) - \mathbf{x}'_i\hat{\beta} = \varepsilon_i - \mathbf{x}'_i(\hat{\beta} - \beta).$$

Thus the residual is the error term plus some additional randomness created by the fact that, for a given sample, $\hat{\beta} \neq \beta$ almost certainly.

The least squares estimator is a rule to generate estimates of the vector β . Most textbooks call these estimates $\hat{\beta}$. Some call them $\mathbf{b}$. When needed, we will add the subscript $\hat{\beta}_{ols}$ to emphasize that these estimates were obtained from ordinary least squares. We write:

$$\mathbf{y} = \mathbf{X}\hat{\beta} + \mathbf{u}.$$

Least squares, as the name suggests is a minimization problem, in which we *minimize the sum of squared residuals* (aka SSR).¹ When we weight all observations equally, we call this **ordinary least squares**.

The minimization problem is:

$$\arg \min_{\hat{\beta}} \text{SSR} := \arg \min_{\hat{\beta}} \mathbf{u}'\mathbf{u},$$

where $\mathbf{u}$ is the vector of residuals defined as the difference between *observed* values of y_i and *predicted* values $\hat{y}_i$ of y for unit i .

In contrast to *weighted* least squares, which we will discuss later.

¹The history of least squares is more than 200 years old. Its first formulation was made by Legendre in 1806 (Legendre 1806, viii): “la méthode qui me paroît la plus simple et la plus générale, consiste à rendre minimum la somme des quarrés des erreurs (...). La méthode dont je viens de parler (...) que j'appelle *Méthode des moindres quarrés* peut être d'une grande utilité.” A few years later, Gauss proved that the ordinary least square solution corresponds to the maximum likelihood estimator (Gauss 1857). See Merriman (1877). See also Aitken (1936).

We can equivalently write:

$$\arg \min_{\beta} \text{SSR} = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\hat{\beta}\|^2$$

You may ask here: why *squared* total residuals? Why not just total residuals, or cubic residuals, or something else? Two answers:

1. We will see that minimizing squared residuals (i.e. using OLS) yields estimates with very desirable properties when necessary assumptions are met: OLS is unbiased, consistent, and (more rarely) efficient.
2. As a minimization problem, OLS will yield a unique solution, which other methods will often fail to do so.

We can solve this minimization problem by using first and second order conditions.

$$\begin{aligned} \text{SSR} &= \mathbf{u}'\mathbf{u} \\ &= (\mathbf{y} - \mathbf{X}\hat{\beta})'(\mathbf{y} - \mathbf{X}\hat{\beta}) && \text{by definition of residuals} \\ &= (\mathbf{y}' - \hat{\beta}'\mathbf{X}')(\mathbf{y} - \mathbf{X}\hat{\beta}) && \text{by properties of transpose} \\ &= \mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{X}\hat{\beta} - \hat{\beta}'\mathbf{X}'\mathbf{y} + \hat{\beta}'\mathbf{X}'\mathbf{X}\hat{\beta} && \text{distributing} \\ &= \mathbf{y}'\mathbf{y} - 2\mathbf{y}'\mathbf{X}\hat{\beta} + \hat{\beta}'\mathbf{X}'\mathbf{X}\hat{\beta} && \text{reorganizing} \end{aligned}$$

At this stage, we can take the first order condition with respect to the vector $\hat{\beta}$:

$$\begin{aligned} \frac{\partial \text{SSR}}{\partial \hat{\beta}} &= 0 - 2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\beta} = 0 \\ 2\mathbf{X}'\mathbf{X}\hat{\beta} &= 2\mathbf{X}'\mathbf{y} && \text{moving things around} \\ \mathbf{X}'\mathbf{X}\hat{\beta} &= \mathbf{X}'\mathbf{y} && \text{dividing by 2} \\ (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} && \text{pre-multiplying both sides; A4} \\ \mathbf{I}\hat{\beta} &= \underbrace{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}}_{k \times 1} && \text{where } \mathbf{I} \text{ is the identity matrix} \end{aligned}$$

Note that $\mathbf{X}$ is of dimension $n \times k$; thus $\mathbf{X}'$ is of dimension $k \times n$ and $(\mathbf{X}'\mathbf{X})$ is of dimensions $k \times k$. Finally, $\mathbf{X}'\mathbf{y}$ is of dimension $k \times 1$, thus the whole thing is $k \times 1$. It all works out.

We reach our first important result: $\hat{\beta}_{ols} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$. Given that $\hat{\beta}$ is a vector of scalars, we refer to $\hat{\beta}_k$ as the **point estimate** of β_k for variable k .²

²We focus on point identification, that is, the estimation of a scalar or a vector. There is a rich literature on **set identification** which derives upper and lower bounds on what the treatment effect could be. See (Manski 1990; Tamer 2010).

The only assumption we needed (**A4**) is for the inverse of $(\mathbf{X}'\mathbf{X})$ to exist.

Assumption 4 (Full rank). *The design matrix $\mathbf{X}$ (of dimension $n \times k$) is of full rank with probability 1.*

⚠ Assumption: full rank/no perfect multicollinearity

The rank of a matrix is the number of linearly independent columns (here: variables). When this assumption is met, there is no perfect multicollinearity across variables in $\mathbf{X}$, and the inverse matrix $(\mathbf{X}'\mathbf{X})^{-1}$ exists. Violations typically occur when the matrix $\mathbf{X}$ includes variables that perfectly predict each other. See below for a discussion of causes.

Note. Some textbooks take a slightly different path to arrive at the same conclusion. Instead of minimizing the *total* sum of squared residuals, we can equivalently minimize the *expected* squared residual for a single observation (the population analog):

$$\min_{\beta} \mathbb{E}[(y_i - \mathbf{x}_i'\beta)^2].$$

The first-order condition $\mathbb{E}[\mathbf{x}_i(y_i - \mathbf{x}_i'\beta)] = 0$ yields the population solution:

$$\beta = [\mathbb{E}[\mathbf{x}_i\mathbf{x}_i']]^{-1}\mathbb{E}[\mathbf{x}_iy_i].$$

Replacing population expectations with sample averages ($\frac{1}{n}\mathbf{X}'\mathbf{X}$ and $\frac{1}{n}\mathbf{X}'\mathbf{y}$) recovers $\hat{\beta}_{ols} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$.

Theorem 3.1 (Best Linear Predictor). *The population OLS coefficient*

$$\beta^* = [\mathbb{E}[\mathbf{x}_i\mathbf{x}_i']]^{-1}\mathbb{E}[\mathbf{x}_iy_i]$$

*is the **Best Linear Predictor** (BLP) of y given $\mathbf{x}$: it solves*

$$\beta^* = \arg \min_{\beta} \mathbb{E}[(y_i - \mathbf{x}_i'\beta)^2].$$

This result holds regardless of the true conditional expectation function $\mathbb{E}[y | \mathbf{x}]$: even if the true CEF is nonlinear, β^* gives the best linear approximation to it (in mean squared error). The sample OLS estimator $\hat{\beta}_{ols}$ consistently estimates β^* . This is why OLS is useful even when we do not believe the linear model is literally correct.

Note. We can think of SSR as the **loss function**, often labeled as $\mathcal{L}$. It is the penalty we are facing when making prediction errors.

Note. We can connect OLS to other estimators, such as **LASSO** or **ridge regression**, by adding a penalty on the size of the coefficients to the loss function:

$$\begin{aligned}\mathcal{L}_{\text{LASSO}} &= \|\mathbf{y} - \mathbf{X}\beta\|^2 + \lambda\|\beta\|_1 && \text{(L1 penalty)} \\ \mathcal{L}_{\text{ridge}} &= \|\mathbf{y} - \mathbf{X}\beta\|^2 + \lambda\|\beta\|_2^2 && \text{(L2 penalty)}\end{aligned}$$

The parameter $\lambda \geq 0$ controls the bias-variance trade-off: as $\lambda \rightarrow 0$ both reduce to OLS; as λ grows, coefficients are pulled toward zero. The two penalties differ importantly in shape: the L1 penalty produces *sparse* solutions (many $\hat{\beta}_k$ exactly zero), making LASSO useful for variable selection. The L2 penalty *shrinks* all coefficients smoothly without zeroing them, and ridge regression has the closed-form solution $\hat{\beta}_{\text{ridge}} = (\mathbf{X}'\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}'\mathbf{y}$. Note how adding $\lambda\mathbf{I}$ also fixes any near-singularity in $\mathbf{X}'\mathbf{X}$.

Note. So far, we have no way to interpret $\hat{\beta}$ causally – it is strictly a solution to an optimization problem. In fact, we can't even interpret population β causally. To do this, we will need additional assumptions to be discussed later.

Prediction vs. causal estimation

OLS can be used for two primary purposes: to do prediction and to estimate causal effect. The estimator is the same, but how they are evaluated varies.

	Prediction	Causal estimation
Goal	Minimize out-of-sample forecast error	Estimate a causal parameter β
Key criterion	Predictive R^2 , cross-validated MSE	Unbiasedness and consistency of $\hat{\beta}$
Model selection	Add variables that reduce prediction error	Remove confounding
Omitting a variable	Fine if it doesn't improve prediction	Dangerous if the variable is a confounder

For **prediction**, the goal is to find $\hat{\beta}$ such that $\hat{y} = \mathbf{X}\hat{\beta}$ is as close to out-of-sample (e.g., future) y 's as possible.

For **causal estimation**, the goal is to recover a causal estimand (like ATE). What we want to avoid is biasedness and inconsistency. That's why, as we will see soon, we care so much whether the **exogeneity assumption** $\mathbb{E}[\varepsilon|\mathbf{X}] = 0$ holds.

Nowadays, prediction (and forecasts) can be done more effectively with machine learning. As a result, we seldom use conventional OLS for that. But it's still a useful starting point if you want to understand how prediction works. For causal estimation, OLS still remains the most commonly used tool.

Definition 3.1 (Projection matrix (hat matrix)). The **projection matrix** (or **hat matrix**) is:

$$\mathbf{P} := \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'.$$

It satisfies $\mathbf{P}\mathbf{y} = \mathbf{X}\hat{\beta} = \hat{\mathbf{y}}$: premultiplying $\mathbf{y}$ by $\mathbf{P}$ projects $\mathbf{y}$ onto the column space of $\mathbf{X}$, yielding fitted values. $\mathbf{P}$ is symmetric ($\mathbf{P}' = \mathbf{P}$) and idempotent ($\mathbf{P}^2 = \mathbf{P}$).

Definition 3.2 (Annihilator matrix (residual maker)). The **annihilator matrix** (or **residual maker**) is:

$$\mathbf{M} := \mathbf{I} - \mathbf{P} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'.$$

It satisfies $\mathbf{M}\mathbf{y} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{u}$: premultiplying $\mathbf{y}$ by $\mathbf{M}$ yields the OLS residuals. Like $\mathbf{P}$, $\mathbf{M}$ is symmetric and idempotent. Crucially, $\mathbf{M}\mathbf{X} = \mathbf{0}$: the annihilator wipes out any vector in the column space of $\mathbf{X}$, which is why it is used in the Frisch-Waugh-Lovell theorem below to partial out regressors.

3.3 Special case: OLS with single regressor

It is useful to write down the special case of a single regressor to capture the intuition behind least square regression.

To begin with, we are now minimizing:

$$\min \sum u_i^2 = \min \sum (y_i - \hat{\alpha} - \hat{\beta}x_i)^2$$

We can solve this by calculus. Now, we could directly take the partial derivatives of this last expression using the chain rule. Or we can do it by hand. By the chain rule, we get:

$$\begin{aligned}\frac{\partial \text{SSR}}{\partial \hat{\alpha}} &= -2 \sum (y_i - \hat{\alpha} - \hat{\beta} x_i) && \text{extracting constant} \\ 0 &= 2 \left(\sum y_i - \sum \hat{\alpha} - \sum \hat{\beta} x_i \right) && \text{distributing} \\ &= \sum y_i - n \hat{\alpha} - \hat{\beta} \sum x_i && \text{dividing by 2}\end{aligned}$$

and:

$$\begin{aligned}\frac{\partial \text{SSR}}{\partial \hat{\beta}} &= \sum -2 (x_i y_i - \hat{\alpha} x_i - \hat{\beta} x_i^2) \\ 0 &= -2 \left(\sum x_i y_i - \hat{\alpha} \sum x_i - \hat{\beta} \sum x_i^2 \right) && \text{distributing} \\ &= \sum x_i y_i - \hat{\alpha} \sum x_i - \hat{\beta} \sum x_i^2 && \text{dividing by 2}\end{aligned}$$

This gives us two equations with two unknowns. We can reshape them as the **normal equations** (I also drop the subscripts for readability):

$$\begin{aligned}n \hat{\alpha} + \hat{\beta} \sum x &= \sum y \\ \hat{\alpha} \sum x + \hat{\beta} \sum x^2 &= \sum xy\end{aligned}$$

There are various ways to solve these equations. One way is to use some matrix algebra. Specifically:

$$\begin{bmatrix} n & \sum x \\ \sum x & \sum x^2 \end{bmatrix} \begin{bmatrix} \hat{\alpha} \\ \hat{\beta} \end{bmatrix} = \begin{bmatrix} \sum y \\ \sum xy \end{bmatrix}$$

We can then isolate the parameters, assuming that the inverse of the first matrix exists. In practice, this requires the absence of perfect multicollinearity. In the bivariate case, it means that the variance of X should be greater than zero.

$$\begin{bmatrix} \hat{\alpha} \\ \hat{\beta} \end{bmatrix} = \begin{bmatrix} n & \sum x \\ \sum x & \sum x^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum y \\ \sum xy \end{bmatrix} \quad \text{by A4}$$

Taking the inverse, we have:

$$\begin{bmatrix} \hat{\alpha} \\ \hat{\beta} \end{bmatrix} = \frac{1}{n \sum x^2 - (\sum x)^2} \begin{bmatrix} \sum x^2 & -\sum x \\ -\sum x & n \end{bmatrix} \begin{bmatrix} \sum y \\ \sum xy \end{bmatrix}$$

But we can simplify this a bit: note that $\sum y = n\bar{y}$ and $\sum x = n\bar{x}$. Define $s_{xx} := \sum (x_i - \bar{x})^2 = \sum x^2 - n\bar{x}^2$, $s_{yy} := \sum (y_i - \bar{y})^2 = \sum y^2 - n\bar{y}^2$, $s_{xy} := \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum xy - n\bar{x}\bar{y}$. Therefore:

$$\begin{aligned}
\hat{\beta} &= \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} && \text{from above} \\
&= \frac{n \sum xy - n^2 \bar{x}\bar{y}}{n \sum x^2 - (n\bar{x})^2} \\
&= \frac{\sum xy - n\bar{x}\bar{y}}{\sum x^2 - n\bar{x}^2} && \text{cancelling n} \\
&= \frac{s_{xy}}{s_{xx}} \\
&= \frac{\widehat{\text{Cov}}(X, Y)}{\widehat{\text{V}}(X)} && \text{dividing both terms by n}
\end{aligned}$$

Therefore, $\hat{\beta}$ is the covariance of X and Y , scaled by the variance of X .

As for $\hat{\alpha}$:

$$\begin{aligned}
n\hat{\alpha} + \hat{\beta} \sum x &= \sum y && \text{from the normal equations} \\
n\hat{\alpha} + \hat{\beta} n\bar{x} &= n\bar{y} \\
\hat{\alpha} + \hat{\beta} \bar{x} &= \bar{y} && \text{removing n} \\
\hat{\alpha} &= \bar{y} - \hat{\beta} \bar{x}
\end{aligned}$$

The intercept $\hat{\alpha}$ is therefore the average value of y minus $\hat{\beta}$ times the average value of x .

3.4 Frisch–Waugh–Lovell Theorem

The OLS estimates of the parameters of a regression with more than one independent variable are quite straightforward: $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$. While this is nicely compact, it doesn't tell us all that much about how we should think of $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$.

To help us, we can use the **Frisch–Waugh–Lovell theorem** (or the **regression anatomy theorem**) (Frisch and Waugh 1933; Lovell 1963) to get a better insight into what is going on.³ The theorem states the following.

³See J. Angrist and Pischke (2008) and Filoso (2013) for a discussion. The presentation here follows B. Hansen (2022a, 82).

Theorem 3.2 (Frisch-Waugh-Lovell). *Suppose we have:*

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon.$$

Partition $\mathbf{X}$ into $\mathbf{X}_1$ and $\mathbf{X}_2$, and likewise β into β_1 and β_2 .

1. *Estimate the following model with OLS:*

$$\mathbf{y} = \mathbf{X}_2\beta_2$$

and estimate the residuals $\hat{\varepsilon}_y$

2. *Estimate the following model with OLS:*

$$\mathbf{X}_1 = \mathbf{X}_2\gamma$$

and estimate the residuals $\hat{\varepsilon}_{\beta_2}$

3. *You can then regress $\hat{\varepsilon}_y$ on $\hat{\varepsilon}_{\beta_2}$ and estimate the model's parameter with OLS:*

$$\hat{\varepsilon}_y = \hat{\varepsilon}_{\beta_2}\psi$$

The Frisch-Waugh-Lovell Theorem shows that:

$$\hat{\psi} = \hat{\beta}_1$$

We can also get Frisch-Waugh-Lovell in a slightly different manner:

1. Suppose you have a model such that:

$$y_i = \alpha + \beta x_{i,1} + \delta w_{i,2} + \dots + \gamma z_{i,k} + \varepsilon_i$$

with k independent variables.

2. Define the residual $\tilde{x}_i = x_i - \hat{x}$ from the regression of x on all other independent variables, such that:

$$x_i = \rho + \lambda w_{i,2} + \dots + \tau z_{i,k} + \mu_i \quad (3.1)$$

Then, we can express $\hat{\beta}$ as:

$$\hat{\beta} = \frac{Cov(Y, \tilde{x})}{Var(\tilde{x})}$$

In other words, $\hat{\beta}_x$ will equal the covariance of Y and the residuals from the regression of x on all other independent variables, scaled by

the variance of these residuals. But note that this is the same as the following three steps:

1. Regress x on all other independent variables
2. Obtain the residuals from this regression ($\tilde{x}$)
3. Regress y on $\tilde{x}$, because $\hat{\beta}_x = Cov(Y, \tilde{x})/Var(\tilde{x})$ in the bivariate case.

One way to think about this is to note that $\tilde{x}$ is *whatever variation is left of x once we explain it with all other confounding variables*. So in a sense we purge from x everything that could be attributed to the other confounders. And then we finally scale it by the residuals' variance.

This is a very useful result for a number of reasons. First, it tells us that $\hat{\beta}_x$ will depend on all other independent variables *as long as they are correlated with x* . We can see this from 2: suppose that w is correlated with x , then $\lambda \neq 0$. But then, $\tilde{x}$ will change depending on whether w is included in the regression or not. And therefore, so will $\hat{\beta}$.

We can therefore see the following key implication: if a variable Z is (causally) correlated with X and Y , then we need to include it in our model. Otherwise, we suffer from **omitted variable bias**. See below for more on this.

A related implication concerns the design of our research. If we can **randomize** our independent variable, then by construction *nothing* is correlated (in expectation) with X (since X is randomly assigned), and therefore we don't *need* to adjust for anything (there are other reasons to adjust for third factors, which we'll discuss later). This is why **randomized controlled trials** (RCTs) are so popular.

It seems that the first such experiment was conducted by James Lind, a Scottish physician, who was trying to find a remedy to scurvy.

3.5 Model specifications and their interpretation

3.5.1 “Vanilla” OLS

The linear model expresses the conditional expectation of y given $\mathbf{X}$ (for k independent variables, including a constant):

$$\mathbb{E}[Y|X, Z, \dots K] = \mathbf{X}\beta = \alpha + \beta X + \gamma Z + \dots + \lambda_k K$$

And for a given sample, using OLS estimates, we can plug in:

$$\hat{\mathbb{E}}[Y|X, Z, \dots K] = \hat{y} = \hat{\alpha} + \hat{\beta}X + \hat{\gamma}Z + \dots + \hat{\lambda}_k K.$$

Thus, the predicted values $\hat{y}$ are conditional expectation (or conditional mean) set at specific values of $X, Z, \dots, K$.

Taking the partial derivative of $\mathbb{E}[Y|X, Z, \dots, K]$ with respect to X yields:

$$\frac{\partial \mathbb{E}[Y|X, Z, \dots, K]}{\partial X} = \beta$$

Thus, β is the effect of a small change in X on the *expected value* of Y . We sometimes refer to β as the *marginal effect* of X on Y (it is clear from the previous equation why it is the “marginal” effect), or just the *effect* of X on Y . A few observations:

- For a dummy variable ($X \in \{0, 1\}$), β is akin of “turning on” the variable from 0 to 1 (since the “slope” from 0 to 1 is constant and thus equals the derivative).
- We write the model under the assumption that X is a *cause* of Y , by which we mean that if we could modify the value of X , $\mathbb{E}[Y]$ would change as well (Pearl 2009). However, $\hat{\beta}$ is not automatically an estimate of that causal effect. For the OLS estimate $\hat{\beta}$ to have a causal interpretation, we will need to understand how to draw inferences with OLS (more in the next chapter).

Example 3.1 (Illustrating OLS: quantifying the effect of working from home). Does working from home increase or reduce employee performance? This is what N. Bloom et al. (2015) wanted to study. They partnered with Ctrip, a Chinese travel agency. Ctrip wanted to see if letting people at their call center work from home would affect performance. If working from home doesn’t hurt, then it would also be an opportunity for Ctrip to reduce the rental space they need for their call centers.

The experiment took place during a period of 9 months. Volunteers were either assigned to working from home (treatment group) or continued to work at the office (control group). The team of economists working with Ctrip recorded a bunch of performance metrics, which would serve as dependent variables. This includes an overall performance score (standardized to have mean zero and standard deviation of 1), the (log of the) number of calls per minute as well as the (log of the) time spent on the phone. Using logs is something we will discuss later. For now, suffice to say that in a log-linear model (logged DV, untransformed treatment), we can interpret it as follows: the treatment changes the expected value of the outcome by $100 \cdot \hat{\beta}$ percent.

The models N. Bloom et al. (2015) estimate are:

$$\text{Employee performance}_{i,t} = \beta \text{Treatment} + \gamma_i + \tau_t + \varepsilon_{i,t},$$

where γ and τ are fixed effects (which we will discuss later) and β is our treatment effect. The models are estimated with OLS. (We include the subscripts i for employees and t for weeks, since the researchers recorded performance over 9 months.)

The results are reported in Table 3.2. The numbers on the *Treatment* row are $\hat{\beta}$ for the effect of the treatment (letting people work from home) on overall performance (column 1), calls per minute (log) (column 2), and minutes on the phone (log) (column 3). We see that the intervention increased overall performance by 0.23 units on the z-scale. More concretely, it increased calls per minute by about 12 percent and minutes spent on the phone by about 9 percent. Thus, working from home increased performance across the board.

The result? Ctrip ended up giving the option of working from home to the entire firm.

	Overall	Calls/min (log)	Min on phone (log)
Treatment	0.23	0.12	0.09
S.E.	(0.06)	(0.03)	(0.03)
N emp.	249	134	134
Obs	17,806	9,426	9,426

Table 3.2: Extract from N. Bloom et al. (2015), Table II. Dependent variables listed at the top of the column. The treatment equals 1 if a worker was allowed to work from home and zero otherwise.

3.5.2 Log transformation

Natural logarithmic transformations are popular in economics (in what follows, $\log()$ always refers to the *natural* log, except otherwise stated). Part of the reason is that there are theoretical quantities of interest that lend themselves well to being expressed as logs. For instance, a Cobb-Douglas model in which a firm i 's production is a

function of labor L and capital K :⁴

$$y_i = L_i^\alpha K_i^\beta.$$

This model isn't linear in parameters (α and β are in the exponent). However, we can transform it:

$$\log(y_i) = \alpha \log(L)_i + \beta \log(K)_i,$$

which can be estimated with OLS. Similarly, economists want to know how geographical distance affects international trade. They developed the *gravity model* of trade precisely to capture the notion that (a) larger countries will trade more with each other, and (b) distant countries will trade less. Mimicking Newtonian models, they could write:

$$\text{Trade}_{ij} = \alpha \text{GDP}_i^\beta \text{GDP}_j^\gamma \text{Distance}_{ij}^\kappa \varepsilon_{ij},$$

where ε_{ij} is a stochastic error term with conditional expected value equals 1:

$$\mathbb{E}[\varepsilon_{ij} | \text{GDP}_i, \text{GDP}_j, \text{Distance}_{ij}] = 1$$

You can make this a linear expression by taking the log:

$$\log(\text{Trade})_{ij} = \alpha + \beta \log(\text{GDP})_i + \gamma \log(\text{GDP})_j + \kappa \log(\text{Distance})_{ij} + \log(\varepsilon_{ij}).$$

Another reason for using logs is that some important economic variables are very (right-)skewed. For instance, some variables have outliers with very large values (e.g., GDP). Logging these variables “tames” them. We will see later that OLS can be vulnerable to outliers (on the covariates), so that using logs reduces that problem.

The challenge lies in the interpretation of the parameters when a variable is log-transformed. We can distinguish four cases:

1. Linear-linear. This is what we saw so far.
2. Linear-log
3. Log-linear
4. Log-log

Linear-log. This corresponds to:

$$y = \alpha + \beta \log(X) + \varepsilon$$

In a multiplicative model, a disturbance term that is “neutral” is one that multiplies the rest on average by one.

~~The additive model is interpreted that~~
~~the error term is a perturbation~~

Silva and Tenreyro (2006) note that, by Jensen's inequality, we have

$\mathbb{E}[\log(X)] \leq \log(\mathbb{E}[X])$. This matters for the error term: if the variance of the error term (pre-log) depends on other variables (here: GDP, distance), then the log-transformed version of the error term might have a non-zero conditional expected value. What we discuss here works if error terms are unrelated to the vector $\mathbf{X}$ of covariates. See Silva and Tenreyro (2006), footnote 2.

⁴See Kenneth Benoit's excellent set of notes: [here](#).

Recall that β is the slope, which here corresponds to the effect of increase $\log(X)$ by 1. But note:

$$1 = \log(e),$$

where e is Euler's number (about 2.72). Thus, by the properties of log:

$$\log(X) + 1 = \log(X) + \log(e) = \log(eX).$$

Thus, β is the same as the change in $\mathbb{E}[Y]$ caused by multiplying X by 2.72, or, equivalently, increasing it by 172%.

Now, this is not typically the kind of change in X we are interested in. More commonly, we are interested in an increase of X by, say, 1% or 10% or 100%. It entirely depends on the context of your treatment. You should always pick a Δ (change in the treatment) that is economically meaningful, even for the linear-linear model.

The solution is straightforward: instead of adding 1 (i.e., $\log(e)$) (which multiplies X by e), we can add $\log(1 + p)$ (which multiplies X by $1 + p$) for whatever p we are interested in (p is the percentage change we want to study, e.g. 0.3 for 30%). For instance, if we want to estimate the increase of X by 20%, we would use $\log(1 + 0.2) = \log(1.2)$ as the increase:

$$\text{Effect of increasing } X \text{ by } 20\% = \beta \cdot \log(1.2).$$

Here, there's a small trick that is useful to calculate this, since we typically don't compute logs in our heads. For small values of p , we can use the following approximation:

$$\log(1 + p) \approx p.$$

Thus, we can straightforwardly capture the effect of a p percentage increase in X :

$$\text{Effect of increasing } X \text{ by } p \approx p \cdot \beta.$$

For instance, if $p = 0.01$ (i.e., an increase of X by 1%), then:

$$\text{Effect of increasing } X \text{ by } 1\% \approx \frac{\beta}{100}.$$

Note that this approximation is off for larger increases.

Log-linear. This corresponds to:

$$\log(y_i) = \alpha + \beta X_i + \varepsilon_i$$

By definition:

$$\beta = \log(y|X + 1) - \log(y|X),$$

which we can rewrite (for readability) as y being a function of X ($y(X)$ reads: the expected value of Y at X):

$$\beta = \log(y(X + 1)) - \log(y(X)),$$

By the properties of logs (difference of logs is the log of their ratio), we can rewrite:

$$\beta = \log\left(\frac{y(X + 1)}{y(X)}\right).$$

Let's exponentiate:

$$\exp(\beta) = \frac{y(X + 1)}{y(X)}.$$

Note that the right-hand side is the ratio of the “new” over the “old” value of Y (“new” meaning the expected value of y when X has been increased by one unit). For instance, if the ratio equals 1.2, then the new value is 1.2 times the old value. If we subtract 1, we can interpret this as a percentage change. For a ratio of, say, 1.2, it means that Y increases on average by 20% ($1.2 - 1 = 0.2$) when X increases by one unit. Thus, β will tell us something about the percentage change in Y .

Again, there is a nifty trick here, which is that:

$$\exp(\beta) \approx 1 + \beta.$$

Thus, a one unit change in X corresponds to multiplying Y by $1 + \beta$, which corresponds to a change of Y by $100 \cdot \beta$ percent. For instance, if $\beta = 0.2$, then a one-unit increase of X multiplies Y by 1.2, that is, Y increases by 20% ($0.2 \cdot 100$).

If we want to use a different change in X (instead of 1 unit), we simply multiply the exponential. For instance: $\exp(2\beta)$ tells us the effect of increasing X by 2 units.

Log-log. This corresponds to:

$$\log(y_i) = \alpha + \beta \log(X_i) + \varepsilon_i.$$

We can combine the logic of the log-linear and linear-log models. If we increase X by p percent, we write:

$$\beta \cdot \log(1 + p).$$

Define $k = 1 + p$ (that is, k is what you multiply X by). Then:

$$\frac{y(kX)}{y(X)} = \exp(\beta \log(k)) = k^\beta.$$

For instance, taking $p = 0.01$ (and thus $k = 1.01$), we obtain:

$$\frac{y(1.01 \cdot X)}{y(X)} = 1.01^\beta$$

So for an increase of X by 1%, Y is multiplied by 1.01^β , meaning Y changes by $100 \times (1.01^\beta - 1)$ percent. For small p (so $k = 1 + p$ is close to 1), we use $k^\beta = e^{\beta \ln k} \approx 1 + \beta \ln k \approx 1 + \beta p$, giving:

$$100 \times (k^\beta - 1)\% \approx 100 \beta p\%.$$

Taking $k = 1.01$ (so $p = 0.01$, a 1% increase in X): the percentage change in Y is approximately $100 \cdot \beta \cdot 0.01\% = \beta\%$. Thus a 1% increase in X corresponds to approximately a $\beta\%$ change in Y .

This is why β is often referred to as an **elasticity**. In economics, the elasticity price of demand is the change in quantity demanded (in percent) when the price increases by 1 percent. The log-log model captures precisely this.

To sum up (keeping in mind that some of these results are approximations valid for small changes only):

- Linear: when X increases by one *unit*, Y changes by β *units* (on average).
- Linear-log: when X increases by one *percent*, Y changes by $\beta/100$ *units* (on average).
- Log-linear: when X increases by one *unit*, Y changes by $100 \cdot \beta$ *percent* (on average).
- Log-log: when X increases by one *percent*, Y changes by β *percent* (on average).

Example 3.2 (Log-log model illustration). In a highly cited paper, Mankiw, Romer, and Weil (1992) try to explain why some countries have higher GDP per capita than others. Building on a famous model by Solow (1956), they test the conjecture that the key driver for economic growth is investments (i.e., capital accumulation).

Using subscript c for country, they estimate a model of the fol-

lowing kind:

$$\log \left(\frac{\text{GDP}}{\# \text{ working-age people}} \right)_c = \alpha + \beta \log \left(\frac{\text{Investment}}{\text{GDP}} \right)_c$$

This is a log-log model. The OLS estimates are reported in Figure 3.1. Looking at non-oil producing countries (column 1), they found that:

- $\hat{\alpha} = 5.48$ (that's *CONSTANT*)
- $\hat{\beta} = 1.42$ (that's $\ln(I/GDP)$)

The latter is what we were looking for: it says that for every additional percent of investments (normalized by GDP), GDP per worker increases by 1.4 percent on average.

TABLE I
ESTIMATION OF THE TEXTBOOK SOLOW MODEL

Dependent variable: log GDP per working-age person in 1985			
Sample:	Non-oil	Intermediate	OECD
Observations:	98	75	22
CONSTANT	5.48 (1.59)	5.36 (1.55)	7.97 (2.48)
$\ln(I/GDP)$	1.42 (0.14)	1.31 (0.17)	0.50 (0.43)
$\ln(n + g + \delta)$	-1.97 (0.56)	-2.01 (0.53)	-0.76 (0.84)
R^2	0.59	0.59	0.01
<i>s.e.e.</i>	0.69	0.61	0.38

Figure 3.1: Source: Mankiw, Romer, and Weil (1992, 414).

3.5.3 Nonlinear treatment

Suppose you have a continuous treatment, or one that takes more than two values. A linear regression model $y = \alpha + \beta X$ offers a linear approximation of the effect of X on Y . But this might not be likely. Maybe we have theoretical reasons to expect nonlinearities that we cannot easily model. Or maybe we just don't want to assume linearity. There are two approaches to handle this. First, it makes sense to **dichotomize** or **bin** the treatment. Second, we can make the model nonlinear by including polynomials of the variable in question.

Regarding the binning approach, we move from:

$$Y_i = \alpha + \beta X_i$$

to:

$$Y_i = \alpha + \beta \mathbb{I}[X_0 \leq X_i < X_1] + \gamma \mathbb{I}[X_1 \leq X_i < X_2] \dots$$

where $\mathbb{I}$ is an indicator that equals 1 if the condition inside the brackets is met. This allows the effect of X to vary based on the range in which it is located. Note that this is akin to making dummies for each group. For instance, if X is years of schooling, we would have a dummy for 10 years, another for 11 years, etc. (with one category omitted to serve as baseline).

3.5.4 Polynomial models

The interpretation of point estimates can become more complicated with nonlinear models and must be adapted accordingly. By “non-linear,” we mean nonlinearity in *variables*. The linear model remains linear in *parameters*.

One type of nonlinear model is the **quadratic model**. This captures situations in which the marginal effect of a variable varies according to its starting value.

Concretely, write:

$$\mathbb{E}[Y|X] = \alpha + \beta X + \gamma X^2.$$

This yields a derivative that is a function of X :

$$\frac{d\mathbb{E}[Y|X]}{dX} = \beta + 2\gamma X.$$

In such a case, the effect of X is not constant and depends on ‘where’ we are with respect to X .

Something that might be interesting is to estimate the max or min, which is:

$$-\frac{\beta}{2\gamma}.$$

Of course, nothing (aside from data) prevents you from adding higher terms. You could have a cubic polynomial, for instance, if you want:

$$\mathbb{E}[Y|X] = \alpha + \beta X + \gamma X^2 + \lambda X^3.$$

Example 3.3 (Illustrating the polynomial model: cubic treatment effect). Fang, Keane, and Silverman (2008) examine how a healthcare insurance called Medigap affects average medical

spending. Their model is:

$$\begin{aligned}\text{Medical spending (USD)}_i &= \alpha + \beta \text{Medigap}_i + \gamma \text{Age}_i \\ &\quad + \lambda \text{Age}_i^2 + \phi \text{Age}_i^3 + \varepsilon_i,\end{aligned}$$

where age is a person's age *minus* 65 (so that the polynomial is centered on zero when people become eligible for this healthcare program). The main treatment effect is to reduce average medical spending by USD 4,393. But what is of interest to us here is the cubic polynomial of age. The results are reported in **?@tbl-fangetal2008**.

As we see (column 1), we have: $\hat{\gamma} = 388$, $\hat{\lambda} = 1.9$, and $\hat{\phi} = 0.12$. The marginal effect of one year is:

$$\frac{\partial \widehat{\text{Spending (USD)}}_i}{\partial \text{Age}_i} = \gamma + 2\lambda \text{Age}_i + 3\phi \text{Age}_i^2$$

Thus, we can compute that for someone who is 67 years old (two years after 65), the marginal effect of getting older is:

$$\frac{\partial \widehat{\text{Spending (USD)}}_i}{\partial \text{Age}_i} \Big|_{\text{Age}=2} = 388 + (2 \cdot 1.9 \cdot 2) + (3 \cdot 0.12 \cdot (2^2)) \approx 397,$$

or an increase in medical spending by USD 397.

In contrast, the marginal effect for someone who is 85:

$$\frac{\partial \widehat{\text{Spending (USD)}}_i}{\partial \text{Age}_i} \Big|_{\text{Age}=20} = 388 + (2 \cdot 1.9 \cdot 20) + (3 \cdot 0.12 \cdot (20^2)) \approx 608,$$

or an increase in medical spending by USD 608 on average.

TABLE 2
ORDINARY LEAST SQUARES REGRESSION RESULTS OF TOTAL MEDICAL EXPENDITURE ON
“MEDIGAP” COVERAGE IN THE MCBS

VARIABLE	A. WITHOUT HEALTH CONTROLS			B. WITH DIRECT HEALTH CONTROLS		
	All (1)	Female (2)	Male (3)	All (4)	Female (5)	Male (6)
Medigap	-4,392.7*** (346.5)	-6,037.4*** (455.5)	-1,863.4*** (538.8)	1,937.0*** (257.2)	1,677.3*** (348.0)	2,420.9*** (395.8)
Female	270.0 (356.2)	. . .	. . .	-751.6*** (283.3)	. . .	. . .
Age - 65	387.5*** (138.0)	460.6*** (175.5)	292.9 (228.5)	394.5*** (117.2)	417.5*** (144.6)	355.4* (196.8)
(Age - 65) ²	1.9 (10.6)	-1.8 (13.2)	5.6 (18.8)	-27.5*** (9.2)	-32.0*** (11.4)	-22.8 (16.2)
(Age - 65) ³	.12 (.22)	.17 (.27)	.07 (.43)	.47** (.21)	.55** (.25)	.47 (.38)
State dummy	Yes	Yes	Yes	Yes	Yes	Yes
Year dummy	Yes	Yes	Yes	Yes	Yes	Yes
Observations	15,945	9,725	6,220	14,129	8,371	5,758
Adjusted R ²	.073	.092	.060	.211	.196	.252

NOTE.—The dependent variable is total medical expenditure. All regressions are weighted by the cross-section sample weights. Health controls included in panel B are described in detail in the Data Appendix under the category Health. A total of 71 health indicators are included. Robust standard errors clustered at the individual level are in parentheses.

* Significant at 10 percent.

** Significant at 5 percent.

*** Significant at 1 percent.

3.5.5 Interaction models

In the linear model $y = \alpha + \beta X$, we are making an implicit assumption: the average treatment effect β is constant. But this seems often implausible. For instance, consider these examples:

- The effect of an ad on a product’s sales might differ for young adults versus older customers.
- The effect of a new technology on a country’s output (GDP) might differ for a democracy versus a dictatorship.
- The effect of a policy announcement on a politician’s level of support might differ for women versus men.
- The effect of a Work-from-Home policy on a firm’s performance might differ for a manufacturing firm versus a financial firm.

For these reasons, we may sometimes want to investigate whether the average treatment effect varies by group, which we call **heterogeneous treatment effects**. Here, there are two approaches:

1. You know the groups you want to study. For instance, you want to see whether your map app creates more engagement for car owners versus people who use bikes. This is what we will discuss below.
2. You don’t know the groups and want the data to reveal them. For instance, it could be that your app works well for men aged 27-35 who live in suburbs, but you don’t know this. To reveal heterogeneity, recent developments in machine learning

have been suggested (Athey, Tibshirani, and Wager 2019; Chernozhukov, Demirer, et al. 2018).

Aside from revealing heterogeneity, interaction models are also useful because they allow the construction of models that avoid assumptions about linearity of the conditional expectation function (i.e., $\mathbb{E}[Y|\mathbf{X}]$). We talk here of **saturated models** and we will return to this later.

The starting point to allow heterogeneity is to build an interactive model in which independent variables enter the model independently (we call these “constituent terms”) and as their joint product (“interaction term”) (Brambor, Clark, and Golder 2006):

$$\mathbb{E}[Y|X, Z] = \alpha + \beta X + \gamma Z + \tau XZ.$$

We are interested in the effect of X , which we obtain by taking the partial derivative with respect to X :

$$\frac{\partial \mathbb{E}[Y|X, Z]}{\partial X} = \beta + \tau Z.$$

Here’s what’s key: the effect of an increase of X on $\mathbb{E}[Y]$ will depend on the value of τZ . This differs from the simpler linear model, in which the effect of X was β and did not depend on anything else. Now, the effect of an increase of X will depend not only on β , but also on τZ . Since τ is a constant (and presumably non-zero), the effect of X will vary depending on the value of Z . If τ is large enough, you can end up in situations in which the effect of X could be, say, positive for some values of Z and negative for others. Using the imaginary examples listed above, you could imagine:

- An ad increases sales among old customers but reduces it for young customers.
- A new technology has zero effect on GDP in dictatorships but a positive effect in democracies.
- etc.

What analysts typically do is to select a few meaningful values of Z and compute the marginal effect of X at these values, using $\hat{\beta}$ and $\hat{\tau}$ estimated via OLS. If Z is a dummy (dichotomous) variable, then this simply boils down to computing the effect of X when $Z = 0$ and then again when $Z = 1$. You can also augment this by computing the value of the marginal effect at a range of values of Z (say, $[Z_0, Z_1]$) and then plotting the marginal effect on the y-axis as a function of Z (x-axis).

Note that you can increase the complexity of these models by interacting more than two variables. You could have:

$$y = \alpha + \beta X + \gamma Z + \lambda W + \tau XZ + \rho ZW + \phi XZW + \varepsilon.$$

The thing to keep in mind is that, once you introduce interaction effects, you want to include all possible combinations. For instance, you would need very good reasons to estimate this model:

$$y = \alpha + \beta X + \tau XZ + \rho ZW + \phi XZW + \varepsilon,$$

where Z and W have been omitted. This imposes weird assumptions that are hard to justify (Brambor, Clark, and Golder 2006).

Example 3.4 (Illustrating interaction effect). In the 1990s, macroeconomist started observing a decoupling between US and European economies. In the US, productivity (measured as output per worker) increased, whereas it didn't in Europe. In the long term, this creates considerable diverging GDP paths.

Is this divergence caused by differing management styles? Or was it caused by the rise of internet and related technologies? N. Bloom, Sadun, and Reenen (2012) looked at the performance of European firms. They asked whether the effect of IT spending was larger for US-owned than for non-US owned firms. One hypothesis is that US managers knew better how to benefit from investments in IT, so that spending one dollar on IT would have a larger effect if done under US than under European (or non-US more generally) management.

After collecting a bunch of data, they estimate the following model:

$$\begin{aligned} \text{Output/worker (log)}_i &= \beta \text{US owner}_i + \gamma \text{IT/worker (log)}_i \\ &\quad + \lambda \beta \text{US owner}_i \cdot \text{IT/worker (log)}_i \\ &\quad + \dots + \varepsilon_i, \end{aligned}$$

where i denotes a firm.

The effect of IT spending then is as follows. For a non-US owned firm based in Europe:

$$\frac{\partial \text{Output/worker (log)}_i}{\partial \text{IT/worker (log)}} = \gamma;$$

and for a US-owned, Europe-based firm:

$$\frac{\partial \text{Output/worker (log)}_i}{\partial \text{IT/worker (log)}} = \gamma + \lambda$$

The results are reported in Table 3.3 in which we add variables one after the other. Given the log, the interpretation is a bit tricky (see logs elsewhere in this chapter). But here, we can say the following:

- A firm that is US-owned is 6-7 percent more productive than a Europe-owned one (models 1 and 2).
- Roughly doubling the amount of IT per worker increases productivity by 5 percent, regardless of the owner's origin (model 2)
- Roughly doubling the amount of IT per worker for a European owner increases productivity by 4 percent for a European-owned firm and by 6 (2+4) percent for a US-owner firm.

From this and from other data, the authors conclude that “Americans do it better”, that is, are more efficient in turning IT into output (at least in the 2000s).

	M1	M2	M3
US owner	0.07	0.06	0.07
IT capital (log)		0.05	0.04
IT (log) * US owner			0.02
Obs	21,746	21,746	21,746

Table 3.3: Extract from N. Bloom, Sadun, and Reenen (2012), Table 2 (only reporting the parameters of three variables). Analysis of Europe-based firm performance. The dependent variable is a firm's output per worker (logged). *IT capital* is IT spending per worker (logged). *US owner* indicates whether a firm is owned by a US firm (yes=1; 0 if the firm is owned by Europeans).

There is another reason to use interaction effect models. Under some conditions, interaction models allow us to relax the assumption of linearity of the *conditional expectation function*. Consider the case in which D is a dummy treatment. If our model is:

$$y_i = \alpha + \beta D_i + \varepsilon_i,$$

then there is no problem with linearity. We can compute $\mathbb{E}[Y|D = 0]$ and $\mathbb{E}[Y|D = 1]$. There is no other values of D (say, $D = 3$) that could introduce nonlinearity.

But now, suppose you want to include an additional independent variable $Z \in \{0, 1\}$:

$$\mathbb{E}[Y|D] = \alpha + \beta D + \gamma Z$$

Is the conditional expectation function still linear? Not necessarily, though we are here assuming it is. The problem is the following. We can compute the following expected values for Y :

$$\begin{aligned}\mathbb{E}[Y|D = 0, Z = 0] &= \alpha \\ \mathbb{E}[Y|D = 1, Z = 0] &= \alpha + \beta \\ \mathbb{E}[Y|D = 0, Z = 1] &= \alpha + \gamma \\ \mathbb{E}[Y|D = 1, Z = 1] &= \alpha + \beta + \gamma\end{aligned}$$

Here's the key: the treatment effect is β regardless of the value of Z . To see this, take the difference between the 4th and the 3rd expression (the treatment effect of X conditional on $Z = 1$): it's β . Then take the 2nd minus the 1st expression (where we condition on $Z = 0$): it's β again. Thus, this *additive* model leaves no room for variation in treatment effect. But this is the same as *assuming* linearity of the conditional expectation function.

To relax this assumption, we would like to allow a unique estimate for each combination of independent variables. We can achieve this by including interaction effects:

$$\mathbb{E}[Y|D] = \alpha + \beta D + \gamma Z + \lambda(D \cdot Z)$$

This allows us to generate distinct treatment effects depending on Z :

$$\begin{aligned}\mathbb{E}[Y|D = 0, Z = 0] &= \alpha \\ \mathbb{E}[Y|D = 1, Z = 0] &= \alpha + \beta \\ \mathbb{E}[Y|D = 0, Z = 1] &= \alpha + \gamma \\ \mathbb{E}[Y|D = 1, Z = 1] &= \alpha + \gamma + \beta + \lambda\end{aligned}$$

Note now that the ATE when $Z = 1$ is $\beta + \lambda$, whereas it is β when $Z = 0$. This allows the treatment effect to vary. We call this a **saturated model**. In such models, there is no problem of linearity *by construction*. Note of course that this can only be done in models with discrete and finite set of categories (like dummy variables). Linearity problems return when the covariates are continuous.

3.6 Multicollinearity

3.6.1 Perfect multicollinearity

So far, the only assumption we made to derive OLS estimates is that $\mathbf{X}'\mathbf{X}$ is invertible. This, as we saw, is the same as saying that $\mathbf{X}$ must be of full rank, which in turn is the same as saying that no variable is a linear combination of others. The assumption we make here is the absence of **perfect multicollinearity**. Without it, we cannot compute a minimum to the least squares problem and we are stuck.

There are two reasons why this assumption may not be fulfilled:

1. Incorrectly specified model.
2. Insufficient data and bad (or unlucky) sampling.

The former is a problem of *theory*. The latter is a data problem. Let's clarify. First, we may face the problem of building up a model that contains variables that, by construction, are perfectly collinear. For instance, assuming a world with two genders for simplicity, we may try to estimate:

$$\text{Wage}_i = \alpha + \beta \text{Female}_i + \gamma \text{Male}_i + \varepsilon_i.$$

Here, $\mathbf{X}$ is not of full rank: if a unit is a female (=1), then we know by construction that male=0, and vice versa. Thus, the two variables perfectly predict each other. Likewise, if we do an analysis of financial flows and want to control for seasonality (and using daily data indexed by t):

$$\text{CAPEX}_t = \alpha + \beta T_t + \gamma \text{Winter}_t + \rho \text{Spring}_t + \phi \text{Summer}_t + \lambda \text{Fall}_t + \varepsilon_t$$

If the observation is measured in winter, you know by construction that the other variables will be zero.

The solution here is easy: modify your model to remove one of the variable. This has no consequence. For instance:

$$\text{Wage}_i = \alpha + \beta \text{Female}_i + \varepsilon_i.$$

Then, β is the difference between $\mathbb{E}[Y|\text{Male}]$ and $\mathbb{E}[Y|\text{Female}]$. Furthermore, note that $\mathbb{E}[Y|\text{Male}] = \alpha$. We call *male* the **omitted category** (or the **baseline category**), but note that while it's omitted, we still recover the expected value of Y for that category: it's the y-intercept.

The second reason for perfect multicollinearity is an “unlucky” sample. For instance, if we tried to get data on wages and gender, we

might randomly end up with a sample that only contains women. Then, we couldn't estimate the model, since there is no variance on that variable. The solution here, typically, is to collect more data.

A version of this story that is less about lack of luck is a poor sampling design. For instance, if you want to study the effect of being above 65 years old (=1) on public opinion about a product (adjusting for having a college degree), then do not run your survey on the grounds of a university, since you might only end up with young people with a college degree.

3.6.2 High multicollinearity

High multicollinearity – a large but not perfect correlation between independent variable(s) – is *not* a violation of the Gauss-Markov or of any assumption in the linear model and OLS. As such, we don't require anything aside from a lack of *perfect* multicollinearity when we work with these models and estimation methods.

This being said, high multicollinearity can still be annoying. The issue is *not* that it will create bias or inconsistency. Nothing about these properties require low multicollinearity. No, the problem is that high multicollinearity will make $(\mathbf{X}'\mathbf{X})^{-1}$ larger. And since this is part of the expression of the variance of $\hat{\beta}$, this means that high multicollinearity will reduce t statistics and make it more difficult to detect a statistical effect. More on this later.

To illustrate the problem, consider the following example. Suppose that we want to measure the effect of people's height on their wages, and we want to adjust for gender. We would build the following model:

$$\text{Wage}_i = \alpha + \beta \text{Height}_i + \gamma \text{Male}_i + \varepsilon_i$$

But suppose that our data almost only contains tall men and short women. The two variables will be highly, though not perfectly, correlated. Then it will be very hard to tease out the effect of height versus the effect of gender.

You can check whether your model suffers from high multicollinearity. The most common test is the **Variance Inflation Factor** (VIF). Consider a model such as:

$$y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + \varepsilon.$$

Then, variable x_i 's VIF is:

$$\text{VIF}_{x_i} = \frac{1}{1 - R_i^2},$$

where R_i^2 is the R-squared of a regression of x_i on all other independent variables x . Typically, people start to worry when VIF is greater than 10. But there's no hard rule about this.

What's the solution to high multicollinearity? To the degree that the problem is a lack of independent variation across variables, the solution will be to add more data.

What you should *not* do is to drop the variable(s) that is (are) causing the problem. A variable's inclusion in a model is based on theoretical grounds. Either it is a confounder or it isn't. The fact that it's "annoying" to have a confounder that's highly correlated with your treatment isn't a good justification to remove it.

Review questions

Exercise 3.1. Derive the ordinary least *cubes* estimator. To simplify, consider the model $y = \beta x$ (no intercept). What is it? Is there a closed-form solution? Bonus: implement this estimator in R or Python.

Exercise 3.2. When we bin variables into dummy indicators (see nonlinear models), why do we omit one category?

Appendix

Matrix algebra

Matrix algebra

A quick refresher on matrices and matrix algebra. Matrices are indexed by two dimensions, row $\times$ columns, and are often referred to in capitalized letters in bold fonts. An $n \cdot k$ matrix $\mathbf{X}$ corresponds to:

$$\mathbf{X} = \begin{bmatrix} x_{1,1} & x_{1,2} & x_{1,3} & \cdots & x_{1,k} \\ x_{2,1} & x_{2,2} & x_{2,4} & \cdots & x_{2,k} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{n,1} & x_{n,2} & x_{n,3} & \cdots & x_{n,k} \end{bmatrix}_{n \times k}$$

A matrix in which one of the dimensions equals 1 is a vector (row vector if it is of length $1 \times k$, column vector if $n \times 1$). Since column vectors are more common, it is typically assumed that a “vector” refers to a the latter. Vectors are sometimes labeled in lower-case bold fonts (e.g., $\mathbf{x}$).

The transpose of a matrix flips the rows and columns, such that entry $x_{i,j}$ becomes $x_{j,i}$ for all i, j . It is denoted $\mathbf{X}^\top$ or $\mathbf{X}'$.

$$\mathbf{X} = \begin{bmatrix} A_{1,1} & B_{1,2} & C_{1,3} \\ D_{2,1} & E_{2,2} & F_{2,4} \end{bmatrix}_{2 \times 3}$$
$$\mathbf{X}' = \begin{bmatrix} A_{1,1} & D_{1,2} \\ B_{2,1} & E_{2,2} \\ C_{3,1} & F_{3,2} \end{bmatrix}_{3 \times 2}$$

Addition. Matrices can be added as long as they are **conformable**, i.e., have the same dimensions.

Multiplication. To multiply matrices, we need the concept of **inner product** (or **scalar product**).⁵ Consider two vectors, $\mathbf{x}$ and $\mathbf{z}$, the inner product is:

$$\mathbf{x}'\mathbf{z} = \sum_i^n x_i z_i = \mathbf{z}'\mathbf{x} = c,$$

where c is a scalar.

The product of two matrices yields a new matrix, whose entry i, j will be the inner product of the i th row of the first matrix and the j th column of the second matrix. (For this to work, we

need the first matrix to have the same number of columns as the second matrix's number of rows.) We then end up with:

$$\mathbf{X}_{n \times k} \mathbf{Z}_{k \times l} = \mathbf{W}_{n \times l}$$

Properties of addition, products, and transposes:

- $\mathbf{X}(\mathbf{W} + \mathbf{Z}) = \mathbf{XW} + \mathbf{XZ}$
- $(\mathbf{XZ})' = \mathbf{Z}'\mathbf{X}'$

Inverse. The inverse of a matrix, written $\mathbf{X}^{-1}$, is a matrix that satisfies the following property:

$$\mathbf{X}\mathbf{X}^{-1} = \mathbf{X}^{-1}\mathbf{X} = \mathbf{I},$$

where $\mathbf{I}$ is the **identity** matrix, a square $(l \times l)$ matrix, with diagonal entries $\mathbf{I}_{i,i} = 1$ and off-diagonal entries $\mathbf{I}_{i,j} = 0$. The logic is the following: if you multiply a scalar by its inverse, you get 1 (e.g., $8 \cdot 1/8 = 1$). Likewise, the inverse is what you need to multiply a matrix $\mathbf{X}$ to obtain the identity matrix. Note that the inverse may not exist.

⁵Not to be confused with **direct product** in which x_{ij} is multiplied by $z_{i,j}$ for all i, j .

4 Inference with OLS

This version: September 29, 2026

Overview

We explain how OLS can be used to make inferences about unknown population parameters.

Readings

Here are references on least squares and (causal) inference: (B. Hansen 2022a) (ch. 6-7, and 9), (Stachurski 2016) (ch. 13.1), (Wooldridge 2009) (ch. 4-5), (Verbeek 2018) (ch. 2.4-2.6 and 4), (Hayashi 2000) (ch2.3-2.4), (Stock and Watson 2011) (ch. 5-7).

4.1 Performance of estimators

Our main result ($\hat{\beta}_{ols} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$) did not require any assumption aside from the matrix $\mathbf{X}$ being of full rank. Yet this only guarantees the *existence* of a unique vector $\hat{\beta}_{ols}$ that solves the minimization problem in a given **sample**. It doesn't tell us anything about whether this rule has desirable properties and allows us to say anything about the true, unobservable, **population** vector β .

The properties we may want to achieve are:

- **Unbiasedness.** Note that $\hat{\beta}_{ols}$ is a random variable, since it is the product of random variables. Unbiasedness refers to the property that the expected value of $\hat{\beta}$ obtained via some estimator equals the true population value of β , namely:

$$\mathbb{E}[\hat{\beta}] = \beta$$

- **Consistency.** We may want the distribution of our estimate $\hat{\beta}$ to get increasingly more precise around the true value β as our sample size goes toward infinity. A consistent estimate is one that converges in probability to β . That is:

$$\text{plim} \hat{\beta} = \beta$$

- **Efficiency.** Due to sampling variation, estimators will generate estimates that vary from sample to sample. What we would like is for the estimator to have the smallest possible variance compared to any alternative estimator $\tilde{\beta}$. Typically, we restrict the competition to *unbiased* alternative estimators.

$$\mathbb{V}[\hat{\beta}] \leq \mathbb{V}[\tilde{\beta}], \forall \text{ unbiased } \tilde{\beta}$$

Note. An estimator can be biased yet consistent. For instance, the estimator for the population mean: $\hat{\mu}_x = \frac{1}{n+1} \sum x$ is biased but consistent: as n increases, the bias disappears.

Note. An estimator can be unbiased but inconsistent. For instance, if you estimate the population mean by picking the first value from your sample, your estimate would be unbiased, but the distribution of the error this estimator isn't converging to the true population mean.

What we want to figure out is: under what conditions is OLS unbiased, consistent, and/or efficient?

Unbiasedness: example

Let us establish that the sample average $\bar{x}$ is an unbiased estimator of the population expectation $\mathbb{E}[X]$. We define:

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

What we need to establish is that $\mathbb{E}[\bar{x}] = \mathbb{E}[X]$. To do so, we

take expectations of the previous expression:

$$\begin{aligned}\mathbb{E}[\bar{x}] &= \mathbb{E}\left[\frac{1}{N} \sum_{i=1}^N x_i\right] \\ &= \frac{1}{N} \mathbb{E}\left[\sum_{i=1}^N x_i\right] && \text{extracting constant} \\ &= \frac{1}{N} \sum_{i=1}^N \mathbb{E}[X] \\ &= \frac{1}{N} N \mathbb{E}[X] && \text{n times } \mathbb{E}[X] \\ &= \mathbb{E}[X]\end{aligned}$$

which completes the proof.

4.2 Assumptions so far

So far, the only assumptions we made to derive the OLS estimator is to state that the design matrix $\mathbf{X}$ is of full rank (no perfect multicollinearity) with probability 1 (**A4**).

Let us add a few additional assumptions, some of which we will relax later.

Throughout, except if otherwise stated, we assume that $\mathbf{y}$ and $\mathbf{X}$ are **random variables**. See earlier notes on measure theory and see **A5**. We will generally also assume that units in our sample are independently drawn from identical distributions (often referred to as the **IID** assumption). We also generally assume that these distributions have finite first moments and finite and positive second moments.

Assumption 5 (IID). *The random variables X, Y are randomly drawn from independent and identical distributions (IID).*

In addition, X, Y have finite first moment and finite nonzero second moments.

Assumption: IID with finite moments

This assumption combines three ideas: random sampling, IID, and properties of the distribution of X and Y . Finite moments

excludes certain types of distributions, such as Cauchy, but it's otherwise not overly restrictive in practice. Later, we will discuss what happens when the variables are not IID (with independence being a focus in particular).

Aside from this, we also start by assuming that $\mathbf{y}$ can be expressed as a linear function of $\mathbf{X}$, where linearity is about the parameters connecting X to Y (A6). For instance:

$$\begin{array}{ll} Y = X^\beta & \text{not linear} \\ Y = \beta X + \gamma X^2 & \text{linear in parameters} \end{array}$$

Later, we will see under what conditions this assumption is plausible.

Assumption 6 (Linearity). *Y can be expressed as a linear function of X , such that:*

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon,$$

where $\mathbf{X}$ can include any transformation (linear or not) of an arbitrary set of random variables.

⚠ Assumption: Linearity

The linearity assumption does not mean that the relation between X and Y is linear. Rather, this relation is linear *in parameters*. This assumption can hold by construction with discrete treatments in saturated models.

4.3 Unbiasedness

We want to show that $\mathbb{E}[\hat{\beta}] = \beta$. To do so, we must find a way to connect $\hat{\beta}$ to the true parameter β . Starting with the estimator, we can write:

$$\begin{aligned} \hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} && \text{from FOC/SOC of OLS, incl. A4} \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \varepsilon) && \text{by A6} \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon && \text{properties of matrices} \\ &= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon && \text{properties of matrices} \end{aligned}$$

We can now take the expectations:

$$\begin{aligned}
\mathbb{E}[\hat{\beta}] &= \mathbb{E}[\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon] \\
&= \mathbb{E}[\beta] + \mathbb{E}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon] && \text{distributing expectations} \\
&= \beta + \mathbb{E}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon] && \text{since } E(c) = c \\
&= \beta + \mathbb{E}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbb{E}[\varepsilon|\mathbf{X}]] && \text{by law of iterated expectations} \\
&= \beta + \mathbb{E}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{0}] && \text{by A7} \\
&= \beta
\end{aligned}$$

which concludes the proof. We needed here to make a new assumption: the conditional expected value of ε given $\mathbf{X}$ equals zero (**A7**) (see Davidson and MacKinnon (2004, 90) or Stock and Watson (2011, 143)).

In sum, under **A6** to **A7**, OLS will yield **unbiased** estimates of the population parameter β .

Assumption 7 (Exogeneity). *Conditional on predictors, the error term equals zero in expectation, that is:*

$$\mathbb{E}[\varepsilon|\mathbf{X}] = 0$$

The expectation could also have gone right through the second expression if we are willing to claim that $\mathbf{X}$ are fixed and not random variables. Then we could just have assumed that $\mathbb{E}[\varepsilon] = 0$. See Davidson and MacKinnon (2004, 90).

⚠ Assumption: exogeneity/zero conditional expectation

This assumption can be built from two ideas. To see this, note that this equivalent to writing:

$$\mathbb{E}[\varepsilon_i|\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_i, \dots, \mathbf{x}_n] = 0.$$

That is, the expected error term for unit i is zero conditional on the values of all independent variables for all units $1, 2, \dots, i, \dots, n$. Now, we can simplify this to the much weaker assumption:

$$\mathbb{E}[\varepsilon_i|\mathbf{x}_i] = 0,$$

but only if we bring in our earlier assumption of independence (**A5**). Recall that we started by assuming that our observations are IID – one of the “I” stands for independence. Thus, our exogeneity assumption, as stated here (i.e., $\mathbb{E}[\varepsilon|\mathbf{X}] = 0$) can be met by combining $\mathbb{E}[\varepsilon_i|\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_i, \dots, \mathbf{x}_n] = 0$ with independence. Note that this assumption implies that $Cor(\varepsilon, \mathbf{X}) = 0$ and that $\mathbb{E}[\mathbf{X}'\varepsilon] = 0$.

4.4 Asymptotic distribution of $\hat{\beta}_{ols}$

To do inference from a sample to the population, we must understand the nature of $\hat{\beta}$. The starting point is that it is a **random variable**, unlike β which is a constant (at least for now). Why does it vary? Because of sampling! From one sample s to another, we will obtain $\hat{\beta}_s$ that are different from each other. See for instance Figure 4.1, which shows how three different samples yield somewhat different values of $\hat{\beta}$ (0.58, 0.87, and 0.9). We call this **sampling variance**.

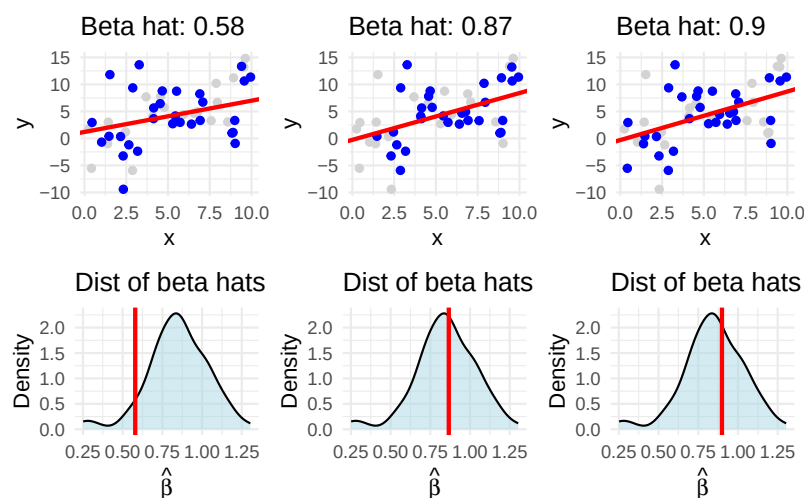


Figure 4.1: **Sampling variance**. Three samples from a common population. 30 observations (blue) are randomly sampled from a population of 50 observations (gray), generating slightly different beta.hat. True model: $y = 0.5 + 0.7x + N(0,5)$. Bottom row: distribution of beta hats from 100 Monte Carlo simulations. The red line reports the beta hat from the figure above.

We are interested in understanding the distribution of $\hat{\beta}$ as N tends to infinity, that is, its **asymptotic distribution**. Take again Figure 4.1: if we could sample over and over again (say, an infinite number of times, or at least a very large number of times), we could build the full distribution of $\hat{\beta}$; this is what the light blue distribution does in the second row of the figure (here: for 100 samples).

The challenge is that we don't know, exactly, what this distribution looks like. We generally cannot re-sample to generate new $\hat{\beta}$.

The workaround will be to see whether we can borrow results from asymptotic statistics to make claims. Concretely (and we will work

through this next), we will borrow the **Lindeberg–Lévy Central Limit Theorem** (CLT) (Theorem 1.7). If you recall, it states that, for a sequence of random IID variables $X_1, X_2, \dots, X_n$, the asymptotic distribution of the centered and scaled sample mean $\bar{X}$ is normal (Gaussian):

$$\frac{\sqrt{N}(\bar{X} - \mu)}{\sigma} \sim \mathcal{N}(0, 1).$$

If we could use this, somehow, for estimates using OLS, then we would know how $\hat{\beta}$ is asymptotically distributed. To top it off, it would be a very nice and well-known distribution, which would allow us to make all kinds of statements regarding the probability of observing ranges of values of $\hat{\beta}$.

The way we will do this is the following: we will try to reformulate $\hat{\beta}$ in a form approaching Lindeberg–Lévy. Then we will see whether we can apply it. In doing so, we will use three theorems from statistics: (1) the Lindeberg–Lévy Central Limit Theory, (2) the Law of Large Numbers, and (3) Slutsky’s Theorem.

Step 1. Let’s reformulate $\hat{\beta}$ in a way that mimics Lindeberg–Lévy. We start with:

$$\begin{aligned} \hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} && \text{by definition} \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \varepsilon) && \text{by A6} \\ &= \beta + (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\varepsilon) && \text{canceling terms} \end{aligned}$$

define the expression $\sqrt{N}(\hat{\beta} - \beta)$ using that last expression:

$$\begin{aligned} \hat{\beta} - \beta &= (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\varepsilon) && \text{shifting terms} \\ \sqrt{N}(\hat{\beta} - \beta) &= \sqrt{N}(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\varepsilon) && \text{multiplying by sqrt N} \\ &= \frac{N}{\sqrt{N}}(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\varepsilon) && \text{sqrt N is same as N/sqrt N} \\ &= N(\mathbf{X}'\mathbf{X})^{-1} \frac{1}{\sqrt{N}}(\mathbf{X}'\varepsilon) && \text{distributing} \\ &= \left(\frac{\mathbf{X}'\mathbf{X}}{N} \right)^{-1} \frac{1}{\sqrt{N}}(\mathbf{X}'\varepsilon) && \text{putting N inside the inverse} \end{aligned}$$

Step 2. Since we want to know what happens in asymptotia, we must study this expression as $N \rightarrow \infty$.

$$\text{plim}[\sqrt{N}(\hat{\beta} - \beta)] = \text{plim} \left[\left(\frac{\mathbf{X}'\mathbf{X}}{N} \right)^{-1} \frac{1}{\sqrt{N}}(\mathbf{X}'\varepsilon) \right]$$

Let us divide this expression in two parts:

$$\left(\frac{\mathbf{X}'\mathbf{X}}{N}\right)^{-1} \quad \text{and} \quad \frac{1}{\sqrt{N}}(\mathbf{X}'\varepsilon)$$

First expression ($\left(\frac{\mathbf{X}'\mathbf{X}}{N}\right)^{-1}$). What will happen to the first expression as N goes to infinity? By the **Law of Large Numbers**, we can write:

$$\frac{\mathbf{X}'\mathbf{X}}{N} \xrightarrow{p} \mathbb{E}[\mathbf{X}'\mathbf{X}] := \mathbf{Q},$$

which is a positive-definite matrix that *converges in probability* to $\mathbf{Q}$.¹ Then, by A4, we know that the inverse of this quantity is defined, that is, $\mathbf{Q}^{-1}$ exists. Informally, this tells us that (a) the firm term converges to something, and (b) it converges in probability.

Second expression ($\frac{1}{\sqrt{N}}(\mathbf{X}'\varepsilon)$). Note that it is a sum of products $\mathbf{X}'\varepsilon$ (scaled by $1/\sqrt{N}$):

$$\mathbf{X}'\varepsilon = \begin{bmatrix} 1_1 & 1_2 & \dots & 1_n \\ x_1 & x_2 & \dots & x_n \\ z_1 & z_2 & \dots & z_n \\ \dots & & & \\ k_1 & k_2 & \dots & k_n \end{bmatrix} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \dots \\ \varepsilon_n \end{bmatrix}$$

At this stage, we can invoke the Lindeberg-Lévy CLT, assuming that ε is IID. The reason is a bit technical, but we can do a short detour to explain it. You may wonder why we can apply Lindeberg-Lévy, since it is generally presented as a result about the sample mean ($\bar{X}$) and not, say, as $\mathbf{X}'\varepsilon$. There are two steps to clarify this:

1. The CLT is actually about the *sum* of random variables, not their (sample) mean. To see this, note that we can rewrite the CLT as:

$$\begin{aligned} \frac{\sqrt{N}(\bar{X} - \mathbb{E}[X_i])}{\sigma} &= \frac{N(\bar{X} - \mathbb{E}[X_i])}{\sqrt{N}\sigma} \\ &= \frac{N\bar{X} - N\mathbb{E}[X_i]}{\sqrt{N}\sigma} \\ &= \frac{\sum(X_i - \mathbb{E}[X_i])}{\sqrt{N}\sigma} \\ &= \frac{\sum X_i - N\mathbb{E}[X_i]}{\sqrt{N}\sigma} \end{aligned}$$

The last expression is equivalent to the one used by Billingsley (1995, 357) in his definition of the Lindeberg-Lévy CLT.

Technically, Lindeberg-Lévy requires ε to be IID. But this assumption can be relaxed. The CLT may also hold if ε is not identically distributed and exhibits dependence. You would need something like the Lindeberg-Feller CLT (with independence across ε) or Billingsley's CLT (with weak dependence)

¹We sometimes write *plim* instead of $\xrightarrow{p}$.

2. From there, take the expression $\sum(X_i - \mathbb{E}[X_i])$ and plug in $\mathbf{X}'\varepsilon$ instead of X . Note that $\mathbf{X}'\varepsilon$ is the same as $\sum^N x_i \cdot \varepsilon_i$ (once you exit matrix notation). Thus, we can write:

$$\sum^N (x_i \cdot \varepsilon_i - \mathbb{E}[x_i \cdot \varepsilon_i])$$

This is a sum of the product of IID variables, centered by the second term! Furthermore, note that $\mathbb{E}[x_i \cdot \varepsilon_i] = 0$ by the exogeneity assumption. This therefore simplifies to:

$$\sum^N (x_i \cdot \varepsilon_i - \mathbb{E}[x_i \cdot \varepsilon_i]) = \sum^N (x_i \cdot \varepsilon_i) \equiv \mathbf{X}'\varepsilon$$

Thus, assuming that the data are IID and that the exogeneity assumption holds, then we can use the CLT on this quantity.

Note here that we can do two extra steps that will be useful. First, we can confirm that, indeed, this second term has a mean of zero (as we would expect based on the CLT). Second, we can express the actual variance of this quantity (to replace the placeholder σ^2 used in the general statement of the CLT). Let's quickly do this and then let's put all pieces back together.

We can obtain the *first moment* of this distribution immediately:

$$\begin{aligned} \mathbb{E} \left[\frac{1}{\sqrt{N}} (\mathbf{X}'\varepsilon) \right] &= \frac{1}{\sqrt{N}} \mathbb{E}[\mathbf{X}'\varepsilon] \\ &= 0 \end{aligned} \quad \text{by A7}$$

This is thus a mean-zero vector.

Let's get to the *second moment*. Ultimately, we will want to find the variance of the whole thing:

$$\mathbb{V} \left[\sqrt{N}(\hat{\beta} - \beta) \right] = \mathbb{V} \left[\mathbf{Q}^{-1} \frac{1}{\sqrt{N}} (\mathbf{X}'\varepsilon) \right]$$

We already noted that $\mathbf{Q}$ is finite and invertible. We can extract it from the variance like a constant by pre- and post-multiplying. Let's do this at the end. Instead, to ensure that this expression is finite (and thus this CLT assumption is met), let us derive the second

part.

$$\begin{aligned}
\mathbb{V}\left[\frac{1}{\sqrt{N}}\mathbf{X}'\varepsilon\right] &= \frac{1}{N}\mathbb{V}[\mathbf{X}'\varepsilon] && \text{property of var} \\
&= \frac{1}{N}\mathbb{V}\left[\sum_{i=1}^N x_i\varepsilon_i\right] && \text{by def of matrix prod} \\
&= \frac{1}{N}\sum_{i=1}^N \mathbb{V}[x_i\varepsilon_i] && \text{by A7} \\
&= \frac{1}{N}N\mathbb{E}[\mathbf{X}'\varepsilon\varepsilon'\mathbf{X}] && \text{back to matrix} \\
&= \mathbb{E}[\mathbf{X}'\varepsilon\varepsilon'\mathbf{X}]
\end{aligned}$$

Here, we need to assume that:

$$\mathbb{E}[\mathbf{X}'\varepsilon\varepsilon'\mathbf{X}] < \infty$$

If so, we can state that:

$$\frac{1}{\sqrt{N}}\mathbf{X}'\varepsilon \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathbf{X}'\varepsilon\varepsilon'\mathbf{X}])$$

Putting it all together. From here, let's put all the pieces back together. Using the Law of Large numbers, we were able to characterize the first expression. Using the CLT, we were able to characterize the second expression. We have:

$$\text{By LLN: } (\mathbf{X}'\mathbf{X}/N)^{-1} \xrightarrow{p} \mathbf{Q}^{-1},$$

and

$$\text{By CLT: } \frac{1}{\sqrt{N}}\mathbf{X}'\varepsilon \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\mathbf{x}_i\mathbf{x}_i'\varepsilon_i^2]).$$

We can combine the two statements using **Slutsky's Theorem** Theorem 4.1 (B. Hansen 2022b, 171).

Theorem 4.1 (Slutsky). *If $A_n \xrightarrow{p} A$ and $B_n \xrightarrow{d} B$ as $n \rightarrow \infty$, then:*

1. $A_n + B_n \xrightarrow{d} A + B$.
2. $A_n B_n \xrightarrow{d} AB$.
3. $\frac{B_n}{A_n} \xrightarrow{d} \frac{B}{A}$ if $A \neq 0$.

The intuition behind Slutsky's Theorem is that a sequence that converges in *distribution* can be combined with a sequence that converges in *probability to a constant*; this combination converges in *distribution* to their sum.

Using the second result from Slutsky, this means that:

$$\begin{aligned}\sqrt{n}(\hat{\beta} - \beta) &= \underbrace{\left(\frac{\mathbf{X}'\mathbf{X}}{N}\right)^{-1}}_{A_n} \underbrace{\frac{1}{\sqrt{N}}(\mathbf{X}'\varepsilon)}_{B_n} \\ &\xrightarrow{d} \mathbf{Q}^{-1} \cdot \mathcal{N}(0, \mathbb{E}[\mathbf{x}_i \mathbf{x}_i' \varepsilon_i^2]) && \text{by Slutsky's theorem} \\ &= \mathcal{N}(0, \mathbf{Q}^{-1} \mathbb{E}[\mathbf{x}_i \mathbf{x}_i' \varepsilon_i^2] \mathbf{Q}^{-1}) && \text{since } \mathbf{A}Z \sim \mathcal{N}(0, \mathbf{A}\Sigma\mathbf{A}') \text{ for } Z \sim \mathcal{N}(0, \Sigma) \\ &:= \mathcal{N}(0, \Omega)\end{aligned}$$

Slutsky's theorem is used constantly in deriving asymptotic distributions. A leading example: when constructing a t -statistic, we divide a quantity with a $N(0, 1)$ limit by a sample standard deviation $\hat{\sigma}$ that converges in probability to the true σ . Slutsky's theorem guarantees the ratio still converges to $N(0, 1)$.

where Ω is defined as $\mathbf{Q}^{-1} \mathbb{E}[\mathbf{x}_i \mathbf{x}_i' \varepsilon_i^2] \mathbf{Q}^{-1}$ and is referred to as the **asymptotic variance-covariance matrix** of $\hat{\beta}$.

We can conclude that under the following assumptions:

- **A4** (full rank): needed to compute $\hat{\beta}$
- **A5** (IID and finite second moment): needed for Lindeberg-Lévy CLT
- **A6** (linearity): needed to link $\hat{\beta}$ to β .
- **A7** (exogeneity): needed for $\hat{\beta}$ to be centered on β (so that $\mathbb{E}[\hat{\beta}] - \beta = 0$)...

that:

$$\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(0, \Omega)$$

Note. $\varepsilon\varepsilon'$ is an $n \times n$ matrix, which is then multiplied by $\mathbf{X}$ (an $n \times k$ matrix). The result is an $n \times k$ matrix. This is then pre-multiplied by $\mathbf{X}'$ (an $k \times n$ matrix), leaving us with a $k \times k$ matrix. Since $(\mathbf{X}'\mathbf{X})$ is a $k \times k$ matrix as well, Ω as a whole is $k \times k$.

Note. We can relax the IID assumption and instead invoke other versions of the CLT, such as the Lindeberg-Feller CLT (independent but not identically distributed ε) or the Billingsley CLT (dependent ε_i , weak dependence, finite variance).

Note. For any individual coefficient $\hat{\beta}_k$, we can standardize: $\frac{\sqrt{N}(\hat{\beta}_k - \beta_k)}{\sqrt{\Omega_{kk}}} \xrightarrow{d} \mathcal{N}(0, 1)$, where Ω_{kk} is the k th diagonal element of Ω . We can also write: $\hat{\beta} \xrightarrow{d} \mathcal{N}(\beta, \frac{\Omega}{N})$.

Note. Sometimes, textbooks use Ω to refer solely to $\mathbb{E}[\mathbf{x}_i \mathbf{x}_i' \varepsilon_i^2]$ and thus write $\mathcal{N}(0, \mathbf{Q}^{-1} \Omega \mathbf{Q}^{-1})$.

From there, we can define the **standard error** of $\hat{\beta}_k$ as the square root of the k th diagonal entry of Ω/N .

4.5 Estimating standard errors

From the asymptotic distribution of $\hat{\beta}$ (or, more precisely, the asymptotic distribution of $\sqrt{N}(\hat{\beta} - \beta)$), we know that:

$$\Omega = \mathbf{Q}^{-1} \mathbb{E}[\mathbf{x}_i \mathbf{x}_i' \varepsilon_i^2] \mathbf{Q}^{-1}$$

The complication comes from ε , which isn't observed, and thus makes the expression $\varepsilon \varepsilon'$ challenging. First, note that this is a $n \times n$ matrix. Taking expectations (and assuming it exists; see **A5**):

$$\mathbb{E}[\varepsilon \varepsilon'] = \begin{bmatrix} \mathbb{E}[\varepsilon_1^2] & \mathbb{E}[\varepsilon_1 \varepsilon_2] & \dots & \mathbb{E}[\varepsilon_1 \varepsilon_n] \\ \mathbb{E}[\varepsilon_2 \varepsilon_1] & \mathbb{E}[\varepsilon_2^2] & \dots & \mathbb{E}[\varepsilon_2 \varepsilon_n] \\ \dots & \dots & \dots & \dots \\ \mathbb{E}[\varepsilon_n \varepsilon_1] & \dots & \dots & \mathbb{E}[\varepsilon_n^2] \end{bmatrix}$$

Note that we can simplify this a tiny bit by noting that $\mathbb{E}[\varepsilon_i^2] = \mathbb{V}[\varepsilon_i]$ (or equivalently σ_i^2):

$$\mathbb{E}[\varepsilon \varepsilon'] = \begin{bmatrix} \mathbb{V}[\varepsilon_1] & \mathbb{E}[\varepsilon_1 \varepsilon_2] & \dots & \mathbb{E}[\varepsilon_1 \varepsilon_n] \\ \mathbb{E}[\varepsilon_2 \varepsilon_1] & \mathbb{V}[\varepsilon_2] & \dots & \mathbb{E}[\varepsilon_2 \varepsilon_n] \\ \dots & \dots & \dots & \dots \\ \mathbb{E}[\varepsilon_n \varepsilon_1] & \dots & \dots & \mathbb{V}[\varepsilon_n] \end{bmatrix}$$

Thanks to the fact that the unconditional error term $\mathbb{E}[\varepsilon] = 0$ by construction.

Note as well that we can similarly replace the off-diagonal entries by using the fact that $\mathbb{E}[\varepsilon_i \varepsilon_j] = \text{Cov}[\varepsilon_i, \varepsilon_j]$:

$$\mathbb{E}[\varepsilon \varepsilon'] = \begin{bmatrix} \mathbb{V}[\varepsilon_1] & \text{Cov}[\varepsilon_1, \varepsilon_2] & \dots & \text{Cov}[\varepsilon_1, \varepsilon_n] \\ \text{Cov}[\varepsilon_2, \varepsilon_1] & \mathbb{V}[\varepsilon_2] & \dots & \text{Cov}[\varepsilon_2, \varepsilon_n] \\ \dots & \dots & \dots & \dots \\ \text{Cov}[\varepsilon_n, \varepsilon_1] & \dots & \dots & \mathbb{V}[\varepsilon_n] \end{bmatrix}$$

This is why $\mathbb{E}[\varepsilon \varepsilon']$ is also called the **variance-covariance matrix of error terms** (it's a matrix of the variances of ε_i and of covariances of ε_i and ε_j).

Note. Ω is also a variance-covariance matrix. Be careful to specify whether you are talking about the variance-covariance matrix of error terms (like $\mathbb{E}[\varepsilon \varepsilon']$) or of parameters (like $\mathbb{V}[\hat{\beta}]$).

But now we are stuck. We have $n(n+1)/2$ unknown parameters to figure out. Here, we may want to simplify this expression via two additional assumptions. The first one is the lack of autocorrelation across units or across time (no autocorrelation); the second is that the error term has a constant and identical variance for all units (homoskedasticity).

Because it's a symmetrical matrix
($Cov(a, b) = Cov(b, a)$).

Assumption 8 (Autocorrelation). *For any two unit $i, j \in N$, where $i \neq j$, we have:*

$$Cor[\varepsilon_i, \varepsilon_j] = 0$$

 Assumption: No autocorrelation

This assumption is typically violated in the following cases:

1. Serial correlation. Error term of a unit measured at time t is correlated with the error term at time $t - 1$. For instance, in finance, a stochastic shock that increases returns of a firm in the first quarter of 2025 may also affect it in the second quarter of 2025.
2. Spatial correlation. Suppose you are doing a study on Swiss municipalities. A high error term in Lausanne may also generate a high error term in Geneva.

We then simplify the variance-covariance matrix of errors as:

$$\mathbb{E}[\varepsilon\varepsilon'] = \begin{bmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & \dots & \dots & \sigma_n^2 \end{bmatrix} \quad \text{via A8}$$

where σ_i^2 is the variance of the error term of unit i .

If we are willing to assume that the variance is identical for all units i (the assumption of **homoskedasticity** (A9), then this simplifies further to:

$$\begin{aligned} \mathbb{E}[\varepsilon\varepsilon'] &= \begin{bmatrix} \sigma^2 & 0 & \dots & 0 \\ 0 & \sigma^2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & \dots & \dots & \sigma^2 \end{bmatrix} & \text{by A9} \\ &= \sigma^2 \mathbf{I} & \text{where } \mathbf{I} \text{ is the identity matrix} \end{aligned}$$

We sometimes refer to error terms ε that are independent (no autocorrelation) and homoskedastic as a **spherical** error terms.

Assumption 9 (Homoskedasticity). *The conditional variance of the error term is a constant:*

$$\mathbb{V}[\varepsilon|\mathbf{X}] = \sigma^2.$$

 Assumption: homoskedasticity

The key here is that σ^2 is a constant and doesn't vary by values of $\mathbf{X}$ (i.e., $\sigma^2 \neq \sigma_i^2$).

Under **A8** and **A9**, we are left with a single unknown, σ^2 . This allows us to simplify the variance:

$$\begin{aligned} \Omega &= \mathbf{Q}^{-1} \mathbb{E}[\mathbf{X}' \varepsilon \varepsilon' \mathbf{X}] \mathbf{Q}^{-1} && \text{from above} \\ &= \mathbf{Q}^{-1} \mathbb{E}[\mathbf{X}' \sigma^2 \mathbf{I} \mathbf{X}] \mathbf{Q}^{-1} && \text{by A9} \\ &= \sigma^2 \mathbf{Q}^{-1} \mathbb{E}[\mathbf{X}' \mathbf{X}] \mathbf{Q}^{-1} \\ &= \sigma^2 \left(\frac{\mathbf{X}' \mathbf{X}}{N} \right)^{-1} \left(\frac{\mathbf{X}' \mathbf{X}}{N} \right) \left(\frac{\mathbf{X}' \mathbf{X}}{N} \right)^{-1} && \text{by definition} \\ &= \sigma^2 N (\mathbf{X}' \mathbf{X})^{-1} && \text{simplifying} \\ &= \sigma^2 \mathbf{Q}^{-1} && \text{equivalently} \end{aligned}$$

Then, since $\mathbb{V}[\hat{\beta}] = \Omega/N$:

$$\begin{aligned} \mathbb{V}[\hat{\beta}] &= \frac{\Omega}{N} \\ &= \frac{1}{N} \sigma^2 \mathbf{Q}^{-1} && \text{from above} \\ &= \frac{1}{N} \sigma^2 \left(\frac{\mathbf{X}' \mathbf{X}}{N} \right)^{-1} && \text{by def of } \mathbf{Q} \\ &= \frac{1}{N} \sigma^2 N (\mathbf{X}' \mathbf{X})^{-1} && \text{extracting } N \\ &= \sigma^2 (\mathbf{X}' \mathbf{X})^{-1} && \text{simplifying} \end{aligned}$$

We can consistently estimate σ^2 with s^2 (some textbooks use $\hat{\sigma}^2$), the (estimated) **standard error of the regression** :

$$s^2 = \frac{\sum_{i=1}^N u_i^2}{N - k}$$

where N is the sample size, k the number of independent variables (including a constant), and u the residual.

This yields:

$$\hat{\mathbb{V}}[\hat{\beta}]_{\text{homoskedastic}} = \frac{\sum_{i=1}^N u_i^2}{N-k} (\mathbf{X}'\mathbf{X})^{-1}$$

Now, in general, it is unlikely for error terms to be homoskedastic. Thus we can just skip this assumption. However, we could also test for it. One such test is called **Breusch-Pagan test** and goes as follows (Breusch and Pagan 1979):

We follow the presentation of the test by Wooldridge (2002, 126).

1. Estimate the linear model

$$y = \mathbf{X}\beta + \varepsilon$$

with OLS. Derive the residuals:

$$u_i = y_i - \hat{y}_i$$

and square them (u_i^2).

2. Estimate the linear model:

$$u^2 = \mathbf{X}\gamma + \mu,$$

and evaluate the F statistic that all parameters $\gamma_1, \gamma_2, \dots, \gamma_k$ are jointly equal to zero (we will see F statistics a bit later). This corresponds to the null hypothesis that we have *homoskedasticity*. If we reject this null hypothesis (at a pre-set critical threshold α , e.g., 0.05), then we face the problem of *heteroskedasticity*.

If we are not willing to make the homoskedasticity assumption, meaning that we expect **heteroskedasticity**, then we are left with the earlier matrix, which we can estimate using the residuals $\mathbf{u}$ instead of the errors. Doing so, and replacing sample averages for expected values yields:

$$\begin{aligned} \frac{\Omega}{N} &= \frac{1}{N} \underbrace{\left(\frac{\mathbf{X}'\mathbf{X}}{N} \right)^{-1}}_{k \times k} \underbrace{\left(\frac{\mathbf{X}'\mathbf{u}\mathbf{u}'\mathbf{X}}{N} \right)}_{k \times k} \underbrace{\left(\frac{\mathbf{X}'\mathbf{X}}{N} \right)^{-1}}_{k \times k} \\ &= \frac{1}{N} \left(\frac{1}{N} \sum_{i=1}^N X_i X_i' \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N X_i X_i' u_i^2 \right) \left(\frac{1}{N} \sum_{i=1}^N X_i X_i' \right)^{-1} \quad \text{NB: } X_i \text{ is a row vector} \\ &= \left(\sum_{i=1}^N X_i X_i' \right)^{-1} \left(\sum_{i=1}^N X_i X_i' u_i^2 \right) \left(\sum_{i=1}^N X_i X_i' \right)^{-1} \quad \text{simplifying} \\ &= (\mathbf{X}'\mathbf{X})^{-1} (\mathbf{X}'\mathbf{u}\mathbf{u}'\mathbf{X}) (\mathbf{X}'\mathbf{X})^{-1} \quad \text{in matrix notation} \end{aligned}$$

These are so-called “**robust**” or **Huber-Eicker-White** estimates of the (co)variance (P. J. Huber 1967; White 1980; Freedman 2006). As before, the standard error of parameter $\hat{\beta}_k$ is the square root of

the diagonal value of row k . Note that this a *biased* but *consistent* estimator of the variance Ω/N (we won't prove it, but see White (1980)).

In practice, we generally avoid assuming homoskedasticity for two reasons. First, we generally don't have good reasons to assume it. Second, if the relation between X and Y isn't linear and thus we are approximating $\mathbb{E}[Y]$, then we will have heteroskedasticity by construction. Whenever our treatment is continuous, it is hard to assume that its effect is linear over its entire support, and thus it seems unlikely that homoskedasticity will hold.

4.6 Standard errors for interaction effects

As discussed in the point estimation discussion, we sometimes believe that the effect of X varies depending on the value of some third variable Z . This requires the inclusion of an interaction effect:

$$y = \alpha + \beta X + \gamma Z + \lambda XZ + \varepsilon,$$

with estimated marginal effect:

$$\frac{\partial \hat{\mathbb{E}}[Y]}{\partial X} = \hat{\beta} + \hat{\lambda}Z.$$

The standard error of this marginal effect is slightly more complicated than that of a single parameter:

$$se(\hat{\beta} + \hat{\lambda}Z) = \sqrt{\mathbb{V}[\hat{\beta}] + Z^2\mathbb{V}[\hat{\lambda}] + 2Z \cdot Cov[\hat{\beta}, \hat{\lambda}]},$$

where $\mathbb{V}[\hat{\beta}]$, $\mathbb{V}[\hat{\lambda}]$, and $Cov[\hat{\beta}, \hat{\lambda}]$ can be obtained from Ω .

Theorem 4.2 (Delta method). *Let $\hat{\beta}$ be an estimator satisfying $\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(0, \Omega)$. Let $g : \mathbb{R}^k \rightarrow \mathbb{R}^m$ be a continuously differentiable function. Then:*

$$\sqrt{N}(g(\hat{\beta}) - g(\beta)) \xrightarrow{d} \mathcal{N}(0, \nabla g(\beta)' \Omega \nabla g(\beta)),$$

where $\nabla g(\beta)$ is the $k \times m$ Jacobian of g evaluated at the true β . In practice, replace β with $\hat{\beta}$ and Ω with a consistent estimate.

For the interaction effect $g(\hat{\beta}) = \hat{\beta} + \hat{\lambda}Z$, the gradient is $[1, 0, \dots, Z, \dots]'$, which recovers the formula above. See the Appendix for a proof sketch.

4.7 Efficiency

4.7.1 Gauss-Markov theorem

If our assumptions A5-A7 are met (which guarantee unbiasedness), and if the error terms are independent (no autocorrelation, A8) and have constant variance (homoskedasticity, A9) (both setting the result that $V[\hat{\beta}] = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$), then we obtain the following result (see (B. Hansen 2022a), p.104).

Theorem 4.3 (Gauss-Markov). *Take any unbiased linear estimator $\tilde{\beta}$ of β . Then:*

$$V[\tilde{\beta}] \geq \sigma^2(\mathbf{X}'\mathbf{X})^{-1}.$$

In other words, the variance of any non-OLS (unbiased) estimator can at best equal the variance obtained by OLS, as long as Assumptions A4 to A9 are met. Some people refer to this as OLS being **BLUE**: best linear unbiased estimator (i.e., the most efficient estimator among unbiased estimators).

This result establishes that, under these assumptions, OLS is **efficient** – another desirable property in addition to unbiasedness.

Proof of Gauss-Markov

Let me sketch a proof of the Gauss-Markov theorem. What we want to show is that any non-OLS linear unbiased estimator of β has a higher variance than $\hat{\beta}_{ols}$. I'll refer to an arbitrary competing estimator $\tilde{\beta}$.

We start with the model:

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon$$

Since we are limiting ourselves to *linear* estimators, we can write such an estimator $\tilde{\beta}$ as a function of y :

$$\tilde{\beta} = m + M\mathbf{y}$$

where m and M are a vector and a matrix of constants, respectively. In OLS, for instance, $M = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

Next, we assume that $\tilde{\beta}$ is unbiased (since we are only considering the category of *unbiased* linear estimators). This implies

that:

$$\begin{aligned}
E(\tilde{\beta}|\mathbf{X}) &= E(m + M\mathbf{y}|\mathbf{X}) \\
&= m + M\mathbb{E}[\mathbf{y}|\mathbf{X}] && \text{(distributing the expect.)} \\
&= m + M\mathbb{E}[\mathbf{X}\beta + \varepsilon|\mathbf{X}] && \text{by A6} \\
&= m + M\mathbf{X}\beta + M\mathbb{E}[\varepsilon|\mathbf{X}] && \text{(distr. expect.)} \\
&= m + M\mathbf{X}\beta && \text{by A7}
\end{aligned}$$

The requirement for $\tilde{\beta}$ to be unbiased, thus, is:

$$\begin{aligned}
m &= 0 \\
M\mathbf{X} &= \mathbf{I}
\end{aligned}$$

Note that OLS satisfies this, since for least squares we have $M = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ and thus $M\mathbf{X} = \mathbf{I}$. The implication is that a linear unbiased estimator will look like:

$$\tilde{\beta} = M\mathbf{y},$$

with M satisfying the above requirement. So, a typical non-OLS estimator will be (wlog):

$$M = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + C$$

where C is any matrix. Since $\tilde{\beta}$ is unbiased, we know that $M\mathbf{X} = \mathbf{I}$ must hold. At the same time:

$$\begin{aligned}
M\mathbf{X} &= [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + C]\mathbf{X} \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} + C\mathbf{X} \\
&= \mathbf{I} + C\mathbf{X}
\end{aligned}$$

The implication is that $C\mathbf{X} = 0$.

So, we can now compute the variance of $\tilde{\beta}$. Let's write:

$$\begin{aligned}
\tilde{\beta} &= M\mathbf{y} \\
&= M(\mathbf{X}\beta + \varepsilon) && \text{(A6)} \\
&= M\mathbf{X}\beta + M\varepsilon && \text{(distr. M)} \\
&= \mathbf{I}\beta + M\varepsilon && \text{(to be unbiased)} \\
\tilde{\beta} - \beta &= M\varepsilon
\end{aligned}$$

The variance-covariance matrix is thus:

$$\begin{aligned}
\mathbb{E}[(\tilde{\beta} - \beta)(\tilde{\beta} - \beta)'] &= \mathbb{E}[M\varepsilon(M\varepsilon)'] \\
&= \mathbb{E}[M\varepsilon\varepsilon'M'] && \text{(properties of matrices)} \\
&= M(\mathbb{E}[\varepsilon\varepsilon'])M' && \text{(distr. expect.)} \\
&= M\sigma^2\mathbf{I}M' && \text{by A5 (independence) and A9} \\
&= \sigma^2MM'
\end{aligned}$$

So what is MM' again? It is:

$$\begin{aligned}
MM' &= [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + C][(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + C]' \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\
&\quad + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'C' + C\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + CC' \\
&= (\mathbf{X}'\mathbf{X})^{-1} + CC' \quad (\text{since } C\mathbf{X} = 0)
\end{aligned}$$

Thus, the variance of $\tilde{\beta}$ and $\hat{\beta}_{ols}$ are:

$$\begin{aligned}
\sigma^2 MM' &= \sigma^2 [(\mathbf{X}'\mathbf{X})^{-1} + CC'] \quad (\text{for alternative estimator}) \\
\sigma^2 MM' &= \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} \quad (\text{for OLS})
\end{aligned}$$

Since CC' is a quadratic term, it is positive. Thus, the variance of the rival estimator $\tilde{\beta}$ is larger than the variance of the OLS estimator, except if CC' happens to be zero. But if it is, then the estimator is actually OLS.

4.7.2 Cramér-Rao lower bound

One may sometimes wonder: can't we find an estimator with a variance of zero or almost zero? The answer is no: since we are looking at stochastic data-generating processes with random samples, estimators will always have a positive variance. This intuition is captured in the **Cramér-Rao Lower Bound theorem**, which characterizes the lowest variance of an unbiased estimator (see (Gailmard 2014), p.307).

The general statement of the Cramér-Rao theorem is the following. Let $X = (X_1, X_2, \dots, X_n)$ be a random sample from a probability distribution with PDF $f(X|\beta)$, where β is our parameter of interest, and let $\hat{\beta}$ be an unbiased estimator of β . The variance of $\hat{\beta}$ is bounded below by the inverse of the Fisher information of the data-generation process, namely:

$$\mathbb{V}[\hat{\beta}] \geq \frac{1}{\mathcal{I}(\beta)}$$

where $\mathcal{I}(\beta)$ is the **Fisher information** given by:

$$\mathcal{I}(\beta) = -\mathbb{E} \left[\frac{\partial^2}{\partial \beta^2} \log f(X|\beta) \right] = \mathbb{E} \left[\left(\frac{\partial}{\partial \beta} \log f(X|\beta) \right)^2 \right]$$

To see how this applies, note that the linear model can be rewritten as a conditional normal distribution centered on $\hat{y}$ (either by assuming

normally distributed error terms or by using our earlier asymptotic results):

$$y \sim \mathcal{N}(\mathbf{X}\beta, \sigma^2 \mathbf{I})$$

The **log-likelihood** function for this model is (see Appendix):

$$\log L(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)$$

The *score* (first derivative of the log-likelihood with respect to β) is:

$$\frac{\partial \log L}{\partial \beta} = \frac{1}{\sigma^2} \mathbf{X}'(\mathbf{y} - \mathbf{X}\beta)$$

The **Fisher information** $\mathcal{J}(\beta)$ is the *negative expected value* of the second derivative of the log-likelihood with respect to β . First, we compute the second derivative:

$$\frac{\partial^2 \log L}{\partial \beta \partial \beta'} = -\frac{1}{\sigma^2} \mathbf{X}' \mathbf{X}$$

Next, take the expected value of this quantity:

$$\mathcal{J}(\beta) = -E \left[\frac{\partial^2 \log L}{\partial \beta \partial \beta'} \right] = \frac{1}{\sigma^2} \mathbf{X}' \mathbf{X}$$

Taking the inverse, we find that:

$$\frac{1}{\mathcal{J}(\beta)} = \sigma^2 (\mathbf{X}' \mathbf{X})^{-1} = \mathbb{V}[\hat{\beta}]$$

The variance of the OLS estimator is exactly equal to the inverse of the Fisher information, and thus it achieves the Cramér-Rao lower bound. This is the best (smallest) that is feasible, another way to say that OLS – under all assumptions listed above, including homoskedasticity – is efficient.

4.8 Consistency

The idea of consistency is that, as $N \rightarrow \infty$, the probability that an estimate is farther from its true value than some arbitrary constant ε goes to zero. Formally, we say that $\hat{\beta}$ converges in probability to β iff:

$$\lim_{N \rightarrow \infty} \Pr(|\beta - \hat{\beta}| \geq \varepsilon) = 0, \forall \varepsilon > 0.$$

More compactly, this is written as:

$$plim \hat{\beta} = \beta$$

or

$$\hat{\beta} \xrightarrow{p} \beta$$

Informally, consistency is the idea that, as the sample size increases, the probability of getting a value of $\hat{\beta}$ beyond some threshold ε goes toward zero. Put another way, we want to obtain increasingly more precise estimates as the sample size increases.

We can use our earlier results on the asymptotic distribution of $\hat{\beta}$ to help us out. The reason is that the logic is the same: we wanted to know what happens to $\hat{\beta}$ as N went to infinity. We want to do the same here. Thus, many of the results below repeat what we had seen earlier.

To establish this result, we can write:

$$\begin{aligned} plim \hat{\beta} &= plim(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \\ &= plim(\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon) && \text{see unbiasedness above} \\ &= plim\beta + plim(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon \\ &= \beta + plim(\mathbf{X}'\mathbf{X})^{-1}plim(\mathbf{X}'\varepsilon) && \text{since } \beta \text{ is a constant} \end{aligned}$$

Now, as N increases, so will $\mathbf{X}'\mathbf{X}$. To avoid blowing up, we normalize by N/N :

$$plim \hat{\beta} = \beta + plim \left(\frac{1}{N}(\mathbf{X}'\mathbf{X}) \right)^{-1} plim \left(\frac{1}{N}\mathbf{X}'\varepsilon \right)$$

What we want is for the last term to disappear. Let's take its components one by one. By the **Law of Large Numbers**, we can claim that:

$$plim \left(\frac{1}{N}\mathbf{X}'\mathbf{X} \right) = \mathbb{E}[\mathbf{X}'\mathbf{X}] := \mathbf{Q},$$

that is, the first term converges to its expected (finite) value, which is a positive definite (and invertible) matrix $\mathbf{Q}$. We now have:

$$plim \hat{\beta} = \beta + \mathbf{Q}^{-1}plim \left(\frac{1}{N}\mathbf{X}'\varepsilon \right)$$

We only have the last term to tackle. To ensure that the probability limit of $\hat{\beta}$ is β , we must show that:

$$plim \left(\frac{1}{N}\mathbf{X}'\varepsilon \right) = 0$$

If we take expectations, we see that:

$$\text{plim} \left(\frac{1}{N} \mathbb{E}[\mathbf{X}'\varepsilon] \right) = 0 \quad \text{by A7}$$

Thus, under assumptions **A5** to **A7**, and thanks to the Law of Large Numbers, **OLS is consistent**.

4.9 Taking a deep breath

Here are some of the main results we have obtained so far:

- **Unbiasedness** of OLS : we need a random sample of IID variables and a linear model (**A5** and **A6**), a full rank matrix of regressors (**A4**), and exogeneity (**A7**).
- **Consistency** of OLS : same assumptions (with an assist from the Law of Large Numbers).
- **Efficiency** of OLS : same assumptions plus the assumption of homoskedasticity (**A9**), which together gives us the Gauss-Markov theorem.

We are not making many assumptions, but we are making strong ones:

- Exogeneity is very hard to meet with observational (i.e., non-experimental) data.
- Homoskedasticity is very rarely justified.
- Independence depends on the issue that you are studying.

4.10 Omitted variable

What are the consequences of specifying our model incorrectly by omitting a variable z ? Without loss of generality, let us go over the case in which we have two regressors, X and Z . We are interested in the effect of X but we can't include Z (e.g., it's not possible to measure it).

Suppose that the true model is:

$$y_i = \lambda + \tau x_i + \gamma z_i + \mu_i,$$

and assume that all assumptions listed so far are met (including $E(\mu|x, z) = 0$). τ is the true treatment effect. Yet we erroneously estimate:

$$y_i = \alpha + \beta x_i + \varepsilon_i.$$

When we do this, we move γz into ε . What are the consequences of such a decision? Since we started with the bivariate case, we can write:

$$\begin{aligned}\hat{\beta} &= \frac{\text{Cov}(X, Y)}{\mathbb{V}[X]} \\ &= \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} && \text{canceling the denominators} \\ &= \frac{\sum y_i(x_i - \bar{x})}{\sum x_i(x_i - \bar{x})} && \text{algebra trick}\end{aligned}$$

We can plug in the true model for y :

$$\begin{aligned}\frac{\sum y_i(x_i - \bar{x})}{\sum x_i(x_i - \bar{x})} &= \frac{\sum (x_i - \bar{x})(\lambda + \tau x_i + \gamma z_i + \mu_i)}{\sum x_i(x_i - \bar{x})} \\ &= \frac{\lambda \sum (x_i - \bar{x}) + \tau \sum x_i(x_i - \bar{x}) + \gamma \sum (x_i - \bar{x})z_i + \sum (x_i - \bar{x})\mu_i}{\sum x_i(x_i - \bar{x})} \\ &= \tau + \gamma \frac{\sum (x_i - \bar{x})z_i}{\sum x_i(x_i - \bar{x})} + \frac{\sum (x_i - \bar{x})\mu_i}{\sum x_i(x_i - \bar{x})}\end{aligned}$$

Note that we can't simplify $\sum (x_i - \bar{x})z_i$ nor $\sum (x_i - \bar{x})\mu_i$ because neither z nor μ are constant.

Taking expectations, we get:

$$\begin{aligned}\mathbb{E}[\hat{\beta}] &= \mathbb{E}[\tau] + \mathbb{E}\left[\gamma \frac{\sum (x_i - \bar{x})z_i}{\sum x_i(x_i - \bar{x})}\right] + \mathbb{E}\left[\frac{\sum (x_i - \bar{x})\mu_i}{\sum x_i(x_i - \bar{x})}\right] \\ &= \tau + \gamma \mathbb{E}\left[\frac{\sum (x_i - \bar{x})z_i}{\sum x_i(x_i - \bar{x})}\right] + \mathbb{E}\left[\frac{\sum (x_i - \bar{x})\mu_i}{\sum x_i(x_i - \bar{x})}\right] && \text{taking constants out} \\ &= \tau + \gamma \mathbb{E}\left[\frac{\sum (x_i - \bar{x})z_i}{\sum x_i(x_i - \bar{x})}\right] && \text{if A7 is met}\end{aligned}$$

Instead of our findings to establish unbiasedness, we now have a new additional term on top of our true value τ . But note that the term in the expectation is simply:

$$\mathbb{E}\left[\frac{\sum (x_i - \bar{x})z_i}{\sum x_i(x_i - \bar{x})}\right] = \frac{\text{Cov}[X, Z]}{\mathbb{V}[X]} := \omega.$$

This is the same as the parameter of a regression of Z on X and thus captures the effect of X on Z . Define this effect ω . We can write that the estimate of the ‘wrong’ regression combines the following terms (where $i \rightarrow j|k$ means the effect of i on j , conditional on k):

$$\mathbb{E}[\hat{\beta}] = \overset{x \rightarrow y|z}{\hat{\tau}} + \underset{z \rightarrow y|x}{\underbrace{\gamma}_{\gamma}} \cdot \overset{x \rightarrow z}{\hat{\omega}}$$

The bias itself equals the difference between the average of the (badly) estimated effect ($\hat{\beta}$) and the true effect (τ):

$$\text{Bias} := \mathbb{E}[\hat{\beta}] - \tau = \gamma\omega$$

The bias will depend on γ (the effect of Z on Y adjusting for X) and on the ω (the effect of X on Z). If these two are *both* nonzero, then **OLS will be biased**.

Things get worse. Not only is OLS biased, but it is also **inconsistent**. Since γ and ω are constants, nothing would make them go away as $n \rightarrow \infty$.

Therefore, a variable Z that is both causing X and Y will make $\hat{\beta}_{\text{ols}}$ biased and inconsistent if it is omitted from the estimated model.

- $\hat{\beta}$ will be **biased** $\mathbb{E}[\hat{\beta}] \neq \beta$.
- $\hat{\beta}$ will be **inconsistent** (increasing the sample size won’t help).

4.11 Null Hypothesis testing

4.11.1 Simple hypothesis

We move on to the world on (social) scientific theories. Recall that OLS yields an estimate $\hat{\beta}$ from a sample. Yet, setting aside asymptotic properties of OLS, we don’t know what the true value of β is.

Hypothesis testing is the process of answering the question: *given a hypothesis that we may have about β , how confident are we about it after seeing $\hat{\beta}$?* For instance, if we hypothesize that $\beta = -5$, and our sample yields $\hat{\beta} = 45.5$, will we start doubting our hypothesis?

In practice, hypothesis testing is conducted as follows.

Step 1. Set your **null hypothesis**, typically called H_0 . If we are interested in a single parameter β_D (for the effect of D on Y), we would then set it as:

$$H_0 : \beta_D = c.$$

We could also use **composite** values (e.g., an interval), but this is rare. In practice, the most common null hypothesis is that $c = 0$, which means that the treatment D has no effect on Y . But other hypotheses exist. For instance, in macroeconomics, you may want to test whether the fiscal multiplier is 1.

Step 2. Set your **rival** or **alternative hypothesis** (H_a or H_1). Commonly, we will either be interested in whether β_D is *different* from c or whether it is larger (or smaller) than c . In the first case, we say that we are conducting a **two-tailed test** and in the latter case we are conducting a **one-tailed test**.

$$H_a : \beta_D \neq c$$

$$H_a : \beta_D > c$$

$$H_a : \beta_D < c$$

Step 3. We then estimate our model and obtain some value $\hat{\beta}_D$. Now, the probability that $\hat{\beta}_D = \beta_D$ exactly is zero (there is zero probability mass at a single point). We will always get an estimate that differs from its true value. The question we are asking is: how ‘different’ must $\hat{\beta}$ be from c for us to doubt $\beta_D = c$?

To answer this, we can rely on our results on the asymptotic distribution of $\hat{\beta}_D$: as $N \rightarrow \infty$, it will be distributed $\hat{\beta}_D \sim \mathcal{N}(\beta, \Omega/N)$. We can replace β with our null hypothesis, yielding:

$$\hat{\beta}_D \sim \mathcal{N}(c, \Omega/N)$$

In other words, *if our null hypothesis were true*, then $\hat{\beta}_D$ would be distributed normal around c . If $c = 0$, then this distribution is centered on zero.

Note that since we need to estimate σ^2 with s^2 , we replace the normal distribution with the Student’s t distribution, whose sole parameter is the number of **degrees of freedom**, i.e., $N - k$. The difference between the two is small; as the degrees of freedom tend to infinity, the t distribution converges to the standard normal distribution $\mathcal{N}(0, 1)$.

Given that we know the distribution, we can then also form rigorous statements regarding the probability of obtaining a value like $\hat{\beta}_D$ or

more extreme.² Concretely, we can standardize our parameter to a value that we shall call t :

$$\tilde{t} = \frac{\hat{\beta}_D - c}{se(\hat{\beta})}$$

I write $\tilde{t}$ to denote the *ex post* t statistic obtained from a given sample.

For instance, consider $c = 0$, $\hat{\beta}_D = 3$, and $se(\hat{\beta}) = 1.5$, we obtain:

$$\tilde{t} = \frac{3}{1.5} = 2$$

The t statistic tells us how many standard errors away $\hat{\beta}$ is from the constant c .

The question is: *is being two standard errors away from c a lot?* To answer this, we build on the fact that we know the exact distribution of t and that we can thus infer the probability of obtaining a given t value or more extreme. Refer back to Student t 's distribution table and see how much mass there is left and right of that t value.

Concretely, we can:

1. Compute the probability of obtaining a t equal to or greater than $\tilde{t}$. This would be done in a one-tailed test that $\beta > c$. (Likewise, we could compute the probability of obtaining a negative t value that is smaller than $\tilde{t}$ if we tested $\beta < c$.)
2. Compute the probability that $|t| > |\tilde{t}|$. This would be done in a two-tailed test.

The probability in question is a **p-value** (or just p). To reiterate, it tells us: *what is the probability of obtaining a t value greater than $\tilde{t}$ (one-tailed test) or what is the probability of obtaining a t value greater than the absolute value of $\tilde{t}$ (two-tailed test).* As stated above, we can compute it by comparing the t statistic we obtained against a t_{N-k} distribution.

Step 4. We started by saying that we wanted to evaluate the plausibility of our null hypothesis. More formally, what we meant is to say: *given some criterion of plausibility, do we have enough evidence to reject the null hypothesis?*

²The probability of $\hat{\beta}$ being a particular value is zero. See the chapter on probability and measure theory.

The criterion is going to be as follows. For an α selected by the analyst ($0 < \alpha < 1$), we define a **Type I** error as:

$$\begin{aligned}\alpha &:= \mathbb{P}[\text{Type I error}] \\ &= \mathbb{P}[\text{reject } H_0 | H_0 \text{ is true}]\end{aligned}$$

You will often see analysts using $\alpha = 0.05$ (i.e., setting the probability of a Type I error at 5%) or smaller.

The **Type II error** is the complementary mistake — failing to reject H_0 when it is false:

$$\mathbb{P}[\text{Type II error}] = \mathbb{P}[\text{fail to reject } H_0 | H_0 \text{ is false}].$$

The complement is the **power** of the test: the probability of correctly rejecting a false null hypothesis. Power depends on three factors: (1) the **effect size** (how far the true β is from c), (2) the **sample size** N (larger samples yield smaller $se(\hat{\beta})$, increasing $|\tilde{t}|$), and (3) the **significance level** α (a less stringent threshold raises power at the cost of more Type I errors). A common benchmark is power ≥ 0.8 , which is the basis for **power analysis** — computing the required N before data collection.

The last step then is to assess whether the probability of obtaining a given t statistic or more extreme is smaller or larger than α . We reject the null hypothesis if $p\text{-value} < \alpha$: in such a case, our estimate $\hat{\beta}_D$ is very unlikely if the null hypothesis is true, leading us to cast doubt over H_0 's truthfulness. If this condition is met, we say that we *reject the null hypothesis*. Equivalently, we say that $\hat{\beta}$ is *statistically significant*.

If, however, $p > \alpha$, then we say that *we fail to reject the null hypothesis*: we don't have enough evidence to cast doubt over it.

Note that instead of comparing our p -value to α , we can equivalently compare our t statistic to a critical value of t^c that corresponds to α . For instance, since we know the distribution of the t statistic, we can identify the critical t^c value beyond which there is less than α of mass under the curve (or equivalently, the set of values t_i that lie in this part of the distribution; we call this the **rejection or critical region** (Casella and Berger 2021), p.374). For large N , for instance, we know that there is 2.5% of the area under the t distribution to the right of 1.96. Likewise, there is 2.5% to the left of -1.96. Thus, we know that t values that are either larger than 1.96 or smaller than -1.96 have a less than 5% of occurring if our null hypothesis is true.

There has been considerable criticism of hypothesis testing in recent years (Gelman and Stern 2006; Gelman 2013; Gelman and Loken 2013; Altman and Krzywinski 2017). Some of the concerns include:

- Hypothesis testing tests hypotheses that are never true: $\beta = 0$ is a strawman that never occurs. Given enough data, we will then always end up rejecting the null hypothesis.
- A corollary to the previous point: a point estimate can be significant even if its effect is tiny and unimportant. We must not confuse *statistical significance* with *practical significance* (or *economic significance*). Likewise, a point estimate could be very large and meaningful, yet if its standard error is large as well, it will be insignificant. But this doesn't mean its effect is zero!
- Hypothesis testing leads to a binary outcome (significant/insignificant) that is too coarse. We might prefer to make statements about the range of plausible values for β .
- Hypothesis testing leads to biases in terms of research that is published: if there is a preference for “positive” findings (i.e., significance), then research will over-produce studies that show effects even when there are none.

Multiple testing

When several hypotheses are tested simultaneously — for instance, testing the effect of a treatment on many outcomes — the probability of at least one false rejection grows. Testing m independent hypotheses each at level α gives a family-wise false-rejection probability of $1 - (1 - \alpha)^m$, which approaches 1 as m grows.

Two standard corrections:

- **Bonferroni:** test each hypothesis at level α/m . This controls the **family-wise error rate** (FWER) — the probability of *any* false rejection. Simple to apply, but conservative when tests are positively correlated.
- **Benjamini-Hochberg:** ranks p -values and applies a graduated threshold. Controls the **false discovery rate** (FDR) — the expected fraction of rejections that are false positives. More powerful than Bonferroni when many true effects exist.

4.11.2 Multiple restrictions

In econometrics and statistics, when testing q restrictions simultaneously, the null hypothesis (H_0) typically takes the form:

$$H_0 : R\beta = r$$

where R is a $q \times k$ matrix whose rows encode the restrictions, β is the $k \times 1$ parameter vector, and r is a $q \times 1$ vector of constants. The alternative hypothesis (H_1) suggests that at least one restriction fails:

$$H_1 : R\beta \neq r$$

For instance, we could compare the following two models:

$$\begin{aligned} y &= \tau + \beta x + \varepsilon \\ y &= \tau + \beta x + \gamma z + \lambda w + \mu \end{aligned}$$

where we wish to test $\gamma = 0$ and $\lambda = 0$ jointly. This requires two rows in R :

$$R = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{2 \times 4}, \quad r = \begin{bmatrix} 0 \\ 0 \end{bmatrix}_{2 \times 1},$$

which corresponds to:

$$H_0 : \gamma = \lambda = 0.$$

To test this, we cannot rely on the t test (which tests a single constraint). Instead, we may use the F test, which allows us to determine if a set of linear restrictions on the parameters of a model is valid. The test statistic compares the restricted model (H_0) to the unrestricted model (under H_1).

The F -statistic is calculated as:

$$\begin{aligned} F &= \left(\frac{\sum (y_i - \hat{y}_r)^2 - \sum (y_i - \hat{y}_u)^2}{(n - k_r) - (n - k_u)} \right) / \left(\frac{\sum (y_i - \hat{y}_u)^2}{n - k_u} \right) \\ &= \frac{\left(\frac{\sum u_r^2 - \sum u_u^2}{k_u - k_r} \right)}{\left(\frac{\sum u_u^2}{n - k_u} \right)} \end{aligned}$$

where k_u and k_r are the number of parameters in the unrestricted and the restricted model (and thus $k_u - k_r = q$ equals the number of restrictions). The residuals u_u and u_r refer to the unrestricted and restricted model, respectively.

The F statistic is distributed with two parameters $F(q, n - k_u)$ distribution. Since we know its distribution, we can compute p values that are evaluated as the mass under the F distribution at or beyond a given F statistic. If F statistic is “large” (using the F distribution), it means that there is little mass beyond this point, indicating that it is implausible to obtain such an F under the null hypothesis. In

that case, we reject the null hypothesis. As in hypothesis testing, the threshold of Type I error (α) is up to the analyst's judgement, but typical values are $\alpha = 0.05$ or smaller.

4.12 Confidence intervals

Confidence interval flips the approach used in hypothesis testing. In hypothesis testing, we try to see if our estimate is plausible compared to a value that is dictated by theory, not by the data. In most cases, we say something like: “we believe that $\beta = 0$, and let's see if the data tell us otherwise.” In confidence interval, we center our analysis on $\hat{\beta}$ and build uncertainty around it (a **confidence interval**).

A confidence interval is an interval that has, *ex ante*, an $1 - \alpha$ probability of containing the true value β . The value $1 - \alpha$ is referred to as the **coverage (probability)** of a confidence interval.

I insist on “*ex ante*”: the statement is about the coverage *before* we draw a random sample, not about a given confidence interval obtained *ex post* once the data is obtained. The interval computed by your software either contains β or it doesn't, so the probability *ex post* is 1 or 0.

We compute confidence intervals as:

$$CI_{1-\alpha}[\beta] = [\hat{\beta} - t_{N-k}^c \cdot se(\hat{\beta}); \hat{\beta} + t^c \cdot se(\hat{\beta})]$$

where t^c is the critical t value at α with $N - k$ degrees of freedom. For instance, if $\alpha = 0.05$ and the sample is large, the critical t value is 1.96.

What happens this is the following: over repeated samples, this interval will capture the true value of β $1 - \alpha$ percent of the time.

This confidence interval can straightforwardly be obtained from two pieces:

1. Our unbiasedness result, which allows us to replace β with $\mathbb{E}[\hat{\beta}]$. Thus, over repeated samples (i.e., in expectation), the distribution of $\hat{\beta}$ is centered on β .
2. Our sampling distribution of $\hat{\beta}$.

To show this, recall:

$$t = \frac{\hat{\beta} - \beta}{se(\hat{\beta})} \sim t_{N-k}.$$

We can then write:

$$\mathbb{P}[-t_{N-k}^c \leq t \leq t_{N-k}^c] = 1 - \alpha$$

where t_{N-k}^c is the critical t value for $\alpha/2$. This result comes from the t distribution's definition. We can then replace:

$$\begin{aligned} 1 - \alpha &= \mathbb{P} \left[-t_{N-k}^c \leq \frac{\hat{\beta} - \beta}{se(\hat{\beta})} \leq t_{N-k}^c \right] && \text{by definition} \\ &= \mathbb{P} \left[-t_{N-k}^c se(\hat{\beta}) \leq \hat{\beta} - \beta \leq t_{N-k}^c se(\hat{\beta}) \right] && \text{moving standard errors} \\ &= \mathbb{P} \left[\hat{\beta} - t_{N-k}^c se(\hat{\beta}) \leq \beta \leq \hat{\beta} + t_{N-k}^c se(\hat{\beta}) \right] && \text{moving } \hat{\beta} \end{aligned}$$

Thus, the confidence interval is centered on β . If we repeatedly did this calculation over multiple samples, and took the expectation of this probability, we end up with a probability of $1 - \alpha$.

Note that there is a link to hypothesis testing. Without loss of generality, let's say that we were earlier interested in testing the hypothesis that $\beta = 0$. Then, if our confidence interval contains 0, we automatically know that we will fail to reject the null hypothesis. Conversely, if our confidence interval doesn't contain 0, then we know that we would reject the null hypothesis. To see this, note that the null hypothesis assumes a normal distribution around 0, whereas confidence intervals build a distribution around $\hat{\beta}$. If $\hat{\beta}$ is larger than, say, two standard errors from 0, then it must also be that $\hat{\beta}$ minus two standard errors is larger than 0.

If: $\hat{\beta} > 0 + 1.96se(\hat{\beta})$ significant under hypothesis testing

Then: $\hat{\beta} - 1.96se(\hat{\beta}) > 0$

and thus: $0 \notin [\hat{\beta} - 1.96 \cdot se(\hat{\beta}); \hat{\beta} + 1.96 \cdot se(\hat{\beta})]$ confidence intervals don't include 0

Thus, confidence intervals and hypothesis testing are two separate ways to do inference, but they are connected.

Example 4.1 (Racial discrimination in hiring). Bertrand and Mullainathan (2004) ask the following question: do employers discriminate based on the race of job applicants? You cannot easily use observational data to answer this question: maybe applicants from different race/ethnic backgrounds are systematically different on some (unobserved) dimension.

To overcome this, Bertrand and Mullainathan (2004) do an **audit study**: they sent fake CVs in response to job ads in Boston and Chicago. The treatment was to send CVs with “Black-sounding” names (Lakisha, Jamal) (control) versus CVs

with “white-sounding” names (Emily, Greg) (treatment). Aside from the name at the top, the CVs were identical. Because names are randomly assigned, any difference in callback rates can only be due to the name (which captures race)

The main result is an OLS regression of a binary callback indicator (dependent variable equals 1 if the employer called back) on a dummy for white-sounding name, controlling for CV quality (Table 4.1). The estimate was $\beta \approx 0.032$: white-sounding names receive about 3.2 percentage points more callbacks. The baseline callback rate was 6.5% for Black-sounding names, which means that the callback rate was 50% larger for White-sounding CVs.

Is this gap statistically distinguishable from zero? The estimated standard error, clustered at the name level to account for multiple CV carrying the same name, is about 0.008. The t -statistic exceeds 4, and the 95% confidence interval [0.016, 0.048] lies entirely above zero. The null hypothesis that CVs names have no effect on callbacks is rejected.

	Callback rate
Black-sounding name	6.45%
White-sounding name	9.65%
Difference (OLS estimate)	0.032***
<i>SE (clustered by name)</i>	<i>(0.008)</i>
Résumé quality controls	
N	4,870

Table 4.1: Effect of a white-sounding name on employer callback. Outcome: binary callback indicator (= 1 if the employer contacted the applicant). Treatment: CV with a White-sounding name (Emily, Greg) versus CV with a Black-sounding name (Lakisha, Jamal). The top two rows show raw callback rates; the OLS estimate controls for résumé quality. Standard errors clustered at the name level. *** $p < 0.01$. Source: Bertrand and Mullainathan (2004).

4.13 Model fit

We are often interested in knowing how our model performed. By “model,” we mean how our independent variables and the parametric assumptions we place on them allow us to explain an outcome. This

is distinct from OLS properties: we are now evaluating our model *ex post* here.

There are several ways to measure **model fit**.

4.13.1 R^2

The R^2 (aka **coefficient of determination**) of a model is the share of the variation in $\mathbf{y}$ that is explained by $\mathbf{X}\hat{\beta}$. Formally:

$$R^2 = 1 - \frac{\sum u_i^2}{\sum (y - \bar{y})^2}$$

The numerator of the second term is the sum of squared residuals (aka residual sum of squares). The denominator is the sum of square total (or total sum of squares).

R^2 will be located on the $[0, 1]$ interval, with larger values denoting a larger share of $\mathbf{y}$ explained by our model. Its advantage lies in (a) it being easy to interpret, and (b) telling us how much variation is left that could affect our results. This has been used by analysts to engage in robustness tests to see the degree to which omitted variables could affect our results (Oster 2019).

Note. In the bivariate case (one regressor and an intercept), $R^2 = r_{XY}^2$, the square of the sample correlation between X and Y . This gives a clean geometric anchor: R^2 is the squared cosine of the angle between the centered $\mathbf{y}$ and the fitted values $\hat{\mathbf{y}}$.

The disadvantage of R^2 is that it is extremely sensitive to the sample you obtain. Holding the model constant, increasing the variance of X generally increases the R^2 , making it very difficult to compare across datasets.

Furthermore, R^2 is weakly increasing in additional random variables: adding variables can only reduce $\sum u_i^2$ and thus increase R^2 . As a result, you may sometimes encounter the **adjusted R^2** :

$$R_a^2 = 1 - (1 - R^2) \frac{n - 1}{n - k},$$

for n observations and k parameters (including a constant). This quantity punishes R^2 for the inclusion of irrelevant variables (by increasing k). The downside is that it is not bounded on the $[0, 1]$ interval anymore.

4.13.2 σ^2

We already encountered the **standard error of the regression** (σ) or **residual variance** (σ^2). This is the unconditional variance of the error term and is estimated by s^2 :

$$s^2 = \frac{\sum u_i^2}{n - k},$$

for n observations and k parameters (including a constant). Take the square root to estimate s , the estimated standard error of the regression (i.e., s is an estimate of σ).

The better the model fit, the smaller s^2 (and s). This indicates that the model has a low level of noise and thus performs well. You can interpret s^2 as follows: 95% of predicted/fitted values ($\hat{y}$) will be located within $+/-2s$ of observed values y . Informally and imprecisely, you can think of this as the average residual in your prediction.

What's nice is (a) that it is an estimate of a population parameter, and (b) is expressed in terms of the dependent variable. Thus, in a study on predicted income in millions of USD, $s = 3$ suggests that our predicted estimate will be off, on average, by USD 3 million. It is thus very interpretable.

4.13.3 F test

The F test, which we encountered above, tests several restrictions on parameters. We can expand this to the special case where we test the restriction that all parameters (minus the constant) are jointly equal to zero. This represents an assessment of the performance of the *overall* model (and not just restrictions on a small subset of parameters).

The null hypothesis is that (ignoring the intercept, since a model with a single parameter is the comparison point):

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_{k-1} = 0,$$

against the alternative hypothesis that some parameter(s) is different from zero. We call the estimated model the “unrestricted” model (i.e., where we don't restrict our parameters to equal zero) which we compare against the “restricted” model (where all parameters are zero, except the constant):

Restricted: $\mathbf{y} = \alpha$

Unrestricted: $\mathbf{y} = \alpha + \beta X + \gamma W + \dots + \varepsilon$

The F statistic is computed as (using r as subscript for restricted and u for unrestricted):

$$\begin{aligned}
F &= \left(\frac{\sum (y_i - \hat{y}_r)^2 - \sum (y_i - \hat{y}_u)^2}{(n - k_r) - (n - k_u)} \right) / \left(\frac{\sum (y_i - \hat{y}_u)^2}{n - k_u} \right) \\
&= \left(\frac{\sum (y_i - \bar{y})^2 - \sum (y_i - \hat{y}_u)^2}{(n - k_r) - (n - k_u)} \right) / \left(\frac{\sum (y_i - \hat{y}_u)^2}{n - k_u} \right) && \text{if testing sig of all parameters} \\
&= \left(\frac{\sum (y_i - \bar{y})^2 - \sum (y_i - \hat{y}_u)^2}{(k_u - k_r)} \right) / \left(\frac{\sum (y_i - \hat{y}_u)^2}{n - k_u} \right) \\
&= \left(\frac{\sum (y_i - \bar{y})^2 - \sum u_u^2}{(k_u - k_r)} \right) / \left(\frac{\sum u_u^2}{n - k_u} \right) && \text{by def} \\
&= \left(\frac{\sum (y_i - \bar{y})^2 - \sum u_u^2}{(k_u - 1)} \right) / \left(\frac{\sum u_u^2}{n - k_u} \right) && \text{restricted model has 1 parameter}
\end{aligned}$$

for k parameters.

Under the null hypothesis, the F statistic is distributed F . As with t tests, larger values are implausible under the null hypothesis.

4.14 Forecasts and predictions

We take a break here to discuss how to make predictions and forecasts. I said that econometrics is often about identifying causal effects. But it can also be about predictions and forecasts. This can be helpful for description (“here’s a what a customer with a given income, gender, etc. would spend on our product”) and for policy purposes (“here’s how a 10 basis point decrease in interest rates would affect house sales”).

We know that the predicted value of y , given a model $\mathbf{X}$ and parameters β , equals:

$$\hat{y} = \mathbf{P}\mathbf{y} = \mathbf{X}\hat{\beta},$$

where $\mathbf{P} := \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. We can think of predictions and forecasts in two ways.

1. We take an arbitrary vector $\mathbf{x}'_f$ that we are interested in and multiply it with β (where f denotes the design vector for which we want a prediction). For instance, if we are trying to identify how much a customer might spend in your shop as a function of their personal features (gender, age, income, etc.), we could create a synthetic customer who is female, 35 years old, earns

CHF50k per year, etc. We would typically call this an **out-of-sample** prediction, at least if we don't already possess units with this profile in our data:

$$[\mathbf{x}'_f \hat{\beta}]_{1 \times 1} = [1 \quad \text{Female} \quad 35\text{yo} \quad \dots \quad \text{CHF50k}]_{1 \times k} \times \begin{bmatrix} \hat{\alpha} \\ \hat{\beta} \\ \vdots \\ \hat{\gamma}_k \end{bmatrix}_{k \times 1}$$

2. We can model $\mathbf{y}$ as a dynamic process, where past values of y predict present values of y . This is a **forecast** and involves dynamic modeling with time series (or panel data), which I will discuss later.

For now, let's focus on making predictions. The point prediction is straightforward: just compute $\mathbf{x}'\hat{\beta}$ for your vector of values of $\mathbf{X}$ that you are interested in. However, this doesn't give you any sense of uncertainty regarding this prediction. What we want is to add confidence intervals around this prediction. How do we do this?

The general idea will be to measure the standard error around $\hat{y}$ and then multiply it by a critical t value (e.g., $t^* = 1.96$ for a type I error rate (or α) of 0.05 and a large sample). Here, there is a subtlety to keep in mind:

1. If you want the confidence interval around the prediction/forecast of the *expected value* of Y , you will need to incorporate the uncertainty of $\hat{\beta}$, $\hat{\gamma}$, etc.
2. If you want the confidence interval around the prediction/forecast of a *new value* Y , you will need in addition the uncertainty around ε (the error term).

Let's start with the expected value. We can write:

$$\mathbb{E}[y|\mathbf{x}_f] = \mathbf{x}_f \beta,$$

where can be estimated with:

$$\hat{y}_f = \mathbf{x}_f \hat{\beta}$$

This quantity is a random variable for two reasons: (a) $\hat{\beta}$ is a random variable, and (b) there is uncertainty from the error term (which we ignore for now, since we focus on the mean value of y).

The variance of $\hat{y}_f$ can then be defined as:

$$\mathbb{V}[\hat{y}_f] = \mathbb{V}[\mathbf{x}_f \hat{\beta}] = \sigma^2 [\mathbf{x}'_f (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_f],$$

and σ^2 (the variance of the regression) can be consistently estimated by s^2 as we saw previously:

$$s^2 = \frac{\sum u_i^2}{N - k}.$$

If we want to measure the uncertainty of around Y , then we must add our error term. Note that the residual is (by construction):

$$u_f = y_f - \hat{y}_f = \underbrace{[\mathbf{x}'_f \beta + \varepsilon]}_{\text{true model at } f} - \hat{y}_f.$$

Since $\mathbf{x}'_f \beta$ is fixed, and since ε shouldn't covary with anything, the variance of u_f equals the sum of the variances of the two remaining terms:

$$\mathbb{V}[u_f] = \mathbb{V}[\varepsilon] + \mathbb{V}[\hat{y}_f].$$

Thus, we have the same variance as above plus the variance of the error term (again). We can write:

$$\mathbb{V}[\mathbf{x}_f \hat{\beta} + u_f] = \sigma^2 [\mathbf{x}'_f (\mathbf{X}' \mathbf{X})^{-1} \mathbf{x}_f] + \sigma^2$$

In either case, take the square root of $\mathbb{V}[\cdot]$ to obtain the standard error and multiply it by t^* .

4.15 Outliers and leverage

Sometimes, people worry about **outliers**, that is, values of variables that are *abnormal* in terms of their value compared to the overall distribution of the variable.

To figure out what the problems might be, we first define **leverage values** (B. Hansen 2022a, 83). The leverage of an observation i (i.e., a row of data) tells us how dependent OLS estimates *could* be on that observation (as we will see: it's a potential threat that may or may not be realized in practice). Each observation has its own leverage value, which is defined as the diagonal of the projection matrix. Recall that the projection matrix (also known as the “hat” matrix) is:

$$\mathbf{P} := \mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}',$$

where $\mathbf{P}\mathbf{y} = \hat{\mathbf{y}}$ gives us the predicted values of y (hence the label “hat” matrix: it ‘puts’ the hat on y).

For unit $i \in \{1, 2, \dots, N\}$, we define the leverage value h_{ii} as:

$$h_{ii} := X_i'(\mathbf{X}'\mathbf{X})^{-1}X_i,$$

where h_{ii} is the i th diagonal entry of the projection/hat matrix.

For instance, with three variables X, Z, W (we ignore the intercept here; add columns of 1 accordingly) and focusing on, say, unit 2, we would have the following leverage:

$$h_{2,2} = [x_2 \quad z_2 \quad w_2]_{1 \times 3} \left(\begin{bmatrix} x_1 & x_2 & x_3 \\ z_1 & z_2 & z_3 \\ w_1 & w_2 & w_3 \end{bmatrix} \begin{bmatrix} x_1 & z_1 & w_1 \\ x_2 & z_2 & w_2 \\ x_3 & z_3 & w_3 \end{bmatrix} \right)_{3 \times 3}^{-1} \begin{bmatrix} x_2 \\ z_2 \\ w_2 \end{bmatrix}_{3 \times 1}$$

which yields a scalar, h_{ii} . This scalar will be relatively large when the vector $\mathbf{x}_i$ for unit i is different from the other vectors $\mathbf{x}_j$.

The leverage h_{ii} has several properties:

1. $h_{ii} \in [0, 1]$
2. $\sum_{i=1}^N h_{ii} = k$, where k is the number of independent variables.

An indicator that can be useful is the max of the leverage across all observations. If you then take several models, you can in principle compare what this max is and see how responsive it is to the inclusion of a given variable.

Now, here's what could be tricky. The leverage h_{ii} only tells us of i 's *potential* influence. Whether it is realized depends on the *residual* u_i attached to it. What will generally happen is that observations that are outliers will have larger influence. To see this, we can use the special case of a single predictor and an intercept, which we saw when we discussed OLS with one treatment. We have:

$$\text{Design matrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \dots & \dots \\ 1 & x_n \end{bmatrix}.$$

Then, you can write:

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} n & \sum x_j \\ \sum x_j & \sum x_j^2 \end{bmatrix},$$

whose inverse is:

$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{n \sum x_j^2 - (\sum x_j)^2} \begin{bmatrix} \sum x_j^2 & -\sum x_j \\ -\sum x_j & n \end{bmatrix}.$$

We can then write the leverage as:

$$h_{ii} = \begin{bmatrix} 1 & x_i \end{bmatrix} \frac{1}{n \sum x_j^2 - (\sum x_j)^2} \begin{bmatrix} \sum x_j^2 & -\sum x_j \\ -\sum x_j & n \end{bmatrix} \begin{bmatrix} 1 \\ x_i \end{bmatrix}.$$

After some algebra, we obtain:

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum (x_j - \bar{x})^2}.$$

Here's where the problem might come from: the values with the largest h are those that are furthest from the mean (numerator of the second term). Values at the mean are not very informative and OLS is not going to "use" them as much as values that are very large (in absolute terms). Thus, values with very large x_i compared to the mean will have more influence on the regression line. If one observation has a particularly large h , then the concern is that the point estimate is not very 'robust' in the sense that another random sample might yield a very different result. Imagine, for instance, a regression of political views on income, and your sample accidentally includes Elon Musk: his leverage h_{Musk} will be very large and therefore affect $\hat{\beta}$. Next sample might then yield a very different $\hat{\beta}$ if you don't randomly get a billionaire in it.

Now, having units with large leverage doesn't invalidate our OLS estimates. We never made any assumption regarding the contribution of individual units. However, from a common-sense perspective, we might worry that our results could be affected by a handful of influent observations. They might represent values that are uncommon and that respond nonlinearly to the treatment. For instance, imagine a sample with 10 units, nine of which having an x between 0 and 10 and one having $x = 1,000,000$. It might well be that x has no effect on y at small values, but the response could be very large at high values. Then, including this observation would dramatically change the results, but do we believe them?

There is something neat that you can do here, in case you are worried about outliers (or more precisely, units with large leverage): the **leave-one-out** regression (LOOR) (B. Hansen 2022a, 84). This consists in estimating $\hat{\beta}$ based on taking out one observation out at a time (e.g., you drop one country or one firm from your sample and regress Y on $\mathbf{X}$). What's very neat is that you can do this without having to run N regressions. B. Hansen (2022a, 84) shows that you can estimate the LOOR estimate $\hat{\beta}_{-i}$ as follows:

$$\hat{\beta}_{-i} = \hat{\beta}_{\text{ols}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i\tilde{u}_i,$$

where:

$$\tilde{u}_i = \frac{u_i}{1 - h_{ii}},$$

where u_i is the residual of the OLS estimate of Y on $\mathbf{X}$. Here's the thing: if h_{ii} are small, then $\tilde{u}_i$ will be similar to u_i . And then, we generally will see that $\mathbf{x}_i \tilde{\mathbf{u}}$ is small, and as a result, the LOOR estimate $\hat{\beta}_{-i}$ will be similar to the OLS estimate $\hat{\beta}$.

Review questions

Exercise 4.1. Are OLS estimates biased, consistent, and efficient with non-normal error terms? Run a Monte Carlo simulation to find out inductively the answer.

Exercise 4.2. Reproduce the proof of unbiasedness with OLS, flagging every assumption necessary along the way.

Exercise 4.3. Run a Monte Carlo to evaluate the information generated by competing measures of fit. Concretely, start with the model:

$$y = 1 + 2x + \mathcal{N}(0, 5)$$

and increase the range of X from $X \sim U[0, 1]$ to $X \sim U[0, 100]$. How does R^2 change? What about σ^2 and F ?

Appendix

Delta method: proof sketch

The delta method follows from a first-order Taylor expansion of g around the true β :

$$g(\hat{\beta}) \approx g(\beta) + \nabla g(\beta)'(\hat{\beta} - \beta).$$

Rearranging and scaling by $\sqrt{N}$:

$$\sqrt{N}(g(\hat{\beta}) - g(\beta)) \approx \nabla g(\beta)' \sqrt{N}(\hat{\beta} - \beta).$$

Since $\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}(0, \Omega)$ by the asymptotic normality of OLS, and since $\nabla g(\beta)$ is a fixed matrix (evaluated at the true β), we apply the linear transformation property of the normal distribution: if $Z \sim \mathcal{N}(0, \Omega)$, then $\mathbf{A}Z \sim \mathcal{N}(0, \mathbf{A}\Omega\mathbf{A}')$. With $\mathbf{A} = \nabla g(\beta)'$:

$$\sqrt{N}(g(\hat{\beta}) - g(\beta)) \xrightarrow{d} \mathcal{N}(0, \nabla g(\beta)' \Omega \nabla g(\beta)).$$

In practice, replace β with $\hat{\beta}$ and Ω with a consistent estimator (homoskedastic or robust, as appropriate). Standard errors for $g(\hat{\beta})$ are then $\sqrt{\nabla g(\hat{\beta})' \hat{\Omega} \nabla g(\hat{\beta})} / N$.

Example: interaction effect. For $g(\hat{\beta}, \hat{\lambda}) = \hat{\beta} + \hat{\lambda}Z$ (the marginal effect of X when Z takes value z_0), the Jacobian is $[1, 0, \dots, z_0, \dots]'$, and the formula reduces to $\mathbb{V}[\hat{\beta}] + z_0^2 \mathbb{V}[\hat{\lambda}] + 2z_0 \text{Cov}[\hat{\beta}, \hat{\lambda}]$.

Log-likelihood of the linear model

Log-likelihoods are useful for the estimation of parameters under probabilistic models. We have used OLS for estimation so far, but other methods, such as maximum likelihood – which relies on maximizing the log of the likelihood – are powerful too. For now, the log-likelihood is useful as a step in establishing the result that OLS has the lowest feasible variance, i.e., it meets the **Cramér-Rao lower bound theorem**.

Consider the linear regression model:

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon$$

We can rewrite this as follows:

$$\mathbf{y} \sim \mathcal{N}(\mathbf{X}\beta, \sigma^2\mathbf{I}),$$

either by assuming that the error term is normally distributed with constant variance, or by using our asymptotic results (and assuming homoskedasticity).

The probability density function of a (multivariate) normal distribution is (using the definition of the normal distribution):

$$f(\mathbf{y}|\beta, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)\right)$$

The **likelihood function** $L(\beta, \sigma^2)$ is the joint probability of observing $\mathbf{y}$ given the parameters β and σ^2 . For the normal distribution, it is the product of the density functions for all observations. We then take the log of this likelihood function (i.e., the **log-likelihood**) because it's easier to differentiate.

$$\log L(\beta, \sigma^2) = \log\left(\frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)\right)\right)$$

We can simplify this expression in two steps.

1. The logarithm of the normalizing constant:

$$\log\left(\frac{1}{(2\pi\sigma^2)^{n/2}}\right) = -\frac{n}{2} \log(2\pi\sigma^2)$$

2. The logarithm of the exponential term:

$$\log\left(\exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)\right)\right) = -\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)$$

We can then write the **log-likelihood function** of the linear model:

$$\log L(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)$$

From here, see the notes to obtain the Cramér-Rao lower bound.

5 Causal point identification

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Overview

What worries applied econometricians the most is the problem of endogeneity. It creates bias and inconsistency. Here, we detail the consequences of endogeneity, what causes it, and the conditions needed to avoid it.

Readings

On the many meanings of identification, see Lewbel (2019). For limits to (point) identification, see Tamer (2010) and Manski (1990). For an introduction to DAGs in general, see Pearl (2009). A shorter introduction to DAG is Pearl, Glymour, and Jewell (2016). For DAGs with panel data, see Reisach et al. (2025).

5.1 Identification

Lower costs of computational power and the realization that econometric analysis has business value has triggered a rapid increase in empirical work in economics and other social sciences.

In parallel, the standards for what qualifies as trustworthy econometric analysis have also increased (Leamer 1983; Joshua D. Angrist and Pischke 2010). This is often referred to as the **credibility revolution**.

This notion captures several ideas, but at its core is a desire for applied econometric analysis to generate (point) estimates that are well **identified** (Lewbel 2019). Here is a useful formal definition of this concept.

Definition 5.1 (Identification). Let $F(Y, D, \mathbf{X})$ denote the population joint distribution of observable variables. An estimand θ (e.g., the ATE or the ATT) is **identified** if it can be expressed as a known function g of F alone:

$$\theta = g(F(Y, D, \mathbf{X})).$$

No other value $\theta' \neq \theta$ is consistent with the same F . When θ cannot be point-identified, the data are still consistent with a set of values (the **identified set**). We then speak of **partial identification** (Manski 1990).

Identification asks whether θ would be uniquely recoverable if you knew F exactly. Here, $F(\cdot)$ is a population object. Think of it as what you would know perfectly with infinite data. With a finite sample you only have an estimate $\hat{F}_n \rightarrow F$, so identification tells you whether, in principle, more data would get you to θ .

An **identification strategy** then does two things. First, at the population level, it states the assumptions that establish which functional form g makes $\theta = g(F)$ hold in a given context. Second, at the estimation level, it provides the implied sample estimator $\hat{\theta} = g(\hat{F}_n)$, the sample analogue of g , which converges to θ as $\hat{F}_n \rightarrow F$. The assumptions do the identification work; the estimator exploits those assumptions to compute θ from data.

What econometricians then look for are *credible* identification strategies. They define an estimand θ that they wish to estimate (e.g., ATE) and then search for a design that allows them to identify this estimand and estimate it consistently. By “credible,” we mean that the assumptions are as plausible as possible, or at the very least transparent. See Table 5.1 for the link between identification, consistency, and unbiasedness.

Concept	Question	Level
Identification	Is the target θ reachable from data?	Population
Consistency	Does the estimator reach θ as $n \rightarrow \infty$?	Asymptotic
Unbiasedness	Does the estimator point at θ in expectation at any n ?	Finite sample

Table 5.1: Identification, consistency, and unbiasedness.

i Identification problems in practice

To see why identification matters, consider the problem of returns to education: does education increase wages? Suppose you want to identify β , the causal effect of a college degree $D \in \{0, 1\}$ on wages Y . The true model is:

$$Y_i = \alpha + \beta D_i + \gamma U_i + \varepsilon_i,$$

where U is unobserved ability (e.g., being school smart) and $\text{Cov}(D, U) > 0$. This captures the idea that people who have high ability also tend to get more formal education.

Now, suppose you cannot measure ability U . Then, the distribution $F(Y, D)$ (no U !) only reveals (using the omitted variable formula):

$$\mathbb{E}[Y \mid D] = \alpha + \underbrace{\left(\beta + \gamma \cdot \frac{\text{Cov}(D, U)}{\text{Var}(D)} \right)}_{\text{one quantity from } F(Y, D)}.$$

This single estimate confounds two unknown parameters, β and $\gamma \cdot \text{Cov}(D, U) / \text{Var}(D)$. The problem then is the following: there are infinitely many pairs (β, γ) that are consistent with the same $F(Y, D)$. Thus, β is not uniquely identified from $F(Y, D)$ alone. Additional information will be needed to pin down $\theta = \beta$. As we will see later, this could be a natural experiment, an instrument, etc.

Why is this so important? Well, imagine that you were tasked to generate the estimate of the effect of an advertisement campaign on sales. How would you proceed? You will need to build a design that yields numbers you trust. And for this, “garbage in, garbage out” regressions (‘garbage in, garbage out’) won’t do.

Those involved in the credibility revolution have, fortunately, developed a set of tools and frameworks to improve the quality of empirical research. This ranges from the application of experimental methods to economic issues (next chapter) to the design of so-called **quasi-experimental** designs. But first, let us briefly investigate the sources of endogeneity (recall that this assumption was essential for OLS to yield consistent estimates). Then, we will do a quick introduction to directed acyclic graphs to have a good framework to engage in causal inference.

5.2 Endogeneity

5.2.1 What is endogeneity?

Suppose we have a linear model such as:

$$Y = \mathbf{X}\beta + \varepsilon,$$

where $\mathbf{X}$ is an $n \times k$ matrix (k parameters, including a constant) and β is a k vector of parameters. Then the **exogeneity assumption** states that:

$$\mathbb{E}[\varepsilon|\mathbf{X}] = 0$$

When this condition is *not* met, the OLS estimates are **biased** and **inconsistent** (see chapter on inference with OLS). Here's one way to build your intuition on this: suppose we want to know the effect of some treatment D on $\mathbb{E}[Y]$ and we have:

$$y = \alpha + \beta_D D + \varepsilon$$

- If D is exogenous, then $\partial\mathbb{E}[Y|D]/\partial D = \beta_D$.
- If D is endogenous ($\mathbb{E}[\varepsilon|D] \neq 0$), then a change in D will operate via β_D (good) but also via ε (bad).

Since at best we can only know the direction of the bias (often we don't even know that!) but not its magnitude, and since more data won't help (why?), we won't know if the estimates we obtain from a given dataset are anywhere close to the true population value of β_D and we can't straightforwardly draw inferences from our sample.

Typically, endogeneity arises from these causes:

1. Omitted variable bias
2. Simultaneity bias
3. Measurement error

But those are not the only reasons why things might go wrong. A non-random sample can invalidate inference about a population, for instance. For instance, if you only survey rich people about their willingness to pay (WTP) for a new product, it's unlikely that you can make valid causal inferences about the population WTP.

5.2.2 Omitted variable bias

Suppose the true model is:

$$Y = \beta D + \gamma W + \varepsilon,$$

such that $\mathbb{E}[\varepsilon|D, W] = 0$. For simplicity, let's assume that all variables have been demeaned (we thus don't need to worry about intercepts). Next, let's suppose that we estimate the “short” regression instead:

$$Y = D\rho + \mu$$

What we are wondering is: $\mathbb{E}[\mu|D] = 0$?

Let's start with the OLS definition of the slope of the “short” regression. Since all variables are demeaned, we can write:

$$\rho = \frac{\mathbb{E}[DY]}{\mathbb{E}[D^2]}.$$

Substitute the true model:

$$\mathbb{E}[DY] = \mathbb{E}[D(\beta D + \gamma W + \varepsilon)] = \beta \mathbb{E}[D^2] + \gamma \mathbb{E}[DW] + \mathbb{E}[D\varepsilon].$$

By assumption ($\mathbb{E}[\varepsilon|D, W] = 0$ from the true model), the structural error is orthogonal to the regressors:

$$\mathbb{E}[D\varepsilon] = 0$$

We can bring back the full expression for ρ (i.e., divide by $\mathbb{E}[D^2]$):

$$\rho = \beta + \gamma \frac{\mathbb{E}[DW]}{\mathbb{E}[D^2]},$$

which is the omitted variable bias formula: the parameter of the “short” regression equals the true causal effect plus the effect of the omitted variable on the outcome (γ) times the effect of treatment on the omitted variable ($\mathbb{E}[DW]/\mathbb{E}[D^2]$).

Knowing the *sign* of the two components allows analysts to anticipate the direction of the bias even without knowing its magnitude. If $\gamma > 0$ (the omitted variable raises Y) and $\mathbb{E}[DW]/\mathbb{E}[D^2] > 0$ (D and W are positively correlated), then $\rho > \beta$: the short regression *overstates* the true effect. If the two components have opposite signs, the bias is toward zero (attenuation). This kind of sign reasoning is a standard first step in assessing the credibility of an estimate.

Next, we can return to μ and define it more precisely:

$$\begin{aligned}\mu &= Y - D\rho \\ &= (\beta D + \gamma W + \varepsilon) - D \left(\beta + \gamma \frac{\mathbb{E}[DW]}{\mathbb{E}[D^2]} \right) \\ &= \gamma \left(W - D \frac{\mathbb{E}[DW]}{\mathbb{E}[D^2]} \right) + \varepsilon.\end{aligned}$$

Use the fact that $\mathbb{E}[\varepsilon \mid D] = \mathbb{E}[\mathbb{E}[\varepsilon \mid D, W] \mid D] = 0$ (law of iterated expectations). We can then take the conditional expectation of μ given D and find out when it equals zero (or not):

$$\mathbb{E}[\mu \mid D] = \gamma \left(\mathbb{E}[W \mid D] - D \frac{\mathbb{E}[DW]}{\mathbb{E}[D^2]} \right).$$

The right-hand side is not zero for a generic joint distribution of (D, W) . Therefore, in general:

$$\mathbb{E}[\mu \mid D] \neq 0,$$

i.e. the residuals from the short regression are mean-dependent on D and we have endogeneity. The conditions for this *not* to happen are:

- $\gamma = 0$ (the omitted variable has no effect on the dependent variable).
- $\mathbb{E}[W \mid D] = D \cdot \mathbb{E}[DW]/\mathbb{E}[D^2]$ almost surely. This happens if $\mathbb{E}[W \mid D] = 0$, since both sides would then be equal to zero. In practice, this means that D is unrelated to W – our other strategy to obtain unbiased estimates.

If *either* happens, then $\mathbb{E}[\mu \mid D] = 0$ and we are ‘saved.’

Sensitivity analysis for OVB

Even when OVB cannot be eliminated, we can ask: *how strong would an omitted confounder need to be to overturn our finding?* This is the logic of **sensitivity analysis** (Cinelli and Hazlett 2020).

The idea is to parameterize the confounding by two quantities: how strongly the confounder is associated with the treatment D , and how strongly with the outcome Y (after controlling for included covariates). These can be expressed as partial R^2 values — the share of residual variance explained by the confounder in each regression. The key output is a **robustness value** (RV): the minimum partial R^2 that an omitted variable would need

to have with *both* D and Y simultaneously to reduce the estimated effect to zero. A large RV indicates the finding is robust to moderate confounding.

This approach lets researchers make statements of the form: “our estimate would be overturned only by a confounder as strong as [the most influential observed covariate], which we regard as implausible.” It is increasingly expected in applied work.

5.2.3 Measurement error in treatment

Suppose you have a treatment D and that the true model is:

$$Y = \beta D + \varepsilon$$

with $\mathbb{E}[\varepsilon|D] = 0$. As above, we demean D and thus skip the need for a y-intercept.

Unfortunately, we find out that D is imprecisely measured. Let’s say you can only observe:

$$\tilde{D} = D + \mu,$$

We assume that $\mathbb{E}[\mu] = 0$ and that μ isn’t correlated with D , ε , nor Y . It’s a random shock and we call this the **classical errors-in-variables** problem. Such measurement error can happen, for instance, when you ask people questions and they can’t remember exactly the correct answer (eg, at what time did you wake up yesterday?). This happens fairly often when asking consumers about their past behavior.

Let’s see what happens then:

$$\begin{aligned} Y &= \beta(\tilde{D} - \mu) + \varepsilon && \text{replacing } D \\ &= \beta\tilde{D} + (\varepsilon - \beta\mu) && \text{rearranging} \\ &= \beta\tilde{D} + \tilde{\varepsilon} && \text{redefining} \end{aligned}$$

where:

$$\tilde{\varepsilon} := \varepsilon - \beta\mu$$

We can now ask: does the following hold...

$$\mathbb{E}[\tilde{\varepsilon}|\tilde{D}] \stackrel{?}{=} 0$$

To show that it doesn’t, it suffices to show that $\text{Cov}(\tilde{\varepsilon}, \tilde{D}) \neq 0$. The logic is: $\mathbb{E}[\tilde{\varepsilon}|\tilde{D}] = 0$ would imply $\mathbb{E}[\tilde{\varepsilon}\tilde{D}] = 0$ (by LIE) and thus

$\text{Cov}(\tilde{\varepsilon}, \tilde{D}) = 0$ (since $\mathbb{E}[\tilde{\varepsilon}] = 0$); the contrapositive gives $\text{Cov} \neq 0 \Rightarrow \mathbb{E}[\tilde{\varepsilon}|\tilde{D}] \neq 0$. The covariance is indeed nonzero: $\tilde{\varepsilon}$ contains μ , and μ is positively correlated with $\tilde{D}$ (when the shock μ is large and positive, so is $\tilde{D}$). Thus, we face an endogeneity problem:

$$\mathbb{E}[\tilde{\varepsilon}|\tilde{D}] \neq 0.$$

OLS will yield biased and inconsistent estimates. We can spell out the bias explicitly.

$$\begin{aligned} \hat{\beta} &= \frac{\text{Cov}(\tilde{D}, Y)}{\mathbb{V}(\tilde{D})} && \text{from def of OLS} \\ &= \frac{\text{Cov}(D + \mu, Y)}{\mathbb{V}(D + \mu)} && \text{from measurement error} \\ &= \frac{\text{Cov}(D, Y) + \text{Cov}(\mu, Y)}{\mathbb{V}(D) + \mathbb{V}(\mu) + 2\text{Cov}(D, \mu)} && \text{properties of var, cov} \\ &= \frac{\text{Cov}(D, Y)}{\mathbb{V}(D) + \mathbb{V}(\mu)} && \text{by assumption about } \mu \\ &= \frac{\mathbb{V}(D)\text{Cov}(D, Y)/\mathbb{V}(D)}{\mathbb{V}(D) + \mathbb{V}(\mu)} && \text{multiply and divide by var of } D \\ &= \beta \frac{\mathbb{V}(D)}{\mathbb{V}(D) + \mathbb{V}(\mu)} && \text{by def of } \beta_{\text{OLS}} \\ &= \beta \frac{\sigma_D^2}{\sigma_D^2 + \sigma_\mu^2} && \text{lighter notation} \end{aligned}$$

Thus, OLS will yield estimates that are biased toward zero, since $\sigma_D^2/(\sigma_D^2 + \sigma_\mu^2) \in (0, 1)$ if $\sigma_\mu^2 > 0$. We often call this **attenuation bias** (the estimate is an *attenuated* estimate of the true treatment effect).

Note: measurement error in Y . Classical measurement error in the *outcome* is harmless. If we observe $\tilde{Y} = Y + \nu$ with $\nu \perp D$ and $\nu \perp \varepsilon$, regressing $\tilde{Y}$ on D still gives a consistent estimate of β : ν is absorbed into the residual without violating exogeneity. The cost is larger residual variance and wider standard errors, but no bias.

Note: non-classical measurement error. The derivation above assumes μ is independent of D , Y , and ε — the *classical* case. In practice, measurement error is often non-classical: respondents systematically over- or under-report D (e.g., income, hours worked), or the error correlates with the true value. Non-classical measurement error can bias in either direction and does not generally produce attenuation. The classical case should be understood as the favorable benchmark.

5.2.4 Simultaneity bias

Some data-generating processes involve a system of equations. For instance, markets are typically characterized by two equations: one for demand and one for the supply side. We may therefore end up with situations in which two variables cause each other, such as price and quantity.

Consider the following two equations:

$$\begin{aligned} Y &= \beta D + \varepsilon \\ D &= \tau Y + \mu. \end{aligned}$$

As in the previous two cases, we demean D ; we also demean now Y as well, so that we don't need any y-intercept. We will assume that $|\tau\beta| \neq 1$.

Suppose we are interested in the effect of D on Y and want to know if we can just estimate the first equation. Can we assume that $\mathbb{E}[\varepsilon|D] = 0$?

We can write:

$$\begin{aligned} D &= \tau Y + \mu \\ &= \tau(\beta D + \varepsilon) + \mu && \text{plugging in for } Y \\ D(1 - \tau\beta) &= \tau\varepsilon + \mu && \text{rearranging} \\ D &= \varepsilon \frac{\tau}{1 - \tau\beta} + \frac{\mu}{1 - \tau\beta} && \text{defined since } |\tau\beta| \neq 1 \end{aligned}$$

But note that D will therefore be correlated with ε . As we stated above: a nonzero correlation implies dependence (though the reverse isn't true). As a result, $\mathbb{E}[\varepsilon|D] \neq 0$.

5.2.5 Selection bias

A fourth threat to valid causal inference arises when the **sample is not drawn randomly from the population of interest**. Even with a perfectly specified model and no simultaneity or omitted variables, inference can fail if the probability of being included in the sample depends on the outcome or on variables correlated with the treatment.

Suppose the true relationship is $Y = \alpha + \beta D + \varepsilon$ with $\mathbb{E}[\varepsilon|D] = 0$ in the population. Now suppose that only individuals with $Y > \bar{Y}$ (say, above a threshold) appear in the dataset — perhaps because they were the ones who responded to a survey, or because the dataset is

drawn from administrative records that only capture a subgroup. The observed sample then has $\mathbb{E}[\varepsilon|D, \text{selected}] \neq 0$, and OLS on the selected sample is biased even though the population model is correctly specified.

A common form of selection bias is **survivorship bias**: only units that “survived” some process of selection (firms that did not go bankrupt, patients who did not drop out of a trial, soldiers who returned from combat) are observed. Estimates based on survivors will be systematically misleading about the full population.

Selection bias is particularly insidious because it can masquerade as a genuine causal relationship: the analyst may not even know that the sample is selected, and more data from the same source will not fix the problem. The DAG perspective makes this transparent: selection is a collider — it is jointly caused by D and Y (or variables correlated with them) — and conditioning on a collider opens a spurious path between D and Y (see the collider bias discussion below).

5.2.6 Endogeneity

To summarize: we have multiple threats that can generate endogeneity. And the exogeneity assumption is critical to obtain the most desirable properties of OLS, namely unbiasedness and consistency. Thus, a lot of research in econometrics has been about ways to handle this problem (Leamer 1983; Joshua D. Angrist and Pischke 2010). You may encounter this in references to concepts such as the **credibility revolution** or the **identification revolution**.

To connect back to Chapter 2’s framework: the **estimand** we are typically after is the ATE, ATT, or LATE — a property of the potential outcome distribution. The endogeneity problems above are precisely the conditions that prevent us from *identifying* that estimand from observational data. Solving the identification problem means constructing a research design (or finding a set of assumptions) under which a consistent estimate of the target estimand is achievable. The rest of this course is largely about such designs. But before we look at this, we first take a detour via directed acyclic graphs as a method to make sense of data-generating processes.

5.3 Directed Acyclic Graphs

Once we move away from the world of RCTs and experimental data, we will have to work with **observational data**. The challenge here is the one discussed earlier: with observational data, we face the risk

of confounding and erroneously attribute causal effects where there are only spurious correlations. That's because we don't control the assignment of treatment.

As a result, we are moving to the world of a single treatment linked to a given outcome. Instead, we must think hard about the **data-generating process** (the process by which observational data is generated).

A useful tool to do so are **directed acyclic graphs** (DAGs). DAGs allow us to construct models rigorously (Pearl 2009; Pearl, Glymour, and Jewell 2016; Pearl and Bareinboim 2024).

5.3.1 Graphs

A **graph** consists of two elements:

1. Vertices (or nodes). These will be variables.
2. Edges. These will be paths of dependence.

There exists different types of graphs:

- Directed or not. A directed graph establishes causal relations of dependence. Edges are then “arrows” from one node to another.
- Among directed graphs (DAG): cyclic or not. A directed graph in which a causal path can start and return to the same variable is *cyclic*.

Since we are interested in quantifying causal estimates, we will be working with directed graphs. And in most cases, but not all, we will be interested in directed acyclic graphs (DAGs).

In principle, a DAG can be expressed as a system of equations. For instance:

$$\begin{aligned} D &= 3Z + f_x(U_x) && \text{with } f_x(U_x) \sim \mathcal{D}_x \\ Z &= f_z(U_z) && \text{with } f_z(U_z) \sim \mathcal{D}_z \\ W &= 2Z + f_w(U_w) && \text{with } f_w(U_w) \sim \mathcal{D}_w \\ Y &= 0.5D + 1.1Z - 0.4W + f_y(U_Y) && \text{with } f_y(U_Y) \sim \mathcal{D}_y \end{aligned}$$

This system contains endogenous and exogenous variables. Exogenous variables are not explained by the model. These are $f_i(U_i)$ which are drawn from some arbitrary distribution and are often called error terms. Endogenous variables are explained by the system.

However, in practice, we often don't know what the functional forms relating all variables are. Thus, we often use a visual representation

Note that the definition of endogenous in a DAG is different from the way this concept is used in econometrics (and later this chapter).

of the graph, as in Figure 5.1. The system of equation above resembles the third DAG (minus the error terms that are removed to save space).

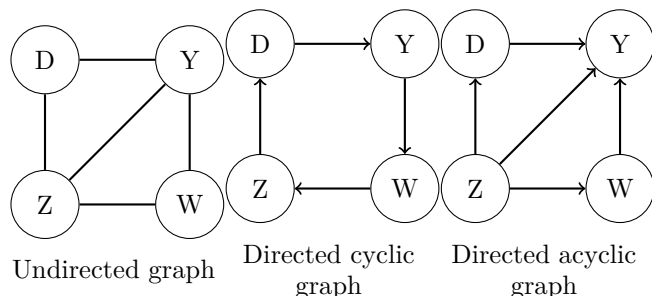


Figure 5.1: Three types of graphs.

5.3.2 Lessons from DAGs

For a DAG to be useful, it must be connected to a probability distribution. The bridge is the following assumption.

Definition 5.2 (Causal Markov condition). A DAG $\mathcal{G}$ and a joint distribution P over its variables satisfy the **Causal Markov condition** if every variable V is conditionally independent of its non-descendants (*NonDesc*) given its parents (*Pa*) in $\mathcal{G}$:

$$V \perp\!\!\!\perp \text{NonDesc}(V) \mid \text{Pa}(V).$$

Under this condition the joint distribution factorizes as a product of local conditionals:

$$P(V_1, \dots, V_k) = \prod_{j=1}^k P(V_j \mid \text{Pa}(V_j)).$$

The Causal Markov condition is what makes graphs informative about distributions: it says that the DAG encodes *all* the conditional independence restrictions implied by the causal structure. In particular, it is the foundation for d-separation (below): every conditional independence statement predicted by d-separation holds in any distribution satisfying the Causal Markov condition.

We can use DAGs to figure out how to build a model to estimate the causal effect of D . We can proceed in three steps.

First, there are three types of paths depending on relations between nodes:

1. **Forks:** a node has two causal arrows *departing* from it. For instance: $D \leftarrow Z \rightarrow Y$ has a fork (Z).
2. **Mediators:** a node has a causal arrow pointing at it and another pointing away. For instance: $D \rightarrow M \rightarrow Y$ has a mediator (M).
3. **Colliders:** a node has two causal arrows *pointing* at it. For instance: $D \rightarrow W \leftarrow Y$ has a collider (W).

These three path structures determine when information flows between nodes in a DAG. The key formal concept is **d-separation** (the “d” stands for directional). It tells us, given a conditioning set, whether two nodes are conditionally independent.

Definition 5.3 (D-separation). A path between nodes D and Y is **blocked** by a conditioning set $\mathbf{S}$ if:

1. The path contains a non-collider node V (a fork or mediator on the path) and $V \in \mathbf{S}$, or
2. The path contains a collider node V and neither V nor any descendant of V is in $\mathbf{S}$.

Two nodes D and Y are **d-separated** given $\mathbf{S}$ if every path between them is blocked. D-separation implies conditional independence:

$$D \perp\!\!\!\perp Y \mid \mathbf{S}.$$

The intuition for the two conditions:

- **Non-colliders (condition 1).** A fork $Z \rightarrow D$ and $Z \rightarrow Y$ transmits correlation between D and Y : knowing Z tells you something about D and something about Y , so D and Y become correlated through Z . Conditioning on Z closes that channel. This is the statistical reason why controlling for a confounder in a regression removes its confounding effect.
- **Colliders (condition 2).** A collider $D \rightarrow W \leftarrow Y$ naturally blocks the path: by default, the causes D and Y are independent within each level of W . But conditioning on W opens the path. If you know the joint effect W and one of the causes, you can partially infer the other cause. This is why adjusting for a collider is harmful: it creates a spurious association where none existed before.

D-separation is a powerful reading tool: all the conditional independence statements a DAG predicts about the data can be read off by checking d-separation, without any parametric assumptions.

Second, the key necessity to estimate causal effects of a treatment D on an outcome Y is that we must *block backdoors*. Backdoors are causal paths that go from the treatment to the outcome via a path that includes an arrow pointing at the treatment. For instance, suppose that you have the following two paths:

1. $D \rightarrow M \rightarrow Y$
2. $D \leftarrow Z \rightarrow Y$

Third, not all paths must be blocked. Only (a) *open* (b) *backdoors* must be blocked. This means that we must only worry about paths from D to Y where an arrow is pointing at D . And among these paths, we must worry about those that are open. What makes them open? The absence of a collider! Thus:

- $D \leftarrow Z \rightarrow W \rightarrow Y$ is an open backdoor and must be closed.
- $D \leftarrow Z \rightarrow W \leftarrow Y$ is a closed backdoor (W is a collider that closes it) and must not be touched.

With OLS, closing a backdoor typically means adjusting for a variable on an open backdoor. In the first example, it would mean including Z or W as a right-hand side covariate ($y = \alpha + \beta D + \gamma Z + \varepsilon$ or $y = \alpha + \beta D + \rho W + \varepsilon$).

Blocking all open backdoors by conditioning on a set $\mathbf{X}$ is precisely the **unconfoundedness** assumption from Chapter 2: $(Y^0, Y^1) \perp D \mid \mathbf{X}$. The DAG makes this assumption transparent: given a causal graph, we can check whether the variables we condition on are sufficient to block every open backdoor path from D to Y .

The DAG framework formalizes this through the **backdoor criterion**. Before stating it, we need to distinguish two types of conditioning.

Definition 5.4 (Observation versus intervention: the $do(\cdot)$ operator). The **conditional expectation** $\mathbb{E}[Y \mid D = d]$ describes the average Y among units that *happen to have* $D = d$ in the observed data. It is a property of the joint distribution $F(Y, D)$ and mixes causal effects with selection effects.

The **interventional expectation** $\mathbb{E}[Y \mid do(D = d)]$ describes the average Y if we were to *force* every unit to take $D = d$ by external intervention, while leaving the rest of the data-generating process unchanged (Pearl 2009). In DAG terms, the $do(\cdot)$ operation removes all arrows pointing into D (eliminating any backdoor paths) and sets $D = d$.

These two quantities coincide — $\mathbb{E}[Y \mid do(D = d)] = \mathbb{E}[Y \mid D = d]$

— only when D has no open backdoor paths (i.e., D is exogenous). Under confounding they diverge, and it is $\mathbb{E}[Y \mid do(D = d)]$ that captures the causal effect of interest.

Definition 5.5 (Backdoor criterion). A set of observed variables $\mathbf{S}$ satisfies the **backdoor criterion** relative to the ordered pair (D, Y) if:

1. No variable in $\mathbf{S}$ is a descendant of D .
2. $\mathbf{S}$ blocks every path between D and Y that contains an arrow pointing into D .

When the backdoor criterion holds, the causal effect of D on Y is identified by the **adjustment formula**:

$$\mathbb{E}[Y \mid do(D = d)] = \sum_s \mathbb{E}[Y \mid D = d, \mathbf{S} = s] \mathbb{P}(\mathbf{S} = s).$$

Why condition 1 matters. Including a descendant of D in $\mathbf{S}$ means conditioning on something that D itself affects. This can block legitimate causal paths from D to Y (if the descendant is a mediator) or open new spurious paths (if it is a collider). Condition 1 prevents this error.

Connection to regression. In a linear model, the adjustment formula is exactly what OLS computes when all variables in $\mathbf{S}$ are included on the right-hand side as “control” variables: the coefficient on D estimates $\partial \mathbb{E}[Y \mid do(D)] / \partial D$, provided the backdoor criterion holds for $\mathbf{S}$.

Note:

- You *don’t* want to adjust for (i.e., “control for”) a variable that is a mediator on a path from D to Y and pointing away from D (Montgomery, Nyhan, and Torres 2018). For instance, in the path $D \rightarrow M \rightarrow Y$, you don’t want to adjust for M . Doing so would “hide” part of the effect of D on Y .
- You *don’t* want to adjust for a variable that is a collider on a path between D and Y (Elwert and Winship 2014). In the path $D \rightarrow W \leftarrow Y$, you don’t want to adjust for W . If you do so, you open an otherwise closed backdoor.
- You *do* want to adjust for a variable that is a fork pointing toward D and toward Y .

Collider bias in practice. A common mistake is restricting a sample based on a collider. Suppose you study the effect of physical fitness (D) on heart disease (Y) using data from hospital admissions.

Being hospitalized (W) is caused by both low fitness and by heart disease: $D \rightarrow W \leftarrow Y$. If you condition on hospitalized patients — i.e., restrict to $W = 1$ — you open the path $D \rightarrow W \leftarrow Y$, inducing a spurious association between fitness and heart disease that does not exist in the general population. This phenomenon, known as **Berkson’s paradox** or **collider stratification bias**, explains why many findings from clinical datasets fail to replicate in population samples (Elwert and Winship 2014).

Example 5.1 (Collider bias in policing research). A big topic in the United States and elsewhere is the (mis)use of force by the police and whether they target suspects of different races differently. Know, Lowe, and Mummolo (2020) examine whether existing studies might be tainted by collider bias.

To see this, the starting point is to note that past studies often ask: “among people who were stopped by police, are Black individuals more likely to be subjected to force?” By doing so, they condition on police contact having occurred. But contact is a **collider**! It is jointly caused by the suspect’s race and by the officer’s assessment of threat level.

We can put this in a DAG. Let D denote race (say, Black and White to keep things simple), Y denote use of force, and W denote police contact. The relevant path structure is $D \rightarrow W \leftarrow \text{threat} \rightarrow Y$, making W a collider. By construction, conditioning on $W = 1$ (i.e., restricting the sample to individuals who were stopped) opens this otherwise closed path. This induces a (spurious) negative correlation between race and the officer’s threat assessment. This can make racially motivated stops appear race-neutral, or even make it look as though Black civilians are treated *more* leniently. This would be exactly the wrong conclusion.

Using administrative data from the Dallas and Houston Police Departments, Knox et al. decompose the use of force across the race of individuals into two pieces:

- A component attributable to the contact decision, and
- A component attributable to behavior conditional on contact.

After correcting for collider bias, they find that **Black civilians are substantially more likely to be subjected to use of force than White civilians at comparable threat levels.**

5.3.3 Frontdoor criterion

The backdoor criterion handles confounding by adjusting for observed covariates. But what if the confounders are unobserved? In some cases, a different approach still allows identification even if we *don't* have access to all variables we would need!

Suppose there is an unmeasured variable U that affects both the treatment D and the outcome Y . That's a backdoor path $D \leftarrow U \rightarrow Y$. Further suppose that U is unobservable and thus we cannot include it (e.g., it is something invisible like 'ability'). It could look like identification is impossible. But if D affects Y entirely through an observable mediator M , and if the path from D to M is itself unconfounded, identification is still achievable! That's the **frontdoor criterion**.

Definition 5.6 (Frontdoor criterion). A set of observed variables $\mathbf{M}$ satisfies the **frontdoor criterion** relative to (D, Y) if:

1. All directed paths from D to Y pass through $\mathbf{M}$.
2. There are no unblocked backdoor paths from D to $\mathbf{M}$.
3. All backdoor paths from $\mathbf{M}$ to Y are blocked by D .

When the frontdoor criterion holds, the causal effect is identified by the **frontdoor formula**:

$$\mathbb{E}[Y \mid do(D = d)] = \sum_m \mathbb{P}(M = m \mid D = d) \sum_{d'} \mathbb{E}[Y \mid M = m, D = d'] \mathbb{P}(D = d').$$

Intuition. The formula decomposes the total effect into two identified legs. Condition 2 says there is no confounding on the $D \rightarrow M$ path, so the effect of D on M is identified by the simple conditional $\mathbb{P}(M \mid D)$. Condition 3 says all backdoors from M to Y run through D , so conditioning on D closes them, identifying the effect of M on Y . Summing over M and over the marginal distribution of D combines the two legs into the total causal effect of D on Y .

Classic example: smoking and cancer. Suppose smoking (D) causes lung cancer (Y) entirely through the accumulation of tar deposits in the lungs (M). One could worry that an unobservable genetic trait U affects both the propensity to smoke and cancer risk. This would open the backdoor $D \leftarrow U \rightarrow Y$. Since U is unobserved we cannot block this path by conditioning. That's a line of defense of the tobacco industry: it's not cigarettes that cause cancer, it's this genetic trait that is a confounder. And we cannot disprove it, since we cannot estimate models including this U !

However, the frontdoor criterion can help us: (1) the only directed path from D to Y goes through M ; (2) there is no confounding on $D \rightarrow M$ since U does not directly affect tar deposits; (3) the backdoor from M to Y via $M \leftarrow D \leftarrow U \rightarrow Y$ is blocked by conditioning on D . The frontdoor formula then identifies the effect of smoking from data on smoking rates, tar levels, and cancer rates alone, without ever observing the genetic confounder.

i Frontdoor versus instrumental variables

The frontdoor criterion and instrumental variables both deal with identification in the presence of unobserved confounders. However, they do so differently. An instrument Z uses an external source of variation in D that affects Y only through D . The frontdoor criterion exploits a mediator M through which D affects Y , where the D -to- M segment is unconfounded. Both require strong knowledge of the DGP (and the DAG). In practice, credible instruments are hard to find, and a credible mediator requires knowing the full mechanism from D to Y .

Still, the problem is that we often don't know with any degree of certainty what the correct DAG is. Think of the following example: we want to measure the effect of an increase in interest rates (D) on consumer confidence (Y). How could we build a DAG? There could be all kinds of variables affecting both D and Y , including business cycles, trade shocks, unemployment, wars, etc. Of course, an expert in the domain can build more convincing DAGs than a random person. But will we ever be sure that we have the right model? Probably not, though **causal discovery** can help (see below).

5.4 Recent developments

The field of causal inference and DAGs is moving *fast*. Here are two areas where there has been a lot of developments in recent years.

Causal discovery. In genera, DAGs are built by researchers on the basis of their domain knowledge. For instance, a medical researcher interested in the effect of smoking will build a DAG based on what they know about public health. But building a convincing DAG is very demanding. Instead, a growing literature on **causal discovery** asks whether the causal structure can itself be learned from data. Under a set of assumptions (mostly: no unmeasured confounders exist and that the joint distribution is Markov with respect to the true DAG), algorithms such as the **PC algorithm** (Spirtes, Glymour, and Scheines 2000) can recover the *Markov equivalence class* of the true

graph (the skeleton and the orientations of some edges) from conditional independence tests. Extensions such as the **FCI algorithm** relax the no-hidden-confounders assumption and can flag the presence of unmeasured common causes. These methods are particularly useful when the number of potential variables grows large and expert knowledge about the causal structure is limited or debated.

DAGs with panel data. DAGs are most naturally applied to cross-sectional data (i.e., each unit is measured once at a given time). However, panel data (repeated observations of units over time; e.g., a dataset of countries measured every year) brings in a temporal dimension that can make identification easier: assuming causes precede their effects, arrows in a panel DAG must respect the time ordering of variables. This rules out certain backdoor paths by construction (a variable measured at time $t + 1$ generally cannot cause a variable at time t), and it is part of the identification logic behind difference-in-differences, fixed effects, and other panel methods that we cover later. Reisach et al. (2025) discuss the extension of DAGs to panel settings, including how to represent time-varying confounders and lagged effects within the graph.

6 Randomized controlled trials

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Overview

We saw that one of the key problems of econometrics is how to avoid omitted variable bias. Simplifying, this is the set of variables that are causing both the treatment and the outcome. To avoid it, we would like to have situations in which the treatment is uncorrelated to confounders. How can we achieve this? By randomizing the treatment: if the treatment is truly randomly assigned, it won't be correlated to confounding factors (in expectation). That's the idea behind randomized controlled trials (RCTs), which have become extremely common in business (e.g., A/B testing) and in research.

Readings

People interested in experimental methods in the social sciences should refer to S. L. Morgan and Winship (2014) (on causality writ large), Imbens and Rubin (2015) (Part II) (on randomized experiments), Athey and Imbens (2016) (on the econometrics of RCTs), and Alan S. Gerber and Green (2012) (on field experiments).

6.1 Randomized Controlled Trials (RCTs)

6.1.1 Overview

To overcome the limitations caused by incomplete knowledge of DAGs, researchers can turn to **research designs that do not require full knowledge of the DAG**. What we need is to find a design in which we can confidently say that there are no open backdoors in our DAG. One way to do so is by taking control over the assignment of the treatment D : if we can decide who gets the

treatment and who doesn't, we can break the correlation between D and other variables Z that could confound D 's effect.

That's where **randomization** comes in. The intuition is simple: if you randomize D (eg, use a coin toss to decide who gets the treatment and who doesn't), then by construction, nothing will be correlated with it (at least in expectation). Thus, by taking control over D 's assignment, we break all the arrows pointing at D : the variation of D is not caused by anything else than our randomization.

Thus, **randomized controlled trials** are very popular: we can ensure that exogeneity is met by design, thereby increasing the odds that our estimates are unbiased and consistent.¹ It is popular in academic research, including many fields in the social sciences. Examples of papers using RCTs include (Duflo, Kremer, and Robinson 2009; Aklın et al. 2017; Asquith 2020). It is also popular in industry, where it's often referred to as **A/B testing**, where two groups receive different versions of an intervention, and outcomes are then compared. Types of RCT experiments include:

- **Lab experiment:** experiments conducted with individuals in a controlled lab environment. For instance, this includes experiments in which people playing with each others in a game theoretic setting.
- **Field experiments:** experiments conducted in a real-life setting. For instance, this includes experiments conducted with teachers in schools.
- **Survey experiments:** experiments conducted within surveys; people receive randomized content. For instance, people might receive competing ads and the analyst wants to know which one makes people more interested in a product.

Unfortunately, RCTs cannot always be used. There are ethical and practical constraints that limit their scope. More on this below. But this explains why people often switch gears and use **quasi-experimental** methods. These methods try, by and large, to find exogenous variation in D that makes a given context *look* like an RCT. This includes methods such as **instrumental variables**. More on this in separate course notes.

6.1.2 The problem

Consider the following starting point. Suppose we had a dataset with a treatment $D \in \{0, 1\}$ and an outcome Y . Suppose that D is not

¹Technically, we will still need our earlier assumptions, including a random sample and no perfect multicollinearity.

exogenous. There is some variable out there that drives some units i to get the treatment and others not, and that this variable also causes Y . This is a typical example of confounding. What would be the meaning of our naive comparison of outcomes for treated units versus control units?

$$\begin{aligned}
\text{Naïve difference} &= E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 0] \\
&= E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 1] \\
&\quad + E[Y_i(0)|D_i = 1] - E[Y_i(0)|D_i = 0] \quad \text{adding and subtracting} \\
&= E[Y_i(1) - Y_i(0)|D_i = 1] \\
&\quad + E[Y_i(0)|D_i = 1] - E[Y_i(0)|D_i = 0] \quad \text{regrouping} \\
&\equiv \text{ATT} + \text{Selection (baseline) bias}
\end{aligned}$$

where **ATT** refers to the average treatment effect on the treated units.

The question is: how to get rid of selection bias?

6.1.3 Randomization of treatment

By randomizing D_i , we make D independent from $Y(0)$, $Y(1)$ and functions thereof. We typically write:

$$(Y(0), Y(1)) \perp D.$$

We call this the **ignorability assumption** (Rosenbaum and Rubin 1983). This means that D is independent of $Y(0)$, $Y(1)$, and of functions of the two (such as $Y(1) - Y(0)$). Thus, learning whether a unit was treated doesn't yield any information about the potential outcomes or the treatment effect.² A useful implication of ignorability is that: $E[Y(0)|D] = E[Y(0)]$ and $E[Y(1)|D] = E[Y(1)]$.

Note that we might sometimes write:

$$(Y(0), Y(1)) \perp D | \mathbf{X},$$

in which case we say that $Y(0)$ and $Y(1)$ are *conditionally* independent of D . This would be an assumption we would need to make if $\mathbf{X}$ is correlated with D and Y . We call this the **conditional ignorability assumption**.

Here's the neat part: randomization guarantees, in expectation, that this assumption is met. I say "*in expectation*" because we could by

²Of course, if the treatment effect is different from zero, then D and (observed) Y will be dependent.

bad luck fail to have independence. For instance, imagine that I randomize assignment of aspirin to a sample of 100 patients, with 50 women and 50 men. I could accidentally randomize it in a way that all the men get the aspirin and all the women are in the control group. If gender is a confounder, then this random assignment won't help us.

Setting this threat aside, we can simplify:

$$\begin{aligned}
\text{Naïve difference} &= E[Y_i(1) - Y_i(0) | D_i = 1] \\
&\quad + E[Y_i(0) | D_i = 1] - E[Y_i(0) | D_i = 0] && \text{from above} \\
&= E[Y_i(1) - Y_i(0) | D_i = 1] && \text{by ignorability} \\
&= E[Y_i(1) - Y_i(0)] && \text{by ignorability} \\
&\equiv \text{Average treatment effect}
\end{aligned}$$

Thus, randomization removes selection bias *and* makes the ATT equal the ATE.

The latter makes intuitive sense: the ATE and ATT will diverge if, for instance, units can decide by themselves to take the treatment based on their knowledge of its effectiveness. But if treatment assignment is random, then there is no reason to expect that the population taking the treatment is different in a meaningful way from the population not taking it, and thus ATE and ATT will coincide.

6.1.4 Estimation

Since we can ignore selection and confounding, we could estimate the ATE by plugging in sample averages and taking the difference in means:

$$\widehat{\text{ATE}} := E[\widehat{Y_i(1)} - Y_i(0)] = \frac{1}{N_1} \sum_{i:D=1} Y_i - \frac{1}{N_0} \sum_{i:D=0} Y_i,$$

where N_1 and N_0 is the number of treated and control units.

We can then compute the standard error of this expression:

$$\begin{aligned}
se(\widehat{\text{ATE}}) &= se \left(\frac{1}{N_1} \sum_{i:D=1} Y_i - \frac{1}{N_0} \sum_{i:D=0} Y_i \right) \\
&= \sqrt{\frac{\sigma_{y1}^2}{N_1} + \frac{\sigma_{y0}^2}{N_0}}
\end{aligned}$$

We will need to plug in sample estimates of σ_{y1}^2 and σ_{y0}^2 (the variance of $Y(1)$ and $Y(0)$, respectively).

Alternatively, we can estimate a linear model with OLS:

$$Y = \hat{\alpha} + \hat{\beta}D + u,$$

where $\hat{\beta}$ is the estimate of the ATE. The advantage of using OLS is that it allows us to include additional covariates in the model. As long as these covariates are not (post-treatment) mediators (Montgomery, Nyhan, and Torres 2018), and as long as they have an effect on Y , their effect will be to reduce the size of the squared residuals and thus reduce standard errors. We can then also use whatever types of standard errors we want depending on the design of our study (see below on clustered standard errors).

A more robust approach to covariate adjustment is the **Lin (2013) estimator** (Lin 2013) : include both demeaned covariates $\tilde{\mathbf{X}} = \mathbf{X} - \bar{\mathbf{X}}$ and their interactions with treatment $D \times \tilde{\mathbf{X}}$ in the regression. Because D is randomized, the OLS point estimate on D is unbiased with or without controls; but the Lin estimator is asymptotically at least as efficient as OLS with controls alone and avoids finite-sample extrapolation bias that can arise when treatment and control groups have different covariate distributions.

Randomization inference. An alternative to the asymptotic t -test is **randomization inference** (R. A. Fisher 1935) : under the sharp null hypothesis that the treatment had no effect on any unit ($Y_i(1) = Y_i(0)$ for all i), the treatment labels are exchangeable across observations. We can permute the treatment assignment vector, recompute $\hat{\beta}$ for each permutation, and build the exact distribution of the test statistic under the null. The fraction of permutations yielding a value at least as extreme as the observed $\hat{\beta}$ is an exact p -value — no asymptotic normality required. This makes randomization inference particularly attractive in small samples or when the normal approximation is suspect.

6.2 RCT in practice

In practice, here are typical steps used to conduct an RCT:

1. **Define the population and unit of analysis.** Are you looking at firms? At consumers? At municipalities? Draw a representative sample from that population.
2. **Define the treatment and outcomes.** Specify the treatment of interest (e.g., a direct change or an encouragement to take action) and measurable outcomes.

3. **Measurement strategy.** Decide how you will gather data on both treatment and outcomes, such as surveys or administrative data.
4. **Hypothesis building .** Formulate hypotheses that state the expected relationship between the treatment and outcome (e.g., $H_0 : \beta = 0$ versus $H_1 : \beta \neq 0$).
5. **Conduct a power analysis .** Calculate the minimum sample size required to detect a meaningful effect. Power analysis helps ensure that the study is neither underpowered (too small to detect effects) nor overpowered (excessively costly). More below.
6. **Measure pre-treatment data.** These data can be used in several ways:
 - Check balance of covariates between treatment and control group: did your randomization really break correlations between D and potential confounders?
 - Use these variables later as controls to increase the precision of our estimates.
7. **Assign treatment.** Use a random number generators to treat units in your sample. Generally: $\mathbb{P}(\text{Treatment}) = 0.5$ maximizes power when outcome variances are similar under treatment and control ($\sigma_{y1}^2 \approx \sigma_{y0}^2$); in general, the optimal split is $p^* = \sigma_{y1}/(\sigma_{y1} + \sigma_{y0})$. You can use Bernoulli trials.
8. **Implement treatment.** If needed, collect data on treatment compliance (more on this below).
9. **Measure outcome(s).**
10. **Estimate treatment effects.** This may happen with or without adjustments for pre-treatment variables.

Multiple testing

Some studies look at several outcomes and whether they are affected by a treatment. This is fine in terms of obtaining a consistent estimate: $\hat{\beta}$ is still consistent and unbiased.

However, it is problematic from an *inferential* standpoint. Imagine you add outcomes after outcomes. You are bound to get some statistically significant effects even when the treatment has no effect. This is the problem of **multiple comparison**.

In RCTs, multiple comparison is especially acute: studies typically collect data on many outcomes. Imagine that you collect a

large number of outcome: even if your treatment has no effect, at least α (significance level) will turn out to be significant by chance.

What you want to do is the following. First, make sure to pre-specify what outcomes you have reasons to model. You want to avoid fishing for results. Second, you will need to correct p values for the number of tests. We saw this earlier when we discussed corrections such as Bonferroni or Benjamini-Hochberg

6.2.1 Stratified randomization

Random assignment of units to treatment generates ignorability *in expectation* (i.e., as n grows large). However, you could still have covariate imbalance (between treatment and control) by chance, especially in small samples. Suppose for instance that you want to estimate the effect of giving students a laptop on their grades. Suppose that your sample is an intro class with 50% mechanical engineers and 50% mathematicians. Your randomization could accidentally yield a study in which most treated units are mechanical engineers and most control units are mathematicians.

Stratified (block) randomization is a strategy to reduce this risk. You first partition the sample into blocks defined by discrete covariate values, then you randomize independently within each block. In our example, instead of randomizing with probability 0.5 over the entire class, we would randomize with probability 0.5 *within* mathematicians and with probability 0.5 *within* mechanical engineers. That way, we ensure that there will be no correlation between students' majors (math or mech) and their treatment status (laptop or not).

The gain is quite intuitive to grasp: by construction, treatment and control are balanced (i.e., have similar levels) on the stratifying variable. As a result, between-strata variation does not contribute to estimation error. The variance of the difference-in-means estimator under stratified randomization is:

$$\text{Var}(\hat{\beta}_{\text{strat}}) = \sum_s w_s^2 \left(\frac{\sigma_{y1,s}^2}{N_{1,s}} + \frac{\sigma_{y0,s}^2}{N_{0,s}} \right),$$

where $w_s = N_s/N$ is the share of stratum s in the sample and subscript s denotes within-stratum quantities. This variance is never larger than the variance under simple randomization and is strictly smaller whenever strata differ in mean outcomes.

In practice: always stratify on variables that strongly predict Y (e.g., geography, income, etc.). During the estimation stage, include the

stratum variable(s) as controls in the final regression to account for the within-strata structure of the design.

A problem you may face: you randomized treatment assignment, but it failed in the sense that you have imbalance on key variables between the treatment and the control group. The implication is that you now have a variable (or more) that is correlated with the treatment (because randomization failed) and with the outcome. For instance, imagine you have a list of 100 customers. You can randomize them into treatment and control. Then, you collect data about them (age, gender, income, etc.). You realize that the randomization didn't work (e.g., treated units are substantially older than control). What can you do? One approach is to re-randomize (K. L. Morgan and Rubin 2015). This requires having well-defined criteria for what constitutes an acceptable randomization (since there will always be some difference on average between treatment and control group on covariates $\mathbf{X}$). Another is to use the Lin (2013) estimator in which you regress Y on D as well as demeaned $\mathbf{X}$ and the interaction of D and these demeaned $\mathbf{X}$'s.

6.3 Power analysis

6.3.1 Statistical power and sample size selection

This step – selecting a sample size – differs from what we have seen so far. We didn't really consider the possibility until now that we could choose the size of our sample: we took whatever data we received as a given. But in RCTs, you can control (within your budget constraint) the size of the sample. This is very interesting: we know that standard errors are responsive to sample size (the larger the sample, the smaller the standard errors). Thus, we can ask ourselves: what N would I need to detect an effect of size β ?

To get to this idea, we need a few definitions:

- **Type I error** : the probability of rejecting the null hypothesis when the null was in fact true, ie:

$$\alpha = \mathbb{P}(\text{reject } H_0 | H_0 \text{ true})$$

and is the probability of a **false positive**. It's often referred to as α (not to be confused with the y-intercept in the linear model).

- **Type II error** : the probability of failing to reject the null hypothesis when the alternative is true, ie:

$$\beta = \mathbb{P}(\text{fail to reject } H_0 | H_0 \text{ false})$$

and is the probability of a **false negative**. It's often referred to as β (not to be confused with the slope of the linear model).

- **Statistical power** : the probability of rejecting the null when the null was in fact false, ie:

$$\kappa = \mathbb{P}(\text{reject } H_0 | H_0 \text{ false}) = 1 - \beta$$

6.3.2 Minimum detectable effect (MDE)

In general, we would like power (κ) to be high: why bother do an RCT if we end up with low probability of detecting an effect? But how can we set it? To do so, we will need to come up with a reasonable guess regarding the anticipated treatment effect.

To see this, let us start with the following idea: given a research design and what we know about a given treatment, what is the required sample size to detect the **minimum detectable effect (MDE)**, which is the smallest effect size that a study can detect with a specified probability of achieving statistical significance? Let's work through an example.³

Consider the example in Figure 6.1. Suppose that we want to test the null hypothesis that $\beta = 0$. Yet suppose that the true effect is $\beta = 3$. If the true effect is $\beta = 3$, then our distribution of $\hat{\beta}$ will be centered on 3 (since it is unbiased). However, some $\hat{\beta}$'s that we will obtain over repeated samples will be smaller and some larger than 3. In fact, some will be so small that a null hypothesis test will deem them statistically insignificant.

In the example in Figure 6.1, all the draws of $\hat{\beta}/se(\hat{\beta})$ below 1.96 (the critical t value assuming we have a large sample) will lead us to commit a Type II error : we will fail to reject the null even though the null is false.

From there, the MDE is the smallest effect that, if true, has an $p\%$ probability of generating a statistically significant effect for a given α (H. S. Bloom 1995). For instance, we might say something like: "assuming we want 80% power (κ) and we want to set the Type I error at $\alpha = 0.05$, then 3.4 is the MDE that can be detected." In (fig:mde1?), this would mean that the unshaded part of the blue

³This section benefitted a lot from Cyrus Samii's notes on MDE.

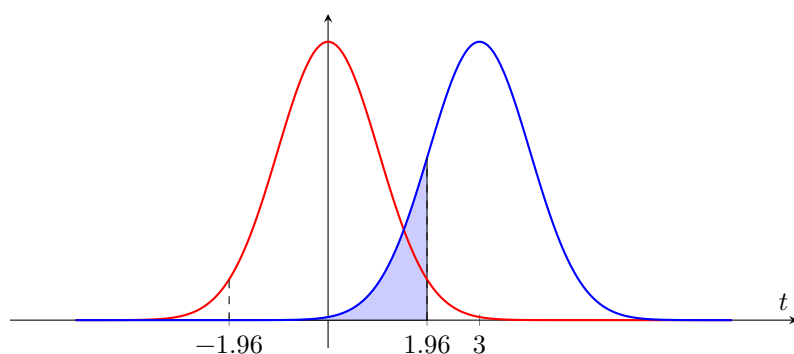


Figure 6.1: Minimum detectable effect problem. The red curve is the distribution of t values ($\hat{\beta}/se(\hat{\beta})$) under the null hypothesis that $\beta = 0$. The blue one is the distribution of t values if $\beta = 3$. Eyeballing, about one-third of the draws from the $\beta = 3$ distribution will be statistically insignificant, that is, will have a t -value below 1.96 (mass under the blue curve below 1.96, shaded in blue). Thus, statistical power is about $2/3$ here (unshaded mass under the blue curve above 1.96).

distribution represents 80% of the mass (which requires to move the blue curve to the right).

Note: it's up to you to decide how to set κ and α . The convention is often to use $\kappa = 0.8$ and $\alpha = 0.05$ (or smaller), but this is a matter of how much you care about being wrong in one way or another.

⚠ Type S and Type M errors

There is another type of errors that could emerge aside from Type I and Type II errors: you could get a significant estimate of β that is of the wrong sign. In (fig:mde1?), this would be the share of outcomes under the blue curve that's smaller than -1.96 (on the left). These draws of $\hat{\beta}$ are significant but negative, even though β is positive. In (fig:mde1?), this probability seems pretty small (we can barely see the share under the curve to the left of -1.96), but it can be large in other settings.

The share of the blue curve below the critical t statistic of the opposite sign to β and is referred to by Gelman and Carlin (2014) as a **Type S** ("sign") error:

Type S error $:= \mathbb{P}(\hat{\beta} \text{ has the wrong sign} | \hat{\beta} \text{ is statistically significant})$

Gelman et al. also define the problem of **Type M** (for

“magnitude”) problem or **exaggeration ratio**. Consider (fig:mde1?): if we condition on statistical significance, the only estimates of the correct sign that we will obtain are those above 1.96. But the distribution of what remains of the blue curve (truncated at 1.96) will have a different mean than 3 (the true treatment effect). We can then define the exaggeration ratio as:

$$\text{Type M ratio} := \frac{\int_{-\infty}^{-t_{\alpha/2}} \hat{\beta} \cdot N(\hat{\beta}; \beta, 1) d\hat{\beta} + \int_{t_{\alpha/2}}^{\infty} \hat{\beta} \cdot N(\hat{\beta}; \beta, 1) d\hat{\beta}}{\beta \cdot \mathbb{P}(|\hat{\beta}| > t_{\alpha/2})}$$

where β is the ATE, ie, the true average treatment effect. Note that we standardized the variable so that $se(\hat{\beta}) = 1$; this is just for convenience.

The numerator integrates $\hat{\beta}$ weighted by the density over the two significant regions: estimates that are significantly negative (first term, from $-\infty$ to $-t_{\alpha/2}$) and estimates that are significantly positive (second term, from $t_{\alpha/2}$ to ∞). Dividing by $\beta \cdot \Pr(|\hat{\beta}| > t_{\alpha/2})$ converts this into the conditional expectation of $\hat{\beta}$ among statistically significant draws, normalized by the true effect β .

In (fig:mde1?), the Type M ratio for $\alpha = 0.05$ corresponds to the expected value of $\hat{\beta}$ conditional on $|\hat{\beta}| > 1.96$, divided by 3 (the true β). When power is low, the significant estimates will on average exaggerate the true effect: the exaggeration ratio will be well above 1.

6.3.3 Computing MDE

Formally, the MDE β^{MDE} is:

$$\text{MDE} \equiv |\beta_{\alpha, \kappa, se(\hat{\beta})}^{\text{MDE}}| = se(\hat{\beta})(t_{\alpha/2} + t_{1-\kappa})$$

In other words, for a given α , κ , and $se(\hat{\beta})$, the MDE will be the point at a distance of $se(\hat{\beta})(t_{\alpha/2} + t_{1-\kappa})$ to 0 (assuming that's our null hypothesis; otherwise, just shift this by a constant).

Think of it this way: if the true effect β were 1.96 standard errors away from 0, then our blue curve in (fig:mde1?) would be centered at 1.96 (a bit to the left of where it is now). Let's assume we set $\alpha = 0.05$. Half the time, we would draw a $\hat{\beta}$ that's insignificant (ie, it would have a t statistic less than 1.96, which is the critical t value

for $\alpha = 0.05$) and half the time it would be significant (t statistic above 1.96). Thus, our power would be 50%. That's the $t_{\alpha/2}$ in the equation of the MDE. If we want to increase our power (increase κ), then we need to move even further from 1.96. That's the additional $t_{1-\kappa}$ of the MDE equation. The higher κ , the smaller we want the blue mass to be, and thus the larger the MDE becomes.

⚠ MDE example

A quick example will clarify this: suppose we decide that we want to run an RCT with the following parameters:

- Type I error: $\alpha = 0.01$
- Type II error: $\beta = 0.1$, which implies that $\kappa = 0.9$.

I can then compute the critical values attached to it. First, our Type I critical t statistic:

```
> qt(0.01/2, 1000, lower.tail=F)
[1] 2.580755
```

Then then same for Type II:

```
> qt(0.1, 1000, lower.tail=F)
[1] 1.282399
```

This then yields:

$$\text{MDE} = se(\hat{\beta})(t_{\alpha/2} + t_{1-\kappa}) = se(\hat{\beta})(2.58 + 1.28) = 3.86 \cdot se(\hat{\beta})$$

Thus, fixing κ and α at 0.9 and 0.01, respectively, means that the MDE will be 3.86 standard errors away from 0. In other words, to achieve a power of 90% and using an α of 0.01 Type I error, then the smallest detectable effect will be 3.86 standard errors away from zero.

Note that this is more demanding than 'just' obtaining a statistically significant effect. You can get the latter with estimates only 2.58 standard errors away from zero (for $\alpha = 0.01$).

As you can see, the MDE will become larger if:

- We reduce our α (reduce the probability of committing a Type I error)
- We increase our κ (increase our power, ie, reduce the probability of a Type II error)
- We increase our standard errors (less precise estimates)

6.3.4 Optimal sample size

We now know what effect size we can detect for a given α , κ , $se(\hat{\beta})$. But we can flip the question and say: what kind of sample size do we need to detect a given MDE?

To do so, we will need to estimate the standard error of $\hat{\beta}$. Recall the definition of $\hat{\beta}$'s standard error:

$$se(\hat{\beta}) = \sqrt{\frac{\sigma_{y1}^2}{N_1} + \frac{\sigma_{y0}^2}{N_0}},$$

where σ_{y1}^2 and σ_{y0}^2 are the variances of the outcome under treatment and control conditions.

In an RCT, we don't know for sure *ex ante* how many units will be in the treatment and in the control group (since it's a random draw) and thus we don't know what N_1 and N_0 are. However, we can write this down in a probabilistic way, with probability of receiving a treatment p and sample size N :

$$se(\hat{\beta}) = \sqrt{\frac{1}{N} \left(\frac{\sigma_{y1}^2}{p} + \frac{\sigma_{y0}^2}{1-p} \right)}$$

Recall that $MDE = se(\hat{\beta})(t_{\alpha/2} + t_{1-\kappa})$. We can plug in the estimate for $se(\hat{\beta})$ and solve for N :

$$N = \frac{(t_{\alpha/2} + t_{1-\kappa})^2 \left(\frac{\sigma_{y1}^2}{p} + \frac{\sigma_{y0}^2}{1-p} \right)}{MDE^2}$$

The implications are straightforward:

- N decreases in the size of the MDE. The larger the effect we anticipate, the fewer observations we will need.
- N increases in $\sigma_{y1}^2, \sigma_{y0}^2$. The noisier our variables, the more observations we will need.
- N decreases when α increases, κ decreases.

Here's the way we can use this knowledge. In practice, we will do as follows.

1. We start by coming up with a plausible guess regarding the treatment effect of our intervention D . This guess can be informed by our knowledge. It could also be a threshold imposed by commercial reasons (eg, "my new website only makes sense if it increases consumer engagements by 2,000 users per month").

2. Then, we would plug in the MDE in the expression above, set $t_{\alpha/2}$ and $t_{1-\kappa}$ depending on our tolerance of Type I and Type II errors, plug in a reasonable estimate for σ_{y1}^2 and σ_{y0}^2 (eg from past data).
3. We would then compute the sample size N needed to detect such an effect, given all the parameters we just fixed.

This would then tell us how to set the sample size of this RCT.

6.4 Group-level treatment

Sometimes, it won't be possible to treat individual units. Instead, you might have to treat entire groups. For instance, imagine you are randomizing electrification programs at the village level: then the entire village will be treated (or not). Even though you might be interested in household- or individual-level outcomes, you won't be able to randomize at that level.

This, in itself, is not a problem. You are still randomizing, and thus ignorability will still hold. You can obtain unbiased and consistent estimates of treatment effects using the estimation approach discussed above. Good.

However, group-level treatment does raise issues for **inference**. Recall that one of the difficulty when writing down the standard errors of a parameter $\hat{\beta}$ is the matrix $E[\varepsilon'\varepsilon]$. If you recall, this is the variance-covariance matrix of error terms. If we assume (a) homoskedasticity (constant variance of error terms conditional on $\mathbf{X}$) and (b) independence of error terms (zero covariance), then we could write:

$$E[\varepsilon'\varepsilon] = \begin{bmatrix} \sigma^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma^2 \end{bmatrix}$$

and $Var(\hat{\beta})$ simplifies to $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$. Then, as we saw earlier (see notes on inference with OLS), we can estimate σ^2 with s^2 :

$$s^2 = \frac{\mathbf{u}'\mathbf{u}}{n - k},$$

where n is the sample size and k the number of parameters that are estimated.

But now, the assumption of independence becomes harder to justify . For instance, individuals within a village may have abnormally high (or low) values of Y and thus larger (or smaller) error terms. Hence, we might have something like this (ignoring heteroskedasticity):

$$E[\varepsilon'\varepsilon] = \begin{bmatrix} \sigma^2 & a & a & 0 & 0 & 0 \\ a & \sigma^2 & a & 0 & 0 & 0 \\ a & a & \sigma^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma^2 & b & b \\ 0 & 0 & 0 & b & \sigma^2 & b \\ 0 & 0 & 0 & b & b & \sigma^2 \end{bmatrix}$$

where a and b are the covariances of error terms within group A and group B . For instance, it could be that each unit is an individual ($i = 1, 2, \dots, n$), and we have two villages in our sample (A and B). We would then say that the errors are correlated within groups (aka there is nonzero **intracluster correlation**) (Moulton 1986, 1990).

Fortunately, we can overcome this problem. We will typically use **clustered standard errors** for inference (Kloek 1981; Liang and Zeger 1986; A. Colin Cameron and Miller 2015; Abadie et al. 2023) . These standard errors will allow within-group (or “within-cluster”) correlation of error terms. And typically, since we assume heteroskedasticity anyway, we will also let the variance vary.

Concretely, we estimate clustered standard errors, using G to indicate groups (or clusters) and u as the residuals:

$$\text{Var}_{\text{cluster}}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{g=1}^G \mathbf{X}'_g \mathbf{u}_g \mathbf{u}'_g \mathbf{X}_g \right) (\mathbf{X}'\mathbf{X})^{-1}$$

Informally, we take cluster by cluster and compute the middle bit using the residuals of that group. We then sum up over all groups and divide by the square of $\mathbf{X}'\mathbf{X}$ (since it appears twice).

6.5 Compliance

6.5.1 Compliance problems

One of the big problems in RCTs is compliance: what happens if the treated unit refuses to take the treatment? We call these units non-compliers. For instance:

- An RCT in which people are asked to watch YouTube videos selected by the analyst: people might watch other videos instead (or none at all).
- An RCT in which people are encouraged to buy a new item: people might not be interested.

This problem affects our analysis. We can think of it this way:

1. Treatment is assigned to all units i ($T \in \{0, 1\}$). Eg: the analyst decides who will be asked to take an aspirin and who won't.
2. Each unit i decides whether to comply or not ($D \in \{0, 1\}$). Eg: the patients decide whether to swallow the aspirin they received or not.
3. If $T_i = 1$ and $D_i = 1$: we say that unit i is a complier. (For now, we assume that nobody in the control group cheated and took the treatment, such that $T_i = 0$ implies $D_i = 0$.)

The problem is that this makes things a bit hazy regarding the average treatment effect (ATE): we now have a form of selection where some units comply and others don't, and we don't have tools to address this. Fortunately, there are other estimands that are of help here: the intent to treat effect (ITT) and the local average treatment effect (LATE) .

6.5.2 Intent to treat and local average treatment effects

The solution is to think of two alternative treatment effects that we might be interested in.

- **Local average treatment effect (LATE)** is the effect of the treatment *among compliers*. That is, for those who complied, how did Y change on average?
- **Intent to treat (ITT)** is the effect of treatment averaging the outcomes Y of those who were treated and accepted the treatment ($T = 1$ and $D = 1$) and those who didn't accept it ($T = 1$ and $D = 0$), compared to the rest ($T = 0$).

Consider the following example from Aklin et al. (2017). We encouraged a random set of Indian households to acquire solar panels. We considered several outcomes of interest, including consumption of kerosene (solar panels should reduce the need for kerosene lamps), household income (electricity can be used to create small businesses), time spent studying (solar panels can be used for night light), etc.

Essentially, we randomized where the sales representatives of the solar company we worked with would go. Whether a household received a visit or not is what we really randomized and this corresponds to T ($T = 1$ if a household received a sales pitch from a representative). Of course, out of all the households they visited, only a small number of households rented these solar panels (thus, we had many $T = 1$ and $D = 0$). This made it natural for us to ask two kinds of questions:

- What is the average effect of solar panels on those who rented them (the compliers)? That's the LATE.
- What is the average effect of solar panels overall regardless of whether they rented them or not? That's the ITT.

You might wonder: why do we care about ITT? We do care about it from a policy (or business strategy) perspective. If my question is "I'm the Prime Minister and I want to roll out solar panels across an entire state. I want to know what kind of effect this program will have!" In such a case, I cannot limit myself to focus solely on the households that actually end up using them. I need to account for the fact that some will not comply.

Thus, we computed both the ITT and the LATE. The results are reported in Figure 2 for the case of kerosene. As you can see, our intervention reduced kerosene consumption a lot for people who complied (-48 rupees per month). But once we take into account non-compliers, then we note that the overall effect is watered down (-14 rupees per month). This makes sense: since they didn't start using solar panels, non-compliers will not experience a decrease of kerosene consumption. Thus, they will drag the effect of solar panels down.

Table 2. Effect of MGP solar microgrids on household kerosene spending. Effects are shown for spending in the private market (A), the PDS (B), and overall (C). The SEs are clustered by habitation and are shown in parentheses. All dependent variables are measured in rupees per month. $n = 3825$; number of households, 1281. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	ITT (OLS)		LATE (IV)	
	(1)	(2)	(3)	(4)
(A) Kerosene spending on private market				
Treatment	-14.01*** (5.28)	-14.49** (6.91)	-47.49** (19.83)	-49.36** (24.62)
Household FE		Yes		Yes
Wave FE	Yes	Yes	Yes	Yes
Pretreatment mean for control group = 72				
First-stage estimate			0.29	0.29
First-stage F statistic			10.15	10.01

Figure 6.2: Source: Aklin et al. (2017).

6.5.3 Estimation of ITT and LATE

The ITT can be estimated with OLS as the following linear model:

$$Y_i = \alpha + \beta T_i + \varepsilon_i,$$

where $\hat{\beta} = \widehat{\text{ITT}}$. Note that we use T (the assignment to treatment) rather than D (the treatment itself). Since T is randomized, this model can be consistently estimated with OLS.

In our example, this means that we used data on whether a household was visited by a sales representative (T) rather than whether they rented solar panels (D).

For the LATE, we must estimate it using an **instrumental variable** approach, in which T serves as an instrument for D . These are then estimated with two-stage least squares. The long story short is that the two stages are:

$$\begin{aligned} D_i &= \phi + \gamma T_i + \mu_i && \text{1st stage} \\ Y_i &= \alpha + \beta_{IV} \hat{D}_i + \varepsilon_i && \text{2nd stage} \end{aligned}$$

where $\hat{\beta}_{IV} = \widehat{\text{LATE}}$. More on this in the section devoted to instrumental variables.

6.6 Limitations of experimental methods

Despite their benefits, RCTs cannot always be used:

- **Cost.** Implementing an RCT can be resource-intensive due to the need for large sample sizes, data collection, and intervention costs.
- **Feasibility.** Some variables cannot be randomized in practice. For instance, I cannot randomize how religious someone feels. Nor can I instruct all the central banks in the world to randomize their interest rates.
- **Ethical concerns.** Some treatments, such as access to life-saving drugs, raise ethical issues around fairness and harm.

See Teele (2014) for a discussion on field experiments, in particular. See Nardini (2014) on ethics in clinical trials (including some horrible cases).

⚠ Hawthorne and John Henry effects

Participants who know they are being observed may change their behavior independently of the treatment. That's called the **Hawthorne effect** (named after studies at Western Electric's Hawthorne plant in the 1920s–30s). The symmetric problem in the control group is the **John Henry effect**: control-group participants aware of being denied the treatment may behave differently. For instance, in a school experiment where some children receive a tablet and others not, those who don't might work extra hard (maybe out of spite!).

Both effects contaminate the treatment–control comparison and can bias estimates in either direction. To solve that, you typically want to avoid having the researcher being too visible and units unaware of whether they are in the treatment or the control group. But that's not always feasible and sometimes it's also unethical.

⚠ Attrition

Attrition occurs when units randomized into the study drop out before the outcome is measured. Attrition is harmless if it is *missing completely at random* (MCAR) — i.e., dropout is

independent of treatment status and potential outcomes. It becomes a threat when it is *differential*: if treated units who feel the treatment is ineffective are more likely to drop out, the remaining treated sample is positively selected and the estimated ATE is biased upward.

Three scenarios are standard:

- **MCAR** (dropout independent of treatment and potential outcomes)
- **MAR** (dropout depends on observed covariates but not on the missing outcome)
- **MNAR** (dropout depends on the missing outcome itself).

Only MCAR guarantees unbiased estimation without correction.

External validity

An RCT can yield a perfectly credible estimate of the ATE for the specific sample studied, yet that effect may not generalize to other populations, settings, or time periods — the problem of **external validity**. Experiments recruit specific samples (volunteers, households in a particular region), operate in particular institutional contexts, and measure outcomes over a fixed horizon. An effect found in a pilot study in rural India says relatively little about the same intervention in urban France or at national scale.

Deaton (2010) and Cartwright (2011) argue that this is a fundamental limitation of the RCT paradigm in the social sciences: an internally valid estimate is not the same as a policy-relevant one. Vivaldi (2020) documents that effect sizes from development economics RCTs vary enormously across replications of nominally identical interventions. At scale, **general-equilibrium effects** (e.g., a training program that raises wages only because it is small relative to the labor market) and implementation capacity constraints can further attenuate or reverse pilot-study findings.

External validity is not a reason to abandon RCTs, but it is a reason to interpret effect-size estimates cautiously and to complement experimental evidence with replication across diverse contexts.

Example 6.1 (Social pressure and voter turnout). One of the

big questions in politics in democracies is how to increase turnout. Democracies depend on people going to vote. And political parties want to know that too, at least for their own voters.

In this context, Alan S. Gerber, Green, and Larimer (2008) conducted a large field experiment. Right before the August 2006 Michigan primary (the moment when members of parties select their candidates to the main election in November), they randomly assigned 180,002 registered voters to one of five treatments:

- A control group receiving nothing
- A civic-duty mailer
- A “Hawthorne” mailer (informing recipients they were being observed)
- A “self” mailer showing the recipient’s own past turnout record
- A “neighbors” mailer showing the recipient’s record alongside the records of nearby neighbors.

Because assignment was random, the ATE is estimated as a simple difference in turnout rates between each arm and the control group.

The civic-duty and Hawthorne conditions raised turnout by small (though not trivial) amounts (1.8 and 2.5 pp respectively). Showing one’s own past record raised turnout by 4.9 pp. But showing neighbors’ records produced an effect of **8.1 percentage points**. This suggests that people really can be socially pressured into going to vote!

The design also illustrates several issues we discussed. First, the unit of randomization is the household, so standard errors are clustered at that level (in a two-adult household, two people are treated at the same time). Second, SUTVA could be a problem: someone who is treated might go chat with someone who isn’t and in a sense treat them as well. Third, the external validity of a social-pressure effect depends heavily on the salience of the norm. The same treatments might not have any effect in other countries.

Condition	Turnout	ATE vs. control	SE
Control	29.7%	—	—
Civic duty	31.5%	+1.8 pp	(0.3)
Hawthorne	32.2%	+2.5 pp	(0.3)
Self	34.5%	+4.9 pp	(0.3)
Neighbors	37.8%	+8.1 pp	(0.3)

Condition	Turnout	ATE vs. control	SE
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Table 6.1: Effect of social-norm mailers on primary turnout. Outcome: voted in the August 2006 Michigan primary (0/1). Treatment: randomly assigned mailer type. Standard errors clustered at the household level in parentheses. Source: Alan S. Gerber, Green, and Larimer (2008). {#tbl-gerber2008}

Example 6.2 (Does management matter? Evidence from India). N. Bloom et al. (2013) apply an RCT to a common management question: do modern management practices causally raise firm productivity? The worry with observational data (such as the correlations in N. Bloom and Van Reenen (2007)) is that well-managed firms may simply be better-run in many ways at the same time. In that case, management quality is a symptom of organizational capability rather than a cause of it.

To solve this problem, the authors partnered with a management consulting firm to randomly assign five months of intensive consulting to a subset of large textile plants near Mumbai. Treated plants received structured training on a range of management skills (inventory control, quality inspection, sales planning, and human resource practices). Control plants received only an initial diagnostic visit. The assignment was random, and therefore any systematic difference between firms should be caused by the treatment.

Treated plants adopted nearly all recommended practices, and total factor productivity rose by approximately **17% relative to controls** over the following year. Most of the gain materialized quickly (within three months). The study also found that treated plants reduced defects, lowered inventories, and increased on-time delivery, providing a mechanism for the productivity gain.

Of course, it's unclear how well these results would travel. But they suggest that good management skills are important factors that can help a firm become more profitable.

Outcome	Treatment effect	SE
Total factor productivity	+17%***	(5%)
Quality defects	−52%***	(14%)

Outcome	Treatment effect	SE
Inventory (days of stock)	−25%***	(9%)
On-time delivery	+27%***	(8%)

Table 6.2: Effect of intensive management consulting on plant outcomes. Outcome: each row is a distinct plant-level measure. Treatment: random assignment of five months of structured consulting (vs. diagnostic visit only). Standard errors in parentheses. *** $p < 0.01$. Source: N. Bloom et al. (2013).

6.7 Recent developments

6.7.1 Heterogeneous treatment effects and causal forests

The methods in this chapter estimate the *average* treatment effect. In many applied settings — targeting, personalization, resource allocation — the more relevant question is: *for whom* does the treatment work? The **conditional average treatment effect** (CATE), $\tau(\mathbf{x}) = E[Y(1) - Y(0) \mid \mathbf{X} = \mathbf{x}]$, captures treatment effect heterogeneity as a function of observed covariates.

Classical approaches (OLS interaction terms, subgroup analysis) struggle when $\mathbf{x}$ is high-dimensional. **Causal forests** (Wager and Athey 2018) adapt the random forest algorithm to the causal problem: trees are grown to maximize heterogeneity in treatment effects rather than prediction accuracy, and sample splitting (“honesty”) prevents overfitting so that valid confidence intervals can be constructed. The **grf** package in R implements this directly. CATE estimates can guide targeting decisions (treat only units with $\hat{\tau}(\mathbf{x})$ above a threshold) or identify which subpopulations benefit most from an intervention — questions directly relevant to industry A/B testing and policy design.

6.7.2 Adaptive experiments

Standard A/B testing fixes the treatment allocation at $p = 0.5$ for the full experiment. **Adaptive experimental designs** update the allocation probability as data accumulate, routing more observations toward the arm that appears more effective. The best-known approach is the **multi-armed bandit**, where **Thompson sampling**

draws the allocation probability from a posterior distribution over arm means, balancing exploration (learning about potentially inferior arms) against exploitation (assigning more units to the current best arm).

Adaptive designs can reach the same statistical power with fewer observations assigned to inferior treatments — a meaningful advantage in medical trials or costly field experiments. The main complication is inference: naive p -values from adaptively collected data are not valid because the stopping time and allocation probability are data-dependent. Corrections based on **anytime-valid inference** address this and are increasingly standard in large-scale industry testing platforms.

7 Instrumental variables

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Overview

The problem of endogeneity is at the core of modern econometrics. One solution relies on a simple idea. Suppose you have an endogenous treatment. What this means is that part of its variation is caused by something else (which we might not be able to measure). What if we could find another variable that we do measure and that causes the treatment? Then, we could identify which part of the treatment's variation can be attributed to that other variable and use this to estimate causal effects. That's the idea behind instrumental variables.

Readings

People interested in instrumental variables may read (Imbens and Rubin 2015), (Joshua D. Angrist, Imbens, and Rubin 1996) (one of the papers that kickstarted the use of IV in economics), (Sovey and Green 2011) (a short practical guide), and (Imbens and Rubin 2015) (ch.24-26) (an in-depth discussion of IVs). Additional useful papers include (Heckman, Urzua, and Vytlačil 2006), (Peter M. Aronow and Carnegie 2013).

7.1 Introduction to instrumental variables

When the exogeneity assumption cannot be met, then OLS estimates are biased and inconsistent. Instrumental variables (IV) provide an alternative estimation method by using an external variable (the **instrument**) that helps isolate the exogenous variation in the endogenous treatment (D) that we care about.

A variable Z must satisfy two critical conditions to serve as an instrument for our endogenous variable of interest, D (for reasons that will shortly become clear):

1. **Relevance:** Z must be correlated with the endogenous explanatory variable ($Cov(Z, D) \neq 0$). This ensures that Z captures the variation in D necessary for estimating its effect.
2. **Exclusion restriction:** Z must not be correlated with the error term in the model ($Cov(Z, \varepsilon) = 0$). This ensures that the instrument does not independently affect the outcome variable, aside from its effect through D .

For example, in studying the effect of education on earnings, the distance to the nearest college can be a valid instrument if it affects the likelihood of attending college (relevance) but does not directly influence earnings other than through education (exclusion restriction).

7.2 Mathematic justification

To see how such a variable Z would help, we must place it in the same equation as β . We can write:

$$\begin{aligned}
 Cov(Y, Z) &= Cov(\alpha + \beta D + \varepsilon, Z) \\
 &= Cov(\alpha, Z) + Cov(\beta D, Z) + Cov(\varepsilon, Z) \\
 &= \beta Cov(D, Z) \qquad \text{by prop of cov and by excl restr}
 \end{aligned}$$

Then, solving for β :

$$\beta_{IV} = \frac{Cov(Y, Z)}{Cov(D, Z)}$$

Note how the two assumptions already came into play:

- **Relevance** is needed for the estimation of β_{IV} ; if $Cov(D, Z) = 0$, then the denominator of β_{IV} wouldn't be defined.
- **Exclusion restriction** is needed for consistency: otherwise, we couldn't get rid of $Cov(\varepsilon, Z)$ (more below).

We can even go a bit further. If we divided both the numerator and denominator by $Var(Z)$, we find that:

$$\begin{aligned}
 \beta_{IV} &= \frac{Cov(Y, Z)/Var(Z)}{Cov(D, Z)/Var(Z)} \\
 &= \frac{\gamma}{\pi}
 \end{aligned}$$

where γ and π are parameters from the following two equations:

$$\begin{aligned}
 Y &= \alpha_0 + \gamma Z + \varepsilon_0 \\
 D &= \alpha_1 + \pi Z + \varepsilon_1
 \end{aligned}$$

This definition allows us to rewrite the IV in a slightly different way:

$$\beta_{IV} = \frac{\text{Reduced form}}{\text{First stage}}$$

💡 Terminology

People often refer to the following expressions as...

Reduced form equation: $Y = \alpha_0 + \gamma Z + \varepsilon_0$

First stage equation: $D = \alpha_1 + \pi Z + \varepsilon_1$

Structural form equation: $Y = \alpha_2 + \beta D + \varepsilon_2$

Note: the content of this note assumes that the treatment effect β is a constant until Section 6, when we explore the problem of *heterogeneous* treatment effects.

7.3 Estimation

7.3.1 Wald IV estimator

Note that the estimated treatment as defined above is a **population** quantity:

$$\beta_{IV} = \frac{Cov(Y, Z)}{Cov(D, Z)}$$

This being said, it can straightforwardly be estimated with the **Wald estimator** (Wald 1940; S. L. Morgan and Winship 2014, 296), which applies if the instrument Z is a dummy variable. Start with the structural equation and then condition on Z :

$$\begin{aligned} Y &= \alpha + \beta D + \varepsilon && \text{structural equation} \\ \mathbb{E}[Y|Z=1] &= \mathbb{E}[\alpha|Z=1] + \mathbb{E}[\beta D|Z=1] + \mathbb{E}[\varepsilon|Z=1] && \text{taking cond expectation} \\ &= \alpha + \beta \mathbb{E}[D|Z=1] && \text{constants and by excl rest} \end{aligned}$$

Likewise, if we condition on $Z = 0$, we obtain:

$$\mathbb{E}[Y|Z=0] = \alpha + \beta \mathbb{E}[D|Z=0]$$

We can then take the difference between the two and rearrange:

$$\begin{aligned} \mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0] &= [\alpha + \beta \mathbb{E}[D|Z=1]] - [\alpha + \beta \mathbb{E}[D|Z=0]] \\ &= \beta [\mathbb{E}[D|Z=1] - \mathbb{E}[D|Z=0]] && \text{simplifying} \\ \beta_{\text{Wald}} &= \frac{\mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0]}{\mathbb{E}[D|Z=1] - \mathbb{E}[D|Z=0]} && \text{isolating } \beta \end{aligned}$$

We can then compute the sample conditional means of all four components:

$$\begin{aligned}\hat{\mathbb{E}}[Y|Z=1] &= \frac{\sum_i Z_i Y_i}{\sum_i Z_i} & \hat{\mathbb{E}}[Y|Z=0] &= \frac{\sum_i (1-Z_i) Y_i}{\sum_i (1-Z_i)} \\ \hat{\mathbb{E}}[D|Z=1] &= \frac{\sum_i Z_i D_i}{\sum_i Z_i} & \hat{\mathbb{E}}[D|Z=0] &= \frac{\sum_i (1-Z_i) D_i}{\sum_i (1-Z_i)}\end{aligned}$$

Then:

$$\hat{\beta}_{\text{Wald}} = \frac{\hat{\mathbb{E}}[Y|Z=1] - \hat{\mathbb{E}}[Y|Z=0]}{\hat{\mathbb{E}}[D|Z=1] - \hat{\mathbb{E}}[D|Z=0]}$$

The limitation is that this doesn't neatly suit itself to the inclusion of additional covariates nor of continuous instruments. Yet we may still want to block backdoor that could generate confounding and we may want to use other types of instruments. As a result, we often use two-stage least squares as an estimation method.

7.3.2 Two-Stage Least Squares (2SLS)

The 2SLS method is a two-step procedure used to estimate the coefficients in an IV setting. As its name indicates, we will use OLS twice.

Let's start with the case of one instrument and one endogenous treatment. We can build the 2SLS from first principles. Recall we said that the IV estimator is the ratio of the effect of Z on Y divided by the effect of Z on D . Thus, we can write this as:

$$\begin{aligned}\beta_{\text{IV}} &= \frac{(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}}{(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D}} \\ &= (\mathbf{Z}'\mathbf{D})^{-1}\mathbf{Z}'\mathbf{y}\end{aligned}$$

Notice the link to the Wald estimator above: the numerator is the effect of Z on y and the denominator is the effect of Z on D . The difference is that $\mathbf{Z}$ can contain several variables.

The 2SLS estimator in case of an over-identified model (a model with more instruments than endogenous variables) is given by:

$$\hat{\beta}_{2\text{SLS}} = [\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D}]^{-1}[\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}]$$

Just-identified model

We write that a model is...

- **under-identified** if the number of instruments is strictly smaller than the number of endogeneous variable(s);

- **just-identified** if the number of instruments is equal to the number of endogenous variable(s);
- **over-identified** if the number of instruments is strictly greater to the number of endogenous variable(s).

💡 Some extra math of IV

Here is how to get from:

$$\beta_{2SLS} = [\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D}]^{-1}[\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}]$$

to:

$$(\mathbf{Z}'\mathbf{D})^{-1}\mathbf{Z}'\mathbf{y}$$

when the model is just-identified (i.e., we have the same number of instruments as endogenous variables; see box).

First, the IV optimization problem is the following minimization problem:

$$\min_{\beta} (\mathbf{y} - \mathbf{D}\beta)' \mathbf{P}_Z (\mathbf{y} - \mathbf{D}\beta),$$

where the **projection matrix** $\mathbf{P}_Z$ is:

$$\mathbf{P}_Z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'$$

(Note: the projection matrix is the same $\mathbf{Z}$ times the OLS estimator once you add something at the end of this expression. See it by yourself: add $\mathbf{y}$ and then you see that it's the same as the OLS estimator of the effect of $\mathbf{Z}$ on $\mathbf{y}$ times $\mathbf{Z}$.)

Solving this problem, we obtain (as above):

$$\beta_{IV} = [\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D}]^{-1}[\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}]$$

Note that $\mathbf{D}'\mathbf{Z}$ is square under just-identified conditions (as is $\mathbf{Z}'\mathbf{D}$). As a result, it is also invertible (which would not be the case when we have an over-identified model). Then we can rewrite:

$$[\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D}]^{-1} = (\mathbf{Z}'\mathbf{D})^{-1}(\mathbf{Z}'\mathbf{Z})(\mathbf{D}'\mathbf{Z})^{-1}$$

so that:

$$(\mathbf{Z}'\mathbf{D})^{-1}(\mathbf{Z}'\mathbf{Z})(\mathbf{D}'\mathbf{Z})^{-1}(\mathbf{D}'\mathbf{Z})(\mathbf{Z}'\mathbf{Z})^{-1}(\mathbf{Z}'\mathbf{y})$$

Simplifying, we obtain:

$$\beta_{IV} = (\mathbf{Z}'\mathbf{D})^{-1}(\mathbf{Z}'\mathbf{y})$$

But this simplification hinges on the invertibility of $\mathbf{D}'\mathbf{Z}$ and $\mathbf{Z}'\mathbf{D}$, which, recall, is only possible because the two matrices are of same dimensions. In substance, this means that we have as many endogenous as instrumental variables. If we had more instruments than endogenous variables, then we are stuck with the earlier, longer expression. Not to worry: this is where 2SLS will help us, since we can move ahead with over-identified models with this estimator.

At last, we can note that:

$$\begin{aligned}
\beta_{IV} &= (\mathbf{Z}'\mathbf{D})^{-1}(\mathbf{Z}'\mathbf{y}) \\
&= [\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D}]^{-1}[\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}] && \text{from above} \\
&= [\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D}]^{-1}\mathbf{D}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y} && \text{multiply/divide by } \mathbf{Z}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1} \\
&= (\hat{\mathbf{D}}'\hat{\mathbf{D}})^{-1}\hat{\mathbf{D}}'\mathbf{y} \\
&:= \hat{\beta}_{2sls}
\end{aligned}$$

The last bit is the consequence of:

$$\hat{\mathbf{D}} = \mathbf{Z} \underbrace{(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D}}_{\Pi_{ols}}$$

where Π_{ols} is the OLS estimate of the effect of $\mathbf{Z}$ on $\mathbf{D}$ (recall the formula $\beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$). This corresponds to the first stage of 2SLS.

In practice, estimation happens as follows.

1. **First stage:** regress the endogenous explanatory variable (D) on the instrument (Z) and other exogenous variables to obtain the predicted values of D ($\hat{D}$) with OLS:

$$D_i = \pi_0 + \pi_1 Z_i + \mu_i$$

In matrix notation, this corresponds to generating estimates to generate the predicted value:

$$\begin{aligned}
\hat{\mathbf{D}} &= \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{D} \\
&= \mathbf{Z}\Pi
\end{aligned}$$

2. **Second stage:** replace D with its predicted values ($\hat{D}$) from the first stage and estimate the outcome equation with OLS:

$$Y_i = \alpha + \beta_{2sls} \hat{D}_i + \varepsilon_i,$$

with the corresponding matrix notation indicated above.

$\hat{\beta}_{2sls}$ is a consistent estimator of β . This procedure ensures that the variation in D used to estimate its effect on Y comes only from the exogenous variation provided by Z .

Note. If you literally run two OLS regressions successively and use the second-stage standard errors, they will be incorrect. The second-stage OLS uses $\hat{D}$ instead of D , so it computes residuals as $y - \hat{D}\hat{\beta}$ rather than the true structural residuals $y - D\beta$. This understates residual variance and produces standard errors that are too small. That's why IV standard errors will tend to be larger. Routines in R and Python should automatically correct them when using 2SLS.

7.4 Inference

Let us check what the performance of IV is with respect to our key criteria of **bias** and **consistency**. We will leave **efficiency** aside.

7.4.1 Consistency

Issue: I said that valid IVs yield **consistent** estimates, not that they yield **unbiased** estimates. Let's work through it, and let us start to check consistency. We had:

$$\beta_{IV} = \frac{Cov(Z, Y)}{Cov(Z, D)}$$

Knowing that $Y = \beta D + \varepsilon$, we can write:

$$\begin{aligned} Cov(Z, Y) &= Cov(Z, \beta D + \varepsilon) \\ &= \beta Cov(Z, D) + Cov(Z, \varepsilon) \end{aligned}$$

Thus:

$$\begin{aligned} \beta_{IV} &= \beta \frac{Cov(Z, D)}{Cov(Z, D)} + \frac{Cov(Z, \varepsilon)}{Cov(Z, D)} \\ \beta_{IV} &= \beta + \frac{Cov(Z, \varepsilon)}{Cov(Z, D)} \quad \text{assuming that } Cov(Z, D) \neq 0 \text{ (relevance)} \end{aligned}$$

Then, taking probability limits to assess consistency:

$$\begin{aligned} plim \hat{\beta}_{IV} &= plim \left[\beta + \frac{Cov(Z, \varepsilon)}{Cov(Z, D)} \right] \\ &= \beta + plim \left[\frac{Cor(Z, \varepsilon)}{Cor(Z, D)} \frac{sd(\varepsilon)}{sd(D)} \right] \quad \text{by def of Cov and Cor} \end{aligned}$$

Under the **exclusion restriction** assumption, $Cov(Z, \varepsilon)$ is zero and the second term disappears. As a result, we get consistency.

We can obtain the same result with matrix notation:

$$\begin{aligned}
\hat{\beta}_{2sls} &= (\mathbf{Z}'\mathbf{D})^{-1}\mathbf{Z}'\mathbf{y} && \text{by definition} \\
&= (\mathbf{Z}'\mathbf{D})^{-1}\mathbf{Z}'(\mathbf{D}\beta + \varepsilon) && \text{by def of } y \\
&= \beta + (\mathbf{Z}'\mathbf{D})^{-1}\mathbf{Z}'\varepsilon && \text{distributing} \\
&= \beta + (N^{-1}\mathbf{Z}'\mathbf{D})^{-1}N^{-1}\mathbf{Z}'\varepsilon && \text{stabilizing by } 1/N
\end{aligned}$$

Then 2SLS is consistent if:

$$\begin{aligned}
\text{plim} N^{-1}\mathbf{Z}'\varepsilon &= 0 && \text{exclusion restriction} \\
\text{plim} N^{-1}\mathbf{Z}'\mathbf{D} &\neq 0 && \text{relevance}
\end{aligned}$$

Note: we could also have assessed consistency by noting that the Wald estimator is a ratio of two sample covariances, which are consistent estimates of population covariances.

Thus: the IV estimator is consistent (under **exclusion restriction** and **relevance**).

7.4.2 Bias

However, is IV unbiased? No (Nelson and Startz 1988; Bound, Jaeger, and Baker 1995; Andrews, Stock, and Sun 2019).

The intuition is the following (see Keane and Neal (2023) for the full derivation of these results): when you estimate β_{2sls} , you are using $\hat{\pi}$ from the first stage instead of the true value π . But by doing so, we are inducing bias. We can show this mathematically. Like before, we have:

$$\begin{aligned}
\text{Structural equation: } y &= \beta D + \varepsilon \\
\text{First stage: } D &= \pi Z + \mu.
\end{aligned}$$

If D is endogenous, and if π and Z are exogenous, then the endogeneity must manifest itself via μ , and as a result $Cor(\mu, \varepsilon) \neq 0$. If $Cor(\mu, \varepsilon) = 0$, then D would actually be exogenous and we would not need IV.

If we plug in the second equation in the first, we obtain:

$$y = \beta\pi Z + (\beta\mu + \varepsilon)$$

If we knew π , we could estimate this expression with OLS, since πZ is exogenous: it is uncorrelated with μ and with ε , the composite error term.

But we don't know π and instead use the estimated $\hat{\pi}$, which must be different from π ($\mathbb{P}(\pi = \hat{\pi}) = 0$). We have:

$$\begin{aligned}
 D &= \hat{D} + (D - \hat{D}) && \text{by definition} \\
 &= \hat{\pi}Z + [(\pi Z + \mu) - \hat{\pi}Z] && \text{plugging in for all terms} \\
 &= \hat{\pi}Z + [(\pi Z + \mu) - \hat{D}] \\
 \underbrace{D - \hat{D}}_{\hat{\mu}} &= Z(\pi - \hat{\pi}) + \mu && \text{rearranging}
 \end{aligned}$$

Thus, the residual of the first stage is the error term μ plus the estimation error. What we are really estimating is (once we replace μ by $\hat{\mu}$, the residual of the first stage):

$$y = \beta \hat{\pi}Z + \underbrace{[Z\beta(\pi - \hat{\pi}) + \beta\mu + \varepsilon]}_{\text{composite error}}.$$

The composite error term is now correlated with $\hat{\pi}Z$, and thus the estimates are biased. However, IV is consistent since the difference $\pi - \hat{\pi}$ goes to zero as $n \rightarrow \infty$.

We can show this in a different way, building off the following intuition. Recall that β_{IV} is a ratio:

$$\beta_{IV} = \frac{\text{Reduced form}}{\text{First stage}} = \frac{\gamma}{\pi}$$

where π is the effect of Z (instrument) on D (endogenous treatment), and γ is the effect of Z on Y . In 2SLS, both parameters are estimated with OLS. As we know, OLS parameters are asymptotically normally distributed. But if $\hat{\pi}$ (or, more accurately, $t_{\pi} = \hat{\pi}/se(\hat{\pi})$) is small, then the numerator will often be 'dangerously' close to zero. As a result, $\hat{\beta}_{IV}$ could have a very uneven distribution and $\mathbb{E}[\hat{\beta}_{IV}] \neq \beta_{IV}$. Think of it this way: assume that $\hat{\gamma}$ is fixed at 10. Now imagine that $\hat{\pi}$ is centered at 1 with (say) a standard error of 1. Imagine you get a bunch of values for $\hat{\pi}$ from five different sample:

$$\begin{aligned}
 \hat{\pi} = 1.99 &\rightarrow \hat{\beta}_{IV} = 5.02 \\
 \hat{\pi} = 1.9 &\rightarrow \hat{\beta}_{IV} = 5.2 \\
 \hat{\pi} = 1 &\rightarrow \hat{\beta}_{IV} = 10 \\
 \hat{\pi} = 0.1 &\rightarrow \hat{\beta}_{IV} = 100 \\
 \hat{\pi} = 0.01 &\rightarrow \hat{\beta}_{IV} = 1000
 \end{aligned}$$

See how small values of $\hat{\pi}$ make $\hat{\beta}_{IV}$ increase massively? The values of $\hat{\pi}$ that are one standard deviation above its true value yield a $\hat{\beta}_{IV}$ that's 50% the size of the true β . But values of $\hat{\pi}$ that are almost one standard deviation below its true value yield estimates that are 10,000% too large. Thus, this distribution is not going to be nicely centered on β .¹ But as N increases, the standard errors will be more precise, and the problem disappears.

We can also visualize this in Figure 7.1.

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.

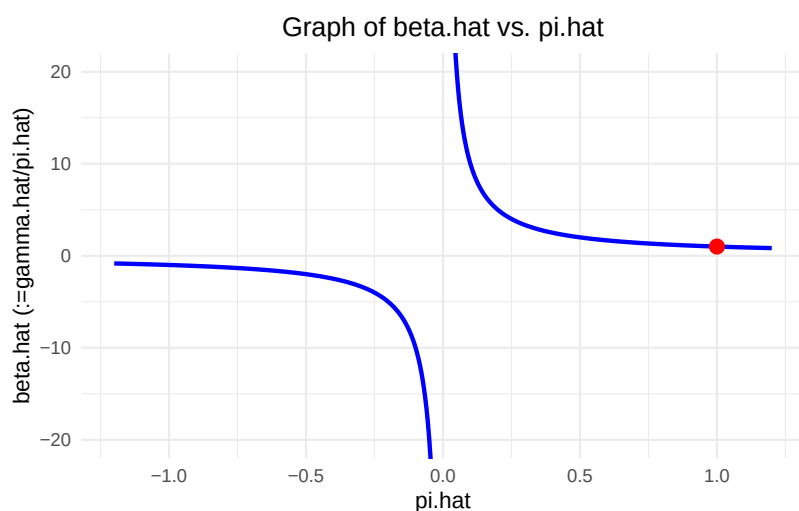


Figure 7.1: Small errors in estimation of π can cause large errors in estimation of β . Red dot: population value of β if $\gamma=1$ and $\pi=1$. Blue lines: set of estimated values β .hat, as π .hat goes from -1.5 to 1.5 (fixing γ at 1). As π .hat gets close to zero, the size of β .hat becomes extremely large (tending toward positive or negative infinity) and far from the true value of β (1).

7.4.3 Standard errors and hypothesis testing

We want to do something similar to our derivation of the distribution of the OLS estimator, namely to get it closer to something we are

¹Note that it may be that β_{IV} has no expected value, depending on the distribution of the numerator and denominator (Keane and Neal 2023).

familiar with: $\sqrt{N}(\hat{\beta} - \beta)$, which we know is *asymptotically* normally distributed.

To do so, we start with:

$$\hat{\beta}_{IV} = (\hat{\mathbf{D}}'\hat{\mathbf{D}})^{-1}\hat{\mathbf{D}}'\mathbf{y}$$

Plug in the structural equation instead of $\mathbf{y}$:

$$\hat{\beta}_{IV} = (\hat{\mathbf{D}}'\hat{\mathbf{D}})^{-1}\hat{\mathbf{D}}'(\mathbf{D}\beta + \varepsilon)$$

This yields:

$$\begin{aligned}\hat{\beta}_{IV} &= \beta + (\hat{\mathbf{D}}'\hat{\mathbf{D}})^{-1}(\hat{\mathbf{D}}'\varepsilon) \\ &= \beta + \left(\frac{1}{N}\hat{\mathbf{D}}'\hat{\mathbf{D}}\right)^{-1} \left(\frac{1}{N}\hat{\mathbf{D}}'\varepsilon\right) \quad \text{multiply/divide by } 1/N \\ \hat{\beta}_{IV} - \beta &= \left(\frac{1}{N}\hat{\mathbf{D}}'\hat{\mathbf{D}}\right)^{-1} \left(\frac{1}{N}\hat{\mathbf{D}}'\varepsilon\right) \\ \sqrt{N}(\hat{\beta}_{IV} - \beta) &= \left(\frac{1}{N}\hat{\mathbf{D}}'\hat{\mathbf{D}}\right)^{-1} \left(\frac{1}{\sqrt{N}}\hat{\mathbf{D}}'\varepsilon\right) \quad \text{multiply by } \sqrt{N}\end{aligned}$$

From here:

$$\begin{aligned}\frac{1}{N}\hat{\mathbf{D}}'\hat{\mathbf{D}} &\xrightarrow{p} \mathbb{E}[\hat{\mathbf{D}}'\hat{\mathbf{D}}] && \text{by the LLN} \\ \left(\frac{1}{\sqrt{N}}\hat{\mathbf{D}}'\varepsilon\right) &\xrightarrow{d} \mathcal{N}(\mathbf{0}, \hat{\mathbf{D}}'\varepsilon\varepsilon'\hat{\mathbf{D}}) && \text{by the CLT}\end{aligned}$$

where LLN is the **law of large number** and CLT the **central limit theorem**.

And we can conclude with:

$$\hat{\beta}_{IV} \sim \mathcal{N}(\beta, \mathbb{E}[\hat{\mathbf{D}}'\hat{\mathbf{D}}]^{-1}\mathbb{E}[\hat{\mathbf{D}}'\varepsilon\varepsilon'\hat{\mathbf{D}}]\mathbb{E}[\hat{\mathbf{D}}'\hat{\mathbf{D}}]^{-1}).$$

This quantity can be estimated by plugging in the residuals $\mathbf{u}$ for ε , just like we did for robust standard errors (Eicker-Huber-White standard errors) with OLS. From there, you can, in principle, do the usual hypothesis testing and build confidence intervals like we did with OLS.

Note. What I just wrote holds *if* the instrument(s) is strong. If not, then inference will not be valid – not just because of the bias stemming from weak instruments, but also because the standard errors will not be accurate. Recent work suggests that usual tests are underpowered if the instrument is weak (Keane and Neal 2023).

In case of weak instruments, then one can use the Anderson-Rubin (robust) confidence intervals instead ((Theodore W. Anderson and Rubin 1949); for a textbook discussion, see Stock and Watson (2011, 472)).

- Compute the residual of the endogenous regression: $u = Y - \beta D$.
- Regress u on the exogenous variables (if any) and the instrument(s).
- The AR statistic is the F-statistic on the parameters attached to the instruments.
- Invert this test: the 95% AR confidence interval is the set of β not rejected at the 5% level by the AR test.

7.4.4 Testing for instrument weakness

As we saw above, the validity of inference *also* depends on the strength of the instrument, at least with finite samples (in asymptotia, things become easier because of the precision of the instrument increases).

The strength of the instrument is measured via the F statistic of the joint effect of all instruments on the endogenous treatment(s).

Now, what is a good F statistic? People used to believe that an F statistic greater than 10 should do the trick (Staiger and Stock 1997).² Unfortunately, recent work suggests that you may need an F statistic that is considerably larger. A recent study suggests an F statistic above 100 to avoid the problems caused by weak instruments (Lee et al. 2022)... This is still a moving field, so let's see how this debate gets settled.

i Beyond the first-stage F : Kleibergen-Paap and Cragg-Donald statistics

The simple first-stage F -statistic is valid only under homoskedastic, serially uncorrelated errors. In practice, applied work almost always uses clustered or heteroskedasticity-robust standard errors. The relevant diagnostics are:

- **Kleibergen-Paap rk Wald statistic** (Kleibergen and Paap 2006): the robust analog of the first-stage F , valid under clustering and heteroskedasticity.
- **Cragg-Donald Wald statistic**: tests the rank of the

²In the special case with a single instrument and a single endogenous variable, the F statistic corresponds to the square of the t statistic. An F statistic of 10 therefore corresponds to a t statistic of about 3.2 in that case.

concentration matrix under iid errors. With multiple endogenous variables, this replaces the scalar first-stage F .

- **Effective F-statistic** (Olea and Pflueger 2013): provides a size-corrected criterion that accounts for the sampling distribution of the instrument under non-iid errors.

Critical values for all three statistics are tabulated by Stock and Yogo (2005). Most applied papers report the Kleibergen-Paap statistic alongside Stock-Yogo critical values.

7.4.5 Testing for instrument exogeneity

It is not possible to test the exogeneity (exclusion restriction) assumption *if* the model is just-identified. If, however, the model is over-identified (more instruments than endogenous treatments), then we can test the exogeneity assumption via the **J-test** (for a textbook explanation, see (Stock and Watson 2011) p.448).

The intuition is the following. Suppose we have two instruments (Z_1 and Z_2) and one endogenous variable (D). Then you should be able to estimate β_{IV} with Z_1 alone, with Z_2 alone, and with both. If all instruments are exogenous, then we should (on average) not see large differences between our estimates. If we do see large differences, then presumably one instrument isn't actually exogenous.

The procedure for this test is, for m instruments:

1. Estimate the equation of interest using 2SLS and all m instruments. Compute the **2SLS residuals** $\hat{u}_i = y_i - D_i \hat{\beta}_{2SLS}$, using the actual values D_i (not the first-stage fitted values $\hat{D}_i$).
2. Regress $\hat{u}_i$ on all instruments $Z_1, Z_2, \dots$ and any other exogenous (non-instrumental) variables $\mathbf{W}$.
3. Compute the F -statistic testing that the coefficients on $Z_1, Z_2, \dots$ are jointly zero. Rejection indicates that at least one instrument appears endogenous.
4. The J -statistic is $J = mF$. Under the null that all instruments are exogenous, $J \sim \chi^2_{m-k}$, where $m - k$ is the number of over-identifying restrictions (instruments minus endogenous variables).

Limitations of the J-test

The J -test is only available in over-identified models ($m > k$). More importantly, it tests for inconsistency *across* instruments, not for their absolute validity. If all instruments are invalid in

the same direction (all correlate with ε in the same way), then the test will not detect the problem. A passing J -test is nice, but not a guarantee of exogeneity.

When the model is just-identified, the exclusion restriction is untestable by construction. A practical solution is to do **sensitivity analysis**: how much deviation from the assumption is needed before overturning out results? Conley, Hansen, and Rossi (2012) say that we can let $Cov(Z, \varepsilon) = \delta$ for a range of small values $\delta \neq 0$. You then report how the estimate changes.

7.5 Heterogeneity and LATE

When we discussed RCTs (see notes on RCT), we had brought up two notions:

1. Treatment effects might vary by subgroups instead of being a universal constant.
2. Units in the treatment group might decide not to take the treatment, and units in the control group might decide not to stay in the control group. Units that don't comply were called "non-compliers."

One natural question is: does the treatment effect vary by groups based on compliance?

To give a motivating example, imagine that the treatment is a TV ad and we have a sample of 1,000 people. Let's say that half is randomly allocated to the treatment, namely watching an ad for 10 minutes per day. Whether people watch or not is beyond our control: some might close their eyes, not listen, be mentally checked out. Likewise, people in the control group might 'cheat' and watch TV anyway.

The good news is that IV can be used to estimate the **local average treatment effect** (LATE): the ATE for compliers.

To refresh your memory:

- Z is the assignment to treatment (i.e. whether unit i is in the treatment or the control group).
- D is the treatment (i.e. whether unit i took the treatment).
- For instance, Z could be the assignment to receive an aspirin or not, and D is whether individuals actually take an aspirin or not.

Then we have four cases:

- A **complier** is a unit i for which: (a) $Z = 1 \rightarrow D = 1$, whereas (b) $Z = 0 \rightarrow D = 0$. We write $D(1) - D(0) = 1$.
- A **defier**: (a) $Z = 1 \rightarrow D = 0$, whereas (b) $Z = 0 \rightarrow D = 1$. We write $D(1) - D(0) = -1$.
- An **always-taker**: (a) $Z = 1 \rightarrow D = 1$, and (b) $Z = 0 \rightarrow D = 1$. We write $D(1) - D(0) = 0$.
- A **never-taker**: (a) $Z = 1 \rightarrow D = 0$, and (b) $Z = 0 \rightarrow D = 0$. We write $D(1) - D(0) = 0$,

where $D(Z)$ is the value of D for $Z = 0$ or $Z = 1$.

We can now see more clearly what is going on (Joshua D. Angrist, Imbens, and Rubin 1996). From above, we had:

$$\beta_{IV} = \frac{\mathbb{E}[Y|Z = 1] - \mathbb{E}[Y|Z = 0]}{\mathbb{E}[D|Z = 1] - \mathbb{E}[D|Z = 0]}$$

We can work through the earlier expression. Let's go bit by bit and start with the denominator. Since D is a dummy variable (as is Z), we can rewrite the expectations as probabilities $P()$:

$$\mathbb{E}[D|Z = 1] = \mathbb{P}(D|Z = 1) \quad \text{and} \quad \mathbb{E}[D|Z = 0] = \mathbb{P}(D|Z = 0)$$

We can thus also write:

$$\beta_{IV} = \frac{\mathbb{E}[Y|Z = 1] - \mathbb{E}[Y|Z = 0]}{\mathbb{P}(D|Z = 1) - \mathbb{P}(D|Z = 0)}.$$

Here it gets a tiny bit technical, so bear with me. $\mathbb{P}(D|Z = 1)$ is the proportion of units taking the treatment when $Z = 1$ and $\mathbb{P}(D|Z = 0)$ is the proportion of units taking the treatment when $Z = 0$.

Here, we will make the simplifying assumption that we don't have defiers (which are kind of a weird bunch of units anyway...). We call this the **monotonicity assumption**: $D_i(Z = 1) - D_i(Z = 0) \geq 0$ for all units i . All units become more likely to take the treatment if they are assigned to it.

As a result, we find that the share of compliers is:

$$\mathbb{P}(D|Z = 1) - \mathbb{P}(D|Z = 0) = \text{Share of compliers}$$

The reason is that always-takers don't matter here: the probability of taking the treatment is invariant in Z . Thus, in the absence of defiers, this difference in probability is entirely driven by compliers.

Then we realize that the denominator of the IV is the same as the share of compliers:

$$\beta_{IV} = \frac{\mathbb{E}[Y|Z = 1] - \mathbb{E}[Y|Z = 0]}{\text{Share of compliers}},$$

where we assume that the share of compliers is strictly positive (which is the same as the relevance assumption).

Now, about the numerator: it's the effect of our assignment to treatment (Z) on the outcome, ignoring whether units complied or not. We encountered this as well: it's the definition of the **intent to treat effect** (ITT). Thus:

$$\beta_{IV} = \frac{\text{ITT}}{\text{Share of compliers}}.$$

Notice that if everyone complies (share of compliers equals 1), then $\text{LATE} = \text{ITT}$. This makes sense: our intent-to-treat ended up treating everyone we wanted, and thus everyone complied, and thus whether I run a regression with Z or D doesn't matter (the two are perfectly correlated).

Conversely, if a small number of people comply (share of compliers smaller than 1), then $\text{ITT} < \beta_{IV}$. This also makes sense: the ITT gets weakened by the fact that some units didn't comply and they didn't actually get treated. In our earlier example, it corresponds to the case where people in the treatment group don't take aspirin. As they won't be affected by it, they will not experience the boost in well-being that those who take the aspirin (the compliers) get.

i Assumptions for LATE identification

The IV Wald estimator recovers the LATE under four assumptions (Joshua D. Angrist, Imbens, and Rubin 1996):

1. **SUTVA**: each unit's potential outcomes depend only on its own treatment, not on others' treatment status.
2. **Independence of the instrument**. Z is as good as randomly assigned: $(D(0), D(1), Y(0), Y(1)) \perp Z$. In an RCT, this holds by design. In observational IV, this is a key identifying assumption.
3. **Exclusion restriction**: Z affects Y only through D ; it has no direct effect on the potential outcomes $Y(0)$ and $Y(1)$.
4. **Monotonicity**. There are no defiers: $D_i(Z = 1) \geq D_i(Z = 0)$ for all i . Assignment to the instrument weakly increases treatment take-up for every unit and certainly does not *decrease it* (nobody is less likely to take the treatment when they are placed in the treatment group).

Together, these four assumptions ensure that the Wald estimator identifies the average treatment effect for the subgroup of **compliers**.

⚠ LATE as a local effect

The LATE is the average treatment effect **for compliers only**. Always-takers and never-takers contribute to neither the numerator nor the denominator of the Wald estimator. This makes sense: the instrument has no leverage over their behavior.

This has two implications. First, $LATE \neq ATE$ in general: whether the two coincide depends on whether compliers are representative of the full population. Second, different instruments for the same treatment D identify treatment effects for *different* complier populations and are therefore not directly comparable across studies.

Whether the LATE is policy-relevant depends on the context. When the instrument directly mimics the policy being implemented (e.g., a randomized subsidy that encourages the exact program of interest), the LATE is the right estimand. When the instrument is a background variable (e.g., geographic distance to a college), the complier group is typically a narrow, self-selected subset whose treatment effect may not generalize (Imbens 2010).

Example 7.1 (The China Syndrome). Autor, Dorn, and Hanson (2013) study how Chinese import competition affected US manufacturing employment over 1990–2007. The identification challenge is that Chinese import penetration is endogenous: regions specializing in industries that were already in structural decline may have seen rising imports for that very reason, not because imports caused the decline.

Their instrument is a **shift-share (Bartik)** instrument. For each US commuting zone (a geographic region that captures a local labor market, i.e. how far people are willing to go to work), they capture its exposure to Chinese imports as the sum across industries of each industry's share in local employment multiplied by growth in Chinese imports nationally.

The key step is to instrument US-specific Chinese import growth with Chinese exports to *other* high-income countries (Germany, Japan, Australia, etc.). The logic is the following: a global surge in Chinese manufacturing capability simultaneously drives exports to all destinations, but the US-specific component (which is driven by US-China trade policy, bilateral demand shocks, or US-specific productivity) does not drive exports to Denmark. Thus, Chinese exports elsewhere serve as a valid proxy for the global component of Chinese comparative advantage.

The first-stage F -statistic is large (around 50–70). The 2SLS estimate implies that a **\$1,000 increase in Chinese import exposure per worker** reduces the manufacturing employment-to-population ratio by about **0.6 percentage points**, with wage effects of similar magnitude. OLS understates the impact, which is consistent with reverse causality.

Outcome	OLS	2SLS
Manufacturing employ- ment/pop. ratio (pp)	−0.21***	−0.596***
<i>SE</i>	(0.05)	(0.136)
Wage growth (log weekly wage)	−0.31***	−0.76**
<i>SE</i>	(0.06)	(0.31)
First-stage F -statistic	—	55.9

Table 7.1: Effect of a \$1,000 increase in Chinese import exposure per worker on US commuting-zone outcomes. Unit of observation: commuting zone $\times$ decade (1990–2000, 2000–2007). Instrument: Chinese exports to other high-income countries. Standard errors clustered by state in parentheses. *** $p < 0.01$, ** $p < 0.05$. Source: Autor, Dorn, and Hanson (2013). {#tbl-autor2013}

Example 7.2 (Enjoying the quiet life: corporate governance and managerial preferences). Does competitive pressure discipline managers, or do insulated managers simply enjoy a “quiet life” — avoiding difficult restructuring, tolerating inefficiency, and distributing rents to workers? Bertrand and Mullainathan (2003) address this with a clean IV strategy based on US state anti-takeover legislation.

In the 1980s, US states passed **Business Combination laws** in waves, sharply reducing the threat of hostile takeovers for firms incorporated in those states. The laws were motivated by political pressure from incumbent managers and legislators, not by the performance or characteristics of individual firms — making their passage plausibly exogenous to any given company. The

instrument is therefore an indicator for whether a firm's state of incorporation passed an anti-takeover law in a given year. The first stage is strong: firms in covered states face significantly lower takeover risk after the law passes.

The IV results show that managerial insulation causes:

- **Higher wages** for white-collar workers (managers distribute rents to themselves and allies rather than restructuring labor costs).
- **Smaller wage cuts** in response to adverse demand or cost shocks — insulated managers avoid the conflict of wage renegotiation.
- **Reduced plant-level efficiency** over time, consistent with managers tolerating slack rather than pushing for productivity improvements.

The OLS estimates of the same relationships are attenuated relative to IV, suggesting that firms subject to competitive pressure sort into better governance — a classic reverse-causality problem that the instrument resolves. The paper is a template for using **legislative changes with staggered state-level adoption** as instruments, a design now widely used in corporate finance, organizational economics, and labor economics.

Outcome	OLS	IV (2SLS)
White-collar wage bill (log)	+0.08	+0.31**
<i>SE</i>	(0.08)	(0.14)
Wage cut after adverse shock (pp)	−1.2**	−0.3
<i>SE</i>	(0.5)	(1.1)
Total factor productivity (log, annual change)	—	−0.024**
<i>SE</i>	—	(0.011)

Table 7.2: Effect of managerial insulation (anti-takeover law passage) on firm-level outcomes. OLS compares covered vs. uncovered firms; IV instruments takeover exposure with staggered state-level business combination law enactment. Standard errors clustered by firm in parentheses. ** $p < 0.05$. Source: Bertrand and Mullainathan (2003).
{#tbl-bertrand2003quiet}

7.6 Recent developments

Let me flag two areas where there is considerable ongoing theoretical and applied work on IVs: Bartik instruments and machine learning.

First, an instrument that has become particularly popular is the so-called **Bartik instrument** (named after the person who thought of it):

$$Z_i = \sum_k s_{ik} \cdot g_k,$$

where s_{ik} is the pre-period share of unit i in category k (e.g., the fraction of a region's workforce in industry k) and g_k is an aggregate growth rate for category k (e.g., national employment growth in industry k). It was used by Bartik (1991) to construct local labor demand shocks. These **Bartik instruments** are now common in labor economics, international trade, and corporate finance.

Goldsmith-Pinkham, Sorkin, and Swift (2020) show that validity of a Bartik instrument depends on the exogeneity of the *shares* and not the aggregate growth rates. This provides a testable implication: pre-period industry shares should be uncorrelated with other determinants of the outcome. In practice, you should verify balance across industries (no pre-trends correlated with high-share industries).

Second, there has been considerable work to use machine learning for instrument selection. When many candidate instruments are available, naive 2SLS that uses all of them suffers from many-weak-instruments bias: the first stage overfits, and the estimator behaves more like OLS. **LASSO-based selection** (Belloni et al. 2012) helps solve this. Here, you use LASSO to select the instruments that predict D particularly well in the first stage. Then, you run 2SLS using only the selected set. You will need to adapt standard errors to account for the selection stage.

The **double/debiased machine learning** (DML) framework (Chernozhukov, Chetverikov, et al. 2018) generalizes this: it allows arbitrary ML methods for both the first stage (predicting D from Z and controls $\mathbf{X}$) and for partialling out the controls from Y , while restoring valid inference via cross-fitting (sample splitting). DML is increasingly used when the control vector $\mathbf{X}$ is high-dimensional.

Appendix

Generalized Method of Moments

Generalized Method of Moments (GMM) is a general estimation framework that unifies OLS, IV, and many other estimators under a single principle. Unlike MLE, GMM does not require specifying a full likelihood; instead, it exploits a set of **population moment conditions**.

The idea. Suppose the true parameter $\theta_0 \in \mathbb{R}^k$ satisfies $q \geq k$ population moment conditions:

$$\mathbb{E}[g(y_i, \mathbf{x}_i; \theta_0)] = \mathbf{0},$$

where $g : \mathbb{R} \times \mathbb{R}^p \times \mathbb{R}^k \rightarrow \mathbb{R}^q$ is a known vector of moment functions. The GMM estimator replaces the population expectation with its sample analog and minimizes the resulting “distance to zero”:

$$\hat{\theta}_{gmm} = \arg \min_{\theta} \bar{g}(\theta)' W \bar{g}(\theta), \quad \bar{g}(\theta) = \frac{1}{n} \sum_{i=1}^n g(y_i, \mathbf{x}_i; \theta),$$

where W is a $q \times q$ positive semi-definite weight matrix. When $q = k$ (exactly identified), the minimizer sets $\bar{g}(\hat{\theta}) = \mathbf{0}$ regardless of W .

Special cases.

- **OLS:** moment condition $\mathbb{E}[\mathbf{x}_i(y_i - \mathbf{x}_i' \beta)] = \mathbf{0}$ (exogeneity). With $W = I$, the GMM objective reduces to the OLS normal equations.
- **IV / 2SLS:** moment condition $\mathbb{E}[\mathbf{z}_i(y_i - \mathbf{x}_i' \beta)] = \mathbf{0}$ (exclusion restriction). When there are more instruments than endogenous regressors ($q > k$), the optimal GMM estimator equals 2SLS.
- **Arellano-Bond** (Ch. 12): moment conditions come from lagged levels being uncorrelated with first-differenced errors; the growing instrument set as t increases is precisely the “many moment conditions” that GMM is designed to exploit.

Efficiency. The asymptotically optimal weight matrix is $W^* = \Sigma^{-1}$ where $\Sigma = \mathbb{V}[g(y_i, \mathbf{x}_i; \theta_0)]$. Using this choice (feasible two-step GMM, replacing Σ with a consistent first-step estimate) achieves the semi-parametric efficiency bound among all estimators that use only the given moment conditions.

Overidentification test. When $q > k$, the extra moment conditions are testable. The **Hansen J -test** (a generalization of the Sargan test used in IV):

$$J = n \bar{g}(\hat{\theta})' \hat{\Sigma}^{-1} \bar{g}(\hat{\theta}) \xrightarrow{d} \chi_{q-k}^2,$$

tests H_0 : all q moment conditions hold at $\hat{\theta}$. Rejection indicates at least one instrument (or moment condition) is invalid.

8 Panel data and fixed effects models

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Overview

Causal inference is hard with observational data, because we struggle to find good counterfactuals that aren't affected by baseline bias. Time can help tackle this problem by allowing us to see trends. The starting point is the *difference-in-differences* design. The diff-in-diff then generalizes into the two-way fixed effects model.

Readings

People interested in panel data should refer to Wooldridge (2002) (a comprehensive textbook), A. Colin Cameron and Trivedi (2005) (ch. 21-22) (another textbook), S. L. Morgan and Winship (2014) (ch.11) (causal inference with repeated observations), J. Angrist and Pischke (2008) (ch.5), B. E. Hansen (2024) (ch.17). For difference-in-differences, see Baker et al. (2026{\S}).

8.1 Baseline bias

You may recall that we tried to understand what, if anything, we can learn from a naïve difference in means of Y between a treatment and a control group. We worked out the following result:

$$\begin{aligned}\text{Naïve difference} &= \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0] \\ &= \mathbb{E}[Y_i(1)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 1] \\ &\quad + \mathbb{E}[Y_i(0)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0] \quad \text{adding and subtracting} \\ &= \mathbb{E}[Y_i(1) - Y_i(0)|D_i = 1] \\ &\quad + \mathbb{E}[Y_i(0)|D_i = 1] - \mathbb{E}[Y_i(0)|D_i = 0] \quad \text{regrouping} \\ &\equiv \text{ATT} + \text{Selection (baseline) bias}\end{aligned}$$

Here, I want to focus a little bit on what this **baseline bias** means. The idea is pretty simple: it might be misleading to compare $\mathbb{E}[Y|D = 1]$ to $\mathbb{E}[Y|D = 0]$ (or rather, their sample analogues). The reason is that we are missing counterfactuals. Consider my treatment group. I don't see what Y would be for these units *had they not been treated*. The naïve difference is only informative if $\mathbb{E}[Y|D = 0]$ among the control group (which I do see) can be used as a counterfactual. If not, then my naïve difference will contain the effect of the treatment on Y for the treatment group plus the difference in baseline between the two groups.

Let me work out a simple example. Suppose that D is access to ChatGPT (0 or 1) and Y is a student's grade. We then have (I'm inventing numbers):

- $\mathbb{E}[Y|D = 1] = 5.3$. On average, students using ChatGPT obtain a grade of 5.3/6.
- $\mathbb{E}[Y|D = 0] = 4.5$. On average, students not using ChatGPT obtain a grade of 4.5/6.

The naïve estimate of the ATE is thus 0.8. But is this accurate? Maybe, but only if students who used ChatGPT would also have had 4.5 on average without ChatGPT. Now, we could imagine a world in which students most tempted to use ChatGPT are those who are struggling in class. Maybe their baseline without ChatGPT might have been a 4.2/6. In that case, our naïve estimate understates the effect of ChatGPT (naïve estimate < ATT):

$$\text{ATT} = 5.3 - 4.2 = 1.1$$

$$\text{Baseline bias} = 4.2 - 4.5 = -0.3$$

$$\text{Naïve estimate} = \text{ATT} + \text{Baseline bias} \Rightarrow 1.1 - 0.3 = 0.8$$

(Or maybe it's the opposite: students who are smarter in the first place are quicker to learn how to use a new tech like ChatGPT. In such case, the naïve estimate would overstate the effect of ChatGPT (naïve estimate > ATT). This would be closer to the image in Figure 1 below.)

Can we get rid of the baseline bias? We don't observe it directly, since one of its constituent term is an unobservable counterfactual (we don't see how the treated units would have behaved without treatment). We saw in earlier notes that RCT equalize the two quantities (in expectation) by randomizing treatment allocation. But we also saw that you can't always use RCTs (whether for ethical or practical reasons).

Fortunately, there is another solution – assuming we can access temporal data with repeated observations.

8.2 Temporal data

Until now, we had models such as:

$$y_i = \beta D_i + \varepsilon_i,$$

where i indicates units of analysis, such as firms, consumers, voters, countries, etc. So far, we could limit ourselves to a single subscript, because i was uniquely identifying each row. We call this **cross-sectional** data .

Sometimes, though, we might be able to get access to **temporal data**. For instance, we could be modeling GDP data in Switzerland. In the simplest case, rows aren't identified by unit (here: Switzerland). Instead, they are identified by time. We talk then of a **time series** data structure . We then would write:

$$y_t = \beta D_t + \varepsilon_t,$$

where t denotes time (days, months, etc.). An example of time dataset, which is entirely structured by time is a year-quarter dataset (here: for Switzerland alone):

Ctry (i)	Year (t)	Y	D	W
Switzerland	2000Q1	13	0	6
Switzerland	2000Q2	12	1	7
...	...	...	...	...
Switzerland	2020Q4	17	1	6

Table 8.1: **Time series data structure**. Here: a year-quarter dataset for Switzerland.

Sometimes, we have variation both across units and across time. Such data is typically called **panel data** or **longitudinal data** . Such a data is structured by repeated observations of the same units (e.g., a set of 100 consumers who receive a survey every quarter).¹

Units i nor time periods t uniquely identify rows anymore; instead, it is both that jointly identify each row:

$$y_{i,t} = \beta D_{i,t} + \varepsilon_{i,t}.$$

¹Sometimes, you might come across studies talking about **cross-section times series**. This is almost identical to a panel dataset, except that the units can change from time period to time period. For instance, think of a dataset of countries with average consumer confidence scores: we have repeated data for Switzerland, France, etc., but who is surveyed might change every year.

In our matrix notation, we now have:

$$\mathbf{y}_{i,t} = \begin{bmatrix} y_{1,1} \\ y_{1,2} \\ \vdots \\ y_{n,t} \end{bmatrix}_{(nt) \times 1} \quad \mathbf{X}_{i,t} = \begin{bmatrix} 1 & D_{1,1} & x_{1,1} & \cdots & w_{1,1} \\ 1 & D_{1,2} & x_{1,2} & \cdots & w_{1,2} \\ 1 & D_{2,1} & x_{2,1} & \cdots & w_{2,1} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & D_{n,t} & x_{n,t} & \cdots & w_{n,t} \end{bmatrix}_{(nt) \times k} \quad \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \vdots \\ \beta_k \end{bmatrix}_{k \times 1} \quad \varepsilon = \begin{bmatrix} \varepsilon_{1,1} \\ \varepsilon_{1,2} \\ \vdots \\ \varepsilon_{n,t} \end{bmatrix}_{(nt) \times 1}$$

Thus, $\mathbf{X}$ is a stacked matrix where each row is a unique combination $i \times t$. For instance, we may have the following dataset. Note how each row is uniquely defined by a country-year. The matrix $\mathbf{X}$ stacks vectors for each country-year, such that $\mathbf{X}$ ($\mathbf{X}$ is constituted of stacked X'_i).

Country	Year	$\mathbf{Y}_{i,t}$	$\mathbf{D}_{i,t}$	$\mathbf{W}_{i,t}$
Switzerland	2000	3	1	6
Switzerland	2001	2	0	6
...	...	...	...	...
Switzerland	2020	5	1	5
France	2000	15	0	4
France	2001	12	0	3
...	...	...	...	...
Zambia	2020	4	1	5

Table 8.2: **Cross-section time-series data structure.** Here: a country-year dataset.

Note that the size of our matrices increases, though the number of parameters remains the same as before.

Here's the thing. Adding time as a structure in our dataset can have three implications.

1. It can make some analysis unreliable if we are not careful. Most straightforwardly: we often assumed independence of error terms ε ; this is unlikely to hold in a temporal setting.
2. We can leverage time to implement **quasi-experimental designs**. This will be the case of difference-in-differences and of fixed effects models.
3. We can leverage time to model **dynamics**. For instance, we can ask interesting questions such as: "how quickly does an increase in interest rates affect bank lending?" We obviously couldn't do this with cross-sectional data.

8.3 Difference-in-differences (DiD)

8.3.1 DiD intuition

Consider Figure 8.1. This represents the problem of baseline bias as stated above. We can do a naïve estimate of the treatment effect by comparing the red (T) minus the blue (C) points. However, we don't know where the units in the treatment group would have ended up if they hadn't been treated ($\mathbb{E}[Y(0)|D = 1]$ in orange). The treatment effect (at least on the treated) is given by the vertical space between the red dot and the orange dot. But of course, the orange dot is not observable.

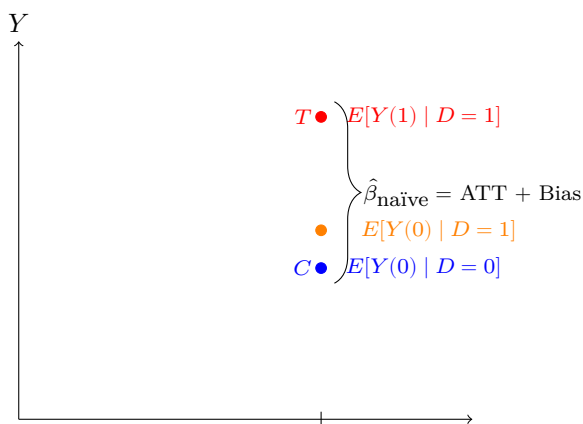


Figure 8.1: Baseline bias with cross-sectional data. Red and blue points are observable average outcomes for the treatment (red) and control (blue) groups. Orange is the unobservable counterfactual for the treatment group, had it not been treated.

Thus the question is: is there a way to estimate the gap between the red dot and the lower black dot? If yes, we could then compute:

$$ATT = \underbrace{[\mathbb{E}[Y(1)|D = 1] - \mathbb{E}[Y(0)|D = 0]]}_{\text{naïve}} - \underbrace{[\mathbb{E}[Y(0)|D = 1] - \mathbb{E}[Y(0)|D = 0]]}_{\text{baseline diff}}$$

The intuition of the **difference-in-differences (DiD)** estimator is the following: if we have data pre-treatment (at time $t - 1$), we could use the difference in $\mathbb{E}[Y]$ between the two groups then as a reasonable guess for that baseline difference (Lechner, n.d.). We could then use this difference to get rid of the baseline difference. This idea is captured by Figure 8.2.

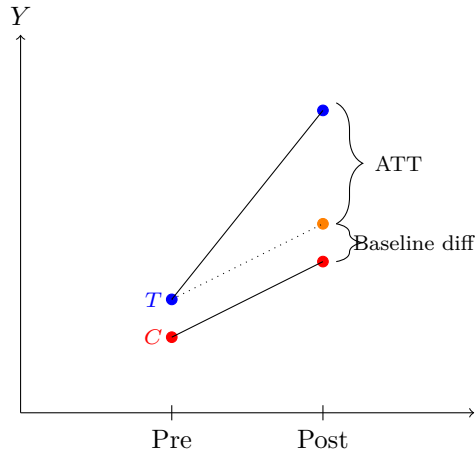


Figure 8.2: Difference-in-difference estimator.

Another way to describe DiD is as the difference of two comparisons:

1. Treated and untreated: as we saw, this is misleading for the ATT if there is a different baseline.
2. Before and after: this is misleading if other variables affect the trend of Y

Instead, by combining the two, DiD removes each source of errors.

Typically, DiD will be useful when:

1. The treatment occurs as an exogenous shock, but we worry that the treatment and the control group were different prior to the shock already.
2. We have panel data both pre- and post-treatment.

8.3.2 Estimation

We start with the simplest case. We have two time periods $T \in \{0, 1\}$. Treatment is assigned between the two periods. We refer to “pre” as $t = 0$ and “post” as $t = 1$. There is an indicator for each unit whether it is in the treatment or the control group. Call this $D \in \{0, 1\}$ (where 1 denotes treatment). Note that this indicator is time-invariant: it simply flags each observation in the dataset.

We can then define the ATT from the DiD framework as:

$$\text{ATT}_{\text{DiD}} = [\mathbb{E}[Y|D = 1, t = 1] - \mathbb{E}[Y|D = 0, t = 1]] - [\mathbb{E}[Y|D = 1, t = 0] - \mathbb{E}[Y|D = 0, t = 0]]$$

There are two ways to estimate DiD. One is to plug in averages and take the differences:

$$\widehat{\text{ATT}}_{\text{DiD}} = \left[\frac{1}{N_{D=1}} \left[\sum Y_i | D = 1, t = 1 \right] - \frac{1}{N_{D=0}} \left[\sum Y_i | D = 0, t = 1 \right] \right] \\ - \left[\frac{1}{N_{D=1}} \left[\sum Y_i | D = 1, t = 0 \right] - \frac{1}{N_{D=0}} \left[\sum Y_i | D = 0, t = 0 \right] \right]$$

There are three reasons why we might want to estimate a linear model with OLS instead:

1. We might want to add additional covariates in the model.
2. In some cases, we might have differences in timing of treatment and more than 2 time periods. See section on fixed effects models below.
3. We might want to use more complicated standard errors (including clustered standard errors; see below).

With a linear model, can then write the DiD model to estimate the ATT as:

$$y_{i,t} = \alpha + \gamma D_{g(i)} + \tau T_t + \beta D_{g(i)} T_t.$$

D is a time-invariant dummy and takes value 1 if unit i is in the treatment group g , 0 otherwise. T is a cross-sectionally-invariant dummy that takes value 1 in the post-treatment period and 0 in the pre-treatment period. The interaction $D_{g(i)} T_t$ can vary both over time and across units.

Note the interpretation of the parameters:

- $\alpha = \mathbb{E}[Y | D = 0, \text{pre}]$: $\bar{Y}$ for untreated, pre-treatment
- $\alpha + \tau = \mathbb{E}[Y | D = 0, \text{post}]$: $\bar{Y}$ for untreated, post-treatment
- $\alpha + \gamma = \mathbb{E}[Y | D = 1, \text{pre}]$: $\bar{Y}$ for treated, pre-treatment
- $\alpha + \gamma + \tau + \beta = \mathbb{E}[Y | D = 1, \text{post}]$: $\bar{Y}$ for treated, post-treatment

So what D really does is to estimate the gap between T and C pre-treatment (the baseline bias). What T does is to estimate the change that happened between the two time periods (the red line in Figure 8.2 isn't flat). And the interaction effect recovers an estimate of the ATT :

$$\widehat{\text{ATT}} = \hat{\beta}$$

To see this:

$$\begin{aligned} \text{Naïve estimate} &= \mathbb{E}[Y | D = 1, \text{post}] - \mathbb{E}[Y | D = 0, \text{post}] \quad \text{by definition} \\ &= [\alpha + \gamma + \tau + \beta] - [\alpha + \tau] \\ &= \gamma + \beta \end{aligned}$$

Then:

$$\begin{aligned}\text{Baseline bias} &= \mathbb{E}[Y|D = 1, \text{pre}] - \mathbb{E}[Y|D = 0, \text{pre}] \\ &= [\alpha + \gamma] - \alpha \\ &= \gamma\end{aligned}$$

Recall that:

$$\text{Naïve estimate} = \text{ATT} + \text{Baseline bias},$$

which implies that the ATT is the naïve estimate minus the baseline bias. We thus get:

$$\begin{aligned}\text{ATT} &= \text{Naïve estimate} - \text{Baseline bias} \\ &= [\gamma + \beta] - \gamma \\ &= \beta\end{aligned}$$

The DiD model can be estimated with least squares. Note that it is a **saturated model**: there is a combination of parameters for each combination of independent variables (here: D and T). As a result, we don't have to worry about linearity.

As for data, here is the structure of a 2-period, municipality-based dataset:

Country	Year	$Y_{i,t}$	$D_{g(i)}$	T_t	$D_{g(i)}T_t$
NYC	2019	3	1	0	0
NYC	2020	2	1	1	1
Pittsburgh	2019	5	0	0	0
Pittsburgh	2020	15	0	1	0
...	...	...	...	...	...
Philadelphia	2019	4	1	0	0
Philadelphia	2020	2	1	1	1

Table 8.3: **DiD data structure.** Here: a city-year dataset, with two periods, time-invariant treatment status, and time-variant interaction effect. New York and Philadelphia are in the treatment group; Pittsburgh isn't.

Example 8.1 (Illustration: minimum wage). Card and Krueger (Card and Krueger 1994) are often credited to develop the DiD

framework. They wanted to study the effect of a minimum wage increase (D) on employment rate (Y) (measured as number of full time-equivalence (FTE) per restaurant). Their design was the following:

- They had data on 410 fast-food restaurants for two US neighboring states: New Jersey and Pennsylvania
- New Jersey experienced an increase in minimum wage from \$4.25 to \$5.05 in 1992.
- Pennsylvania did not experience a minimum wage change during that time.

Thus, we have two groups: 331 restaurants in the treatment group (i.e., in New Jersey) and 79 in the control group (Pennsylvania). We don't necessarily expect the two to have the same level of employment per restaurant pre-treatment (baseline bias). Maybe New Jersey restaurants employ more people on average than restaurants in Pennsylvania. Thus a comparison after the change in minimum wage would not be informative. Instead, a DiD might be particularly suited.

The data they had included the average number of full-time employees per restaurant. From their paper, we can identify each component of the analysis.

	$\bar{Y}_{PA} (D=0)$	$\bar{Y}_{NJ} (D=1)$	Difference
Pre-treatment	23.33	20.44	-2.89
Post-treatment	21.17	21.03	-0.14
ATT			2.75

We can reconstruct the ATT as:

$$\widehat{ATT} = (21.03 - 21.17) - (20.44 - 23.33) = 2.75$$

Thus, minimum wage seems on average to have slightly *increased* employment (instead of decreasing it, as many feared).

You can already guess what some of the identifying assumptions will be. Here, we must be sure that the minimum-wage hike in New Jersey had no effect on Pennsylvania. We must also ponder whether Pennsylvania is a good counterfactual for New Jersey. This will lead us to the **parallel trend** assumption. Finally, we must also be sure that New Jersey's restaurants didn't fire people *before* the law came into force – the problem of **anticipation**.

Example 8.2 (Robots and employment: automation and the panel FE estimator). Acemoglu and Restrepo (2020) study the effect of industrial robots on employment and wages across US commuting zones between 1990 and 2007. The core challenge is that robot adoption is endogenous: regions with an industrial composition that lends itself to automation may be precisely those where employment was already under pressure from other forces (globalization, offshoring, capital deepening).

Their strategy layers two identification approaches. First, they use **first differences** (changes within commuting zones over time), which absorbs time-invariant zone characteristics such as regional industrial culture or geographic factors. Second, they instrument US robot adoption in each industry with **robot adoption in those same industries in other advanced economies** (Germany, Italy, Japan, Sweden). The argument mirrors Autor-Dorn-Hanson: the worldwide drop in robot prices and improving capabilities drives automation globally, but US-specific labor market shocks do not cause German manufacturers to adopt robots.

The IV-FE estimate finds that **one additional robot per 1,000 workers reduces the employment-to-population ratio by 0.18–0.34 percentage points and wages by 0.25–0.5%**. The effect is concentrated in manufacturing but spills over to non-manufacturing employment in the same commuting zone. OLS understates the employment effect, consistent with zones that adopted robots having favorable pre-existing trends. The paper is a clear illustration of how panel FE and IV address different threats to identification: FE removes time-invariant confounders; IV removes time-varying endogeneity that FE alone cannot handle.

Outcome	OLS (FD)	IV-FD
Employment/Pop	−0.09**	−0.34***
<i>SE</i>	(0.04)	(0.07)
Log weekly wages	−0.10**	−0.50***
<i>SE</i>	(0.04)	(0.12)

Table 8.5: Effect of industrial robots on US commuting-zone labor market outcomes, 1990–2007. Unit: commuting zone $\times$ decade. Instrument: robot adoption in the same industries in other advanced economies (Germany, Italy, Japan, Sweden). Standard errors clustered by commuting zone in parentheses. *** $p < 0.01$, ** $p < 0.05$. Source: Acemoglu and Restrepo (2020).

Example 8.3 (Managing with style: do individual managers affect corporate policy?). Does it matter *who* runs a firm? Or do competitive pressures and governance constraints make managers largely interchangeable? This matters: firms tend to pay well for senior managers like CEOs. If it doesn't matter who is the manager, then why bother?

Bertrand and Schoar (2003) answer this with a manager-firm panel dataset that covers large US firms (1969–1999). By tracking the same manager across different firms over time, they can separate a manager's personal “style” from the characteristics of the firms they happen to run.

The identifying variation comes from manager job changes: when a CEO leaves one firm to join another, they carry their style with them. By comparing outcomes in the destination firm before and after the new CEO arrives (and comparing them with firms that kept the same CEO) the authors estimate manager fixed effects (the component of corporate policy attributable to the individual rather than the firm). This is the within-manager variation that a simple cross-sectional regression cannot get at.

The results show that manager FE are jointly significant for all major corporate policies (leverage, acquisition activity, capital expenditures, and return on assets). CEO effects explain 4–5% of the total variance in these outcomes after absorbing firm FE, year FE, and observable CEO characteristics. The effects are economically large. Moving from a manager at the 25th percentile of “style” to the 75th percentile is associated with a change in leverage of roughly 3–4 percentage points. Managers with MBA degrees and younger managers tend toward more aggressive (high-leverage, high-acquisition) styles.

Corporate DV	R^2 without manager FE	R^2 with manager FE	Manager FE contribution	F -test (manager FE jointly = 0)
Leverage	0.512	0.560	4.8%	***
Capital expenditures	0.461	0.514	5.3%	***
Acquisitions	0.388	0.421	3.3%	***
Return on assets	0.541	0.583	4.2%	***

Aside from its substantive contribution, the paper shows that FE can be used not just to control for unit-level confounders but to directly *estimate* the magnitude of individual effects.

8.3.3 Identifying assumptions

Figure 8.2 makes also very obvious what one of the key identifying assumption of the difference-in-difference estimator is: for the gap between treatment and control pre-treatment to be a valid estimate of the baseline difference, it must be that the two groups would have evolved in a similar way if no treatment had been assigned. This is the dotted line in Figure 8.2. Formally, define the baseline bias:

$$\Delta_{D,post} := \mathbb{E}[Y(0)|D = 1, \text{post-T}] - \mathbb{E}[Y(0)|D = 0, \text{post-T}]$$

I use Δ as the difference operator. Here, it's the difference across group, post-treatment.

We then estimate $\Delta_{D,post}$ with $\widehat{\Delta}_{D,pre}$, that is:

$$\widehat{\Delta}_{pre} = \widehat{\mathbb{E}}[Y(0)|D = 1, \text{pre-treatment}] - \widehat{\mathbb{E}}[Y(0)|D = 0, \text{pre-treatment}]$$

Finally, we compute two sets of differences (hence the name):

$$\begin{aligned} \text{Population ATT} &= \underbrace{\mathbb{E}[Y(1)|D = 1] - \mathbb{E}[Y(0)|D = 0]}_{\text{naïve diff}} - \Delta_{D,post} \\ \widehat{ATT} &= \widehat{\mathbb{E}}[Y(1)|D = 1] - \widehat{\mathbb{E}}[Y(0)|D = 0] - \widehat{\Delta}_{D,pre} \end{aligned}$$

Thanks to the longitudinal data, both of the right-hand side quantities are observable. But that gap is a valid estimate as long as it

doesn't change over time. In other words, Δ_D has to be a constant. Or, put another way, the two quantities must co-move for us to use it as a way to get rid of the baseline bias. We call this the **parallel trend assumption**:

$$\mathbb{E}[\Delta_t Y(0)|D = 1] = \mathbb{E}[\Delta_t Y(0)|D = 0],$$

that is, the change in Y over time (Δ_t) is the same for the treatment and the control group, on average.

The parallel trend assumption cannot be tested when you have data for a single pre-treatment period. However, if we do have multiple pre-treatment periods, we can in principle see whether this assumption is plausible. To be clear: we cannot actually test it, because we will never observe the counterfactual of the treatment group had it not be treated. But if we have multiple pre-treatment periods, and if Δ remains roughly constant throughout, then we might be reassured enough to project this difference in the future.

Another assumption that we need is the absence of **anticipation effect**: Y shouldn't change prior to the assignment of D . Here's a motivating example. As an analyst, you want to model household spending as a function of wages. Suppose that you are told that a bunch of salary of a bunch of people in your sample will increase next year. These people might start spend more now. But in a DiD framework, the treatment only starts in January of next year, yet it has already had an effect in months prior.

8.4 One-Way and Two-Way Fixed Effects models

8.4.1 Fixed effect models

The canonical DiD uses two groups (treated versus control) and two time periods (pre and post-treatment). We can generalize this in two ways:

- We can use multiple time periods $t = 1, 2, \dots$, if data are available.
- We can define groups in a more fine-grained way.

Concretely, we can write:

$$y_{i,t} = \kappa_i + \tau_t + \beta D_{i,t} + \varepsilon_{i,t}.$$

Note. κ_i is short for $\sum_i^n \kappa_i$, a set of n dummy indicators (where n is the number of units, not the total dataset). For instance, κ_1 would equal 1 for all observations from Switzerland, and zero otherwise. Next, κ_2 would equal 1 for observations from France, and zero otherwise. We can then include as many parameters κ 's as there are units, at least as long as we don't include a y-intercept. (If we want a y-intercept, we need to suppress one κ – otherwise we would have perfect multicollinearity. The suppressed unit would then serve as baseline and be captured by α , the y-intercept ($\mathbb{E}[Y|\text{baseline group}] = \alpha$).)

This is different from the standard DiD model, where we had a *single* time-invariant indicator $D_{g(i)}$ that was equal to 1 for *all* units in the treatment group. In the generalized model, we have n (or $n - 1$) indicators.

In the same spirit, we have t dummy indicators τ for each time period. If our dataset is structured as country-years, with years from 2010 to 2020, we would have an dummy indicator for 2010 (equal 1 across all countries for observations measured in 2010), another for all observations in 2011 (equal 1 across all countries for observations measured in 2011), etc.

An illustrative dataset is reproduced here.

Ctry	Year	$Y_{i,t}$	$D_{i,t}$	SWI	FRA	2019	2020	2021
Swi	2019	3	0	1	0	1	0	0
Swi	2020	2	1	1	0	0	1	0
Swi	2021	5	1	1	0	0	0	1
Fra	2019	15	0	0	1	1	0	0
Fra	2020	12	0	0	1	0	1	0
Fra	2021	12	0	0	1	0	0	1
...	...	...	...	...	...	...	...	...

Table 8.7: **TWFE data structure.** Here: a country-year dataset, with three periods, with country and year dummies. Switzerland is treated in 2020 and 2021. France is never treated.

Finally, note that D is now varying over time and over units. As such, it plays the role of the interaction term in the standard DiD model (where we interacted a time-invariant $D_{g(i)}$ with a unit-invariant T_t).

With this hypothetical country-year data, a TWFE model would be something like:

$$y_{i,t} = \beta D_{i,t} + \kappa_1 \mathbb{I}(\text{CH})_i + \kappa_2 \mathbb{I}(\text{FRA})_i \\ + \tau_1 \mathbb{I}(2019)_t + \tau_2 \mathbb{I}(2020)_t + \tau_3 \mathbb{I}(2021)_t + \varepsilon_{i,t},$$

where $\mathbb{I}$ is an indicator operator (ie, a dummy that equals yes if the condition is met). (Note the absence of intercept. Why?)

We call this generalized model the **(two-way) fixed effects** model (or **TWFE**). The “two-ways” refers to the two sets of parameters: n parameters (or fixed effects) for units and t parameters (fixed effects) for time. And in the case in which D is a dichotomous treatment, the TWFE will estimate the DiD ATT.

Aside from the fact that there might a lot of parameters κ and τ (for instance: 150 countries measured over ten years would mean 160 such parameters), there is nothing magical about them. These are dummies like other dummies we saw before. However, they get at something interesting from a design perspective. Here’s the key: when you include a fixed effect, then the identification of β stems from variation coming from *within* the fixed effect.

To see what this means, consider a country-year dataset and a two-way fixed effects model. Focus on the country fixed effects. As stated, these will be a series of dummy indicators for each country (Switzerland, France, Germany, etc.). What this means for β is: $\hat{\beta}$ will be based on changes in D *within* countries rather than variation in D *across* countries. Think of this from a counterfactual perspective: when I introduce country fixed effects, I will no longer compare Y between Switzerland and France. Instead, I will compare Switzerland when it’s in the control group to Switzerland when it’s in the treatment group. Then I’ll do the same in France, in Germany, etc. The resulting β will be a weighted comparison of these *within-unit* changes in Y caused by changes in D .

To show what kind of payoff can be obtained from within-variation, consider Figure 8.3, with a continuous treatment x and two units (A, B) measured at three levels of x . For instance, think here of each unit as a customer and y the amount they spend after watching x minutes of ads. The figure depicts **Simpson’s paradox**: if we pool all observations without fixed effects, then the estimated treatment effect is negative (black dotted line). However, it’s clear that *within* units, the treatment effect is positive.

The dotted black line is what you obtain if you estimate:

$$y_i = \alpha + \beta x_i$$

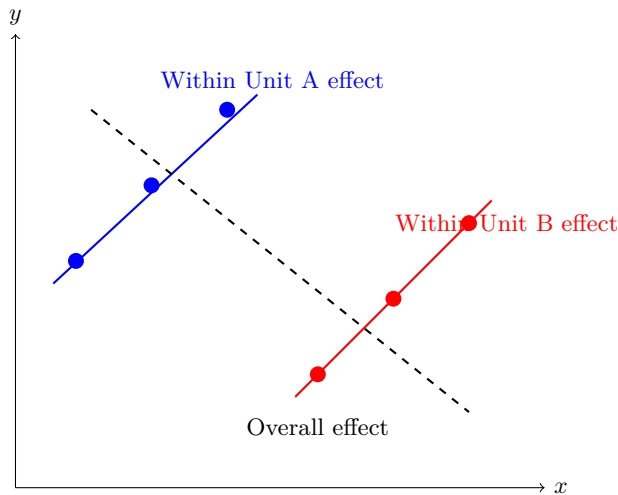


Figure 8.3: Simpson's paradox. Here: each unit has its own y-intercept (i.e., fixed effect), which changes the slope of the treatment effect from negative (dotted line) to positive (red and blue lines).

If you use a (one-way) fixed effect model, instead, you'd estimate (more on one-way FE models in a second):

$$y_i = \phi_i + \gamma x_i.$$

What you would find is that $\hat{\gamma} > 0$ (the slope of the red and blue lines) whereas $\hat{\beta} < 0$ (the slope of the dotted line). The FE model is intuitively more convincing in general. Think of it this way: imagine you were asked to guess what the treatment effect would be for a new 3rd unit C. Would you put money on the treatment effect to be positive or negative?

What's very nice about this is that the fixed effects do what we were looking for in DiD: they remove the baseline differences across units (i.e., they give each unit a different y-intercept). In contrast to the canonical DiD, where we only wanted to remove the differences between treatment and control, here we will remove the differences between each unit (eg, countries).

Likewise, time fixed effects will remove time-specific effects. For instance, if all countries in a given year suffer from a common shock, this shock will be absorbed by the τ for this year.

Note that we could also estimate a **one-way fixed effect model**:

$$y_{i,t} = \beta D_{i,t} + \kappa_i + \varepsilon_{i,t}.$$

We would still be relying on within-unit variation to estimate β (since we include unit fixed effects, i.e., n parameters κ_i). However, this specific model would not take into account time fixed effects. Thus, we wouldn't know if the results are driven by something correlated with time and with our treatment. We could try to get at this by including a linear or nonlinear parameter(s) for time – this would capture **time trends**. For instance, here's a one-way fixed effect model that adjusts for a quadratic time trend:²

$$y_{i,t} = \beta D_{i,t} + \tau_1 T + \tau_2 T^2 + \kappa_i + \varepsilon_{i,t}.$$

But as a general rule, it's hard to justify using time trends (linear or not) over time fixed effects. Maybe the most convincing reason is if you want to do **forecasting**: a time fixed effect is not predictable (since it captures something stochastic about time). Thus, if you want to make a forecast for 2027, you won't be able to account for time trends. However, with a time trend, one can make out-of-sample forecasts that account for time.

Another nice feature of panel data is that we can compute how the treatment effect evolves over time using an **event study** specification. Imagine we have t from -2 to 3 , with treatment beginning at $t = 1$ (so $t \leq 0$ are pre-treatment). Let D_i be a time-invariant indicator equal to 1 for eventually-treated units. We omit the last pre-treatment period ($t = 0$) as the baseline and estimate:

$$\begin{aligned} y_{i,t} = & \alpha + \underbrace{\beta_{-2} D_i \mathbb{I}(t = -2) + \beta_{-1} D_i \mathbb{I}(t = -1)}_{\text{pre-treatment leads}} \\ & + \underbrace{\beta_1 D_i \mathbb{I}(t = 1) + \beta_2 D_i \mathbb{I}(t = 2) + \beta_3 D_i \mathbb{I}(t = 3)}_{\text{post-treatment lags}} \\ & + \tau_{-2} + \tau_{-1} + \tau_1 + \tau_2 + \tau_3 + \kappa_1 + \dots + \kappa_n + \varepsilon_{i,t} \end{aligned}$$

The post-treatment coefficients $\beta_1, \beta_2, \beta_3$ estimate the treatment effect at each lag; plotting them (together with their confidence intervals) against event time gives the **event study plot**. This relaxes the assumption that treatment effects must be immediate and constant over time. For instance, the effect of a new patent on a firm's profit will likely build gradually rather than jumping to a permanent level Y at once.

The pre-treatment coefficients β_{-2} and β_{-1} test whether treated and control units were already diverging *before* treatment was assigned. Under the parallel trend assumption, they should be zero. Testing $H_0 : \beta_{-2} = \beta_{-1} = 0$ jointly (e.g., with an F -test) is the standard

²Instead of a counter T , you could also use years (eg $\tau_1 \text{Year}_t + \tau_2 \text{Year}_t^2$).

pre-trend test . It is considered best practice when multiple pre-treatment periods are available, and graphical inspection of the event study plot is the most common way to present this evidence.

8.4.2 Random effects model and the Hausman test

The **random effects** (RE) model is an alternative to fixed effects. Rather than treating unit-specific intercepts α_i as free parameters to be estimated, RE treats them as *random draws* from some distribution:

$$y_{i,t} = \beta D_{i,t} + \alpha_i + \varepsilon_{i,t}, \quad \alpha_i \stackrel{iid}{\sim} (0, \sigma_\alpha^2).$$

The key identifying assumption is that the unit-specific component α_i is **uncorrelated with the regressors**:

$$\mathbb{E}[\alpha_i | D_{i,t}] = 0.$$

When this holds, RE is a generalized least squares (GLS) estimator that exploits both *within*-unit and *between*-unit variation. This makes it more efficient than FE, which discards between variation entirely. A further practical advantage: because RE does not absorb unit dummies, it can estimate the effect of time-invariant regressors (such as gender, country, or geography), which the FE estimator cannot identify.

When the RE assumption fails (i.e., when unobservable α_i is correlated with $D_{i,t}$) the RE estimator is inconsistent, while the FE estimator remains consistent. In practice, selection into treatment is often driven by unit-specific characteristics, making RE assumptions hard to justify. That's why they are rare in economic analysis.

Hausman test

The **Hausman test** (Hausman 1978) is a diagnostic for choosing between fixed effects (FE) and random effects (RE) models. The logic: if the RE assumption holds, both FE and RE are consistent, but RE is more efficient; if it fails, FE is consistent but RE is not. A large difference between $\hat{\beta}_{\text{FE}}$ and $\hat{\beta}_{\text{RE}}$ thus signals that the RE assumption is violated.

The test statistic is:

$$H = (\hat{\beta}_{\text{FE}} - \hat{\beta}_{\text{RE}})' [\hat{V}(\hat{\beta}_{\text{FE}}) - \hat{V}(\hat{\beta}_{\text{RE}})]^{-1} (\hat{\beta}_{\text{FE}} - \hat{\beta}_{\text{RE}}) \stackrel{a}{\sim} \chi_k^2,$$

where k is the number of time-varying regressors. Under H_0 (RE assumption holds), H is asymptotically chi-squared. Rejecting H_0 favors FE.

Now, in practice, very few people in economics trust random effects. You will generally find FE models to be the main modeling approach.

8.4.3 First-difference estimator

A second alternative to within-FE is the **first-difference** (FD) estimator. Instead of demeaning within units, we difference consecutive observations:

$$\Delta y_{i,t} := y_{i,t} - y_{i,t-1} = \beta(D_{i,t} - D_{i,t-1}) + (\varepsilon_{i,t} - \varepsilon_{i,t-1}),$$

where the unit-specific term α_i cancels out exactly. OLS applied to this first-differenced equation is the FD estimator.

When $T = 2$, FD and FE (within estimator) give identical point estimates. For $T > 2$, they differ: FE pools all within-unit deviations from the unit mean, while FD uses only consecutive changes. Under standard assumptions (homoskedastic and serially uncorrelated errors), FE is more efficient because it uses all available variation. However, FD can be preferable if errors follow a random walk ($\varepsilon_{i,t} = \varepsilon_{i,t-1} + u_{i,t}$ with u iid), because differencing then eliminates the persistent component and yields iid differenced errors.

In practice, the choice between FD and FE matters most when T is small and the serial correlation structure of errors is uncertain. The Wooldridge (2002) test for serial correlation in FD residuals (testing whether $\Delta \hat{u}_{i,t}$ and $\Delta \hat{u}_{i,t-1}$ correlate) can help diagnose this.

8.4.4 Complications with FE and TWFE models

8.4.4.1 Time-invariant treatments

The most obvious shortcoming of TWFE is its inability to estimate parameters of time-invariant variables. Intuitively, it's pretty clear why: since we are relying on *within-unit* variation, and since such a variable does not exhibit within variation, it cannot be estimated. Another way to see this is to note that a time-invariant parameter will be perfectly correlated with unit fixed effects. Likewise (but that's rarer), a treatment that is assigned to all units in the same year cannot be estimated either.

Thus, important treatments cannot be estimated with TWFE. For instance, imagine a model trying to estimate whether a country has

access to the sea as a predictor of its growth rate (landlocked countries might struggle to capitalize on trade). Assuming no change in borders, a country either has access to the sea or it doesn't. And thus, a variable $D_i \in \{\text{landlocked, sea access}\}$ could not be estimated using a TWFE model (or even a single unit FE model).

8.4.4.2 Staggered treatment and heterogeneous treatment effects

Worse, in the last few years, several papers raised concerns about the validity of the TWFE model as a way to estimate the DiD estimator (Goodman-Bacon 2021; Imai and Kim 2021; Callaway and Sant'Anna 2021; De Chaisemartin and d'Haultfoeuille 2020; Arkhangelsky et al. 2024; Baker et al. 2025). The concerns (and solutions) vary, but one of the problem relates to the staggered timing of treatment as well as its heterogeneity (i.e., the notion that the treatment effect may change across units or over time).

Let's start with the following case. Imagine that you have a set of $N = 10$ units and $t = 10$ time periods. Suppose that 4 units get treated at time $t = 2$, 4 units get treated at time $t = 6$, and 2 units never get treated. Thus, we have three groups:

- A: Never treated
- B: Treated at time $t = 2$
- C: Treated at time $t = 6$

The TWFE/DiD model will combine four types of comparisons:

- B with A at $t = 2$
- B with C at $t = 2$.
- C with A at $t = 6$
- C with B at $t = 6$.

This is the 'simple' case with two different treatment slots. You could easily imagine that the treatment could happen any time in your data (for instance, some countries might be treated at in 2000, others in 2001, others in 2003, etc.). Then, the TWFE model yields a weighted combination of all 2x2 comparisons possible (Goodman-Bacon 2021).

There are several problems with this. First, some comparisons don't necessarily make sense. In the example above, is group B really a good comparison group for C at time $t = 6$? By that time, B has already been treated. Thus, these comparisons could be weird.

The bigger problem is that these comparisons are only valid if the treatment effect is constant over time. That is: once a group of units

are treated, their average converges immediately to the new value of Y and stays there. This is often unlikely: many treatments will take time to have an effect. Drugs may necessitate months or years to change health. A change in Central Bank interest rate might take months before having an effect on real estate investments. Etc.

Time-varying treatment effects — a form of heterogeneity — will create problems in the kind of comparisons shown above. When B gets treated, it first moves from Y to Y' (immediate effect of the treatment at time $t = 2$). Then let's suppose that it continues to increase year after year to $Y'' > Y'$. Then, when B is used again as a comparison with C (this time, at $t = 6$), its value will be much higher.

⚠ Negative weights in TWFE

Goodman-Bacon (2021) shows that, with staggered treatment timing, the TWFE estimator $\hat{\beta}_{\text{TWFE}}$ is a weighted average of all possible 2×2 DiD comparisons:

$$\hat{\beta}_{\text{TWFE}} = \sum_{(k,l)} w_{kl} \cdot \hat{\beta}_{kl}^{2 \times 2},$$

where the pairs (k, l) represent every combination of a treated group k and a comparison group l (either never-treated, not-yet-treated, or already-treated), and the weights w_{kl} depend on group sizes and timing.

The central problem: **some weights w_{kl} are negative**. This happens when already-treated units (like B above) serve as the control group for later-treated units (like C). At $t = 6$, B is already treated and potentially has an accumulated treatment effect $Y'' \gg Y$. Using B as the control inflates the estimated effect for C — and after the TWFE algebra, this enters the overall estimate with a negative weight. As a result, $\hat{\beta}_{\text{TWFE}}$ can be a poor estimate of any meaningful average treatment effect, and can even have the *wrong sign* if treatment effects grow over time.

The practical consequences of this:

1. If you think that you may have heterogeneous or time-varying treatment effects, conventional TWFE estimates should be interpreted with a grain of salt.
2. Use diagnostic tools: the Goodman-Bacon decomposition decomposes $\hat{\beta}_{\text{TWFE}}$ into its 2×2 components and displays the corresponding weights. Negative-weight components are a red flag.

3. Use estimators such as those developed by Callaway and Sant'Anna (2021) and De Chaisemartin and d'Haultfoeuille (2020), or Borusyak, Jaravel, and Spiess (2024).

Warning: this literature has generated dozens of papers in a short span of time. Things haven't completely settled just yet. The latest review is Baker et al. (2025).

8.4.5 Inference

In principle, we can do the same kind of inferential work as before. But the temporal and grouped nature of the dataset creates new problems that we didn't have so far. Recall that we assumed generally so far that error terms were heteroskedastic and yet independent across units. This made generally sense in the context of conventional random sampling.

However, now, our sample isn't random in the same way and thus observations are not independent. We repeatedly get information from the same units. We may thus worry that error terms could be correlated.

Not taking such temporal correlation into account generally leads us to have overly tight confidence intervals, thereby committing more Type I errors (falsely rejecting the null hypothesis that $\beta = 0$) (Bertrand, Duflo, and Mullainathan 2004). Fortunately, there is a way out of this by allowing correlation of error terms *within* units. Each unit (eg, each country) represents a cluster or group. We can then write the variance-covariance matrix of $\mathbb{E}[\varepsilon\varepsilon']$ as (here, with two units a and b , each measured over 3 time periods):

$$\mathbb{E}[\varepsilon\varepsilon'] = \begin{bmatrix} \sigma_{a,1}^2 & \varepsilon_{a,1}\varepsilon_{a,2} & \varepsilon_{a,1}\varepsilon_{a,3} & 0 & 0 & 0 \\ \varepsilon_{a,2}\varepsilon_{a,1} & \sigma_{a,2}^2 & \varepsilon_{a,2}\varepsilon_{a,3} & 0 & 0 & 0 \\ \varepsilon_{a,3}\varepsilon_{a,1} & \varepsilon_{a,3}\varepsilon_{a,2} & \sigma_{a,3}^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_{b,1}^2 & \varepsilon_{b,1}\varepsilon_{b,2} & \varepsilon_{b,1}\varepsilon_{b,3} \\ 0 & 0 & 0 & \varepsilon_{b,2}\varepsilon_{b,1} & \sigma_{b,2}^2 & \varepsilon_{b,2}\varepsilon_{b,3} \\ 0 & 0 & 0 & \varepsilon_{b,3}\varepsilon_{b,1} & \varepsilon_{b,3}\varepsilon_{b,2} & \sigma_{b,3}^2 \end{bmatrix}$$

For instance, $\varepsilon_{a,2}\varepsilon_{a,1}$ denotes the correlation of the error terms of unit a at time $t = 1$ and $t = 2$. What we have here is a form of **serial correlation** :

$$Cor(\varepsilon_{i,t-1}, \varepsilon_{i,t}) \neq 0.$$

Note: in the DiD case, the clusters are defined by the *temporal* structure of the data. That is, we expect correlation to happen within

units. For instance, for a dataset of consumers over two periods, a consumer's error term at time t might be predictive of their error term at time $t + 1$. But clusters could be driven by other forces. They could be geographic: households from the same municipalities might have correlated error terms. Or it could be based on industries: firms from the same sector might have correlated errors. And sometimes, it could be both: there is dependence of firms within sectors and years.

Thus, the more general way to write down the problem is to state that there is a nonzero correlation within groups g :

$$\text{Cor}(\varepsilon_{g,j}, \varepsilon_{g,k}) \neq 0 \text{ for } j \neq k.$$

Ignoring these nonzero correlations may lead to seriously biased estimated standard errors (Moulton 1990). In general, estimated standard errors will be too small, leading to an excessive rate of rejection of the null hypothesis.

We can relax the independence assumption by acknowledging the grouped nature of the data and using what we call **clustered standard errors** (allowing you to make **cluster-robust inference**) (Liang and Zeger 1986; Bertrand, Duflo, and Mullainathan 2004; A. Colin Cameron and Miller 2015; Abadie et al. 2023; De Chaisemartin and Ramirez-Cuellar 2024). Clustered standard errors can then be used for hypothesis testing or to build confidence intervals. For a review on various ways in which standard errors could be computed for DiD and FE models more generally, see A. Colin Cameron and Miller (2015) and B. E. Hansen (2024). See Abadie et al. (2023) for in-depth guidance to select the optimal clustering strategy.

What's happening under the hood is the following. Concretely, using residuals u from the OLS regression, and assuming we have G groups (such as countries or firms):

$$\hat{\mathbf{V}}_{\text{clus}}(\hat{\beta}) = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{g=1}^G \mathbf{X}'_g \mathbf{u}_g \mathbf{u}'_g \mathbf{X}_g \right) (\mathbf{X}'\mathbf{X})^{-1}$$

All three terms yield $k \times k$ matrices.³ The new part is the middle term, where we go cluster/group by cluster/group. For a given group

³The first and last terms $(\mathbf{X}'\mathbf{X})^{-1}$ are $k \times k$. The middle term contains: $(\mathbf{u}_g \mathbf{u}'_g)$ (which is $n \times n$). It is first pre-multiplied by an $k \times n$ matrix $\mathbf{X}'_g$ and second it is post-multiplied by an $n \times k$ matrix $\mathbf{X}_g$. We thus obtain a $k \times k$ matrix for the middle term.

g of size N_g , we estimate $\varepsilon\varepsilon'$ with the sample residuals u :

$$\mathbf{u}_g \mathbf{u}_g' = \begin{bmatrix} u_{g,1}^2 & u_{g,1}u_{g,2} & \cdots & u_{g,1}u_{g,N_g} \\ u_{g,2}u_{g,1} & u_{g,2}^2 & \cdots & u_{g,2}u_{g,N_g} \\ \vdots & \vdots & \ddots & \vdots \\ u_{g,N_g}u_{g,1} & u_{g,N_g}u_{g,2} & \cdots & u_{g,N_g}^2 \end{bmatrix}$$

You then pre- and post-multiply this by $\mathbf{X}_g$ (the data for group g). Next, you sum this up for all groups $g \in G$. This gives you the middle part, which is then pre- and post-multiplied by $(\mathbf{X}'\mathbf{X})^{-1}$. This gives the estimate of clustered standard errors. You would then read something like: “the standard errors are clustered by firms” (assuming groups are defined by firms). Note: if the number of groups is small, then the estimates of these variances can be quite poor (Donald and Lang 2007).

9 Constructing Counterfactuals: Matching, Weighting, and Synthetic Control

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Overview

If we think back to the fundamental problem of causal inference, the issue is that we don't observe counterfactuals. For instance, for each treated unit, we don't observe what its Y would have been had it not been treated. And vice versa for control units.

So here's an idea: why don't we figure out a way to produce a plausible number for that missing observation (i.e., the counterfactual)? Two main approaches try to do this: propensity score matching and weighting, and synthetic controls.

Readings

For propensity score methods, the foundational reference is Rosenbaum and Rubin (1983). Imbens and Rubin (2015, pt. II) provides a textbook treatment of matching and weighting. Stuart (2010) reviews matching approaches. For doubly robust estimation, see Robins, Rotnitzky, and Zhao (1994) and Bang and Robins (2005). For synthetic control, see Abadie, Diamond, and Hainmueller (2010) and Abadie (2021). Arkhangelsky et al. (2021) introduce synthetic DiD.

9.1 Propensity score methods

The quasi-experimental designs covered earlier (IV, DiD, RDD) all exploit exogenous variation in treatment assignment. A different strategy applies when all relevant confounders are *observed*: once we condition on the right set of background variables $\mathbf{X}$, treatment assignment is as good as random (by assumption). This is the **selection on observables** assumption (aka unconfoundedness; see Chapter 2):

$$(Y^0, Y^1) \perp\!\!\!\perp D \mid \mathbf{X}.$$

The challenge is that conditioning on many covariates simultaneously is hard. If $\mathbf{X}$ contains many variables, the covariate space grows large and many cells contain no observations.

This is a problem for a few reasons; one is that you make causal claim over values of control variables that aren't available. For instance, consider the following example. You want to study the effect of giving students a license to an AI ($D \in \{\text{No license, license}\}$). Now, suppose that you think that height is a confounder (I don't know why; just bear with me). Finally, imagine that short people are in the treatment group, whereas virtually no tall people is. What is the treatment effect for tall people? We don't know! We are going to extrapolate from short people, which may or may not be appropriate.

Propensity score methods solve this dimensionality problem by collapsing all covariate information into a single scalar.

9.1.1 The propensity score

Definition 9.1 (Propensity score). The **propensity score** is the conditional probability of receiving treatment given observed covariates:

$$e(\mathbf{X}_i) := \mathbb{P}(D_i = 1 \mid \mathbf{X}_i).$$

The propensity score has a key **balancing property**: conditional on $e(\mathbf{X})$, treatment assignment is independent of the full covariate vector $\mathbf{X}$ (Rosenbaum and Rubin 1983):

$$D \perp\!\!\!\perp \mathbf{X} \mid e(\mathbf{X}).$$

This means that among units *with the same propensity score*, treated and control units have on average the same distribution of background characteristics. That's exactly as in a randomized experiment! It

follows that unconfoundedness holds conditional on the scalar $e(\mathbf{X})$ alone, rather than on the full vector $\mathbf{X}$.

In practice, the propensity score is unknown and must be estimated. The standard approach is to use a logistic regression of D_i on $\mathbf{X}_i$ (for logit regressions, see the relevant chapter in our notes). This being said, more flexible estimators (LASSO, random forests) are getting increasingly common when $\mathbf{X}$ is high-dimensional.

After estimation, the first diagnostic is always **overlap**: plot the distribution of estimated propensity scores separately for treated and control units and verify substantial common support. Scores near 0 or 1 signal units with almost no counterpart on the other side, making comparisons unreliable.

9.1.2 Matching

How can you use propensity scores? Recall that you can recreate a setting that looks like an RCT. To do so, the simplest approach is **propensity score matching**: for each treated unit, find one or more control units with a similar estimated propensity score and use their outcomes as the counterfactual. In other words, *match* a treated unit with other(s) that look very similar based on their propensity score. In principle, you could then take their difference as an estimate of treatment effects (which you would then average over all observations). In practice, we use regression.

Note here that we are trying to find the counterfactual of the treated units. For instance, in our example of students, we are taking the group of treated students as given, and we try to find good units in the control group that corresponds to them. This means that we are looking for an estimate of the Average Treatment on the Treated (ATT), not the ATE (which would be the average over the entire population).

How do you identify good matches? There are several tools that can be used:

- **Nearest-neighbor matching**: each treated unit is matched to the control unit with the closest propensity score (with or without replacement).
- **K-nearest neighbor matching**: same as nearest-neighbor, but you take the k closest units.
- **Caliper matching**: matches are only formed when the propensity score difference is below a threshold (e.g., 0.2 standard deviations of the log-odds). Unmatched treated units are discarded, so the estimand becomes the ATT for the matched sample.

- **Kernel matching:** each treated unit is matched to a weighted average of all control units, with weights that decay as propensity score distance increases.

After matching, the ATT is estimated as the difference in mean outcomes between matched treated and control groups. You should adapt standard errors to account for the uncertainty introduced by the matching step. See Abadie and Imbens (2016) for guidelines.

Note. Matching only removes bias due to *observed* covariates. If important confounders are unobserved, matching cannot correct for them. A matched sample that looks balanced on $\mathbf{X}$ may still be confounded by variables outside $\mathbf{X}$. From that perspective, matching doesn't address endogeneity.

Example 9.1 (Can propensity score matching recover an experimental benchmark?). Dehejia and Wahba (2002) address a fundamental question about matching: can it reproduce what a randomized experiment would have found, using only observational data? Their test case is the **National Supported Work (NSW) Demonstration**, a job-training program from the 1970s for which a genuine randomized evaluation exists. LaLonde (1986) had previously shown that a wide range of standard econometric methods — OLS, DiD, instrumental variables — all fail to replicate the experimental estimate of the program's earnings impact when applied to observational comparison groups drawn from large surveys. This was a sobering finding for the credibility of observational methods.

Dehejia and Wahba apply propensity score matching to the same observational comparison groups. The propensity score is estimated by logit regression of program participation on pre-treatment earnings, age, education, ethnicity, and marital status. After matching treated participants to comparison individuals with similar propensity scores, the matched sample is balanced on all these covariates — mimicking what randomization would have achieved.

The key finding: **PSM estimates of the earnings impact (\$1,500–\$1,800) are close to the experimental benchmark (\$1,794)**, while unmatched OLS estimates range from \$-16,000 to \$+2,000 depending on the comparison group used. The table below summarizes:

Method	Earnings impact estimate
Randomized experiment (benchmark)	\$1,794
OLS (CPS comparison group)	\$-\$15,205
OLS (PSID comparison group)	\$-\$8,867
PSM (nearest-neighbor, CPS)	\$1,608
PSM (kernel, PSID)	\$1,940

The result is influential but also sobering in another direction: PSM only works because the authors carefully selected a comparison group with substantial overlap on earnings histories. When the comparison group is far from the treated group in covariate space, matching cannot rescue observational inference.

9.1.3 Inverse probability weighting

A more common way to use propensity scores that doesn't rely on finding matches is to use **inverse probability weighting** (IPW). In IPW, we weight each observation by the inverse of the probability of receiving its actual treatment status. In practice, the standard implementation is the **normalized IPW** estimator, which is equivalent to running a weighted regression of Y on D :

$$\widehat{\text{ATE}}_{\text{IPW}} = \frac{\sum_{i=1}^n \frac{D_i Y_i}{\hat{e}(\mathbf{X}_i)}}{\sum_{i=1}^n \frac{D_i}{\hat{e}(\mathbf{X}_i)}} - \frac{\sum_{i=1}^n \frac{(1 - D_i) Y_i}{1 - \hat{e}(\mathbf{X}_i)}}{\sum_{i=1}^n \frac{1 - D_i}{1 - \hat{e}(\mathbf{X}_i)}}.$$

This is a difference of reweighted means: the treated mean under weights $1/\hat{e}(\mathbf{X})$, minus the control mean under weights $1/(1 - \hat{e}(\mathbf{X}))$. Normalizing by the sum of weights (rather than by n) makes the estimator more stable when propensity scores are near 0 or 1.

When we weight both treatment and control groups, we obtain the ATE. If we only reweight the control group (using weights $\hat{e}(\mathbf{X})/(1 - \hat{e}(\mathbf{X}))$ for controls and leaving treated units unweighted), we get the ATT.

The intuition is the following. A treated unit with a low propensity score was unlikely to be treated *ex ante*. As such, it is informative about what the treatment does for typically-untreated units and should receive a large weight. In the reweighted sample, treatment is uncorrelated with $\mathbf{X}$, mimicking randomization. Units that were highly likely to be treated, on the other hand, don't have much information to offer, and they receive low weights.

The main practical issue with IPW is that weights, which enter the denominator, can have large effects: propensity scores near 0 or 1 can be overly influential. To reduce that problem, some researcher trim extreme weights.

9.1.4 Doubly robust estimation

Matching and IPW each require the propensity score model (or an outcome model) to be correctly specified. The **augmented IPW** (AIPW) estimator combines both in a way that is referred to as **doubly robust**: it remains consistent if *either* the outcome model or the propensity score model is correct, even if the other is wrong (Robins, Rotnitzky, and Zhao 1994; Bang and Robins 2005):

$$\widehat{\text{ATE}}_{\text{AIPW}} = \frac{1}{n} \sum_{i=1}^n \left[\hat{\mu}_1(\mathbf{X}_i) - \hat{\mu}_0(\mathbf{X}_i) + \frac{D_i(Y_i - \hat{\mu}_1(\mathbf{X}_i))}{\hat{e}(\mathbf{X}_i)} - \frac{(1 - D_i)(Y_i - \hat{\mu}_0(\mathbf{X}_i))}{1 - \hat{e}(\mathbf{X}_i)} \right],$$

where $\hat{\mu}_d(\mathbf{X}_i) = \mathbb{E}[Y \mid D = d, \mathbf{X} = \mathbf{X}_i]$ is the predicted outcome from a regression model for each treatment arm. The first two terms are a direct outcome-regression estimate of the CATE at each unit's covariates. The remaining **augmentation terms** correct for residual imbalance using the propensity score. When the outcome model is correct, they average to approximately zero. When the outcome model is wrong, they use the propensity score to remove the bias.

This is almost miraculous: we only need to get one of the two expressions right!

Connection to double/debiased machine learning

AIPW is the semiparametric foundation for the **double (debiased) machine learning** (DML) framework (Chernozhukov, Chetverikov, et al. 2018). DML applies the same doubly robust logic with cross-fitting (sample splitting) to allow flexible ML estimators for both the propensity score and the outcome model. This makes doubly robust estimation practical in high-dimensional settings.

9.2 Synthetic control

The idea of **synthetic control** is intuitive and is best illustrated by an example (Abadie, Diamond, and Hainmueller 2010; Doudchenko

and Imbens 2016; Ferman and Pinto 2019; Athey et al. 2021). Consider a given unit that received a treatment. For instance, it could be a canton: Vaud, alone, decided to implement a new taxation policy in 2020 (this is an imaginary example). For years 2021, 2022, ..., the question is: what would have happened to Vaud's tax revenues (Y) in the absence of the new tax policy (D)? What we are asking is to identify the ATT (average treatment effect on the treated), namely the effect of the tax policy on tax revenues for the canton that implemented it (i.e., Vaud). The intuition of synthetic control is to say: *the counterfactual for Vaud after 2020 is likely some kind of mix of other cantons*. What we need to do is to create a *synthetic* treated unit that imputes its outcome *had it not been treated*, using data from control units.

The question then becomes: how can we weight other cantons to recreated the counterfactual outcome Y after the treatment was assigned? Probably, Vaud is going to be more like Geneva than, say, Niedwald. But how can we identify the right sets of weights?

To solve this, we need a bit of notation. Consider a panel dataset with N units measured T periods, an outcome $Y_{i,t}$, and a dichotomous treatment D . WLOG, we say that the treated unit is N , and that there are $[t, t^*)$ periods until the treatment, and $[t^*, T]$ periods after. We can write the matrix of outcomes *under control* as:

$$\begin{bmatrix} y_{1,1} & y_{1,2} & \cdots & y_{1,T-1} & y_{1,T} \\ y_{2,1} & y_{2,2} & \cdots & y_{2,T-1} & y_{2,T} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ y_{N,1} & y_{N,2} & \cdots & y_{N,T-1} & y_{N,T} \end{bmatrix}$$

Using our example, we can write:

$$\begin{bmatrix} y_{ZH,1990} & y_{ZH,1991} & \cdots & y_{ZH,2019} & y_{ZH,2020} & y_{VD,2021} \\ y_{GE,1990} & y_{GE,1991} & \cdots & y_{GE,2019} & y_{GE,2020} & y_{VD,2021} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ y_{VD,1990} & y_{VD,1991} & \cdots & y_{VD,2019} & y_{VD,2020} & y_{VD,2021} \end{bmatrix}$$

But note that some of these observations are missing, as they are the unobserved counterfactual after t^* :

$$\begin{bmatrix} y_{ZH,1990} & y_{ZH,1991} & \cdots & y_{ZH,2019} & y_{ZH,2020} & y_{ZH,2021} \\ y_{GE,1990} & y_{GE,1991} & \cdots & y_{GE,2019} & y_{GE,2020} & y_{GE,2021} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ y_{VD,1990} & y_{VD,1991} & \cdots & y_{VD,2019} & ? & ? \end{bmatrix}$$

where ZH is Zurich, GE is Geneva, VD is Vaud, etc. You'd have 26 rows in this dataset.

Now, synthetic control suggests to replace the missing observations by a weighted average of the control units. The challenge, then, is to

obtain the weights. Using our example: if tax revenues of Vaud are a combination of the tax revenues of Geneva, Valais, Zurich, etc., then how should we decide what the weights are?

One way to do so is to use pre-treatment data and regress $Y_{N,t < t^*}$ on $\mathbf{Y}_{D=0, t < t^*}$:

$$y_{N,t} = \gamma_0 + \sum_{i=1}^{N-1} \gamma_i y_{i,t} | t < t^*.$$

Estimate this model via OLS to obtain $\hat{\gamma}_0, \hat{\gamma}_1, \dots, \hat{\gamma}_{N-1}$ (note that each control unit receives a time-invariant weight γ_i ; these are, in effect, fixed effects based on pre-treatment data). Then, use these estimates (multiplied by the observed control outcomes $y_{i,t > t^*}$) to generate predicted values $\hat{y}_N$ for post-treatment periods $t^*, \dots, T$ for the treated unit N .

$$\begin{bmatrix} y_{ZH,1990} & y_{ZH,1991} & \cdots & y_{ZH,2019} & y_{ZH,2020} & y_{ZH,2021} \\ y_{GE,1990} & y_{GE,1991} & \cdots & y_{GE,2019} & y_{GE,2020} & y_{GE,2021} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ y_{VD,1990} & y_{VD,1991} & \cdots & y_{VD,2019} & \hat{y}_{VD,2020} & \hat{y}_{VD,2021} \end{bmatrix}$$

The ATT, then, is the averaged difference between y_N (observed outcome for treated unit N) and $\hat{y}_N$ (synthetic counterfactual value). In the case of a single post-treatment period (T), this boils down to:

$$\widehat{\text{ATT}}_{\text{synth}} = y_{N,T} - \hat{y}_{N,T}$$

If we have multiple post-treatment period, the treatment effect can be plotted on a period-per-period basis to see how the effect materializes (or not) over time. Strictly speaking, the ATT would then be the average over all post-treatment periods (from t^* to T):

$$\widehat{\text{ATT}}_{\text{synth}} = \frac{1}{T - t^*} \sum_{t=t^*}^T [y_{N,t} - \hat{y}_{N,t}]$$

Note. The quality of this approach depends on how well you can model weights. This, in turn, typically depends on the amount of pre-treatment data you have. The more observations, the better. Typically, you want many pre-treatment observations. You could presumably also improve the quality of predictions by including covariates $\mathbf{X}$ in the pretreatment weight equation. We then typically plot the predicted Y 's and the observed Y 's for the pre-treatment period and hope that they are good approximation. If not, it means that our treated unit isn't well predicted by a combination of other units.

This is a generalized version by Doudchenko and Imbens (2016) of the original approach by Abadie, Diamond, and Hainmueller (2010). Because of this matrix structure, people sometimes refer to this as a *vertical regression* (Athey et al. 2021).

Note. It's not self-evident how to construct confidence intervals or do hypothesis testing. One solution, suggested by Abadie, Diamond, and Hainmueller (2015), is to reassign treatment to each of the other $N - 1$ units, obtain $\widehat{ATT}$ for each permutation. In other words, you create fake treated units (hence them calling this “in-space placebo”) and then see if similar ATTs obtain in these cases. This gives us a distribution of N $\widehat{ATT}$ (including $\widehat{ATT}_N$, the ATT for the ‘real’ treated unit N). We can then use percentiles of this distribution as p values and can assess how likely our $\widehat{ATT}_N$. Note that when treatment is randomized, this boils down to randomization inference. If D isn't randomized, then this p value only tells us the likelihood of obtaining $\widehat{ATT}_N$ or larger among the distribution of all placebo $\widehat{ATT}$. This is different from p values evaluating the plausibility of the data given a null hypothesis. Other approaches for inference have been developed in the meantime (Xu 2017).

Note. The underlying assumption is that the weights γ_i don't change over time. This may or may not hold.

Note. A different approach would be to run a “horizontal” regression, in which you take the set of control units, regress the last period of the control outcomes are regressed on their lagged values, creating a set of time effects. These can then be plugged in with pre-treatment y for the treated unit(s) to generate forecasts of what y would have been.

Example 9.2 (Synthetic control: illustration). The literature on synthetic controls took off with Abadie, Diamond, and Hainmueller (2010). They use the following example: California launched a tobacco control policy in 1988 (aka “Proposition 99”) (that's the treatment). Did it have an effect on per-capita cigarette sales (outcome)?

If we plot the trend lines, it's hard to tell, as Figure 9.1 shows. There's a general downward trend both in California and in the rest of the US.

But the question is: what would have happened to California in the absence of this tobacco-control policy? That's where we can use the synthetic control approach. What we need to do is to create a synthetic California. Once we do this, we obtain Figure 9.2. We see that tobacco sales would have remained much

higher, had California not passed Proposition 99 (assuming all underlying assumptions are met).

We can then plot the implied ATT, namely the vertical difference between the two curves (Figure 9.3). We see that the effect grew from about 1988 to about 1995 and has remained stable since.

And we can also check which states are used to generate this combination by looking at the largest weights (Table 9.2). This is a good way to ensure that the mix makes sense. In our earlier example, I would raise an eyebrow if Vaud was a combination of Niedwald and Glaris with zero weight on cantons that are similar (geographically, language-wise, in terms of size, etc.).

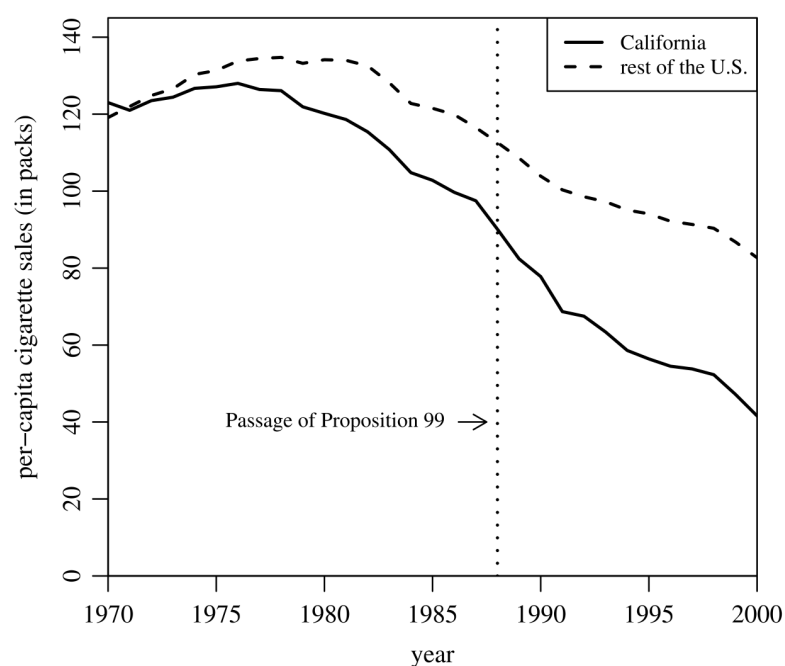


Figure 9.1: Trend lines in cigarette sales (per capita). Source: Abadie, Diamond, and Hainmueller (2010).

9.3 Synthetic difference-in-differences

Synthetic difference-in-differences (SDiD) (Arkhangelsky et al. 2021) is a recent estimator that combines the logic of synthetic control with the logic of DiD. The goal is the same as synthetic control: estimate the ATT for one (or a small number of) treated unit(s)

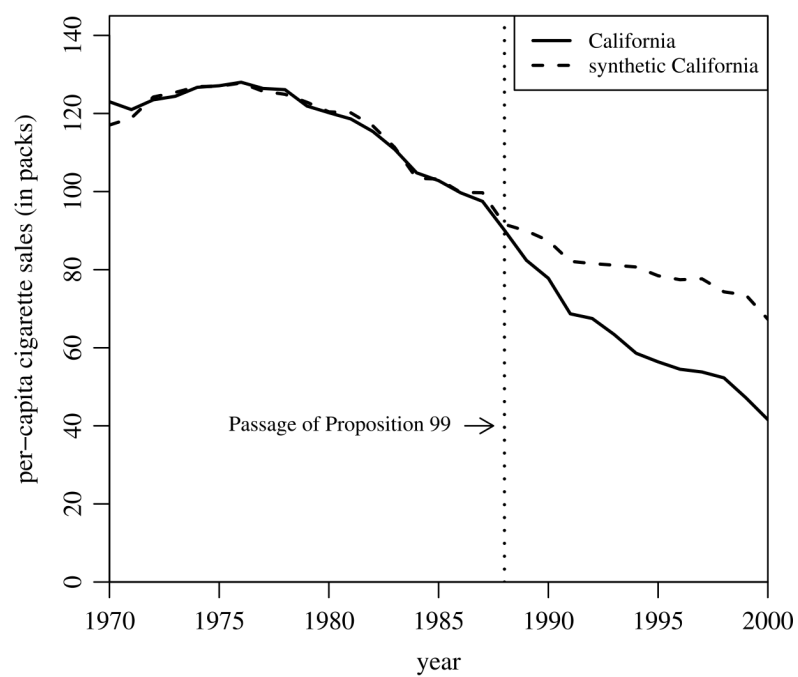


Figure 9.2: The path of observable and of synthetic California.
Source: Abadie, Diamond, and Hainmueller (2010).

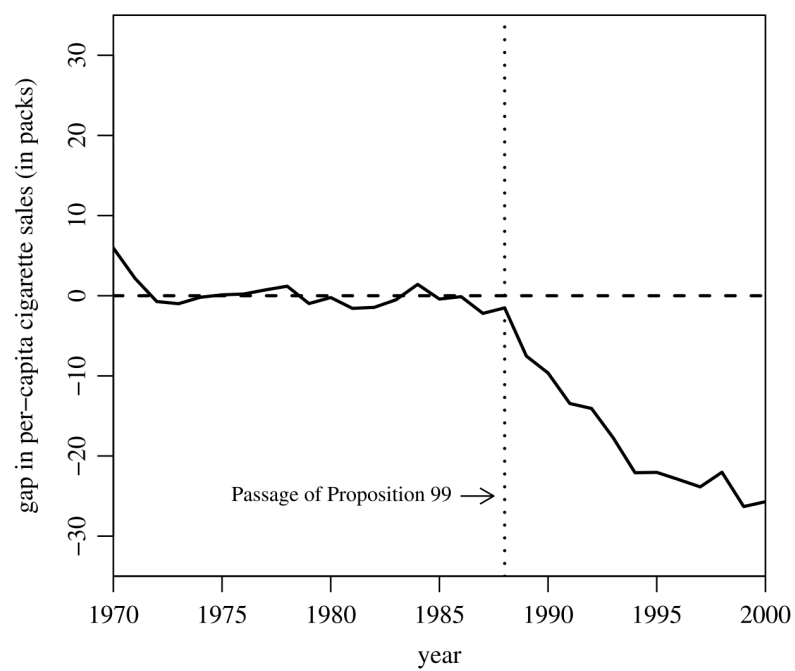


Figure 9.3: Plotting the ATT year-by-year. Technically, the ATT would be the average of the treatment effects after 1988. Note that we want a flat line pre-treatment. Source: Abadie, Diamond, and Hainmueller (2010).

State	Weight	State	Weight
Alabama	0	Montana	0.199
Alaska	–	Nebraska	0
Arizona	–	Nevada	0.234
Arkansas	0	New Hampshire	0
Colorado	0.164	New Jersey	–
Connecticut	0.069	New Mexico	0
Delaware	0	New York	–
District of Columbia	–	North Carolina	0
Florida	–	North Dakota	0
Georgia	0	Ohio	0
Hawaii	–	Oklahoma	0
Idaho	0	Oregon	–
Illinois	0	Pennsylvania	0
Indiana	0	Rhode Island	0
Iowa	0	South Carolina	0
Kansas	0	South Dakota	0
Kentucky	0	Tennessee	0
Louisiana	0	Texas	0
Maine	0	Utah	0.334
Maryland	–	Vermont	0
Massachusetts	–	Virginia	0
Michigan	–	Washington	–
Minnesota	0	West Virginia	0
Mississippi	0	Wisconsin	0
Missouri	0	Wyoming	0

Table 9.2: Table of weights used to generate a synthetic California. Smoking-wise, California appears to be a mix of Utah, Nevada, Montana, Colorado, and Connecticut (in order of importance). Source: Abadie, Diamond, and Hainmueller (2010).

using a weighted combination of control units. But SDiD adds a DiD adjustment that makes it substantially more robust.

Recall that synthetic control solves for unit weights $\hat{\omega}_i$ such that the weighted control outcome tracks the treated unit's pre-treatment outcome as closely as possible. SDiD modifies this in two ways:

1. **Unit weights** $\hat{\omega}_i^{\text{sdid}}$: chosen to match not just the pre-treatment level but also the pre-treatment *trend* of the treated unit. This parallels the DiD parallel-trend assumption.
2. **Time weights** $\hat{\lambda}_t^{\text{sdid}}$: chosen to upweight pre-treatment periods that are most informative about the post-treatment counterfactual — i.e., the periods whose cross-sectional pattern of outcomes most closely resembles the post-treatment periods. This downweights distant pre-treatment periods that are less relevant.

The estimator then takes the form of a doubly-weighted two-way regression. Formally, for a panel with N_0 control units and N_1 treated units, observed over T_{pre} pre- and T_{post} post-treatment periods, the SDiD estimator solves:

$$(\hat{\tau}^{\text{sdid}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg \min_{\tau, \mu, \alpha, \beta} \sum_{i=1}^N \sum_{t=1}^T \hat{\omega}_i^{\text{sdid}} \hat{\lambda}_t^{\text{sdid}} (y_{i,t} - \mu - \alpha_i - \beta_t - \tau D_{i,t})^2,$$

where α_i and β_t are unit and time fixed effects. The weight $\hat{\omega}_i$ is set to zero for treated units in the regression (they only appear in the outcome being modeled), and $\hat{\lambda}_t$ is set to $1/T_{\text{post}}$ for post-treatment periods and is estimated for pre-treatment periods.

Why is SDiD better than pure synthetic control? Two reasons:

1. The DiD-style fixed effects absorb level differences and aggregate time trends, so the unit weights only need to match idiosyncratic trends — a much weaker requirement than matching levels.
2. Because both unit and time weights are involved, SDiD has a *doubly robust* flavor: the estimator remains consistent if either the unit weights or the time weights are correctly specified.

i SDiD versus alternatives

In practice, SDiD, standard DiD/TWFE, and synthetic control often give similar results when treatment effects are homogeneous and pre-treatment fit is good. SDiD is most valuable when:

- The treated unit(s) are outliers in levels (making raw DiD unreliable), and
- There are several pre-treatment periods but the number of control units is large relative to the number of treated units.

For inference, Arkhangelsky et al. (2021) recommend a placebo-based bootstrap: re-assign treatment to control units, compute the SDiD estimate for each permutation, and use the resulting distribution to calibrate p -values. This is analogous to the “in-space placebo” approach used for synthetic control.

10 Maximum likelihood

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Overview

So far, we have focused on OLS. OLS does very well when estimating linear models. However, there is going to be a range of data-generating processes in which the dependent variable isn't defined on the real line. You can still run OLS, but it will sometimes give you weird results. For instance, if the outcome is a dummy indicator, it will give predicted probabilities that are negative.

Another estimator (recall that this is a rule that transforms data into estimates of parameters) is **maximum likelihood**. This is very often used in a range of applications (classification, etc.), and it's therefore worth getting to know its statistical properties, before applying it to conventional models like logit, Poisson, etc.

Readings

People interested in maximum likelihood and nonlinear models should refer to Eliason (1993) or A. Colin Cameron and Trivedi (2005).

10.1 Theory of maximum likelihood

The origins of **maximum likelihood** (ML; not to be confused with *machine learning*) are linked to the work of R.A. Ronald A. Fisher (1922). Suppose we have a set of data and an unknown vector of parameters γ that characterizes their probability density function (PDF) (for a continuous variable) or probability mass function (for a discrete variable). The idea of ML, in a nutshell, is to select as estimate(s) of γ the vector $\hat{\gamma}_{ml}$ that maximizes the likelihood of observing this particular set of data.

For instance, suppose we want an estimate the probability of obtaining a head p in a coin toss, without knowing what this probability is (the coin may not be fair). Suppose further we obtain a dataset of 10 coin tosses, 7 of which are head. Then (trivially, I admit), the estimate p that maximizes the likelihood of obtaining such a dataset is $\hat{p}_{ml} = 0.7$.

Formally, for a sample of N *independent* observations of variable y , we define the **likelihood function**:

$$L(\gamma) = \prod_{i=1}^N f(y_i; \gamma)$$

where L is the likelihood, $\prod$ the product (over N observations) of probability density functions $f()$ of y , as a function of k parameters γ . This looks very much like the expression for the probabilities of a series of independent events (see first chapter). For instance, the probability of five tails in a row with a fair coin is 0.5^5 . The difference is that we have now an unknown parameter γ that we can pick.

Since L can grow too large quickly (imagine 100 coin tosses...) and thus make computation difficult, we often take the log of the likelihood function instead. Call this the **log-likelihood** $\mathcal{L}$ (or sometimes LL):

$$\mathcal{L}(\gamma) = \sum_{i=1}^N \log [f(y_i; \gamma)].$$

Then, as an estimator (i.e., a rule), ML selects $\hat{\gamma}_{ml}$ that maximizes $\mathcal{L}$:

$$\hat{\gamma}_{ml} := \arg \max_{\gamma} \mathcal{L}(\gamma).$$

The maximization is characterized by the **score function** (the gradient of the log-likelihood):

$$s(\gamma) := \frac{\partial \mathcal{L}(\gamma)}{\partial \gamma} = \sum_{i=1}^N \frac{\partial \log f(y_i; \gamma)}{\partial \gamma}.$$

The MLE is the solution to $s(\hat{\gamma}_{ml}) = 0$: at the maximum, the data provide no further information about the direction in which to update γ . The score function also plays a key role in the three tests for ML models (Wald, likelihood-ratio, and score tests; see the *Goodness of fit* section below).

The key challenge will be (a) to define the likelihood function, and (b) to obtain a solution to this maximization problem. Part (a) will primarily be a matter of theory combined with the nature of the

data you are looking at. Part (b) will sometimes involve closed-form solutions, sometimes not. It depends on the probability function. As such, ML estimation may have involve iteration over the space of potential values for γ until a maximum is reached (hopefully a global one).

💡 ML with a linear model

While we often use ML to overcome the shortcoming of OLS when outcomes are nonlinear, you can use for linear models too. Let's start with the usual model $y = \alpha + \beta D + \varepsilon$. In contrast to OLS, in the ML approach, we must specify the likelihood function of y . To do so, we must make an assumption about ε , which is the only source of variation (the other parameters and the data are fixed). Let us assume that ε is a iid stochastic noise distributed normal: $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. Note: this is not a free assumption: the validity of our ML analysis here depends on this assumption!

From this starting point, we can state that:

$$y \sim \mathcal{N}(\alpha + \beta D, \sigma^2)$$

The likelihood function, given n observations, is:

$$L(\alpha, \beta, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \alpha - \beta D_i)^2}{2\sigma^2}\right)$$

That is, it is the product n times of normal distributions. Note that we derive this statement from our assumption about the error term, which we *assume* to be normal (maybe we are wrong!). This is distinct from OLS, where we didn't make any such assumption. Instead, we relied on asymptotic properties of OLS estimators to do inference; and we didn't need anything aside full rank to estimate β .

We take the log to simplify, generating the log-likelihood $\mathcal{L}$ function (with three arguments):

$$\mathcal{L}(\alpha, \beta, \sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \alpha - \beta D_i)^2$$

Maximum likelihood implies the joint maximization of the log-likelihood $\mathcal{L}(\cdot)$ over all three parameters $(\alpha, \beta, \sigma^2)$. Concentrating on α and β : only the last term involves them, and it is negative. Thus, to *maximize* $\mathcal{L}$ over α and β , we *minimize* $\sum_i (y_i - \alpha - \beta D_i)^2$ (independently of the value of σ^2).

But here, you should note that we have seen $\sum_{i=1}^n (y_i - \alpha - \beta D_i)^2$ before: it's the sum of the squared residuals, which is what we

already minimize with OLS. We can minimize this quantity with the general expression $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$. And then, if we take the second derivative, we obtain:

$$\frac{\partial^2 \mathcal{L}}{\partial \beta \partial \beta} = (\mathbf{X}'\mathbf{X})(-\sigma^{-2}),$$

which is negative definite, and thus $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ is the max of the likelihood.

Note, however, that the ML and OLS estimators of σ^2 differ: maximizing over σ^2 gives $\hat{\sigma}_{ml}^2 = \frac{1}{n} \sum_i \hat{\varepsilon}_i^2$, whereas OLS uses the bias-corrected $\hat{\sigma}_{ols}^2 = \frac{1}{n-k} \sum_i \hat{\varepsilon}_i^2$. The ML estimator of σ^2 is therefore biased downward in finite samples. This is the one place where OLS and ML diverge for the linear model. But in larger samples (when n is large), the difference is generally trivial (dividing by 100 or by 99 is barely perceptible).

Thus, ML coincides with OLS for the linear model (for β): $\hat{\beta}_{ols} = \hat{\beta}_{ml}$. ML uses a different starting point — it assumes a distribution for ε and derives a likelihood for Y — but arrives at the same result.

10.1.1 Numerical optimization

For simple models (like the linear model above), the score equations have a closed-form solution. For most nonlinear models, including logit, probit, Poisson, survival models, they do not, and the MLE must be found by iterative numerical methods.

The most common algorithm is **Newton-Raphson**:

$$\gamma^{(t+1)} = \gamma^{(t)} - [\mathbf{H}(\gamma^{(t)})]^{-1} s(\gamma^{(t)}),$$

where $\mathbf{H}(\gamma) = \frac{\partial^2 \mathcal{L}}{\partial \gamma \partial \gamma'}$ is the Hessian matrix and $s(\gamma^{(t)})$ is the score at the current iterate. The step scales the gradient by the local curvature, ensuring large steps when the log-likelihood is flat and small steps near the maximum. The **Berndt–Hall–Hall–Hausman** (BHHH) algorithm (Hall, Hall, and Hausman 1974) replaces the Hessian with the outer product of scores $\sum_i s_i s_i'$, which is cheaper to compute and is positive semi-definite by construction.

Most ML commands in R or Python handle optimization internally. You rarely will need to code Newton-Raphson yourself. In any case, for iterative cases, the algorithm will need an initial guess $\gamma^{(0)}$ to get started. If you're unlucky, your starting choices can lead to non-convergence or convergence to a local maximum. A common strategy

is to start from OLS estimates or from zeros. But in any case, always check that the optimizer converged. Most software will flag a failure to converge.

10.2 Inference with maximum likelihood

10.2.1 Standard error with ML

Asymptotically, ML estimators exhibit all kinds of nice properties. First, and useful for inference, we can establish that the MLE is asymptotically normally distributed:

$$\hat{\beta}_{ml} \sim \mathcal{N}(\beta, \mathbf{I}(\beta)^{-1}),$$

where $\mathbf{I}(\beta)$ is the **information matrix**. It is defined as the negative of the matrix of the expected values of the second derivative (or **Hessian**) of the LL (i.e., $\mathcal{L}$) with respect to β :

$$\mathbf{I}(\beta) = -E \left[\frac{\partial^2 \mathcal{L}(\beta)}{\partial \beta \partial \beta'} \right]$$

We can then conduct similar kind of procedures, including hypothesis testing, as in the linear world.

10.2.2 Confidence intervals for quantities of interest

In many cases, we are not that interested in classical hypothesis testing, mostly because the point estimates don't tell us much. Instead, we compute a quantity of interest, such as a change in probability. But then, we need to find a way to assess our confidence in these quantities. In other words, we need a confidence interval. How can we build them? They don't always exist analytically. Instead, we can simulate them. The most common procedure is the following:

- Obtain the ML estimate of β . Define $\hat{\beta}$ the vector of point estimates and $\hat{V}$ the estimated variance-covariance matrix of the parameters.
- Repeat the following procedure 1,000 times (or more):
 - Randomly draw β 's from the multivariate normal distribution $\mathcal{N}(\hat{\beta}, \hat{V})$.
 - Compute the quantity of interest (e.g. average $\hat{P}_2 - \hat{P}_1$). That's generally one number based on that particular set of parameters.

- Save the quantity of interest
- You then have 1,000 observations of the quantity of interest. You can then construct a 95% confidence interval.

This procedure is the **parametric bootstrap** (Efron and Tibshirani 1994; King, Tomz, and Wittenberg 2000).

The **non-parametric bootstrap** (**efron1979bootstrap?**) is a broader alternative that does not require assuming a parametric model for the errors. The procedure is:

1. Resample n observations $(y_i^*, \mathbf{x}_i^*)$ with replacement from the original dataset.
2. Re-estimate the model on the bootstrap sample to obtain $\hat{\gamma}^*$.
3. Compute the quantity of interest (e.g., a marginal effect or predicted probability) from $\hat{\gamma}^*$.
4. Repeat steps 1–3 many times (typically $B = 999$ or $B = 1,999$).
5. The empirical distribution of the replicated quantities approximates the sampling distribution. Use its standard deviation as the bootstrap standard error, or take the 2.5th and 97.5th percentiles as a 95% confidence interval.

The non-parametric bootstrap is valid whenever the estimator is consistent and the data are IID. It does not require correct specification of the full likelihood and is therefore more robust than the parametric version. It is also the default method for computing confidence intervals for nonlinear quantities derived from OLS or MLE when analytic formulas (such as the delta method) are unavailable or complex. Its main limitation is computational cost; it can also fail for non-smooth estimators or for time-series data with dependence (where a block bootstrap should be used instead to preserve the autocorrelation structure).

10.2.3 Properties of MLE

What about MLE's general properties? We can establish two results.

1. MLE is **consistent**:

$$\lim_{n \rightarrow \infty} \mathbb{P}[|\hat{\beta}_{ml} - \beta| > \varepsilon] = 0$$

2. MLE is **efficient**, in the sense that it achieves the **Cramér-Rao lower bound**, which is the lowest variance one can achieve with random data. We can see this directly from

its asymptotic distribution. Assuming we have an unbiased estimator, the Cramér-Rao inequality indicates that:

$$\mathbb{V}(\hat{\beta}) \geq \frac{1}{-\mathbb{E} \left[\frac{\partial^2 \mathcal{L}(\beta)}{\partial \beta \partial \beta'} \right]},$$

that is, $\mathbb{V}(\hat{\beta})$ cannot be smaller than the quantity of the left-hand side. But that's exactly the variance of the the MLE. Thus, MLE is efficient.

10.3 Goodness of fit

Goodness of fit can be assessed in various ways with maximum likelihood.

- Lagrange multiplier test (LM)
- Wald Test
- Likelihood-ratio test (LR)
- Pseudo- R^2
- AIC and BIC

Other tests can be used on an *ad hoc* basis. For instance, in the case of a nonlinear model with a dichotomous outcome, we may use the percentage of correctly predicted values as a measure of quality of model. I'll discuss these wherever relevant.

10.3.1 Wald test

The **Wald test** assesses whether q restrictions on the parameters hold, using only the unrestricted model. For the null hypothesis $H_0 : R\gamma = r$ (where R is a $q \times k$ restriction matrix), the test statistic is:

$$W = (R\hat{\gamma}_{ml} - r)' [R\mathbf{I}(\hat{\gamma}_{ml})^{-1}R']^{-1} (R\hat{\gamma}_{ml} - r) \sim \chi_q^2.$$

For a single restriction ($q = 1$), $\sqrt{W}$ is approximately standard normal. That's the ML analog of the t -test. The Wald test's advantage is that it requires estimating only the unrestricted model.

10.3.2 Lagrange multiplier (score) test

The **Lagrange multiplier test** (LM, or **Score test**) asks: if we impose the null hypothesis and estimate the restricted model $\hat{\gamma}_r$, how far is the score from zero? Under H_0 , the restricted estimate is near the truth, so $s(\hat{\gamma}_r) \approx 0$:

$$LM = s(\hat{\gamma}_r)' \mathbf{I}(\hat{\gamma}_r)^{-1} s(\hat{\gamma}_r) \sim \chi_q^2.$$

The LM's advantage is that it requires **only the restricted model**. That could be useful if the unrestricted model is computationally expensive.

10.3.3 Likelihood-ratio test

Likelihood-ratio test (LR). This test is similar in spirit to the F test in linear models. The test compares the difference in log-likelihood of two models: one that is unrestricted (including all independent variables) and one that is restricted (where we remove q variables). This test is computed as follows:

$$LR = 2(\mathcal{L}_u - \mathcal{L}_r) \sim \chi_q^2,$$

where $\mathcal{L}_u$ is the unrestricted model and $\mathcal{L}_r$ is the restricted model (the one removing q parameters). LR is distributed χ with q degrees of freedom. We can then apply the usual hypothesis testing with the null being that these q parameters are indeed equal to zero.

Note. Wald, LR, and LM tests are asymptotically equivalent under H_0 but can differ in finite samples. One rule of thumb (that I don't have a source for): use Wald when only the unrestricted model is available; use LM when only the restricted model is available; use LR when both are available and the log-likelihoods are easy to obtain.

10.3.4 Pseudo- R^2

Pseudo- R^2 . There are various measures that mimic the linear model's R^2 . One is to compute:

$$\text{pseudo-}R^2 = 1 - \frac{\mathcal{L}_u}{\mathcal{L}_o}$$

where $\mathcal{L}_u$ is the log-likelihood of the full model and $\mathcal{L}_o$ is the log-likelihood of the intercept-only (null) model. For discrete-outcome

models, both are non-positive. Since the full model always fits at least as well as the null ($\mathcal{L}_u \geq \mathcal{L}_o$, both ≤ 0), the ratio $\mathcal{L}_u/\mathcal{L}_o \in [0, 1]$ and pseudo- $R^2 \in [0, 1]$. Limiting cases: if the model predicts no better than the intercept alone, $\mathcal{L}_u = \mathcal{L}_o$ and pseudo- $R^2 = 0$; if predictions are perfect, $\mathcal{L}_u = 0$ and pseudo- $R^2 = 1$.

10.3.5 AIC and BIC

The **Akaike Information Criterion** (AIC) and **Bayesian Information Criterion** (BIC) are the most widely used criteria for comparing non-nested models and selecting among candidate specifications. Both penalize the log-likelihood for model complexity:

$$\text{AIC} = -2\mathcal{L}(\hat{\gamma}) + 2k, \quad \text{BIC} = -2\mathcal{L}(\hat{\gamma}) + k \ln(n),$$

where k is the number of estimated parameters. Lower values are better. The BIC penalty is heavier than AIC whenever $\ln(n) > 2$, i.e., whenever $n > 7$ — almost always in practice.

The two criteria target different objectives. AIC (asymptotically) minimizes the Kullback-Leibler divergence between the fitted model and the true DGP. As such, it optimizes **predictive accuracy**. BIC approximates the log marginal likelihood. Thus, it focuses on **model selection**. In practice, BIC favors more parsimonious models.

10.4 Recent developments

10.4.1 Penalized MLE

When the number of parameters k is large relative to n , standard MLE can overfit: the model matches the data closely but performs poorly out-of-sample. **Penalized MLE** adds a regularization term to discourage large coefficients:

$$\hat{\gamma}_\lambda = \arg \max_{\gamma} [\mathcal{L}(\gamma) - \lambda P(\gamma)],$$

where $P(\gamma)$ is a penalty and $\lambda \geq 0$ controls its strength. Two common choices that are often used:

- **Ridge** (ℓ_2 penalty): $P(\gamma) = \sum_j \gamma_j^2$. It shrinks all coefficients toward zero but retains all predictors.
- **LASSO** (ℓ_1 penalty): $P(\gamma) = \sum_j |\gamma_j|$. It sets some coefficients exactly to zero (thus selecting variables that will end up in the model).

LASSO logistic regression is widely used for binary classification with many predictors. The penalty parameter λ is typically chosen by cross-validation. Note that penalized estimators are generally biased, since the penalty pushes coefficients toward zero. The estimates are therefore not suitable for *causal inference* without further corrections. However, they can be very useful for *prediction*.

10.4.2 Quasi-MLE and the sandwich variance

When the assumed density $f(y_i; \gamma)$ is **misspecified** (i.e., it does not match the true data-generating process) the standard ML variance estimator $\mathbf{I}(\hat{\gamma})^{-1}$ is incorrect, even if the point estimate is still useful. The **quasi-MLE** (QMLE) framework studies this situation: the estimator converges to the parameter that minimizes the Kullback-Leibler distance between the assumed and true distributions, but inference requires a corrected variance.

The fix is the **sandwich variance estimator**:

$$\hat{\mathbf{V}}_{\text{sandwich}}(\hat{\gamma}) = \mathbf{I}(\hat{\gamma})^{-1} \left(\sum_i s_i s_i' \right) \mathbf{I}(\hat{\gamma})^{-1},$$

where the middle term is the outer product of individual score contributions. Under correct specification the information matrix equality holds ($\sum_i s_i s_i' \approx \mathbf{I}(\hat{\gamma})$) and the sandwich collapses to $\mathbf{I}^{-1}$. Under misspecification, the sandwich remains valid. This is the ML analog of heteroskedasticity-robust standard errors in OLS.

QMLE is for instance used with count data modeled as **Poisson**. Even if the outcome is not truly Poisson distributed, the Poisson log-likelihood yields a consistent estimator of the conditional mean parameters as long as the mean function is correctly specified (Gourieroux, Monfort, and Trognon 1984; Silva and Tenreiro 2006).

11 Nonlinear Models

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Overview

Outcomes aren't always neatly defined on the real line. While this doesn't prevent OLS from being applied, it raises problems of interpretation. We can now use maximum likelihood to estimate more flexible models that are not linear in parameters.

Readings

People interested in nonlinear models should refer to Maddala (1983) (on limited dependent variables), Eliason (1993) (on maximum likelihood), A. Colin Cameron and Trivedi (2005) (ch.5 for maximum likelihood), Section IV for nonlinear models). See also McFadden (1976) (on random utility models) and Train and Weeks (2005) (for discrete choice models in general).

11.1 Linear probability model

11.1.1 Logic of the linear probability model

So far, we wrote down linear models under the assumption that...

- Treatments could be continuous or categorical (with dummy variables being a special case of the latter with two categories)
- Outcomes were defined on the real line (or at least a sizeable subset thereof; for instance, GDP is defined on the nonnegative segment of the real line, so we treat it 'as if continuous').

We did not raise the question of: **what happens if the outcome is discrete?**

Let us start with the special case where $y \in \{0, 1\}$. Some examples:

- Does treatment D determine whether a Central Bank uses a fixed or a floating exchange rate?
- Does treatment D determine whether a consumer buys an app or not?
- Does treatment D determine whether a firm enters a market or not?

We could wonder whether this creates any kind of issues? Can we still use OLS to estimate such a model?

The answer is yes and no. Let's start with the motivation for a 'yes.' Consider the following model:

$$y_i = \alpha + \beta D_i + \varepsilon_i.$$

Now, *nothing* prevents us from estimating these population parameters with OLS (assuming no perfect multicollinearity). We would then obtain $\hat{\alpha}$, $\hat{\beta}$.

The interpretation of these parameters is slightly different but quite intuitive. Note that until now, we modeled the expected value of Y , $\mathbb{E}[Y|D]$. But note that we can also rewrite this as the probability of $y = 1$ given D :

$$\mathbb{E}[Y|D] = \mathbb{P}[Y = 1|D],$$

and thus:

$$\text{Probability: } \mathbb{P}[Y = 1] = \alpha + \beta D_i + \varepsilon_i$$

$$\text{Predicted probability: } \mathbb{P}[\widehat{Y} = 1] = \hat{\alpha} + \hat{\beta} D_i$$

We call this the **linear probability model** (LPM). In this case, we can now interpret our parameters as follows:

- α : probability that $y = 1$ when all covariates are set at zero.
- β : the marginal effect of D on the probability that $y = 1$.

For instance, suppose we have a dataset on consumer choices over a period of one year:

$$\mathbb{P}[\text{Buy new laptop (=1)}_i] = 0.3 + 0.2 \cdot \text{Has a job}_i + \varepsilon_i,$$

then the probability that someone buys a new laptop in the last twelve months is 0.3 (or 30%) for people without a job, and 0.5 (or 50%) for those who have a job.

11.1.2 Inference with LPM

There are several challenges with LPM.

First, it will sometimes generate predicted probabilities that are inconsistent with probability theory. For instance, imagine a model:

$$\mathbb{P}[\text{Buy new laptop } (=1)_i] = 0.4 + 0.1 \text{Monthly income in kCHF} + \varepsilon_i.$$

Suppose that you have someone with an income of CHF7k, then we would have:

$$\mathbb{P}[\text{Buy new laptop } (=1)_i] = 0.4 + 0.1 \cdot 0.7 = 1.1.$$

This is not the end of the world *per se*: if you don't care about predictions, and instead only care about marginal treatment effects, then you can probably ignore this. Instead, you will only wonder if the estimate of the treatment effect ($\hat{\beta}$) is meaningful. And here, some believe that it's a perfectly fine approximation in many cases (J. Angrist and Pischke 2008).

Second, LPM will be construction be heteroskedastic. As X gets very large (in either direction), the average error will increase, and thus the variance of the error term cannot be constant. See Figure 11.1. Thus, we must use robust standard errors (White 1980).

Beyond this, the model is just fundamentally misspecified when treatments are continuous. The expected value of the error term, when X becomes very large or very small, cannot be zero. For very large negative values of X , and assuming a positive relation between X and Y , the values of ε will all be positive, since the predicted line will be below all values, whether they are 0 or 1. Likewise, for very large positive values of X (and still assuming a positive relation), the regression line will be above all observations, and thus $\varepsilon < 0$.

Note that what I just wrote applies to continuous treatments. If we had a dummy treatment D , then all values $\hat{P}$ would be bounded between 0 and 1, and there would be no problem of the kind I just discussed.

11.1.3 OLS' shortcomings

OLS has several shortcomings when analyzing discrete outcomes.

- If Y is a dummy, it may generate predictions outside the $[0,1]$ interval. It is thus poorly suited for predictions.

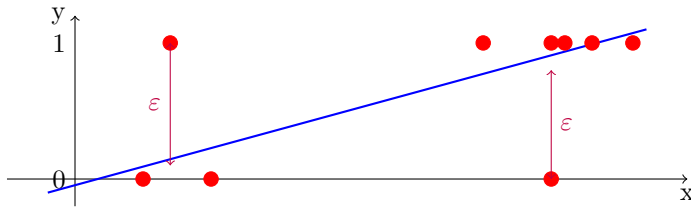


Figure 11.1: Heteroskedasticity with linear probability models.

- If Y is ordinal with >2 categories, it implicitly assumes that Y is continuous. And it can again generate predictions beyond the interval of values that Y can take.
- OLS cannot handle categorical variables that are not ordinal. For instance, imagine a model in which the outcome is the color of the car being bought. How do you translate $\{\text{Red, Green, Blue}\}$ into a numerical variable? There is no order.

11.2 Logit and probit

Linear probability models, by definition, will eventually lead to predictions that are outside the $[0, 1]$ interval (except in the trivial case where parameters equal 0). What we want is a model that constrains predicted values on the $[0, 1]$ interval.

If we want a model between 0 and 1, there are in theory an endless number of models that could do the trick. But not all models are principled. Instead, what we will do is start with a **random utility model**. This allows us to justify the models that we often use: the **logit** and the **probit**.

11.2.1 Models

So how do logit/probit work? There are a number of ways to get to a nonlinear model, but the best starting point is to think it terms of **latent variable model** (or **random utility model**). Instead of beginning with an observable outcome y (which equals to zero or 1), we start with a latent version of that variable y^* . This variable captures the underlying decision process that eventually leads y to be zero or 1. What we want, then, is to write this latent variable as a linear model:

$$y^* = \alpha + \beta x + \varepsilon$$

As of now, the only assumption we make (besides that we have the *correct model*) is that ε is *symmetrically* distributed and *uncorrelated* with X . Note that ε is the only random stuff in there, the rest (α , β , X) being fixed.

We claim that:

$$\begin{aligned} y &= 1 \text{ iff } y^* > 0 \\ &= 0 \text{ iff } y^* \leq 0 \end{aligned}$$

That is, if the latent variable is above some arbitrary cutoff (zero), then the outcome will be 1. Otherwise, it's zero. We can then write:

$$\begin{aligned} \mathbb{P}[y = 1] &= \mathbb{P}[y^* > 0] \\ &= \mathbb{P}[\alpha + \beta x + \varepsilon > 0] \\ &= \mathbb{P}[\varepsilon > -(\alpha + \beta x)] \\ &= 1 - \mathbb{P}[\varepsilon < -(\alpha + \beta x)] \end{aligned}$$

We can write this because ε is random and therefore the model is not deterministic.

Now, we need to think a bit more about ε . The second expression asks us something like: what is the likelihood that ε is smaller than some quantity $-(\alpha + \beta x)$. We have seen similar questions fairly regularly, for instance when we talked about things such as t tests. What we need to answer them is the cumulative distribution function for ε . For the time being, let's call it $F(\cdot)$. We can then continue:

$$\begin{aligned} \mathbb{P}[y = 1] &= 1 - \mathbb{P}[\varepsilon < -(\alpha + \beta x)] && \text{(from above)} \\ &= 1 - F[-(\alpha + \beta x)] \\ &= F(\alpha + \beta x) && (\varepsilon \text{ is symmetric}) \end{aligned}$$

So we have made some progress. Now the question is: what is F ? Remember that it is the cumulative distribution function. The corresponding *density function* tells us how ε is distributed. As said, so far we only assumed that ε is uncorrelated to $\mathbf{X}$ and that it is symmetric. Now, we need to go a bit further and decide what its distribution is.

- If F is the logistic distribution, then we have the **logit** model.
- If F is the (cumulative) normal distribution, then we have the **probit** model.

The logit looks like this:

$$F(x) = \frac{1}{1 + \exp(-x)}$$

In our case, this yields:

$$\begin{aligned} \mathbb{P}[y = 1|X] &= F(\alpha + \beta x) && \text{(from above)} \\ &= \frac{1}{1 + \exp(-(\alpha + \beta x))} \\ &= \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)} \end{aligned}$$

For the probit:

$$\begin{aligned} \mathbb{P}[y = 1|X] &= F(x) \\ &= \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{v^2}{2}\right) dv \end{aligned}$$

11.2.2 Estimation

The challenge comes when we try to estimate these models. As of right now, we only have β , not $\hat{\beta}$. We will use **maximum likelihood** to obtain these estimates. Recall that the idea behind maximum likelihood is the following: given a set of assumptions about the underlying model, and given the data we have, what are the parameters that are the most likely?

For a sequence of draws of the dichotomous outcome $y_1, y_2, \dots, y_n$ ($y \in \{0, 1\}$), what we want to maximize is the joint probability of y given $\mathbf{X}$:

$$\mathbb{P}[y_1, y_2, \dots, y_n | \beta, \mathbf{X}]$$

If the draws are IID, then we can rewrite this as:

$$\mathbb{P}[y_1, y_2, \dots, y_n | \beta, \mathbf{X}] = \prod_{y_i=1} \mathbb{P}[y_i = 1 | \beta, \mathbf{X}] \cdot \prod_{y_i=0} \mathbb{P}[y_i = 0 | \beta, \mathbf{X}]$$

But we know what $\mathbb{P}(y_i = 1)$ is! We can write:

$$\begin{aligned} \ell &= \prod_{y_i=1} F(\mathbf{X}_i \beta | \beta, \mathbf{X}) \cdot (1 - F(\mathbf{X}_i \beta | \beta, \mathbf{X})) \\ &= \prod_{y_i=1} F(\mathbf{X}_i \beta | \beta)^{y_i} \cdot (1 - F(\mathbf{X}_i \beta | \beta))^{1-y_i}, \end{aligned}$$

where ℓ is called the **likelihood function**. Since multiplicative stuff becomes quickly cumbersome (think of what happens if $n = 1,000$), we generally take the log, i.e. the **log-likelihood**:

$$\ell \ell = \sum y_i \ln(F(\mathbf{X}_i \beta)) + \sum (1 - y_i) \ln(1 - F(\mathbf{X}_i \beta))$$

The last step is to maximize this function. First, what function $F()$ is; for instance, it could be the logit or the probit distribution. Then, we take the first derivative with respect to β , set it to zero, and solve for our maximum likelihood estimate $\hat{\beta}_{ml}$.

Note that ML's estimation of these models is, *under regularity conditions*:

- Consistent
- Asymptotically normal (more on this below)
- Asymptotically efficient
- But it is biased in small samples
- But it is inconsistent in the presence of heteroskedasticity.
- ML is also biased+ inconsistent when using fixed effects in panel data (**incidental parameters** problem).

11.2.3 Interpretation of logit and probit

The interpretation of logit/probit is tricky. In a linear additive model, the partial derivative with respect to a variable x was simply its parameter β (assuming no interaction effect). Easy enough. That's not the case anymore. To see this, just visualize the logit/probit curve: the derivative is not constant. In full notation:

$$\frac{\partial \mathbb{P}[y = 1]}{\partial x_j} = f(\mathbf{X}\beta)\beta_j,$$

where $f(z) = \frac{F}{dz}(z)$.

Note. F is a cumulative distribution function (e.g. normal), and f is the probability density function.

To find the marginal effect of x , then, we need to compute the following:

$$\begin{aligned}\Delta \hat{\mathbb{P}} &= \hat{\beta} f(\mathbf{X}\beta) \Delta x \\ &= \hat{\beta} f(\mathbf{X}\beta) \quad (\text{if } \Delta x = 1)\end{aligned}$$

In other words, the marginal effect is $\hat{\beta}$ times a function of “where” we are (where the departure Δx starts from). Note that the effect will be higher “in the middle” (where we go from $\hat{P}$ close to zero to close to one quite rapidly) and smaller when $\mathbf{X}\beta$ is very small or very large. Generally, we often pick values of $\mathbf{X}$ that are meaningful, such as the sample median or the mean.

The incidental parameter problem is the following: suppose your model is $y = f(\alpha_i + \beta x_i)$. The α 's are fixed effects and β is a parameter common to all units. Now, suppose that T is fixed and N varies. As $N \rightarrow \infty$, β will be better estimated; however, that's not true of the α 's, because we still only have the finite T to estimate each of them. In other words: as the number of units goes up, this does nothing to help estimating our α 's, because we are just adding more units. What is worse: the bias can spill over to the estimation of the β 's. See A. Colin Cameron and Trivedi (2005) (p.781).

For our cases, we have:

$$\begin{aligned}
 \Delta \hat{\mathbb{P}} &= \hat{\beta} f(\mathbf{X}\beta) && \text{from above} \\
 &= \hat{\beta} \frac{\exp(\mathbf{X}\beta)}{(1 + \exp(\mathbf{X}\beta))^2} && \text{logit} \\
 &= \hat{\beta} \frac{1}{\sqrt{2\pi} \exp\left(\frac{-(\mathbf{X}\beta)^2}{2}\right)} && \text{probit}
 \end{aligned}$$

(You can see why people who do this manually prefer logit.)

An identical approach would be to do compute:

- Obtain the ML estimate of $\hat{\beta}$.
- Compute $\hat{\mathbb{P}}_1$ for the case you're interested in, such as the median observer.
- Compute $\hat{\mathbb{P}}_2$ when x is replaced by $x + c$ (where c is a constant, generally 1).
- Compute confidence intervals (see below).

Another (and different!) possible quantity of interest would be the **average partial effect** (APE):

- Obtain the ML estimate of $\hat{\beta}$.
- Compute $\hat{\mathbb{P}}_1$ for each existing observation using $\hat{\beta}$.
- Replace variable x_i by $z = x_i + c$ (where c is a constant, generally 1).
- Compute $\hat{\mathbb{P}}_2$ using the new dataset (where z replaces x).
- Compute $\hat{\mathbb{P}}_2 - \hat{\mathbb{P}}_1$ and take the average over the entire sample.
- Compute confidence intervals (see below).

This is fairly rare, partly because the effect will ultimately depend on the entire sample and will not be based on a specific individual.

Example 11.1 (Marginal effect with probit with probit). Suppose we have two independent variables Z and W (and call $\mathbf{X}$ the joint vectors of Z and W as well as a column of 1's). We obtain ML estimates of the probit model:

$$\begin{aligned}
 \mathbb{P}[y_i = 1 | \mathbf{X}] &= \Phi(\mathbf{X}\beta) \\
 &= \Phi(\alpha + \beta_1 Z + \beta_2 W) \\
 &= \Phi(2.3Z - 1.1W) \quad \text{setting } \alpha = 0 \text{ for simplicity}
 \end{aligned}$$

Let us then assume that $Z = 1.5$ and $W = 2$ (imagine that these

are the values of a particularly interesting unit). We then have:

$$\mathbf{X}\beta = 2.3 \cdot 1.5 + (-1.1)(2) = 1.25.$$

Next, we want to find out the marginal effect. We need to compute the PDF ϕ of the normal distribution evaluated at $\mathbf{X}\beta$:

$$\phi(\mathbf{X}\beta) = \phi(1.25) \approx 0.183$$

We can then compute the marginal effects of Z and W for this unit as:

$$\begin{aligned} \frac{\partial \mathbb{P}}{\partial Z} &= 0.183 \cdot 2.3 = 0.42 \\ \frac{\partial \mathbb{P}}{\partial W} &= 0.183 \cdot (-1.1) = -0.2 \end{aligned}$$

Concretely:

- A one-unit increase in Z increases the probability that $Y = 1$ by 0.42 (or 42 percentage points).
- A one-unit increase in W reduces the probability that $Y = 1$ by 0.2 (or 20 percentage points).

💡 Misleading APE

Curran, Meyer, and Wu (2026) flag a complication with average partial effects (APE). Suppose you have a model with three mutually exclusive categories (e.g., the level of educational attainment: no education, secondary school, or university degree) predicting employment status. You want to estimate:

$$\mathbb{P}(\text{Employed})_i = F(\alpha + \beta \text{Secondary School}_i + \gamma \text{University}_i),$$

where “No school” is the omitted (baseline) category, and each of these variable is a dummy. The APE of *secondary school* (compared to *no school*) is:

$$\text{APE}(\text{secondary school}) = \frac{1}{n} \sum_i^n [F(\alpha + \beta + \gamma \text{University}_i) - F(\alpha + \gamma \text{University}_i)]$$

The APE is therefore computed over the entire sample. But we probably would rather condition on respondents who either completed secondary school or none at all, removing people who completed university. Let *ns* mean *no school* and *ss* mean

secondary school, then we take the relevant subsample for our calculation:

$$\text{APE}(\text{secondary school}) = \frac{1}{n_{ns} + n_{ss}} \sum_{i \in \{ns, ss\}}^n [F(\alpha + \beta) - F(\alpha)]$$

Example 11.2 (Racial discrimination in hiring: probit vs. linear probability model). We met the Bertrand and Mullainathan (2004) audit study in Chapter 4, where it illustrated hypothesis testing and confidence intervals for an OLS coefficient. Here the same paper serves a different purpose: showing *when* the linear probability model (LPM) is adequate and when probit is preferable.

The outcome is binary — the résumé either received a callback (1) or did not (0). The LPM simply regresses this binary indicator on a white-name dummy and controls. The probit model instead specifies $\Pr(\text{callback} = 1 \mid \mathbf{X}) = \Phi(\mathbf{X}\beta)$, ensuring predicted probabilities lie in $[0, 1]$ and allowing the marginal effect of any covariate to vary with the baseline probability.

In this application the two approaches give nearly identical **average marginal effects** (AMEs). The white-name effect from the LPM is $\hat{\beta} = 0.032$. The probit AME — $\frac{1}{n} \sum_i \phi(\hat{\mathbf{X}}_i \hat{\beta}) \hat{\beta}_{\text{white}}$ — is approximately 0.030. The difference is negligible because callback rates are neither very low nor very high (the baseline is about 6.5%), placing observations in the middle of the probit S-curve where it is nearly linear.

The lesson is practical: when the outcome probability is far from 0 or 1, probit and LPM give similar average marginal effects, and LPM’s simplicity is a virtue. Probit becomes materially different — and more reliable — when the outcome is rare (e.g., predicting corporate bankruptcy, where baseline rates are below 1%) or when the researcher needs predicted probabilities that stay in $[0, 1]$ for all covariate values.

	LPM	Probit (AME)
White-sounding name	+0.032***	+0.030***
<i>SE (clustered by name)</i>	<i>(0.008)</i>	<i>(0.008)</i>
Résumé quality (high)	+0.036***	+0.034***
<i>SE</i>	<i>(0.004)</i>	<i>(0.004)</i>
Baseline callback rate	6.45%	—
Predicted prob. range	−2% to 14%	0% to 14%

N	4,870	4,870
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11.2.4 Hypotesis testing

Once we have $\hat{\beta}_{ml}$, we can construct standard errors and confidence intervals. We can then conduct t tests, just like with OLS. The standard errors are given by the asymptotic variance-covariance matrix:

$$V[\widehat{\beta}_{ml}] = \left[\sum_i \frac{[f(\mathbf{X}\hat{\beta}_{ml})]^2 \mathbf{X}'\mathbf{X}}{F(\mathbf{X}\hat{\beta}_{ml})[1 - F(\mathbf{X}\hat{\beta}_{ml})]} \right]^{-1}$$

The standard errors are given by the square root of the diagonals.

11.2.5 Goodness of fit

Goodness of fit can be assessed in various ways. See the notes on maximum likelihood for a general overview (likelihood-ratio test, etc.).

In the specific case of a nonlinear model with dichotomous outcomes, we can add a few additional tests.

Percentage correctly predicted (PCP). This is straightforward:

- Estimate the ML model of $\mathbb{P}[y = 1|\mathbf{X}]$.
- Compute the predicted probability $\hat{y}$.
- Generate a variable that is the prediction of your model; the variable equals 0 if $\hat{y} < 0.5$ and 1 if $\hat{y} > 0.5$.
- Calculate the percentage of correct prediction.

Note. The issue is that if the sample proportion of 1's, $\tilde{p}$ is very low or very high, and that we end up predicting only zeros or only 1's, then our predictions will be outstanding, but we will be wrong on the few important cases (these rare 1's or 0's). One solution is to replace the threshold of 0.5 ($\hat{y} < 0.5$) by $\tilde{p}$, such that $\hat{y} < p \rightarrow 0$ and 1 otherwise.

Expected PCP. See Herron (2000). The idea here is to reward good predictions. If you predict $y = 1$ based on $\hat{p} = 0.9$, that's better than if you had done so with $\hat{p} = 0.51$. Herron developed the notion of expected PCP:

$$\text{ePCP} = \frac{1}{N} \left(\sum_{y_i=1} \hat{\mathbb{P}}_i + \sum_{y_i=0} (1 - \hat{\mathbb{P}}_i) \right)$$

Example 11.3 (PCP example). Suppose you are a consultant for a US political party. You get a sample of 100 adults with information about their partisanship (Democrat or Republican), their gender, and their income. You estimate a probit or logit model:

$$\text{Republican}_i = f(\alpha + \beta \text{Gender}_i + \gamma \text{Income}_i)$$

You then obtain estimated probabilities of being a Republican, $\widehat{\text{Republican}}$. And since anyone who is not a Republican is a Democrat in this sample, you also obtain $\widehat{\text{Democrat}} = 1 - \widehat{\text{Republican}}$. You can then build a cross-tab to see if your model is good at making predictions. Here, the percent of correct predictions is 91% $((48 + 43)/100)$. See Table Table ??.

	Democrat	Republican
$\widehat{\text{Democrat}}$	48	6
$\widehat{\text{Republican}}$	3	43
Total	51	49

Example 11.4 (Forecasting corporate bankruptcy with logit). An example of why logit (estimated with ML) can be preferable over liner models (estimated with OLS) comes from Shumway (2001). In this paper, he studies firm bankruptcy and what predicts it using annual accounting and market data on US publicly listed firms from 1962 to 1992.

The baseline approach in the literature to predict bankruptcy at the time was Altman's Z-score: a weighted linear combination of financial ratios estimated by linear discriminant analysis, which is related to OLS. The problem is that neither OLS nor discriminant analysis is designed for binary outcomes: both can produce predicted probabilities below 0 or greater than 1.

Shumway estimates a logit model where the dependent variable is whether a firm files for bankruptcy in a given year ($= 1$). The covariates include firm size (log of market equity relative to total market), net income/total assets, total liabilities/total assets, past stock returns, and stock return idiosyncratic volatility. The model also captures duration to account for the fact that the risk of bankruptcy is time dependent.

The logit model correctly identifies three-quarters of actual

bankruptcies among the firms it flags as highest-risk, compared to barely half for the linear model. This shows the downside of using OLS for a (rare) binary outcome: the linear model cannot capture the non-linearity in how financial distress accumulates, particularly in the tails where the risk of bankruptcy is actually concentrated.

Model	% of bankruptcies in top decile of predicted risk
Altman Z-score (linear)	53%
Logit hazard model	75%

Table 11.3: Out-of-sample bankruptcy prediction performance. Outcome: binary bankruptcy indicator. The “top decile” refers to the 10% of firm-years assigned the highest predicted bankruptcy probability. Source: Shumway (2001).

11.3 Poisson regression

Poisson regression is used to model **count outcomes** (non-negative integers) such as the number of patents filed by a firm, emergency room visits per year, or crimes per neighborhood. A linear model is tricky: it ignores the non-negativity of counts, can predict negative values.

The Poisson model assumes that Y is distributed Poisson:

$$Y_i \mid X_i \sim \text{Poisson}(\lambda_i), \quad \lambda_i = \exp(\mathbf{X}_i\beta),$$

where $\lambda_i = \mathbb{E}[Y_i \mid X_i] > 0$ is the conditional mean. The exponential link guarantees $\lambda_i > 0$ for any β . The log-likelihood is:

$$\mathcal{L}(\beta) = \sum_{i=1}^n [y_i \mathbf{X}_i \beta - \exp(\mathbf{X}_i \beta) - \log(y_i!)] .$$

This is globally concave and solved numerically.

The coefficient β_j is the change in $\log \lambda$ for a one-unit increase in x_j . Exponentiating gives an **incidence rate ratio**: $\exp(\beta_j)$ is the multiplicative factor by which the expected count changes. For instance, $\exp(\hat{\beta}_j) = 1.2$ means that the expected count of Y will be multiplied by 1.2, which is equivalent to saying that it will increase by 20%.

One problem with Poisson is that this distribution assumes that the mean equals the variance: $\mathbb{V}[Y | X] = \mathbb{E}[Y | X] = \lambda$. In practice, count data are often **overdispersed** ($\mathbb{V} > \text{Mean}$), which leads to understated standard errors. Fortunately, there are easy fixes:

1. **Negative binomial regression**: adds a dispersion parameter κ so that $\mathbb{V}[Y|X] = \lambda + \kappa\lambda^2$.
2. **Poisson QMLE with sandwich SEs** (see ML chapter): the Poisson MLE remains consistent as long as the conditional mean is correctly specified, regardless of the true variance structure. Replace standard errors with the sandwich estimator (Silva and Tenreiro 2006).
3. **Quasi-Poisson**: scales all SEs by a dispersion factor estimated from the residuals.

11.4 Ordered probit and logit

Ordered probit and **ordered logit** generalize the binary latent variable model to outcomes with $J \geq 2$ **ordered** categories. For instance, consumer surveys will include variables measured on the so-called Likert scale (“strongly disagree” to “strongly agree”). Likewise, credit ratings are scored in discrete increments (AAA down to D). These are **ordinal variables**. Unlike dependent variables defined on $\mathbb{R}$, here we cannot say that, say, an outcome of 3 (e.g., “strongly agree”) is three times larger than 1 (“strongly disagree”).

As in logit/probit, we have a latent variable $y_i^* = \mathbf{X}_i\beta + \varepsilon_i$ that is linked to the observed ordinal category $y_i \in \{1, \dots, J\}$. We now have $J-1$ unknown **cut points** $\mu_1 < \mu_2 < \dots < \mu_{J-1}$ which split observed outcomes (cutoff between 0 and 1, 1 and 2, etc.):

$$y_i = k \iff \mu_{k-1} < y_i^* \leq \mu_k,$$

with $\mu_0 = -\infty$ and $\mu_J = +\infty$. The probability of each category is:

$$\mathbb{P}[y_i = k | X_i] = F(\mu_k - \mathbf{X}_i\beta) - F(\mu_{k-1} - \mathbf{X}_i\beta),$$

where F is the logistic CDF (ordered logit) or normal CDF (ordered probit). Both the β coefficients and the cut points are estimated jointly by MLE. Note that no intercept is included in $\mathbf{X}_i\beta$; the cut points absorb the location.

Interpretation is again a bit tricky. The sign of β_j gives the direction: $\beta_j > 0$ means that the latent y^* increases, and this in turn increases the probability of higher categories and decreasing lower ones.

For this to be valid, the key identifying assumption is **proportional odds** (or parallel regression): the same coefficient vector β governs the transition between every pair of adjacent categories.¹

11.5 Multinomial logit

Some variables are nominal: they cannot be ordered in any meaningful way. For instance, imagine predicting the mode of transportation that people use: cars, buses, or bikes. It doesn't make sense to say that 'cars' is larger than 'bikes' (whereas above, it made sense to say that 'strongly agree' is larger than 'neutral' which is larger than 'strongly disagree').

Such problems can be analyzed with **multinomial logit** with $J \geq 2$ **unordered** categories.

The model follows from a **random utility framework** that we already saw earlier (McFadden 1976). Individual i chooses alternative j if $U_{ij} > U_{ik}$ for all $k \neq j$, where:

$$U_{ij} = \mathbf{X}_i \gamma_j + \varepsilon_{ij},$$

and the ε_{ij} are iid Type I extreme value (Gumbel) errors. This yields the **softmax** choice probabilities:

$$\mathbb{P}[y_i = j \mid \mathbf{X}_i] = \frac{\exp(\mathbf{X}_i \gamma_j)}{\sum_{k=1}^J \exp(\mathbf{X}_i \gamma_k)}.$$

One category is the **baseline** (set $\gamma_J = 0$) for identification; the remaining $J - 1$ coefficient vectors are estimated by MLE.

The coefficient γ_{jl} gives the change in $\log\left(\frac{\mathbb{P}(y=j)}{\mathbb{P}(y=J)}\right)$. These are the log-odds of alternative j relative to the baseline for a one-unit increase in x_l . Exponentiating gives relative risk ratios.

The Gumbel error assumption implies **independence of irrelevant alternatives** (IIA): the relative probability of choosing j over k does not depend on the presence or attributes of other alternatives. The classic failure case is the “red bus/blue bus” problem: if you add an identical blue bus to a choice set containing a red bus and a car, multinomial logit predicts that both the bus and the car lose share equally, when in reality only the red bus should lose half its

¹This can be tested by fitting $J - 1$ separate binary logits and checking whether the slopes are parallel.

share. IIA is testable (Hausman-McFadden test) and often restrictive. To relax IIA, we can use alternatives such as **nested logit** (groups alternatives into correlated nests) and **mixed logit** (allows random taste variation).

11.6 Mixed effects models

Mixed effects models extend nonlinear models (logit, probit, Poisson, etc.) to **grouped or longitudinal data** by incorporating unit-specific random effects. They generalize the linear random effects estimator to non-Gaussian outcomes, and are also called random effects models or multilevel models.

Consider binary outcomes for unit i observed over $t = 1, \dots, T$ periods. A **random effects logit** specifies:

$$\mathbb{P}[y_{it} = 1 \mid X_{it}, \alpha_i] = F(\mathbf{X}_{it}\beta + \alpha_i), \quad \alpha_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_\alpha^2),$$

where α_i is an individual-level random intercept capturing persistent unobserved heterogeneity (e.g., a consumer's general propensity to buy). Because α_i is unobserved, the likelihood integrates it out:

$$\mathcal{L}(\beta, \sigma_\alpha^2) = \sum_i \log \int \prod_t F(\mathbf{X}_{it}\beta + \alpha_i)^{y_{it}} [1 - F(\mathbf{X}_{it}\beta + \alpha_i)]^{1-y_{it}} \phi\left(\frac{\alpha_i}{\sigma_\alpha}\right) d\alpha_i.$$

This integral doesn't have a closed form. It is generally approximated by Gauss-Hermite quadrature or simulation.

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